Models, Molecules, and Meaning

Uncertainty Quantification, Assay Context, and Structural Representations in AI-Driven Drug Discovery

Bola A. A. Khalil

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Story Behind the Cover

The cover of this thesis brings together its central themes through a symbolic scene inspired by Ancient Egyptian visual language. At the center stands Set/Seth (-stsh), a figure associated in Egyptian mythology with disorder, instability, uncertainty (-isft), and disruption (-hb). In the context of this thesis, Seth represents the uncertainty that permeates modern drug discovery: incomplete knowledge, heterogeneous data, imperfect measurements, and the persistent challenge of deciding when a computational prediction can truly be trusted. Rather than presenting uncertainty as something purely negative, the cover frames it as a fundamental condition of scientific inquiry (-shenw), one that must be understood, quantified, and carefully navigated. Seth forms a main pillar of this visual narrative (-sjd), embodying the theme of Chapter 2: Trusting the Model.

The composition is intentionally asymmetrical. In the left hand, Seth holds the *ankh* , the ancient Egyptian sign of life. Within the scientific logic of the cover, this symbol carries a dual meaning. It evokes the biological foundation of the work, since all predictive modeling in drug discovery ultimately serves life and health. At the same time, it offers a subtle contemporary echo of modern protein representation learning, recalling the protein computational embeddings model, Ankh [1] and the effort to encode biological meaning into forms that models can interpret.

Raised in Seth's other hand is a single offering plate (-shenshw hotp). This plate serves as the central platform of the thesis narrative, bringing together structure, context, and validation in one unified image. Resting within it is a peptide–protein complex, inspired by a GLP-1 receptor bound to semaglutide [2], represented in coloured stone. This molecular assembly symbolizes the structural dimension of the thesis: the challenge of representing chemically and biologically rich systems in ways that preserve meaning while remaining suitable for machine learning. Coiled around the base of the complex is a short beaded peptide motifs, included to make the peptide character of the offering explicit and to allude to the broader theme of structural representation through interpretable building blocks/fragments, referring to the Chapter 4: Thinking in Fragments (-spw).

The most distinctive element of the bowl is the miniature balance placed on its rim. Designed in the spirit of an ancient Egyptian *Ma'at* scale, it introduces experimental evaluation into the scene. One side refers to the recorded readout of an experiment with this molecular system, while the other invokes the Feather ( ) of Ma'at, the

ancient symbol of truth and right measure. In the language of this thesis, the balance represents the assay: not merely as an experimental procedure, but as the framework through which evidence is produced, interpreted, and judged. It is therefore a direct reference to the assay-aware modeling developed in Chapter 3: Reading Between the Assays, and to the broader theme of uncertainty quantification. Bioactivity values do not exist independently of their measurement context; their meaning depends on how, where, and against what standard they are obtained, and trust must be earned through careful weighing.

Ancient symbolic systems condensed complex ideas into a limited set of forms, gestures, and objects on ancient papyrus –subtly alluding to the Papyrus bioactivity dataset [3]–. That principle resonates strongly with computational drug discovery, where rich biological and chemical reality must be translated into representations that algorithms can process. The cover therefore places an ancient tradition of symbolic knowledge-making in dialogue with a modern scientific effort to extract trustworthy insight from complex data.

Taken together, the cover reflects the overarching theme of this thesis: bringing models, molecules, and meaning into dialogue. The major threads of the work, uncertainty quantification, assay-aware modelling, and (peptide) structural representation, are not presented as isolated contributions, but as interconnected efforts to understand how reliable knowledge can be built from complex and imperfect data.

Design credits. Cover concept and final artwork by Bola Khalil.

Revision and Hieroglyphs by Egyptologist, Marina Maher

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Chapter 1

General Introduction

About This Thesis

This thesis investigates computational approaches for predicting molecular bioactivity, with a focus on improving the reliability, interpretability, and transferability of machine learning models used in drug discovery. It addresses three interconnected challenges: quantifying prediction uncertainty to support informed decision-making, accounting for assay heterogeneity in public bioactivity databases through proteochemometric modeling, and developing molecular representations for complex therapeutic modalities such as cyclic peptides. Across these themes, the thesis emphasizes open-source and statistically rigorous benchmarking—including challenging split strategies, repeated model refits, and effect-size-aware inference—to make performance claims both reproducible and practically meaningful for pharmaceutical research.

1.1 Background Information

1.1.1 Difficulties of the Drug Discovery Process

Bringing a new medicine from initial concept to market approval is a lengthy, expensive, and failure-prone process. A typical drug development program spans 10 to 15 years and involves multiple stages of preclinical research, clinical trials, and regulatory review. Recent estimates place the median capitalized cost of developing a single new drug at approximately $1 billion, with mean estimates reaching up to $2.6 billion when accounting for the cost of failed projects and the time value of money [4, 5]. These figures reflect not only the direct expenses of laboratory work and clinical studies but also the high attrition rates that characterize pharmaceutical research and development (R&D).

Attrition is driven by multiple factors, including insufficient efficacy, unacceptable toxicity, and poor pharmacokinetic properties of drug candidates. Many compounds that appear promising in early stages fail to meet the stringent safety and efficacy requirements imposed by regulatory agencies such as the U.S. Food and Drug Administration (FDA) and the European Medicines Agency (EMA). The overall probability that a compound entering clinical trials, starting from an initial hit finding campaign, will eventually receive approval has been estimated at roughly 10% [6]. This low success rate, combined with the enormous financial and temporal investments involved, has motivated the pharmaceutical industry to seek more efficient approaches for identifying and prioritizing drug candidates.

The challenges of drug discovery are further compounded by the increasing complexity of the diseases being targeted. Many contemporary programs focus on multifactorial conditions such as cancer, neurodegenerative disorders, and autoimmune diseases, where the underlying biology is only partially understood. In this context, computational methods have emerged as valuable tools for accelerating the discovery process, reducing costs, and improving the likelihood of success [7–10].

1.1.2 Computer-Aided Drug Discovery and Artificial Intelligence

Computer-aided drug discovery (CADD) refers to the application of computational techniques to assist in the identification, optimization, and evaluation of potential drug candidates. CADD encompasses a broad range of methods, from physics-based molecular docking and molecular dynamics simulations to data-driven approaches rooted

in machine learning (ML) and, more recently, deep learning (DL) [10]. These methods can be applied at virtually every stage of the drug discovery pipeline, including target identification, hit finding, lead optimization, and the prediction of absorption, distribution, metabolism, excretion, and toxicity (ADMET) properties.

The growing availability of large-scale chemical and biological datasets, combined with advances in computational hardware and algorithmic design, has driven the rapid adoption of ML and DL techniques in pharmaceutical research. Deep neural networks, in particular, have demonstrated strong performance in tasks such as molecular property prediction, virtual screening, and de novo molecular design. Models based on graph neural networks (GNNs), recurrent neural networks (RNNs), and transformer architectures have been applied to learn complex structure-activity relationships directly from molecular data [10].

Despite their promise, these computational approaches also face important limitations. ML models are only as good as the data they are trained on, and biases in training data can propagate into model predictions. Furthermore, the interpretation of complex models remains challenging, and the reliable estimation of prediction confidence is an open problem that is central to this thesis. Addressing these challenges is essential for computational models to be meaningfully integrated into real-world drug discovery workflows.

1.1.3 Bioactivity Data and Public Databases

The development of predictive models for drug discovery depends critically on the availability of high-quality bioactivity data. Bioactivity data capture the measured interactions between small molecules and biological targets, typically expressed as potency values such as IC_{50} (half-maximal inhibitory concentration), EC_{50} (half-maximal effective concentration), K_i (inhibition constant), and K_d (dissociation constant). These values are commonly converted to the negative logarithmic pChEMBL scale to facilitate comparison and modeling.

Several large-scale public databases serve as primary data sources for computational drug discovery. ChEMBL is one of the most widely used bioactivity databases, aggregating curated experimental data from published medicinal chemistry literature [11, 12]. It contains millions of bioactivity measurements spanning thousands of targets and provides standardized annotations for compounds, assays, and targets. Other notable resources include PubChem, ExCAPE-DB [13], and BindingDB, each offering complementary coverage of chemical and biological space.

The Papyrus dataset represents a more recent effort to create a large-scale, curated resource specifically designed for bioactivity modeling [3]. Papyrus integrates data from multiple sources, including ChEMBL, and applies systematic standardization and quality filtering to produce a unified dataset suitable for training ML models across diverse targets and endpoints.

A persistent challenge with public bioactivity data is the heterogeneity introduced by combining measurements from different experimental assays. Even when measuring the same compound-target interaction, different laboratories, protocols, and assay technologies can produce systematically different results. Landrum and Riniker demonstrated that nearly half of repeated compound measurements across separate assays deviate by more than 0.3 log units, corresponding to a roughly twofold difference in potency [14]. This inter-assay variability places an effective ceiling on achievable model accuracy and can distort uncertainty estimates if not properly accounted for. Addressing this assay heterogeneity is one of the core motivations for the work presented in Chapter 3 of this thesis.

1.1.4 Quantitative Structure-Activity Relationships and Proteochemometric Modeling

Quantitative structure-activity relationship (QSAR) modeling is one of the foundational approaches in computational drug discovery. QSAR models relate the chemical structure of molecules to their biological activity using mathematical or statistical functions [15]. The underlying premise is that structurally similar molecules tend to exhibit similar biological properties, an assumption known as the similarity principle. In a typical QSAR workflow, molecules are represented by numerical descriptors that encode structural and physicochemical properties, and a supervised learning algorithm is trained to map these descriptors to experimentally measured bioactivity values.

While traditional QSAR models are built for a single biological target, proteochemometric (PCM) modeling extends this paradigm by simultaneously incorporating descriptors of both the ligand (compound) and the target (protein) [16]. By training on data from multiple targets at once, PCM models can leverage shared information across related proteins, enabling predictions for novel target-ligand combinations and supporting multi-target drug design [17, 18]. PCM has been successfully applied to various protein families, including G protein-coupled receptors (GPCRs), kinases, and proteases, often outperforming single-target QSAR models when sufficient cross-target data are available.

A key extension explored in this thesis (Chapter 3) is the integration of assay-level information into the PCM framework. By incorporating descriptors of the experimental assay alongside compound and protein features, the resulting proteochemometric assay-aware modeling (PCAM) approach can capture systematic differences between assays, potentially improving both prediction accuracy and uncertainty calibration in heterogeneous datasets.

1.1.5 Molecular Representations and Descriptors

The choice of molecular representation is a fundamental decision in any computational modeling workflow, as it determines what structural information is available to the learning algorithm. Molecular representations can be broadly categorized into expert-designed (handcrafted) descriptors and learned representations [19]. In this context, representation learning refers to methods that automatically infer useful features from data, rather than relying exclusively on manually engineered descriptors [20].

Among handcrafted representations, molecular fingerprints are the most widely used in cheminformatics. Extended-connectivity fingerprints (ECFPs), also known as Morgan fingerprints, encode the local chemical environment around each atom as a fixed-length binary or count vector [21]. Other approaches include physicochemical descriptor sets such as Mordred [22], which compute hundreds of molecular properties ranging from simple atom counts to topological and geometric indices. String-based encodings are also central: SMILES (Simplified Molecular Input Line Entry System) provides a compact textual representation of molecular graphs and is widely used in sequence-based models [23]. To improve robustness in generative and sequence modeling, SELFIES (Self-Referencing Embedded Strings) was introduced as a molecular string formalism in which every valid token sequence decodes to a chemically valid structure [24]. Group-SELFIES extends this concept to chemically meaningful fragments, enabling robust fragment-level tokenization that better captures larger structural motifs [25].

On the protein side, recent advances in protein language models have enabled the generation of rich, information-dense embeddings from amino acid sequences. Models such as ESM [26], ProtTrans [27], and Ankh [1] capture evolutionary and structural information that can serve as powerful protein descriptors in PCM settings.

More recently, molecular representation learning has increasingly relied on self-supervised pretraining on large chemical corpora, where models are trained with proxy objectives and then transferred to downstream property-prediction tasks. Approaches

such as continuous and data-driven molecular descriptors (CDDD) [28] illustrate how learned latent embeddings can provide compact, task-adaptive molecular features. The work in Chapter 4 of this thesis builds on this paradigm by exploring a novel molecular graph representation based on Group-SELFIES fragments, which offers a flexible and topology-aware alternative to atom-level molecular graphs, with particular relevance for complex modalities such as cyclic peptides.

1.1.6 Uncertainty Quantification in Machine Learning

In high-stakes applications such as drug discovery, knowing how confident a model is in its predictions is often as important as the predictions themselves. Uncertainty quantification (UQ) provides a framework for estimating the reliability of model outputs, enabling practitioners to distinguish between confident and uncertain predictions and to allocate experimental resources accordingly [29].

Uncertainty in ML predictions is commonly decomposed into two categories [30]. Aleatoric uncertainty arises from inherent noise in the data, such as measurement error or biological variability, and cannot be reduced by collecting more training data. Epistemic uncertainty, in contrast, reflects gaps in the model's knowledge due to limited or unrepresentative training data, and can in principle be reduced by improving data coverage.

Several methodological families have been developed to estimate these uncertainties. Deep ensembles, introduced by Lakshminarayanan et al., estimate epistemic uncertainty by training multiple models with different random initializations and aggregating their predictions [31]. The variance across ensemble members serves as a proxy for prediction uncertainty. While effective, ensembles incur significant computational overhead due to the need to train and maintain multiple models.

Monte Carlo (MC) dropout provides a more computationally efficient alternative by applying dropout at inference time and performing multiple stochastic forward passes through a single trained network [32]. The resulting output variability is interpreted as an approximation to Bayesian posterior uncertainty.

Evidential deep learning offers a fundamentally different approach by parameterizing a higher-order distribution over possible output distributions in a single forward pass [33, 34]. This enables simultaneous estimation of aleatoric and epistemic uncertainty without the computational cost of multiple model evaluations, though it may underperform in extrapolative settings.

Despite the availability of these methods, achieving well-calibrated and practi-

cally useful uncertainty estimates in molecular property prediction remains challenging [35, 36]. Chapter 2 of this thesis proposes hybrid approaches that combine Bayesian and evidential paradigms to obtain more robust uncertainty estimates for bioactivity modeling.

1.1.7 G Protein-Coupled Receptors as Drug Targets

G protein-coupled receptors (GPCRs) constitute the largest and most pharmacologically important superfamily of membrane proteins in the human genome. They mediate cellular responses to a wide variety of extracellular signals, including hormones, neurotransmitters, and sensory stimuli. GPCRs are characterized by a conserved architecture of seven transmembrane helices and a signaling mechanism that involves coupling to intracellular G proteins and other effector molecules [37].

The therapeutic relevance of GPCRs is immense. According to a comprehensive analysis by Hauser et al., approximately 34% of all FDA-approved drugs act on GPCRs, targeting 108 unique receptors across diverse therapeutic areas including cardiovascular disease, psychiatric disorders, pain management, and metabolic conditions [37]. The continued identification of new GPCR targets and the development of novel modulators, including allosteric ligands and biased agonists, ensure that this receptor family remains at the forefront of drug discovery.

GPCRs also represent an ideal test case for computational modeling methods. Their large pharmacological datasets, structural diversity, and well-characterized pharmacology make them well-suited for benchmarking approaches such as PCM and PCAM. The work presented in Chapter 3 of this thesis uses a curated dataset of 167 GPCR targets to evaluate assay-aware proteochemometric modeling.

1.1.8 Peptide Therapeutics, Cyclic Peptides, and Membrane Permeability

Peptide therapeutics occupy an important middle ground between small molecules and large biologics. They can offer high target specificity and are increasingly explored for difficult targets, including protein-protein interaction interfaces that are often less tractable for classical small molecules [38, 39]. Within this modality, cyclic peptides are especially attractive because macrocyclization can improve conformational preorganization, proteolytic stability, and, in selected cases, oral exposure [38–40].

A central developability challenge, however, is membrane permeability. For many cyclic peptides, passive translocation across lipid bilayers depends strongly on confor-

mational dynamics rather than on static bulk descriptors alone. In particular, transient intramolecular hydrogen bonding and environment-dependent "chameleonic" behavior can reduce exposed polarity during membrane transit [41–43]. This creates a difficult modeling problem in which topology, local chemistry, and conformational flexibility all influence measured permeability.

Progress in this area has accelerated through public data resources and benchmark studies. CycPeptMPDB aggregates cyclic peptide permeability measurements across multiple assay systems, including PAMPA, Caco-2, MDCK, and RRCK [44]. Subsequent modeling work has shown that performance and transferability can vary substantially across these endpoints, highlighting the need for representations that remain robust across assay contexts and chemically diverse peptide scaffolds [45, 46].

These considerations motivate the focus of Chapter 4, which develops Group-SELFIES Graph Embeddings (GSGE), a fragment-centric representation designed to improve permeability prediction for structurally diverse cyclic peptides without relying on fixed monomer vocabularies or proprietary descriptor pipelines.

1.2 Outline of This Thesis

Chapter 1 provides the necessary background for the remainder of the thesis. It introduces key concepts in machine learning, molecular featurization, and uncertainty quantification that are used throughout the subsequent chapters.

Chapter 2 presents a systematic benchmarking study of uncertainty quantification methods for bioactivity modeling. It compares Bayesian approaches (deep ensembles, MC dropout) with evidential deep learning and proposes hybrid methods that combine both paradigms. The resulting ensemble of evidential models (EOE) achieves the best overall performance across multiple metrics and data splitting strategies on the Papyrus++ dataset, offering a robust and computationally practical default for uncertainty-aware predictions. A hardware-agnostic compute analysis highlights favorable performance-efficiency trade-offs.

Chapter 3 introduces proteochemometric assay-aware modeling (PCAM), an extension of proteochemometric modeling that integrates assay descriptors alongside compound and protein features. Using a curated ChEMBL-derived benchmark spanning 167 GPCR targets, 77,081 compounds, and 5,161 assays, it evaluates the impact of assay context on both prediction accuracy and uncertainty calibration under three generalization regimes of increasing difficulty. The results show that explicit assay representations can improve performance under interpolation conditions and enable meaningful extrapolation to unseen assay types.

Chapter 4 introduces Group-SELFIES Graph Embeddings (GSGE), a fragment-centric molecular graph representation that constructs compound graphs from data-driven vocabularies rather than fixed monomer lists. On the CycPeptMPDB PAMPA benchmark (6,889 cyclic peptides), a graph convolutional network operating on GSGE graphs and fused with seven open-source descriptors achieves a test MAE of 0.329 and an R^2 of 0.802 without proprietary descriptor software or SMILES enumeration. Focused contrasts indicate a modest MAE advantage over CycPeptMP, while nonparametric tests at three-fold resolution still show overlap among top-performing models. The chapter also compares graph, Transformer, and recurrent sequence architectures, and evaluates transfer from PAMPA to Caco-2, MDCK, and RRCK endpoints.

Chapter 5 provides a general discussion that synthesizes the findings from the preceding chapters, examines common themes and methodological considerations, and

identifies limitations and open questions.

Following Chapter 5 , the thesis closes with summaries in English and Dutch that recapitulate the main contributions and discuss future research directions in uncertainty quantification, assay-aware modeling, and molecular representation learning for drug discovery.

Chapter 2

Trusting the Model

Ensemble-Evidential Hybrid Methods
for Uncertainty-Aware Bioactivity Prediction

Based on the publication

Combining Bayesian and Evidential Uncertainty Quantification for Improved Bioactivity Modeling

Bola Khalil, Kajetan Schweighofer, Natalia Dyubankova, Gerard J. P. van Westen, Herman van Vlijmen

Journal of Chemical Information and Modeling, 2025.

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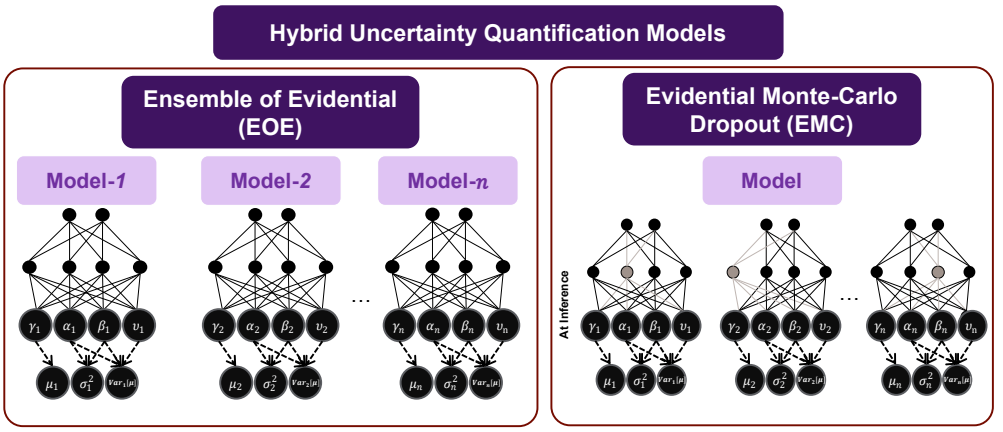


Figure 2.1: Graphical Abstract

Uncertainty Quantification (UQ) has been recognized as prerequisite for reliable and trustworthy computational modeling in drug discovery. Two widely considered paradigms, Bayesian methods (deep ensemble, MC dropout) and evidential learning, differ in their computational demands and expressivity of uncertainties, excelling in complementary settings.

Here, we propose hybrid approaches that combine both paradigms and benchmark them on the Papyrus++ dataset across two endpoints (xC_{50}, K_x) and multiple split strategies. Our ensemble of evidential models (EOE) consistently achieves the best overall performance, yielding the lowest RMSE and leading CRPS and Interval scores, including under the most challenging distributional shifts. While large ensembles often excel in rejection-based utility, EOE matches or surpasses them at a fraction of the computational cost. Statistical tests confirm its advantage, and a hardware-agnostic compute analysis highlights favorable performance–efficiency trade-offs.

These results demonstrate that combining evidential and Bayesian principles yields more accurate and informative uncertainties for bioactivity modeling, with **EOE** offering a robust–and computationally practical–default for uncertainty-aware decision-making in drug discovery.

2.1 Introduction

Machine learning (ML) has become integral to computational sciences, providing powerful tools for data-driven discovery across domains such as bioinformatics and cheminformatics [29]. In pharmaceutical research, ML models are widely applied to assess molecular interactions, screen compounds, and assist in identifying candidates for drug development. While these models demonstrate high accuracy, their application in real-world settings demands an understanding of confidence in model outputs. In high-stakes decision-making, such as early-stage drug discovery, reliance on uncertain or poorly calibrated models can lead to costly experimental failures, resource misallocation, and erroneous conclusions. As a result, UQ has emerged as an important component of ML for scientific applications [36, 47–49].

Uncertainty in ML models falls into two principal categories: *aleatoric* and *epistemic* uncertainty [30]. Aleatoric uncertainty arises from inherent stochasticity, such as measurement noise or biological heterogeneity, and cannot be reduced by acquiring more data. In contrast, epistemic uncertainty reflects lack of knowledge due to insufficient training data or structural biases and can be reduced by improving data coverage or model design. The ability to assess these uncertainties can determine the reliability of ML-driven assessments, guiding experimental validation efforts and optimizing resource allocation. This challenge is exacerbated by variability in public bioactivity data, where combining measurements from different assays can introduce substantial noise [14]; while Papyrus++ [3] applies rigorous standardization, residual assay variability may still impact model reliability.

Various UQ methodologies have been developed to address these challenges [34, 35, 50]. For modeling aleatoric uncertainty, **Probabilistic Neural Networks** (PNNs) [51] parametrize a Gaussian distribution over outputs, enabling the estimation of input-dependent variance. Other approaches include **deep ensembles**, which estimate epistemic uncertainty by aggregating outputs from multiple independently trained models [31], and **Monte Carlo (MC) dropout**, which approximates Bayesian inference by randomly masking network weights during inference [32]. These approaches approximate Bayesian modeling of uncertainty by sampling different model parameterizations. While deep ensembles are widely regarded as effective, they incur significant computational costs. However, they have been shown to be very faithful to the Bayesian ideal in deep learning settings [52, 53]. MC dropout provides a more computationally efficient alternative but may yield inconsistent uncertainty estimates depending on dropout rates and the number of stochastic passes.

2.2. Introduction

A more recent alternative, **evidential deep learning**, reformulates uncertainty estimation by parameterizing a second-order distribution over possible distributions [33, 34, 54]. Unlike ensemble-based or sampling-based methods, evidential learning estimates aleatoric and epistemic uncertainty in a single deterministic forward pass, making it computationally attractive. However, empirical studies indicate that it may struggle in extrapolative settings where data sparsity limits generalization [55].

To address these limitations, we introduce two novel *hybrid* UQ architectures that integrate evidential deep learning with ensemble and dropout-based techniques: (1) an **ensemble of evidential models (EOE)**, which aggregates multiple evidential networks to improve epistemic uncertainty estimation, and (2) an **evidential MC dropout model (EMC)**, which combines evidential learning with stochastic dropout sampling to capture variance in model outputs. These hybrid approaches aim to enhance the robustness of uncertainty estimation while balancing computational efficiency.

Previous work has explored combining evidential and ensemble models. Shen et al. [56] trained an evidential model on outputs from physiological and regression models, while Shao et al. [57] used a multi-modal ensemble of evidential models aggregated via Dempster–Shafer rules. In contrast, we perform Bayesian estimation of evidential model parameters, yielding a unified Bayesian–Evidential hybrid approach.

We evaluate these UQ methodologies in the context of proteochemometric modeling (PCM), a framework that jointly models interactions between ligands and protein targets [16–18]. PCM provides a structured approach to bioactivity assessment by integrating information from both ligand and protein domains, offering a broader context for ML-based assessments. By systematically comparing deep ensembles, MC dropout, evidential deep learning, and our proposed hybrid models, we assess their ability to estimate uncertainty, improve calibration, and provide well-calibrated confidence measures in bioactivity assessment. In addition to accuracy and calibration, we assess the statistical significance of model differences and quantify training cost in hardware-agnostic GPU-hours to provide a rigorous and practically relevant comparison. The broader significance of this study lies in its potential to refine the ML methodologies for drug discovery in the chemical sciences. By identifying limitations in current UQ techniques and proposing hybrid approaches, we provide a framework for enhancing the reliability of ML-driven decision-making in cheminformatics. These findings contribute to ongoing efforts in uncertainty-aware modeling, helping to establish more robust and interpretable ML frameworks for real-world scientific applications.

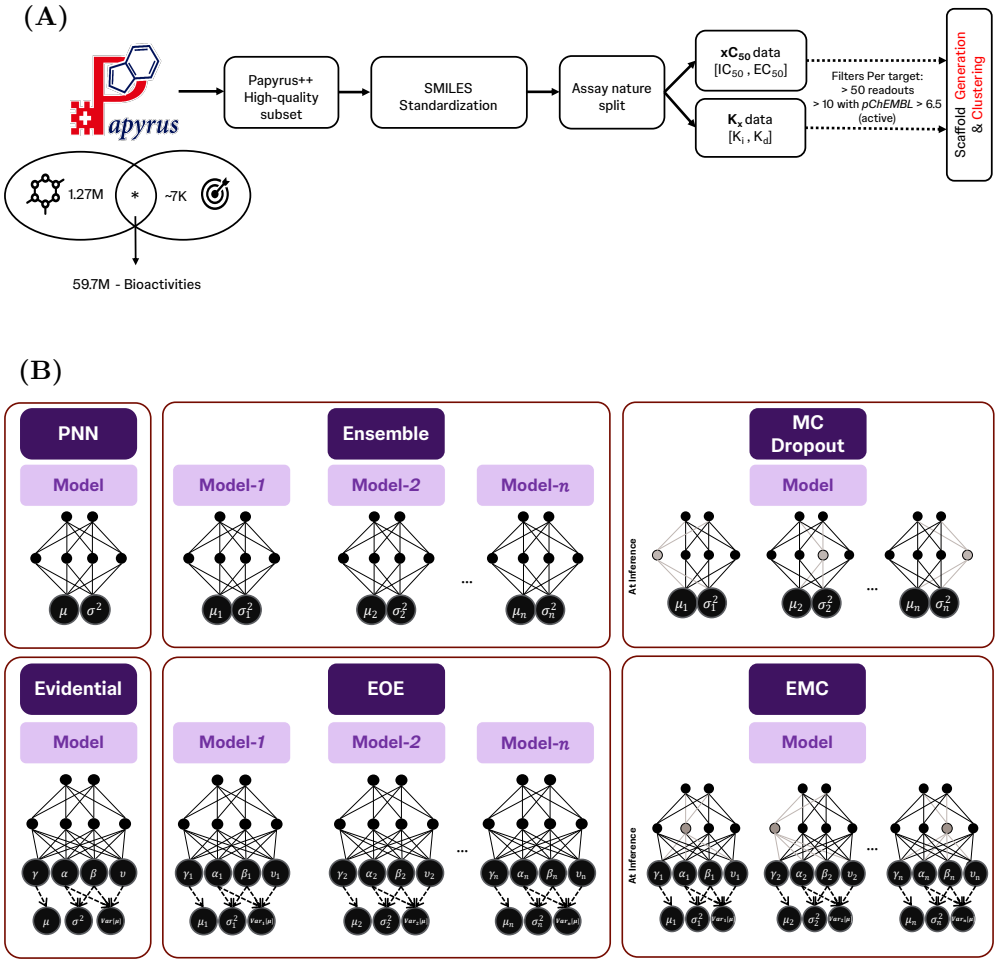


Figure 2.2: Overview of (A) **Data Preparation Workflow:** Starting with the Papyrus++ dataset, SMILES standardization and assay-based splitting yield xC_{50} and K_x subsets. Filtering ensures a minimum of 50 readouts per target and excludes under-represented targets, with at least 10 active data points, i.e., pChEMBL values > 6.5. (B) **UQ Approaches:** Ensemble models aggregate outputs from multiple PNN models. MC dropout estimates uncertainty through stochastic forward passes. Evidential models output distribution parameters (α, β, γ, υ) to derive mean, aleatoric, and epistemic uncertainty. Hybrid approaches (EOE and EMC) integrate ensemble diversity or MC dropout stochasticity with evidential uncertainty estimation. Papyrus logo is reproduced with permission from Ref. Béquignon et.al. [3], Copyright 2021 Springer Nature.

2.2. Results

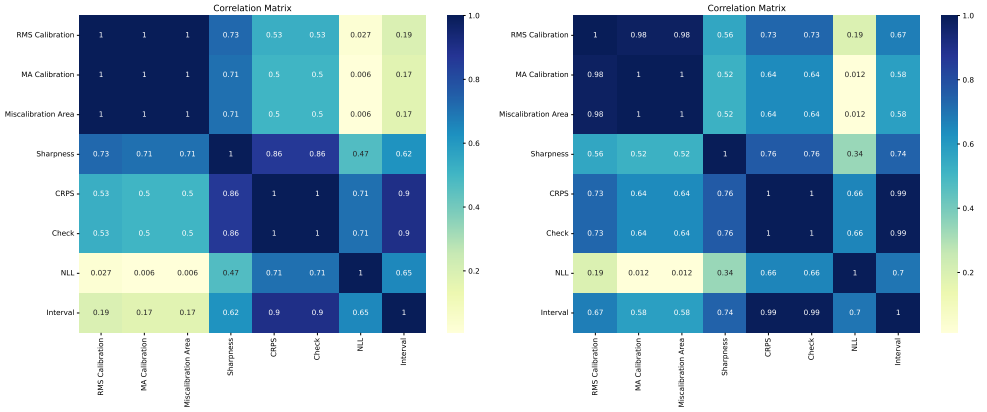
Model	Activity	Performance Metrics	Uncertainty Metrics					Rank Metrics
		RMSE ↓	Miscalib. Area ↓	NLL ↓	CRPS ↓	Interval ↓	Sharpness ↓	RRC-AUC ↓
Stratified								
PNN_1	xC_{50}	.732 (.004)	.133 (.009)	1.478 (.063)	.409 (.003)	2.264 (.031)	.465 (.012)	.986 (.003)
$Ensemble_{100}$	xC_{50}	.728 (.003)	.129 (.011)	1.441 (.065)	.406 (.000)	2.234 (.030)	.469 (.021)	**.825 (.009)**
MC $Dropout_{100}$	xC_{50}	.763 (.008)	**.026 (.016)**	1.140 (.014)	.418 (.004)	2.111 (.015)	.669 (.052)	.920 (.004)
$Evidential_1$	xC_{50}	.682 (.010)	.103 (.035)	1.005 (.015)	.378 (.009)	2.247 (.121)	.414 (.053)	.907 (.011)
EOE_{10}	xC_{50}	**.646 (.002)**	.074 (.008)	**.943 (.003)**	**.352 (.002)**	**2.028 (.029)**	.422 (.012)	.900 (.006)
EMC_{10}	xC_{50}	.693 (.007)	.125 (.034)	1.033 (.011)	.389 (.008)	2.335 (.125)	**.402 (.052)**	.909 (.009)
Random Rejection	xC_{50}	-	-	-	-	-	-	.945 (.008)
PNN_1	K_x	.639 (.011)	.127 (.005)	1.757 (.030)	.341 (.013)	2.059 (.084)	.699 (.383)	.982 (.011)
$Ensemble_{100}$	K_x	.595 (.001)	.079 (.001)	1.261 (.010)	.311 (.001)	1.809 (.005)	.655 (.037)	**.607 (.001)**
MC $Dropout_{100}$	K_x	.641 (.004)	.063 (.011)	1.230 (.083)	.330 (.001)	1.874 (.025)	**.439 (.022)**	.679 (.005)
$Evidential_1$	K_x	.624 (.003)	**.051 (.001)**	.531 (.005)	.313 (.001)	1.754 (.006)	.787 (.005)	.615 (.003)
EOE_{10}	K_x	**.573 (.001)**	.054 (.001)	**.428 (.002)**	**.291 (.000)**	**1.673 (.005)**	.749 (.002)	.613 (.001)
EMC_{10}	K_x	.626 (.003)	.086 (.004)	.560 (.005)	.317 (.002)	1.842 (.011)	.725 (.004)	.620 (.002)
Random Rejection	K_x	-	-	-	-	-	-	.949 (.005)
Scaffold Cluster								
PNN_1	xC_{50}	.907 (.006)	.049 (.010)	1.403 (.024)	.509 (.004)	2.608 (.030)	.777 (.018)	1.008 (.003)
$Ensemble_{100}$	xC_{50}	.878 (.001)	**.031 (.008)**	**1.320 (.009)**	.490 (.000)	2.474 (.009)	.795 (.023)	**.836 (.006)**
MC $Dropout_{100}$	xC_{50}	.928 (.006)	.085 (.008)	1.522 (.031)	.525 (.004)	2.735 (.030)	**.738 (.017)**	.938 (.004)
$Evidential_1$	xC_{50}	.897 (.003)	.049 (.005)	1.360 (.004)	.502 (.002)	2.514 (.006)	.992 (.013)	.906 (.005)
EOE_{10}	xC_{50}	**.871 (.001)**	.062 (.001)	1.323 (.001)	**.488 (.001)**	**2.455 (.005)**	1.000 (.004)	.900 (.003)
EMC_{10}	xC_{50}	.909 (.003)	.042 (.008)	1.383 (.004)	.508 (.001)	2.545 (.007)	.989 (.022)	.909 (.006)
Random Rejection	xC_{50}	-	-	-	-	-	-	.948 (.006)
PNN_1	K_x	1.102 (.012)	**.034 (.006)**	1.531 (.014)	.614 (.006)	3.060 (.034)	1.059 (.022)	.982 (.001)
$Ensemble_{100}$	K_x	1.097 (.002)	.038 (.001)	1.527 (.002)	.611 (.001)	3.045 (.005)	1.038 (.004)	.842 (.005)
MC $Dropout_{100}$	K_x	1.101 (.003)	.044 (.010)	1.596 (.022)	.620 (.002)	3.092 (.017)	1.136 (.031)	.851 (.004)
$Evidential_1$	K_x	1.061 (.016)	.035 (.009)	1.539 (.028)	.589 (.010)	3.072 (.041)	.939 (.018)	.844 (.008)
EOE_{10}	K_x	**.998 (.007)**	.047 (.004)	**1.464 (.009)**	**.553 (.004)**	**2.933 (.017)**	**.889 (.006)**	**.831 (.002)**
EMC_{10}	K_x	1.064 (.023)	.056 (.009)	1.555 (.027)	.591 (.013)	3.133 (.065)	.904 (.011)	.837 (.007)
Random Rejection	K_x	-	-	-	-	-	-	.950 (.003)

Table 2.1: Performance, Uncertainty, and Ranking Metrics for Evaluated Models on xC_{50} and K_x Datasets across Stratified and Scaffold Cluster Splits. Comparative analysis of different uncertainty quantification models, reporting one performance metric; RMSE, five uncertainty quantification metrics; Miscalibration Area, NLL, CRPS, Interval Score, and Sharpness, and one ranking-based metric; RRC-AUC. Lower values indicate better outcomes for all listed metrics. The best-performing model for each metric is highlighted in bold while the second-best is underlined. Random rejection is included as a baseline reference for ranking performance. The results show that EOE_{10} achieves consistently competitive performance, excelling across most uncertainty metrics and demonstrating strong overall accuracy, with significance confirmed across most metrics, highlighting the benefit of combining evidential and Bayesian approaches for bioactivity modeling tasks.

2.2 Results

We evaluate the performance and uncertainty quantification capabilities of all considered models across two bioactivity endpoints; xC_{50} (e.g., IC_{50}, EC_{50}) and K_x (e.g., K_i, K_d), using both stratified and scaffold-cluster data splits. Root Mean Squared Error (RMSE) is used as the primary performance metric, with other values, e.g. Coefficient of Determination (R^2) and Pearson Correlation Coefficient (PCC), provided in the supporting information.

To assess uncertainty quality, we adopt five non-redundant metrics from the Uncertainty Toolbox [58]–Miscalibration Area, Negative Log-Likelihood (NLL), Continuous Ranked Probability Score (CRPS), Interval Score, and Sharpness–. These were se-



(a) xC_{50} Dataset **(b)** K_x Dataset

Figure 2.3: Correlation matrix of uncertainty metrics for models evaluated on the (a) xC_{50} and (b) K_x datasets. The figure presents the correlation structure among uncertainty metrics from the Uncertainty Toolbox [58]. A strong correlation is observed among calibration-related metrics, as well as between CRPS and Check metrics. To ensure a non-redundant selection of uncertainty metrics, we retain five representative metrics: Miscalibration Area, Sharpness, CRPS, NLL, and Interval, which capture distinct aspects of model uncertainty.

lected based on their low redundancy (Figure 2.3) and collectively reflect calibration, probabilistic accuracy, and dispersion of uncertainty estimates. To evaluate the practical utility of uncertainty estimates for selective outputs, we also use RMSE-Rejection Curve Area Under Curve (RRC-AUC). All metrics are computed on the aleatoric uncertainty component, except RRC-AUC, which utilizes total uncertainty. Detailed information on all evaluation metrics can be found in the supporting information.

Table 2.1 summarizes quantitative results, and a complementary visual summary is provided in Figure 2.4. Detailed results can be found in the supporting information. Across all experimental conditions, EOE_{10} consistently achieves the lowest RMSE, despite using only 10 ensemble members–significantly fewer than Ensemble_{100}, which comprises 100 models. The choice of 10 members for EOE strikes a deliberate balance between computational efficiency and diversity among ensemble members. This configuration was empirically determined to provide sufficient variance reduction while keeping training and inference costs manageable, particularly given the added complexity of evidential parameter estimation. Notably, EOE_{10} also performs best on CRPS and Interval Score across all splits and tasks, underscoring its effective trade-off between accuracy and uncertainty expressiveness.

2.2. Results

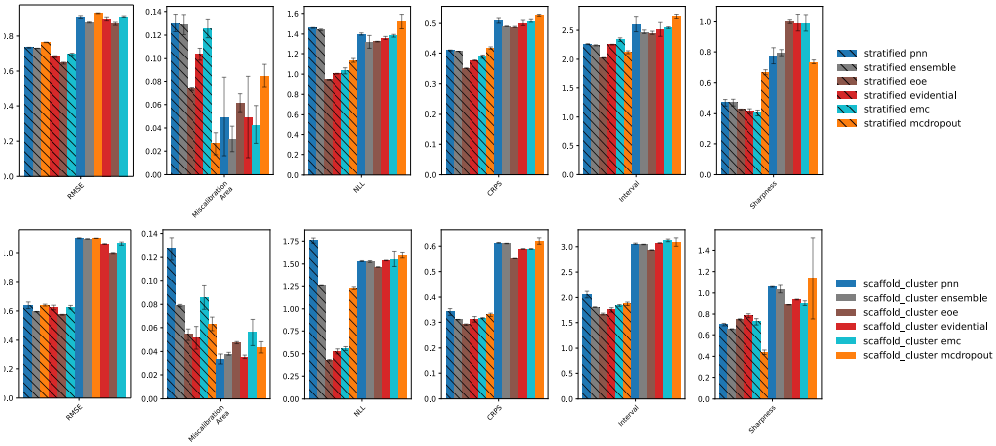


Figure 2.4: RMSE and Uncertainty Metrics for xC_{50} and K_x. Bar plots summarizing model performance (RMSE) and uncertainty estimation metrics (Miscalibration Area, NLL, CRPS, Interval, and Sharpness) across the xC_{50} (top row) and K_x (bottom row) datasets. All metrics are plotted such that lower values indicate better performance. This visualization complements the numerical results reported in Table 2.1 by illustrating trends and differences in model behavior across metrics and datasets.

Stratified Splits

In the more familiar stratified data splits, models were generally able to learn well-calibrated estimations.

$\mathbf{xC_{50}}$ Endpoint: EOE_{10} shows strong performance across uncertainty metrics, obtaining the best NLL, CRPS, and Interval Score, while being second best method for Miscalibration Area and RRC-AUC. EMC_{10} has the narrowest estimated distributions with the lowest Sharpness. Ensemble_{100} outperforms all other methods in the RRC-AUC metric, reflecting strong practical ranking utility.

$\mathbf{K_x}$ Endpoint: Similarly, EOE_{10} leads in NLL, CRPS, and Interval Score and is again second-best in Miscalibration Area and RRC-AUC. MC Dropout_{100} achieves the sharpest uncertainty estimates, while Ensemble_{100} ranks highest in ranking utility as measured by RRC-AUC. EMC_{10} shows robust sharpness and competitive probabilistic calibration.

Scaffold Cluster Splits

This setting represents a more challenging scenario with substantial distributional shift between training and test scaffolds.

xC_{50} Endpoint: Ensemble_{100} achieves the lowest Miscalibration Area, NLL, and best RRC-AUC, and also performs second-best on CRPS and Interval Score. EOE_{10} obtains best CRPS and Interval Score, and is second-best on NLL and RRC-AUC. EMC_{10} achieves the lowest miscalibration, suggesting potential underconfidence. MC Dropout_{100} again produces the sharpest distributions.

K_x Endpoint: This setting represents one of the most challenging evaluation scenarios, as evidenced by the higher RMSE values observed across all models. The increased difficulty stems from the combination of the K_x endpoint and scaffold cluster-based splitting, which introduces substantial distributional shift and limits the model's ability to generalize to unseen chemical scaffolds. In this context, **EOE_{10}** demonstrates robust performance, outperforming all other models on NLL, CRPS, Interval Score, Sharpness, and RRC-AUC. **EMC_{10}** ranks second on both Sharpness and RRC-AUC. These findings highlight the effectiveness of the hybrid evidential-Bayesian modeling approach under distributional stress and low-data generalization scenarios.

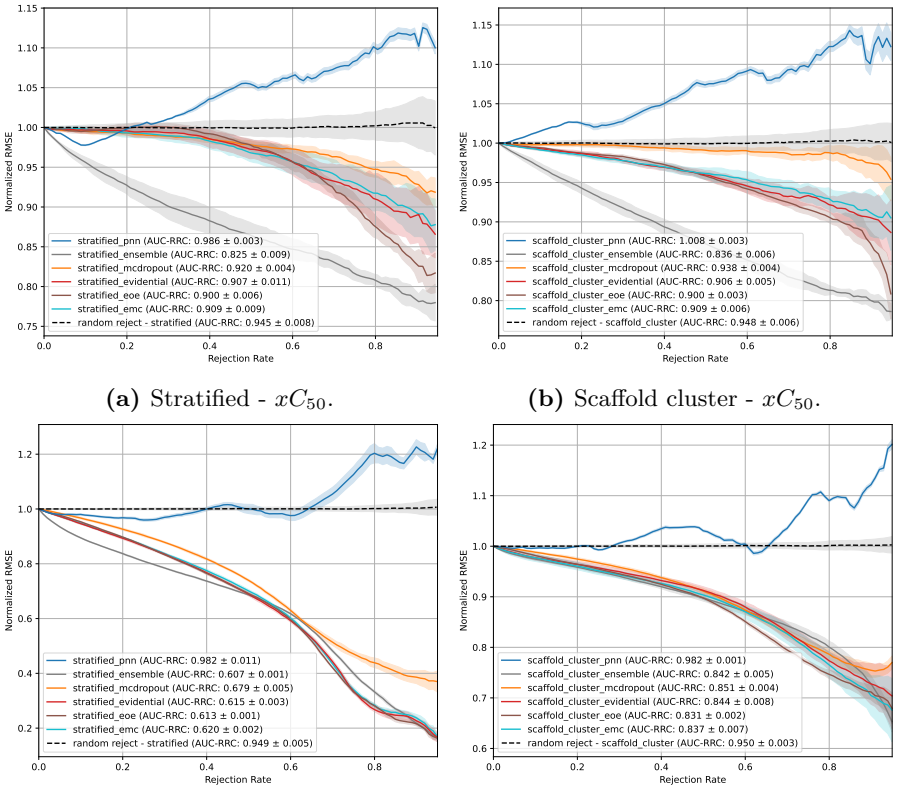
Uncertainty-Based Rejection Performance

RMSE-Rejection Curves (RRC), shown in Figure 2.5, provide a practical measure of uncertainty effectiveness by quantifying how well models identify uncertain instances. The Area Under the Curve (AUC) (lower is better) is reported in Table 2.1. Ensemble_{100} generally achieves the lowest AUCs in stratified settings and in xC_{50} under scaffold split. However, in the most challenging setting-K_x with scaffold splitting-EOE_{10} outperforms all others, achieving the best RRC-AUC, indicating superior capability for filtering low-confidence instances in high-uncertainty regimes. Other rank-based metrics from [35] were also computed and are available in the supporting information. However, we observed that they did not yield consistent conclusions across tasks and were therefore not included in the main body of results.

Miscalibration Direction Analysis

To complement the numerical evaluation of calibration error, we analyzed the direction of miscalibration using calibration curves (Figure 2.6). While the miscalibration

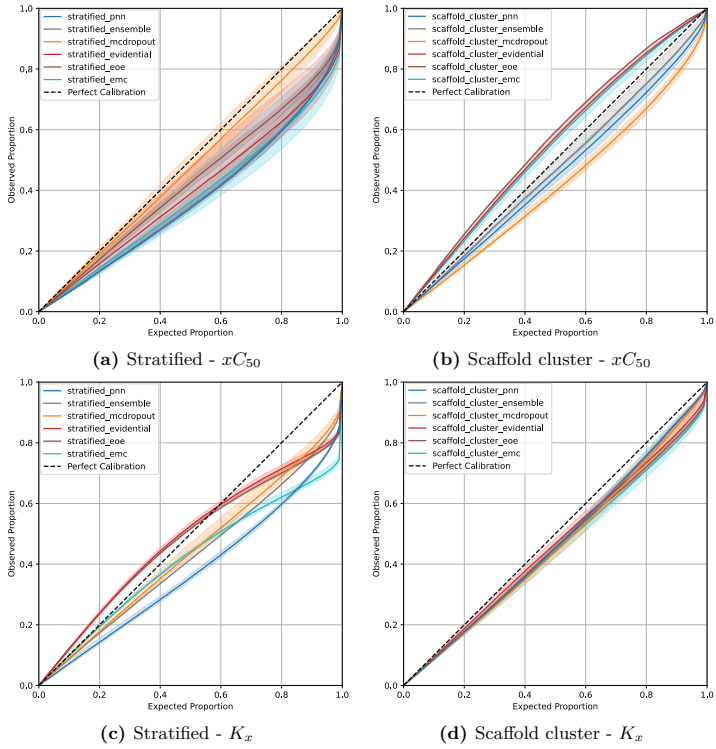
2.2. Results



(a) Stratified - xC_{50}. **(b)** Scaffold cluster - xC_{50}.

(c) Stratified - K_x. **(d)** Scaffold cluster - K_x.

Figure 2.5: RRC for xC_{50} and K_x under stratified and scaffold cluster splits. (a) Stratified-xC_{50}. (b) Scaffold cluster-xC_{50}. (c) Stratified-K_x. (d) Scaffold cluster-K_x. RRC illustrates the utility of uncertainty estimates in identifying unreliable instances. For each model, the RMSE is computed after progressively removing samples with the highest estimated total uncertainty. A lower curve indicates that uncertain outputs are effectively ranked and filtered, leading to improved overall estimation quality. EOE_{10} and Ensemble_{100} demonstrate consistently strong rejection performance, with EOE_{10} particularly excelling in the most challenging setting (K_x, scaffold cluster split).



(a) Stratified - xC_{50} **(b)** Scaffold cluster - xC_{50}

(c) Stratified - K_x **(d)** Scaffold cluster - K_x

Figure 2.6: Calibration curves of models evaluated on xC_{50} and K_x datasets, assessing the alignment between estimated confidence levels and observed accuracy. (a) Stratified–xC_{50}. (b) Scaffold cluster–xC_{50}. (c) Stratified–K_x. (d) Scaffold cluster–K_x. The dashed line represents perfect calibration, where the observed proportion matches the expected proportion.

area metric captures the magnitude of deviation from perfect calibration, the curves reveal whether models are overconfident (below the diagonal) or underconfident (above the diagonal). Across settings, EOE_{10}, Evidential, and EMC tend to exhibit slight overconfidence, particularly under scaffold-based splits, whereas MC Dropout_{100} is consistently underconfident. Ensemble-based models generally have better calibration, especially in the scaffold split, benefiting from model averaging. These findings illustrate that directionality of miscalibration varies across models and settings and should be considered alongside quantitative scores to fully assess calibration behavior.

Overall, EOE_{10} consistently demonstrates top-tier performance in both RMSE and uncertainty estimation metrics. It strikes a particularly strong balance between accuracy, calibration, and uncertainty informativeness–despite being significantly more computationally efficient than the full Ensemble_{100}. EMC_{10}, while slightly less dom-

inant overall, shows favorable sharpness and robustness in certain high-difficulty scenarios. These findings support the effectiveness of combining evidential and Bayesian principles for practical and scalable bioactivity modeling.

2.2.1 Statistical Significance of Model Comparisons

To assess whether observed performance differences between uncertainty quantification methods are statistically meaningful, we first evaluated the distributional assumptions underlying repeated-measures analyses. Figure 2.7 shows the normality diagnostics for the xC_{50} stratified split, where residuals across seeds appear broadly symmetric and bell-shaped, although formal Shapiro–Wilk tests [59] reject strict normality. This aligns with previous findings [60], which caution against over-reliance on normality tests in repeated-measures settings and recommend complementing parametric analyses with robust non-parametric alternatives. Accordingly, we applied both repeated-measures ANOVA with Tukey HSD and Friedman tests with Conover–Holm post-hoc analysis. Full diagnostic plots for the other data splits (xC_{50} scaffold cluster, K_x stratified, K_x scaffold cluster) are provided in the supporting information.

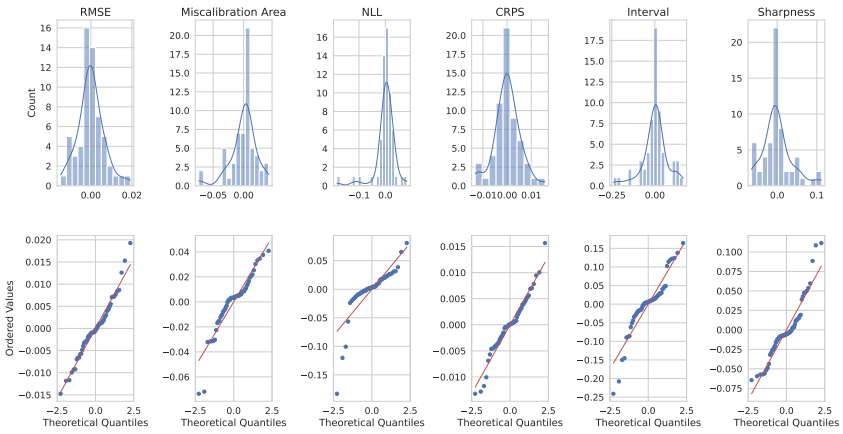


Figure 2.7: Normality diagnostics for repeated-measures residuals on xC_{50} (stratified split). Residual histograms and Q–Q plots for RMSE, Miscalibration Area, NLL, CRPS, Interval, and Sharpness, used to guide the choice between parametric (RM-ANOVA + Tukey HSD) and non-parametric (Friedman + Conover–Holm) tests. Residuals appear broadly symmetric, though Shapiro tests reject normality, underscoring the importance of complementary non-parametric tests.

To complement the distributional diagnostics, we assessed statistical significance of pairwise model differences using the non-parametric Friedman test with Conover–Holm post-hoc correction [60–62]. Figure 2.8 shows the resulting sign plot for the xC_{50} stratified split as a representative example. Three robust patterns emerge. *First*, differences in performance/error (RMSE) are statistically significant across *all* model pairs. *Second*, the same holds for the probabilistic scoring metrics NLL and CRPS, where every pairwise contrast is significant. *Third*, for Miscalibration Area most model pairs do not reach significance–consistent with the qualitative conclusions from the miscalibration direction analysis–yet the difference between EOE and ensemble remains statistically significant. Importantly, EOE also demonstrated statistically significant improvements compared to the full ensemble model in nearly all metrics. While the strength of evidence varied (with p-values ranging from very strong, $p < 0.001$, to more moderate, $p < 0.01$), the consistency of significance highlights the robustness of EOE's advantage.

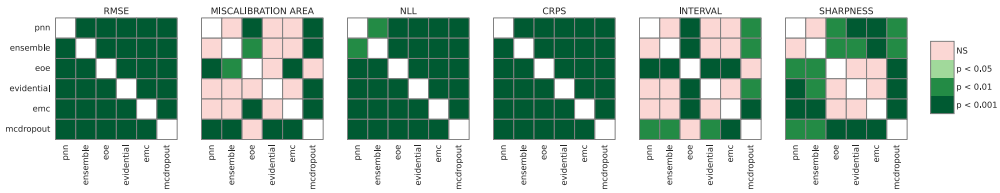


Figure 2.8: Conover–Holm sign plot for xC_{50} (stratified split). Pairwise post-hoc comparisons after Friedman tests across seeds for RMSE and uncertainty metrics. Overall, the EOE vs. ensemble difference is significant across *all* metrics in this setting, indicating a consistent advantage for EOE under stratified conditions.

Beyond the sign plots, we employed Multiple Comparison of Scores (MCS) plots and critical difference (CD) diagrams to provide complementary perspectives on statistical significance. MCS plots summarize mean differences and adjusted p-values in a heatmap-style format, enabling direct inspection of effect sizes. CD diagrams, in contrast, depict average ranks across models and highlight cliques of methods that cannot be statistically distinguished. Together, these complementary statistical visualizations emphasize not only the detectability but also the robustness and practical significance of EOE's advantage. Full sets of MCS and CD plots for all metrics and splits are provided in the supporting information.

2.2.2 Compute Budget and Training Cost Summary

To contextualize performance and UQ results with computational demand, training cost was quantified in hardware-agnostic units as GPU-hours, defined as wall-clock hours multiplied by the number of concurrently used GPUs. In our setup, each run exclusively occupied a single GPU; therefore, GPU-hours are numerically equal to wall-clock hours. Runtimes were exported from Weights & Biases logs and aggregated across all seeds, endpoints, and split types.

Table 2.2: Training cost aggregated by model family

Model	Mean GPU-hours	Total GPU-hours
PNN	1.34	53.57
Ensemble	111.44	4457.58
EOE	8.75	350.15
Evidential	4.74	189.74
EMC	7.05	281.88
MC Dropout	2.45	97.86

Interpretation. Training cost varied markedly across methods (Table 2.2). The full ensemble baseline required by far the largest compute (mean 111.44 GPU-h per run; 4457.58 GPU-h total), reflecting the multiplicative expense of training many independent members. The proposed hybrids were substantially more economical: EOE averaged 8.75 GPU-h per run (350.15 GPU-h total) and EMC 7.05 GPU-h per run (281.88 GPU-h total). Single-model approaches were naturally most compute efficient (Evidential: 4.74 GPU-h; MC Dropout: 2.45 GPU-h; PNN: 1.34 GPU-h).

2.3 Discussion

This study presents a systematic evaluation of UQ approaches for bioactivity modeling, with a focus on combining evidential and Bayesian paradigms. The proposed hybrid models, **EOE** and **EMC**, aim to synergize the complementary strengths of these frameworks: the data-driven regularization and expressivity of evidential methods, and the principled variance estimation offered by Bayesian techniques.

Efficacy of Hybrid Approaches

Across all experimental settings, $\mathbf{EOE_{10}}$ consistently achieved the lowest RMSE values, while requiring nearly an order of magnitude fewer GPU-hours than Ensemble_{100}.

Statistical tests confirmed that these improvements were significant across most metrics, underscoring that the gains are not only empirical trends but also robust to variation across seeds. This balance of accuracy, significance, and efficiency highlights the hybrid approach as a scalable alternative to large ensembles. Furthermore, EOE_{10} achieved the best scores on CRPS and Interval across all conditions, suggesting that it not only performs well but also provides uncertainty estimates with high informativeness and tight estimated distributions–key attributes in real-world drug discovery workflows.

Uncertainty Quality and Calibration

Uncertainty metrics from the Uncertainty Toolbox revealed important differences in the quality and characteristics of the estimated uncertainties. Following correlation analysis to minimize redundancy, we focused on five complementary metrics, enabling a balanced evaluation across calibration, probabilistic accuracy, and dispersion. This selection, combined with statistical tests, ensures that the reported trends are both interpretable and statistically sound. Although $\textbf{Ensemble}_{\textbf{100}}$ and $\textbf{Evidential}_{\textbf{1}}$ occasionally outperformed EOE_{10} on specific metrics (e.g., Miscalibration Area or NLL), EOE_{10} remained consistently competitive across all axes. Calibration curves further revealed that $\textbf{MC Dropout}_{\textbf{100}}$ tended to be underconfident, while $\textbf{EOE}_{\textbf{10}}$, $\textbf{Evidential}_{\textbf{1}}$, and $\textbf{EMC}_{\textbf{10}}$ exhibited slight overconfidence, particularly under scaffold-cluster splits. These observations highlight the importance of jointly assessing calibration directionality and magnitude to understand model behavior more comprehensively.

Practical Utility and Significance of Uncertainty Estimates

From a practical perspective, uncertainty-based rejection analysis demonstrated that $\textbf{EOE}_{\textbf{10}}$ and $\textbf{Ensemble}_{\textbf{100}}$ were particularly effective at identifying unreliable outputs. RRC-AUC values showed that $\textbf{EOE}_{\textbf{10}}$ achieved the best rejection performance in the most challenging experimental setting–K_x with scaffold cluster split–outperforming the full 100-member ensemble. While $\textbf{Ensemble}_{\textbf{100}}$ consistently delivered strong ranking utility across other scenarios, its training cost was more than twelve-fold higher than $\textbf{EOE}_{\textbf{10}}$. Under realistic compute budgets, $\textbf{EOE}_{\textbf{10}}$ therefore provides a more favorable cost–benefit profile, retaining competitive ranking performance while substantially reducing computational demand.

Although the numerical differences in RMSE, calibration, or RRC-AUC may ap-

pear modest, they are practically consequential in drug discovery pipelines where models are used to triage *large* candidate sets. Even incremental gains in calibration or rejection utility can reduce false positives admitted to assay, improve hit enrichment among prioritized compounds, and lower experimental costs. Reliable UQ further enables practitioners to set decision thresholds (e.g., accept, defer, or confirm) that preferentially retain high-confidence outputs and route uncertain cases to follow-up testing. Prior studies confirm that principled UQ frameworks improve success rates in virtual screening and active learning campaigns by coupling probabilistic confidence with ranking [34, 35, 50]. Consequently, the statistically significant advantages observed for $\mathbf{EOE_{10}}$ imply *operational* gains: at fixed experimental budgets, better-calibrated UQ admits fewer low-quality candidates and concentrates resources on higher quality molecules.

Insights on Model Behavior

Although $\mathbf{EOE_{10}}$ demonstrated the most robust overall performance, the analysis revealed that no single method was uniformly optimal across all metrics. For example, $\mathbf{MC\ dropout_{100}}$ consistently provided sharp uncertainties, albeit with reduced calibration and ranking utility. In contrast, $\mathbf{EMC_{10}}$ exhibited strong sharpness and favorable calibration in more difficult settings despite being slightly behind EOE in RMSE and probabilistic scoring metrics.

The results underscore the promise of hybrid UQ architectures in computational drug discovery. While deep ensembles remain a strong baseline, their computational overhead can be prohibitive. Hybrid models like EOE_{10} offer a compelling alternative, achieving competitive or superior performance at lower computational cost.

2.4 Conclusion

In this study, we investigated the integration of Bayesian and evidential uncertainty quantification techniques for bioactivity modeling. We introduced two hybrid models, $\mathbf{EOE_{10}}$ and $\mathbf{EMC_{10}}$, and benchmarked them against established baselines across multiple datasets and experimental splits. Our results demonstrate that $\mathbf{EOE_{10}}$ achieves superior performance in both estimation accuracy and uncertainty expressiveness, outperforming traditional ensembles with significantly fewer members. Statistical tests confirmed the robustness of these improvements across multiple uncertainty metrics, while benchmarking in GPU-hours revealed that hybrids achieve this performance at

a fraction of the computational cost. Notably, **$\mathbf{EOE_{10}}$** excelled under challenging distributional shifts, highlighting its robustness and practical utility.

Overall, this study demonstrates that EOE provides the best balance of performance, uncertainty calibration, and rank-based reliability, making it the most suitable model for bioactivity assessment tasks. MC dropout is a strong alternative when uncertainty calibration is the primary concern, while ensembles remain effective for robust uncertainty ranking at substantially higher computational expense. These insights guide model selection depending on application priorities and highlight hybrid approaches as scalable, trustworthy tools for decision-making in computational drug discovery.

2.5 Methods

2.5.1 Data Preprocessing and Splitting

In computational drug discovery, selecting and processing high-quality datasets is important to developing models with improved generalisability and robustness [3]. Proper data curation mitigates noise and biases, enhancing model reliability and facilitating informed decision-making. This study leverages Papyrus++, a refined subset of the Papyrus dataset [3], which aggregates standardized bioactivity data from multiple sources, including ChEMBL [11]. Papyrus++ retains only high-confidence measurements exhibiting agreement across multiple experimental assays, ensuring consistency and reproducibility in bioactivity assessments. The data preprocessing pipeline is illustrated in Figure 2.2A.

To ensure compatibility with machine learning workflows, we performed rigorous data standardization. Chemical structures were curated using RDKit [63] to correct stereo-chemical inconsistencies, neutralize formal charges with SMARTS patterns [64], and unify isotopic representations. Only wild-type human proteins (UniProt-mapped) were retained. The dataset was filtered to include only IC_{50} and EC_{50} measurements, referred to as the xC_{50} dataset, with targets restricted to those having at least 50 bioactivity measurements, including a minimum of 10 above a threshold of 6.5 on the pChEMBL scale.

To transform molecular information into machine-readable representations, we employed featurization techniques optimized for deep learning. Protein sequences were encoded using Ankh embeddings [1], a transformer-based protein language model demonstrating state-of-the-art performance in structural and functional tasks. Chemical

structures were represented using Extended Connectivity Fingerprints (ECFP) [21], a widely used molecular fingerprinting approach based on the Morgan algorithm. ECFP was computed with a radius of 2 and a vector length of 2048, ensuring a fixed-length representation capturing sub-structural patterns relevant for bioactivity modeling.

Robust data partitioning strategies were employed to evaluate uncertainty estimation techniques under different generalization scenarios. Two splitting strategies were applied: (1) a **stratified split**, ensuring proportional representation of scaffold-based clusters across training (70%), validation (15%), and test (15%) sets to evaluate model performance in a controlled setting, and (2) a **scaffold cluster split**, where entire scaffold clusters were assigned to distinct subsets to assess model generalization to novel chemical scaffolds. The data splitting strategy is depicted in Figure 2.2B.

To implement scaffold clustering, we followed the methodology of He et al. [65]. Scaffolds were extracted using the Bemis-Murcko algorithm [66], which isolates core cyclic frameworks of molecules. Similarity between scaffolds was computed using Maximum Common Substructure (MCS) similarity [67], quantified by the Tanimoto coefficient. Hierarchical clustering with Ward linkage [68] was then applied to assign scaffolds into clusters, optimizing within-cluster variance. The optimal number of scaffold clusters was determined using the silhouette coefficient [69], ensuring meaningful chemical partitioning. These strategies provide a robust framework for evaluating uncertainty quantification methods under both interpolation and extrapolation settings.

2.5.2 Hyperparameter Optimization

For hyperparameter optimization, we employed a Bayesian Optimization and Hyperband (BOHB) search approach [70]. Specifically, we leveraged the sweep functionalities provided by the "Weights and Biases" (`Wandb`) package [71] to efficiently explore the hyperparameter space and identify the configurations that yielded the best performance on the validation set. Bayesian optimization utilizes a probabilistic model to guide the selection of hyperparameter values through an iterative process of testing on a surrogate function before evaluating the actual objective function.

The hyperparameters we optimized included the number and size of hidden layers and featurizers including protein layer (for protein embeddings) and chemical layers (for chemical features), the type of optimizer, weight decay, learning rate (LR), dropout rate, batch size, and the maximum norm for gradients' clipping. Supplementary Table S2, outlines the various values we tested per each of the hyperparameters. The final

chosen hyperparameters are provided in the run configurations of the accompanying code.

2.5.3 Uncertainty Estimation Models

Uncertainty estimation is used in deep learning models for bioactivity assessment, enabling a more informed decision-making process in drug discovery. In this study, we investigate five uncertainty estimation approaches built upon a shared base neural network model. These include Bayesian approaches, i.e., Deep Ensembles and MC dropout, as well as Evidential Deep Learning, and two hybrid models that integrate Bayesian modeling with evidential learning. While Bayesian approaches capture epistemic uncertainty through sampling different models according to a posterior distribution, evidential learning aims to directly learn a second-order distribution of possible estimated distributions.

Each method extends the core model architecture to provide distinct approaches to quantifying aleatoric and epistemic uncertainties. While the classical PNN serves as the foundation for ensemble and MC dropout models, the evidential models (including the hybrid approaches) utilize a modified last layer that outputs different distribution parameters. Nevertheless, all models share the same architecture for feature extraction and regression, ensuring consistency in hyperparameter tuning.

PNN

PNNs [51] assume a Gaussian distribution for the target variable, thus the model outputs both a mean $\mu(\boldsymbol{x};\boldsymbol{\theta})$ and variance $\sigma^2(\boldsymbol{x};\boldsymbol{\theta})$.

The estimated distribution is then given by

$$p(y \mid \mu(\boldsymbol{x};\boldsymbol{\theta}), \sigma^2(\boldsymbol{x};\boldsymbol{\theta})) = \frac{1}{\sqrt{2\pi\sigma^2(\boldsymbol{x};\boldsymbol{\theta})}} \exp\left\{-\frac{(y-\mu(\boldsymbol{x};\boldsymbol{\theta}))^2}{2\sigma^2(\boldsymbol{x};\boldsymbol{\theta})}\right\} \tag{2.1}$$

Training is performed by minimizing the negative log-likelihood of Equation (2.1) [51]. The PNN model architecture comprises separate feature extractors for chemical structures and protein sequences, each feeding into a shared regression layer that estimates the output distribution (Figure 2.2B). This approach allows for the estimation of aleatoric uncertainty in deep learning models for molecular property assessments.

2.5. Methods

Deep Ensemble

Deep Ensembles [31] leverage multiple independently trained models to enhance accuracy and uncertainty estimation. Each model is initialized with different random weights and trained independently. Then, the final output is obtained by aggregating outputs across the ensemble. Deep Ensembles have been shown to be very faithful to the Bayesian ideal [52] and are often considered as gold standard for approximate Bayesian inference [72]. The expected mean output is computed as:

$$\mathbb{E}[\mu(\boldsymbol{x})] \approx \bar{\mu}(\boldsymbol{x}) = \frac{1}{M} \sum_{i=1}^{M} \mu(\boldsymbol{x}; \boldsymbol{\theta}_i) \tag{2.2}$$

where $M = 100$ is the number of ensemble members. Following [73], the uncertainty in the regression setting is usually captured as the variance of the outputs $\mathrm{Var}[y]$, which can the decomposed using the law of total variance:

$$\mathrm{Var}[y] = \mathbb{E}[\sigma^2(\boldsymbol{x})] + \mathrm{Var}[\mu(\boldsymbol{x})] \tag{2.3}$$

Here, the first term is usually associated with aleatoric, the second term with epistemic uncertainty.

The aleatoric uncertainty is given by:

$$\mathbb{E}[\sigma^2(\boldsymbol{x})] \approx \bar{\sigma}^2(\boldsymbol{x}) = \frac{1}{M} \sum_{i=1}^{M} \sigma^2(\boldsymbol{x}; \boldsymbol{\theta}_i) \tag{2.4}$$

while the epistemic uncertainty is given by the variance of the output means:

$$\mathrm{Var}[\mu(\boldsymbol{x})] \approx \frac{1}{M} \sum_{i=1}^{M} \left(\mu(\boldsymbol{x}; \boldsymbol{\theta}_i) - \bar{\mu}(\boldsymbol{x})\right)^2 \tag{2.5}$$

It is worth noting that recent work has proposed Bayesian Optimized Deep Ensembles (BODE), where each ensemble member is hyperparameter-optimized separately to increase diversity and improve calibration [74]. In our approach, the ensemble members were also trained with hyperparameters obtained via Bayesian Optimization; however, the same optimized configuration was applied to all members rather than optimizing each one individually. This strategy provides a computationally efficient compromise while still yielding strong model performance.

MC Dropout

MC dropout [32] is a Bayesian approximation technique that introduces stochasticity into deep neural networks by activating dropout layers during both training and inference. Multiple forward passes yield a distribution of outputs, from which the mean, aleatoric uncertainty, and epistemic uncertainty are estimated using Equations (2.2)–(2.5). We employ $M = 100$ stochastic passes in our implementation.

Evidential Deep Learning

Evidential deep learning [33, 34] models estimate the parameters of a Normal Inverse-Gamma (NIG) distribution, allowing for direct estimation of aleatoric and epistemic uncertainties. The network outputs four parameters: γ, υ, α, and β (we omit expressing them as functions of $\boldsymbol{x}$ and $\boldsymbol{\theta}$ for brevity), which define the NIG distribution:

$$p(\mu, \sigma^2 \mid \gamma, \upsilon, \alpha, \beta) = \frac{\beta^\alpha \sqrt{\upsilon}}{\Gamma(\alpha)\sqrt{2\pi\sigma^2}} \left(\frac{1}{\sigma^2}\right)^{\alpha+1} \exp\left\{-\frac{2\beta + \upsilon(\gamma-\mu)^2}{2\sigma^2}\right\} \tag{2.6}$$

The expected mean, aleatoric, and epistemic uncertainties can be directly estimated from the parameters of the NIG distribution and are given by:

$$\mathbb{E}[\mu(\boldsymbol{x})] \approx \gamma, \qquad \mathbb{E}[\sigma^2(\boldsymbol{x})] \approx \tfrac{\beta}{\alpha-1}, \qquad \mathrm{Var}[\mu(\boldsymbol{x})] \approx \tfrac{\beta}{\upsilon(\alpha-1)}. \tag{2.7}$$

Hybrid Models: EOE and EMC

To further enhance uncertainty quantification, we offer two novel hybrid approaches: **EOE** (Ensemble of Evidential) and **EMC** (Evidential MC dropout). Both methods combine the expressivity of evidential learning with the variance reduction benefits of Bayesian sampling.

The **EOE model** extends ensemble learning by replacing each individual model with an evidential network. This approach retains ensemble diversity while leveraging evidential learning's capacity to jointly model aleatoric and epistemic uncertainty. In contrast to the standard ensemble using $M = 100$ members, our EOE implementation employs only $M = 10$ evidential models for computational efficiency. The **EMC model** applies evidential learning within the MC dropout framework. Each forward pass with activated dropout layers outputs the parameters of a NIG distribution. As in EOE, we use $M = 10$ stochastic passes.

In both EOE and EMC, the outputs of the M networks are first aggregated by computing the mean of each NIG parameter $\bar{\alpha}, \bar{\beta}, \bar{\gamma}, \bar{\upsilon}$. These averaged parameters define a new Normal Inverse-Gamma distribution, from which we compute the final output and uncertainty estimates using the same closed-form expressions as in standard evidential learning (Similar to Equation 2.7):

$$\mathbb{E}[\mu(\boldsymbol{x})] = \bar{\gamma}(\boldsymbol{x}), \qquad \mathbb{E}[\sigma^2(\boldsymbol{x})] = \frac{\bar{\beta}(\boldsymbol{x})}{\bar{\alpha}(\boldsymbol{x}) - 1}, \qquad \mathrm{Var}[\mu(\boldsymbol{x})] = \frac{\bar{\beta}(\boldsymbol{x})}{\bar{\upsilon}(\boldsymbol{x})(\bar{\alpha}(\boldsymbol{x}) - 1)} \tag{2.8}$$

2.5.4 Metrics for Model Evaluation

To rigorously evaluate both the accuracy and reliability of uncertainty estimation, we adopt a focused set of metrics that reflect the key objectives of our study: reliable estimation quality, well-calibrated uncertainty, and practical utility for selective decision-making. All metrics used in the main analysis are computed on the test sets and averaged across multiple seeds. Formal definitions and additional metrics not discussed here are provided in supporting information.

Performance Metric

We report the **RMSE** as the primary performance metric, quantifying the average squared deviation between estimated and observed values. RMSE provides a robust measure of estimation error and is particularly sensitive to large deviations. Additional performance metrics such as $\mathbf{R^2}$ and **PCC** are presented in the supporting information.

Uncertainty Metrics

To assess the quality of the estimated uncertainty distributions, we employ five non-redundant metrics from the Uncertainty Toolbox [58], selected for their complementarity and empirical relevance. The **miscalibration area** quantifies the discrepancy between estimated and observed confidence levels using calibration curves. The **NLL** evaluates the likelihood of the observed values under the estimated distributions, penalizing overconfident or misaligned estimates. The **CRPS** assesses the compatibility between the estimated cumulative distribution and the empirical distribution. The **interval score** jointly evaluates the width and coverage of output intervals, penalizing both over- and underconfident outputs. Finally, **sharpness** captures the concentration or narrowness of the estimated distribution, regardless of its calibration.

Additional metrics, including RMS Calibration, MA Calibration, and Check Score, are reported in the supporting information.

Uncertainty-Based Rejection Performance

To evaluate the practical utility of uncertainty estimates for decision-making, we use the **RRC-AUC**. This metric summarizes how RMSE changes when samples with the highest estimated total uncertainty are iteratively removed. Lower RRC-AUC values indicate that the model's uncertainty estimates are effective in identifying unreliable outputs. Other rank-based metrics from [35], such as Spearman's Rank Correlation between uncertainty and error, or are reported in the supporting information for completeness.

2.5.5 Statistical Significance Testing

To determine whether observed differences in performance and uncertainty quantification metrics were statistically significant, we employed a two–step procedure combining normality diagnostics with both parametric and non–parametric repeated–measures tests, following the methodology outlined by Ash et al. [60].

First, we assessed the distribution of residuals across seeds using visual inspection of histograms and Q–Q plots, complemented by Shapiro–Wilk tests. As noted by Ash et al. [60], such tests can be overly sensitive in small–sample regimes, and normality assessment should not rely solely on hypothesis tests but also on graphical diagnostics. Given the inconclusive results from Shapiro–Wilk across some metrics, we applied both parametric and non–parametric approaches to ensure robustness.

When approximate normality could be assumed, we used a repeated–measures ANOVA with seed as the blocking factor, followed by Tukey's Honest Significant Difference (HSD) post–hoc test to obtain pairwise adjusted p–values. This analysis also quantified the magnitude and direction of mean differences, which we summarized in Multiple Comparison of Scores (MCS) plots.

In parallel, we applied the Friedman test with seeds as blocks to account for potential non–normality, followed by Conover post–hoc pairwise comparisons with Holm correction for multiple testing [61, 62]. These non–parametric results were visualized using sign plots, which directly encode significant pairwise contrasts, and critical difference (CD) diagrams, which summarize average ranks across models and identify cliques of methods without significant differences [75].

For each bioactivity endpoint (xC_{50} and K_x) and each split type (stratified and

scaffold cluster), we conducted tests separately across models and across the selected performance and uncertainty metrics (RMSE, Miscalibration Area, NLL, CRPS, Interval, Sharpness). This dual–testing strategy ensured that conclusions did not depend on a single assumption about distributional form, and aligns with recent recommendations for practically significant model comparisons in molecular machine learning [60]. Importantly, this ensures that observed advantages, such as those of **EOE** over ensembles, are statistically robust and not artifacts of a single analysis choice.

Supplementary Information

S2.1 Mathematical Formulations of Metrics

For completeness, we present the detailed mathematical definitions of all evaluation metrics used in this study. Below are the full formulations for the performance and uncertainty estimation metrics:

S2.1.1 Root Mean Squared Error (RMSE)

This is a standard accuracy metric, measuring the average errors squared, thus increased sensitivity to large deviations. It is defined as

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2} \ . \tag{2.9}$$

S2.1.2 Coefficient of Determination (R^2)

This metric measures how well a regression model explains the variability of the dependent variable y. It is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \ . \tag{2.10}$$

where $\bar{y}$ is the mean of the observed values $\bar{y} = 1/n(\sum_{i=1}^{n} y_i)$. An R^2 value of 1 indicates perfect estimation, while a value of 0 indicates that the model does not explain any of the variability in the response data.

S2.1.3 Miscalibration Area (MCA)

Defined as the area between the calibration line of the regression model and the perfect calibration line, thus essentially its mismatch in the reliability diagram.

It is formally defined as

$$\text{MCA} = \int_0^1 |\hat{p} - p| dp \ . \tag{2.11}$$

The MCA ranges from 0 (perfect calibration) to 0.5 (maximum miscalibration), with values closer to 0 indicating better calibration. To compute the calibration metrics, the expected probabilities are discretised from 0.01 to 0.99 in increments of 0.01,

and the corresponding observed probabilities are then calculated for each of these 99 expected probability values. Lower values of MCA are better, indicating better alignment between estimated and observed uncertainties.

S2.1.4 Negative Log-Likelihood (NLL)

In this work, we assume probabilistic estimators that yield a conditional density for an input x, where the model outputs the distribution parameters. Depending on the nature of the output distribution, we can calculate the negative log-likelihood. For the example of the standard PNN, we obtain the Gaussian NLL

$$\mathrm{NLL}_{\mathrm{Gaussian}} = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{2}\left(\log(2\pi\sigma_i^2) + \frac{(y_i - \hat{y_i})^2}{\sigma_i^2}\right) , \tag{2.12}$$

where $\hat{y}_i$ and σ_i^2 are the outputs of the PNN.

S2.1.5 Continuous Ranked Probability Score (CRPS)

This metric measures the accuracy of the probabilistic forecast, thus the output distribution. Specifically, it utilizes the cumulative distribution $F(x)$ and is formally defined as:

$$\mathrm{CRPS}(F, y) = \int_{-\infty}^{\infty}(F(x) - \mathbb{1}_{\{x \geq y\}})^2 dx \ . \tag{2.13}$$

where $\mathbb{1}_{\{x \geq y\}}$ is an indicator function that equals 1 when $x \geq y$ and 0 otherwise.

S2.1.6 Interval score

The interval score is commonly used for evaluating estimation intervals, which are pairs of quantiles with a prescribed expected coverage. It assesses both the width of the interval and the coverage probability, rewarding narrow intervals while penalizing those that fail to contain the observed value. This score is negatively oriented, meaning lower values indicate better performance. The interval score balances the trade-off between the interval width and the coverage, promoting intervals that are both accurate and concise.

S2.1.7 Sharpness

Sharpness is a crucial metric in evaluating the quality of probabilistic forecasts, as it measures the concentration of the estimated distributions. Unlike calibration, which assesses the agreement between estimated probabilities and observed frequencies, sharpness is concerned solely with the estimation itself, independent of the true outcomes. It reflects the confidence of the model in its estimations. A sharper estimated distribution signifies that the model is more confident, typically resulting in narrower estimation intervals.

In practical terms, sharpness is often quantified–as in equation 2.14– as the mean of the square root of expected variance (standard deviations) on the held-out (test) set. This provides an aggregate measure of how peaked or concentrated the estimations are. A lower sharpness value indicates a tighter and more confident estimation distribution, which is generally desirable if the model is well-calibrated. However, it is important to note that sharpness should be evaluated in conjunction with calibration. A model that is overly sharp but poorly calibrated may exhibit overconfidence, leading to unreliable estimations. Therefore, achieving a balance between sharpness and calibration is essential for effective uncertainty quantification. Mathematically, sharpness can be expressed as follows:

$$\text{Sharpness} = \frac{1}{n}\sum_{i=1}^{n}\sqrt{\sigma_i^2} \tag{2.14}$$

S2.1.8 Normalized RMSE Rejection Curve (RRC)

Furthermore, we are interested in the utility of using the uncertainty information for ranking model outputs. The quality of the uncertainty information is better, if the outputs that are ranked higher have lower RMSE. In the following, we describe the algorithm we used to normalize the RMSE-Rejection curve to make different models more easily comparable.

S2.1.9 Spearman's Rank Correlation Coefficient (ρ)

ρ measures the monotonic relationship between uncertainty estimates and observed errors. It evaluates how well the ranking of uncertainties aligns with the ranking of actual errors. Given two vectors $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$, their rankings are represented as $\boldsymbol{r_{v1}}$ and $\boldsymbol{r_{v2}}$. ρ is calculated using the following formula:

Algorithm 1 Calculate *Normalized* RMSE-Rejection Curve

Input: Dataset D, uncertainty column u, true label column y, estimated label column $\hat{y}$, normalize nrm
Output: Rejection rates r, RMSE values e, AUC a
Sort D by u in descending order
Define rejection steps $S \leftarrow \{0, .01|D|, ..., .95|D|\}$
Compute rejection rates $r \leftarrow S/|D|$
Initialize e as an empty list []
Compute initial RMSE $e_0 \leftarrow \sqrt{\frac{1}{|D|}\sum_{i=1}^{|D|}(D[y]_i - D[\hat{y}]_i)^2}$
for each i in S **do**
    Select data $D_i \leftarrow D[i:]$
    Compute RMSE $e_i \leftarrow \sqrt{\frac{1}{|D_i|}\sum_{j=i}^{|D|}(D_i[y]_j - D_i[\hat{y}]_j)^2}$
    if nrm **then**
        Normalize $e_i \leftarrow e_i/e_0$
    end if
    Append e_i to e
end for
Compute AUC $a \leftarrow \text{AUC}(r, e)$
return r, e, a

$$\rho(\boldsymbol{v_1}, \boldsymbol{v_2}) = \frac{\text{cov}(\boldsymbol{r_{v1}}, \boldsymbol{r_{v2}})}{\text{std}(\boldsymbol{r_{v1}})\ \text{std}(\boldsymbol{r_{v2}})} \tag{2.15}$$

S2.2 Hyperparameter Optimization

We provide an overview over the hyperparameter search space in Table S2.1. The finally obtained hyperparameters for each setting (endpoint and splitting) for the PNN and evidential model can be found in the configuration files in the publicly released repository:
https://github.com/CDDLeiden/uqdd/config.

Parameter	Values
chemical layers	3 layers: [1024, 512, 256]
	2 layers: [1024, 512] or [1024, 256]
	1 layer: [1024] or [512] or [256]
protein layers	3 layers: [1024, 512, 256] or [512, 256, 128]
	2 layers: [512, 128] or [512, 256] or
	1 layer: [512] or [256]
regressor layers	2 layers: [256, 128] or [128, 64]
	1 layer: [256] or [128] or [64]
batch size	128, 256, 512
dropout	0.1, 0.2
optimizer	Adagrad , AdamW , Adam , SGD
learning rate	0.01, 0.001, 0.0005
weight decay	0, 0.01, 0.0001
max norm	null, 50.0, 100.0

Table S2.1: Hyper-parameters and Tested Values for Machine Learning Model Tuning. This table lists the hyper-parameters and their corresponding values explored during the hyper-parameter optimization process. Each row represents a distinct hyper-parameter, detailing the range or set of values tested to identify the optimal configuration for the machine learning model. This overview facilitates the understanding of the search space.

S2.3 Statistical Significance Analyses

To keep the main text concise while preserving methodological transparency, this section provides the complete set of statistical significance visualizations across all four settings: xC_{50}-stratified, xC_{50}-scaffold-cluster, K_x-stratified, and K_x-scaffold-cluster. For each setting, we report (where available): (i) normality diagnostics for repeated-measures residuals, (ii) non-parametric sign plots (Conover–Holm) following Friedman tests, (iii) multiple-comparison heatmaps showing Tukey HSD mean differences and adjusted p-values (parametric path), and (iv) critical difference diagrams summarizing average ranks and non-significant cliques. Seeds act as within-subject blocks throughout; tests are performed per metric and per split type, in line with the analysis plan described in Section: Statistical Significance Testing.

S2.3.1 Normality Diagnostic Plots

To assess whether residuals across models followed approximate normal distributions, we generated diagnostic plots per activity–split setup and per evaluation metric. Residuals were computed by subtracting the per-seed mean across models, following the recommendations of Ash et al. Each plot combines histograms and Q–Q plots of residuals to provide both distributional and quantile-based views.

Visual inspection indicates that residuals are often approximately symmetric and bell-shaped, but formal Shapiro–Wilk tests frequently reject normality. This discrepancy reflects the well-known limitations of strict normality tests in repeated-measures settings.

Consequently, both parametric (RM-ANOVA with Tukey HSD) and non-parametric (Friedman with Conover–Holm) tests were employed for robustness, as described in the Methods.

S2.3. Statistical Significance Analyses

xC_{50} - Scaffold-cluster split

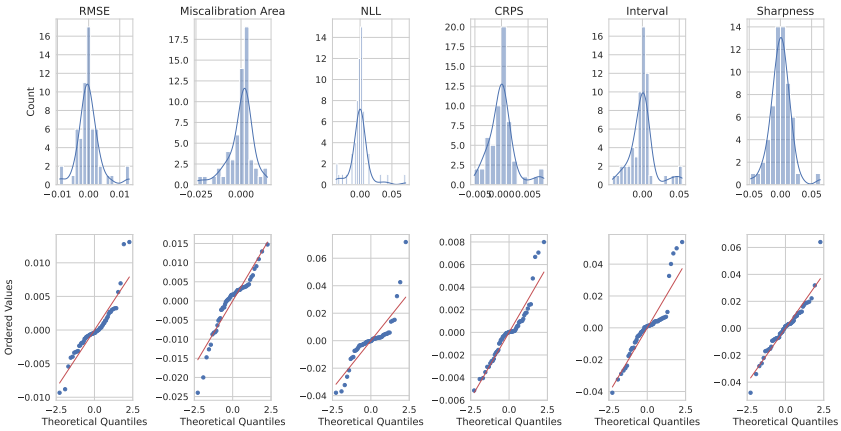


Figure S2.1: Normality diagnostics for xC_{50} (scaffold-cluster split). Histograms and Q–Q plots of residuals across metrics and seeds. These plots illustrate the distributional shape relative to normality; deviations are visible for some metrics. As residuals do not consistently satisfy strict normality (see Ash et al.), both parametric (ANOVA) and non-parametric (Friedman) tests were applied in subsequent analyses.

K_x – Stratified split

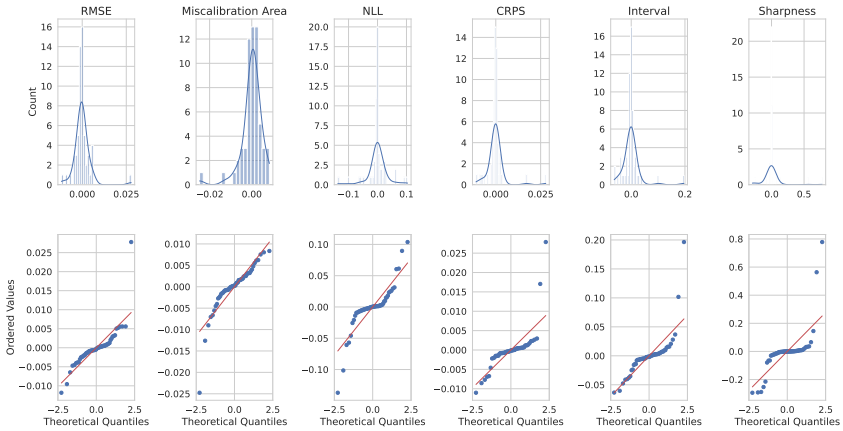


Figure S2.2: Normality diagnostics for K_x (stratified split). Histograms and Q–Q plots of residuals across metrics and seeds. These plots illustrate the distributional shape relative to normality; deviations are visible for some metrics. As residuals do not consistently satisfy strict normality (see Ash et al.), both parametric (ANOVA) and non-parametric (Friedman) tests were applied in subsequent analyses.

S2.3. Statistical Significance Analyses

K_x - Scaffold-cluster split

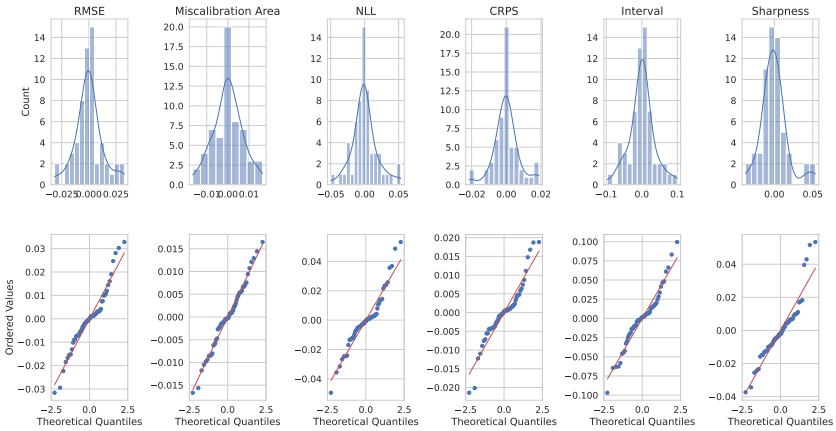


Figure S2.3: Normality diagnostics for K_x (scaffold-cluster split). Histograms and Q–Q plots of residuals across metrics and seeds. These plots illustrate the distributional shape relative to normality; deviations are visible for some metrics. As residuals do not consistently satisfy strict normality (see Ash et al.), both parametric (ANOVA) and non-parametric (Friedman) tests were applied in subsequent analyses.

S2.3.2 Non-parametric sign plots (Conover–Holm)

xC_{50} - Scaffold-cluster

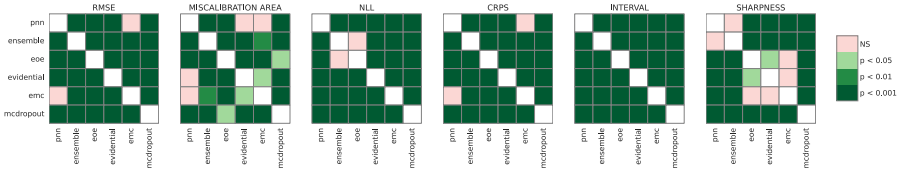


Figure S2.4: Conover–Holm sign plots for xC_{50} (scaffold-cluster split). Pairwise post-hoc comparisons after Friedman tests across seeds. Each heatmap cell corresponds to a pair of models and one evaluation metric. Shaded (marked) cells indicate that the null hypothesis of equal central tendency between the two models was rejected at the family-wise significance level; unshaded (light) cells indicate no significant difference.

K_x - Stratified

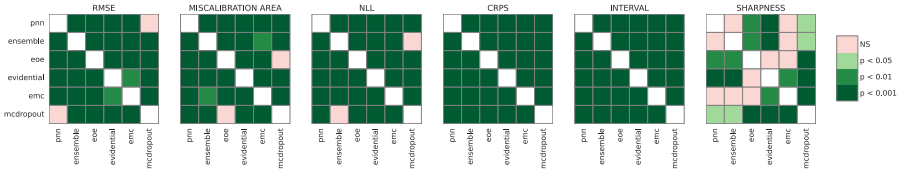


Figure S2.5: Conover–Holm sign plot for K_x (stratified split). Pairwise post-hoc comparisons after Friedman tests across seeds. Each heatmap cell corresponds to a pair of models and one evaluation metric. Shaded (marked) cells indicate that the null hypothesis of equal central tendency between the two models was rejected at the family-wise significance level; unshaded (light) cells indicate no significant difference.

K_x - Scaffold-cluster

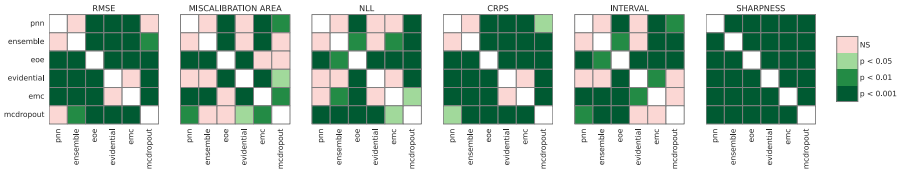


Figure S2.6: Conover–Holm sign plot for K_x (scaffold-cluster split). Pairwise post-hoc comparisons after Friedman tests across seeds. Each heatmap cell corresponds to a pair of models and one evaluation metric. Shaded (marked) cells indicate that the null hypothesis of equal central tendency between the two models was rejected at the family-wise significance level; unshaded (light) cells indicate no significant difference.

S2.3.3 Multiple comparisons of scores / means (Tukey HSD)

To complement the non-parametric analyses, we performed repeated-measures ANOVA followed by Tukey's Honest Significant Difference (HSD) test. These plots visualize pairwise mean differences between models across evaluation metrics, with significance levels indicated after controlling for family-wise error. In contrast to the sign plots, which highlight the existence of significant differences, the MCS plots additionally convey the magnitude and direction of the mean differences, providing a richer picture of where the strongest improvements are achieved.

xC_{50} **- Stratified**

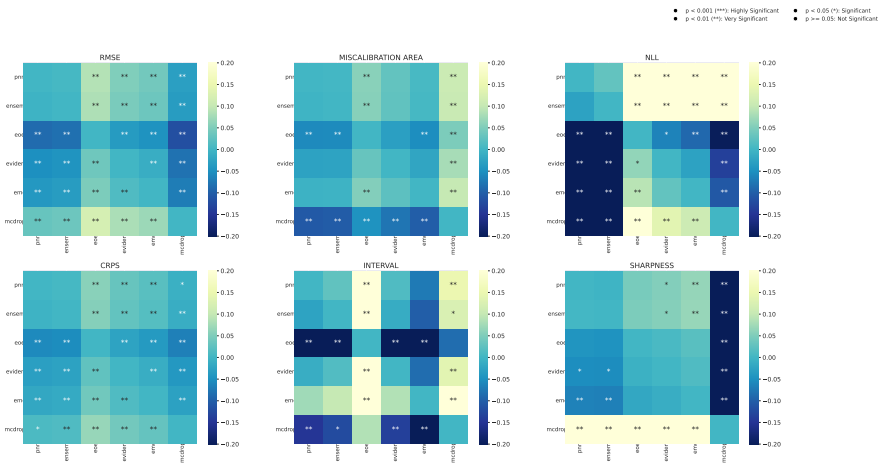


Figure S2.7: Multiple comparisons of means for xC_{50} (stratified split). Heatmaps show pairwise differences in mean score between methods from a repeated-measures ANOVA followed by Tukey's HSD (family-wise error controlled within each metric). Cells are annotated with the mean difference and significance ($p < 0.05$, $p < 0.01$, $p < 0.001$); unstarred cells are not significant at the stated α. Color encodes the magnitude and sign of the mean difference (see colorbar); for metrics to be minimized, the colormap is reversed so that "better" differences appear darker per metric. Methods are ordered by their average performance; the diagonal is blank by construction.

xC_{50} - Scaffold-cluster

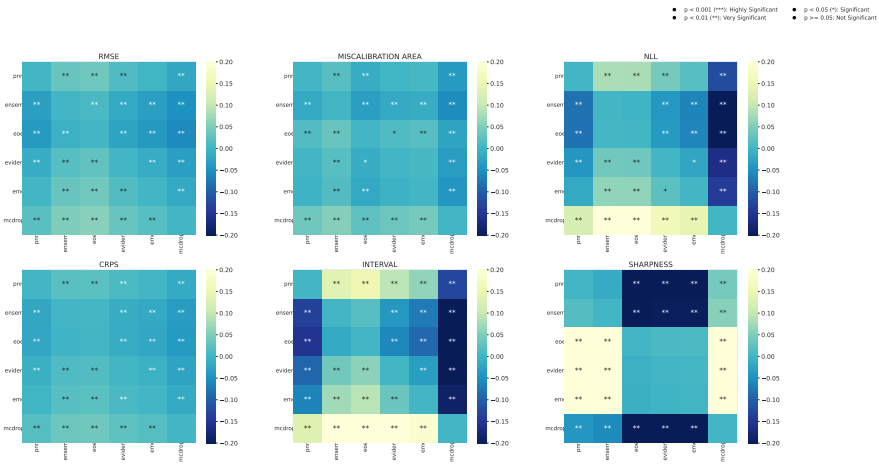


Figure S2.8: Multiple comparisons of means for xC_{50} (scaffold-cluster split). Same visualization and conventions as above: Tukey's HSD post hoc after repeated-measures ANOVA, effect sizes as pairwise mean differences, star annotations for significance, reversed colormap for minimize metrics, methods ordered by average performance, and blank diagonal.

S2.3. Statistical Significance Analyses

K_x - Stratified

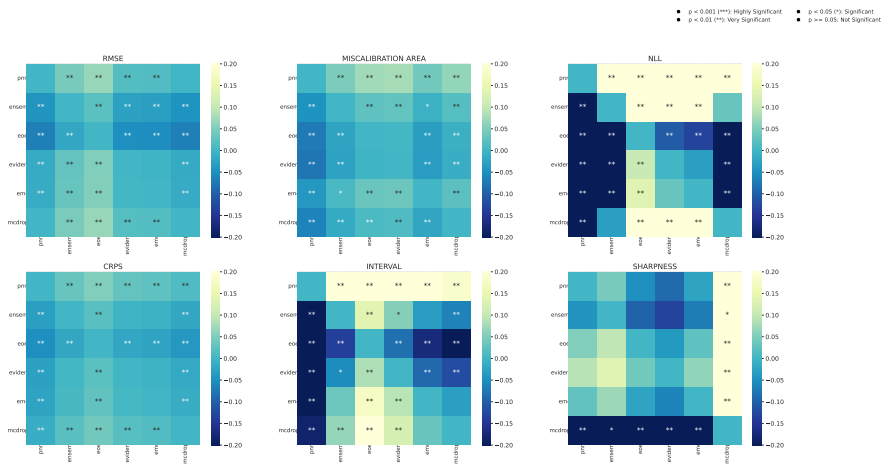


Figure S2.9: Multiple comparisons of means for K_x (stratified split). Same visualization and conventions as above: Tukey's HSD post hoc after repeated-measures ANOVA, effect sizes as pairwise mean differences, star annotations for significance, reversed colormap for minimize metrics, methods ordered by average performance, and blank diagonal.

K_x - Scaffold-cluster

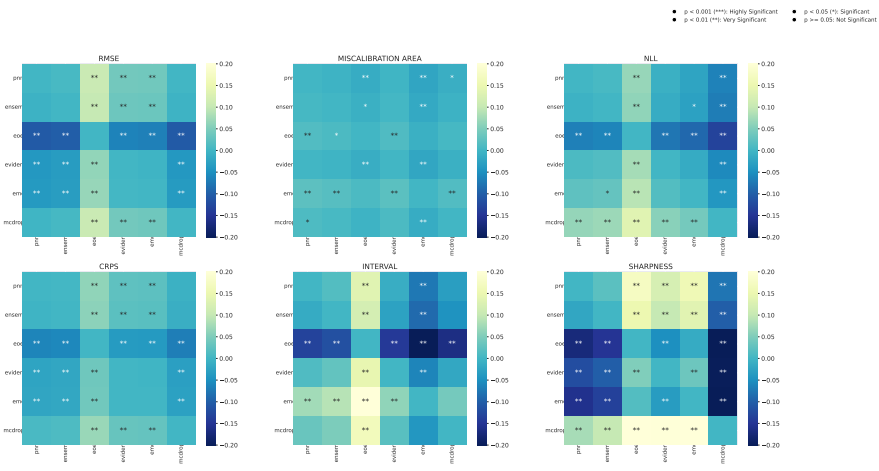


Figure S2.10: Multiple comparisons of means for K_x (scaffold-cluster split). Same visualization and conventions as above: Tukey's HSD post hoc after repeated-measures ANOVA, effect sizes as pairwise mean differences, star annotations for significance, reversed colormap for minimize metrics, methods ordered by average performance, and blank diagonal.

S2.3.4 Critical difference (CD) diagrams

CD diagrams summarize the rank structure of methods, per metric, across seeds. For each metric panel, the horizontal position is the average rank of each method (lower is better). Horizontal bars ("cliques") connect methods that are not significantly different at the chosen family-wise level, based on a post-hoc p-value matrix computed with the Conover–Friedman procedure (seeds as within-subject blocks) and Holm's multiplicity adjustment. CD diagrams convey rank relationships and indistinguishable groups; they do not report effect sizes.

xC_{50} **- Stratified**

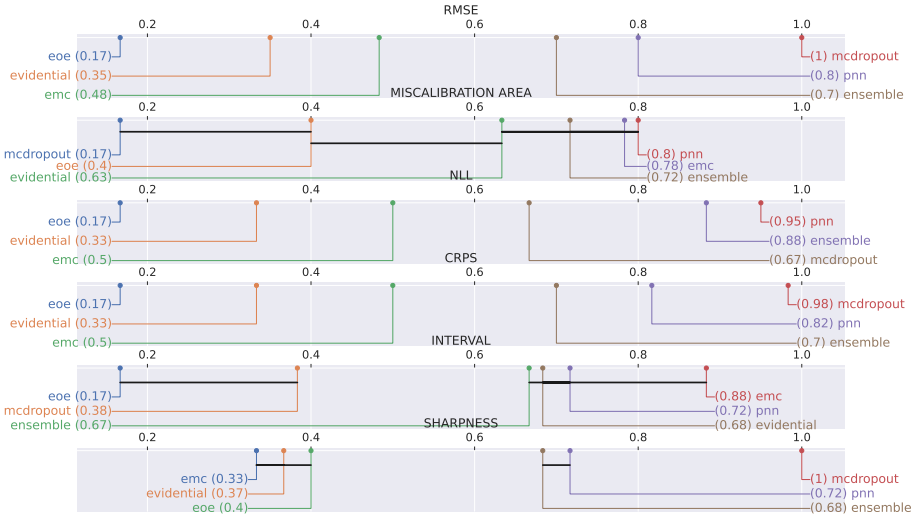


Figure S2.11: Critical difference diagrams for xC_{50} (stratified split). Points show average ranks (lower is better) for each method on the given metric across seeds. Horizontal bars connect methods that are not significantly different at the family-wise level according to the post-hoc p-value matrix used to draw the cliques. The diagram summarizes rank structure and indistinguishable groups; effect sizes are not shown.

xC_{50} - Scaffold-cluster

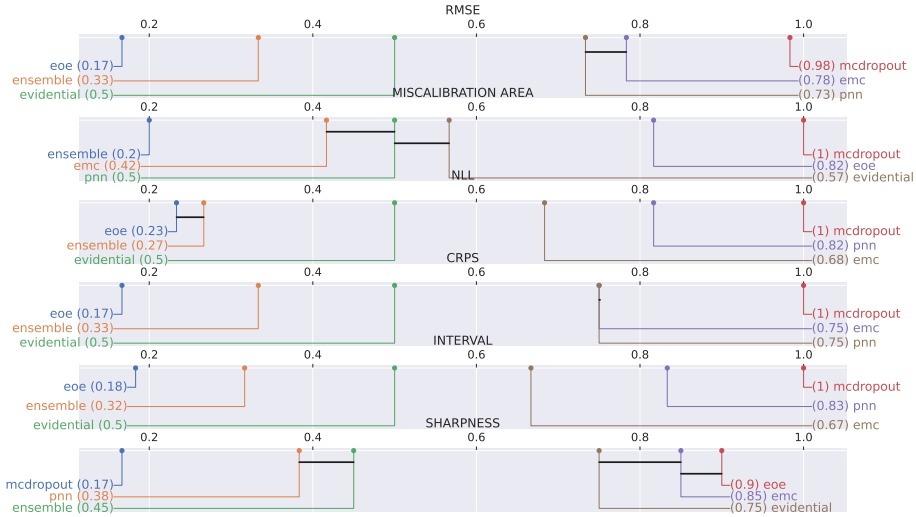


Figure S2.12: Critical difference diagrams for xC_{50} (Scaffold-cluster split). Same conventions as above: average ranks across seeds (lower is better) and cliques indicating pairs/groups without significant differences at the family-wise level according to the post-hoc p-value matrix. The plot summarizes rank structure; effect sizes are not shown.

S2.3. Statistical Significance Analyses

K_x - Stratified

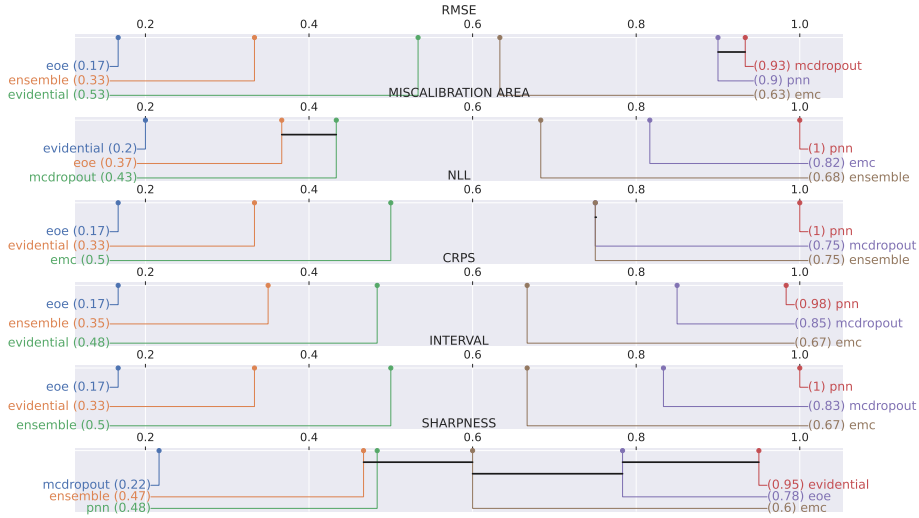


Figure S2.13: Critical difference diagrams for K_x (stratified split). Same conventions as above: average ranks across seeds (lower is better) and cliques indicating pairs/groups without significant differences at the family-wise level according to the post-hoc p-value matrix. The plot summarizes rank structure; effect sizes are not shown.

K_x - Scaffold-cluster

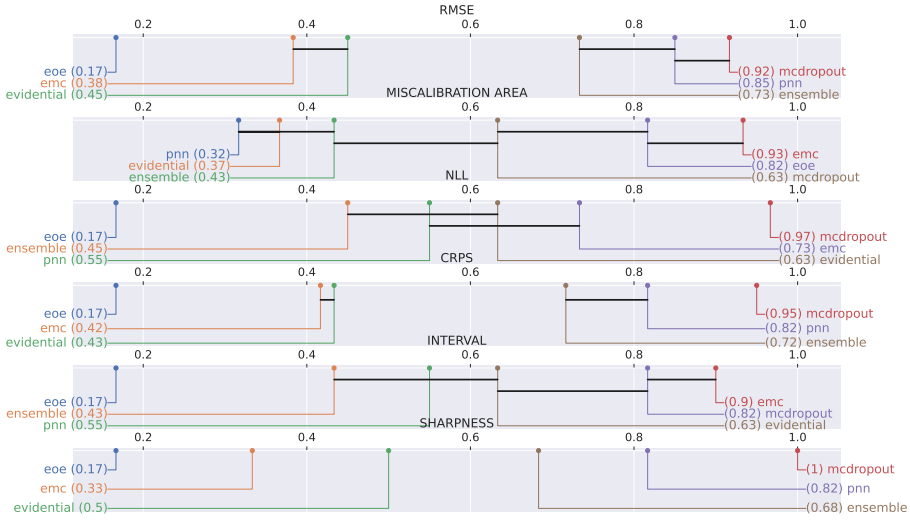


Figure S2.14: Critical difference diagrams for K_x (Scaffold-cluster split). Same conventions as above: average ranks across seeds (lower is better) and cliques indicating pairs/groups without significant differences at the family-wise level according to the post-hoc p-value matrix. The plot summarizes rank structure; effect sizes are not shown.

S2.4 Additional Figures

S2.4.1 Pairplots showing Correlations among Metrics

To further investigate relationships among evaluation metrics, we provide pairplots illustrating the pairwise correlations between both accuracy and uncertainty metrics across all models and datasets (Figures S2.15–S2.18). These visualizations help uncover potential metric redundancies and complementarities. For accuracy metrics, we observe generally strong positive correlations. In contrast, the uncertainty metric pairplots reveal more diverse relationships. These findings highlight the value of a multidimensional uncertainty assessment framework, justifying the selection of five non-redundant UQ metrics in the main analysis.

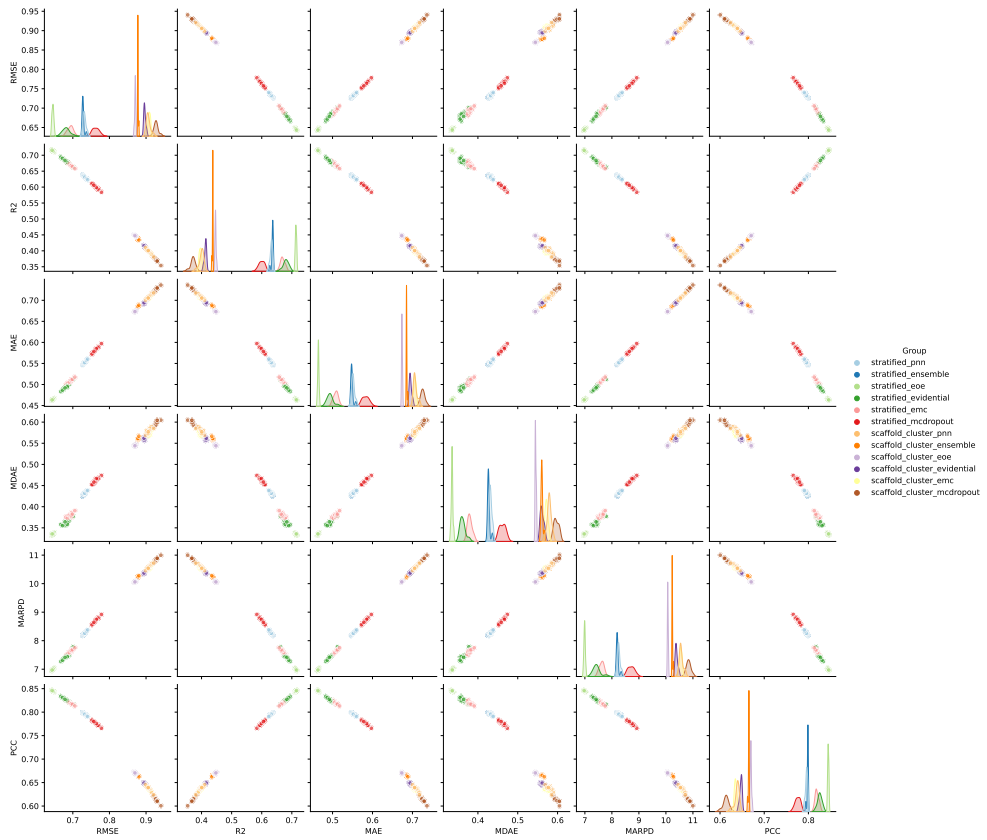


Figure S2.15: Pairplot of accuracy metrics comparing their pairwise relationships for models trained on the xC_{50} dataset.

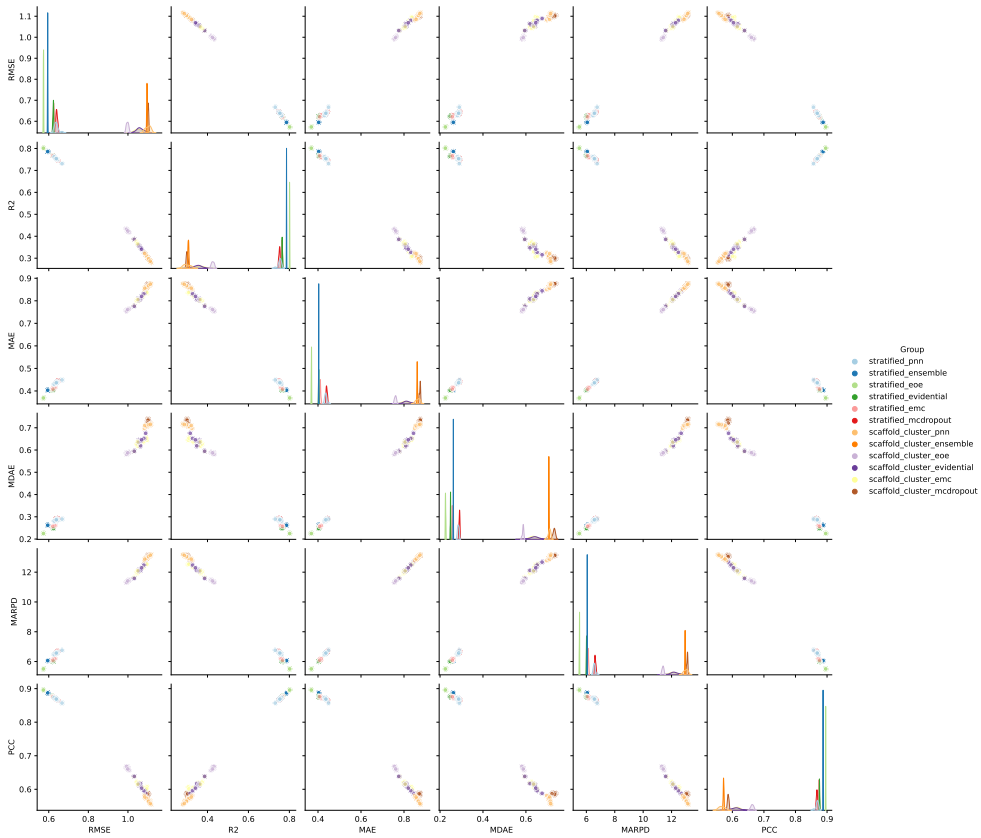


Figure S2.16: Pairplot of accuracy metrics comparing their pairwise relationships for models trained on the K_x dataset.

S2.4. Additional Figures

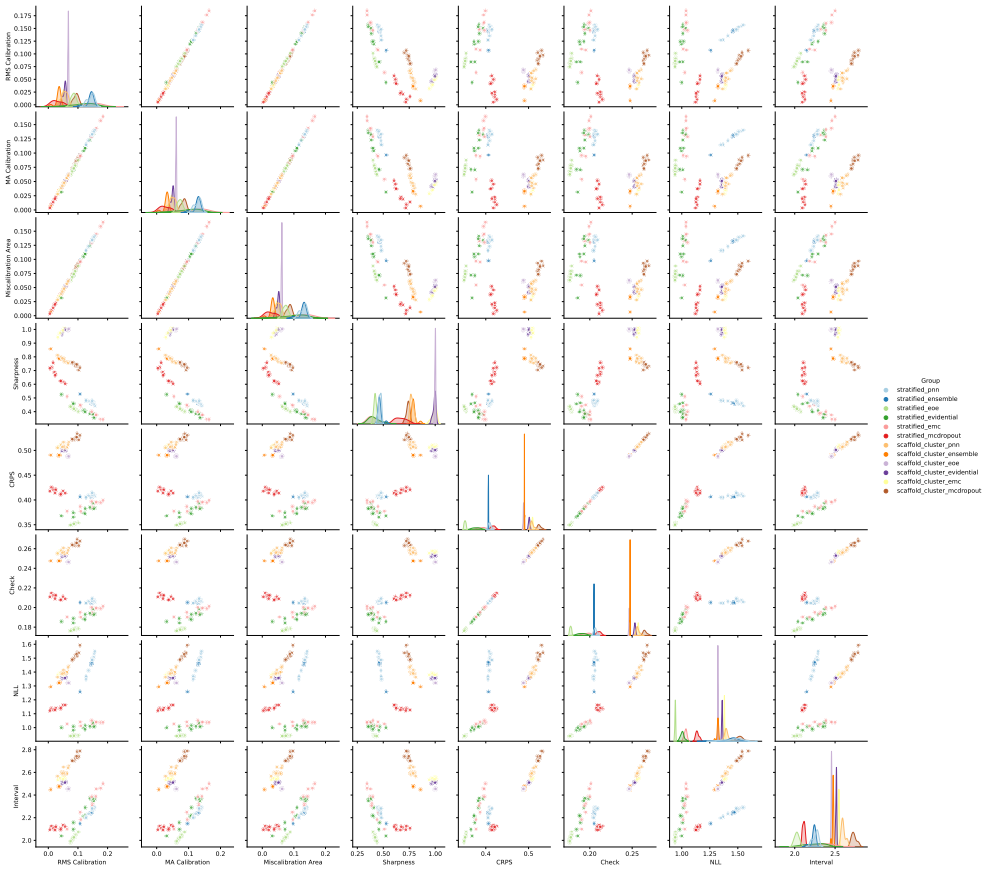


Figure S2.17: Pairplot of uncertainty metrics showing pairwise relationships for models evaluated on the xC_{50} dataset.

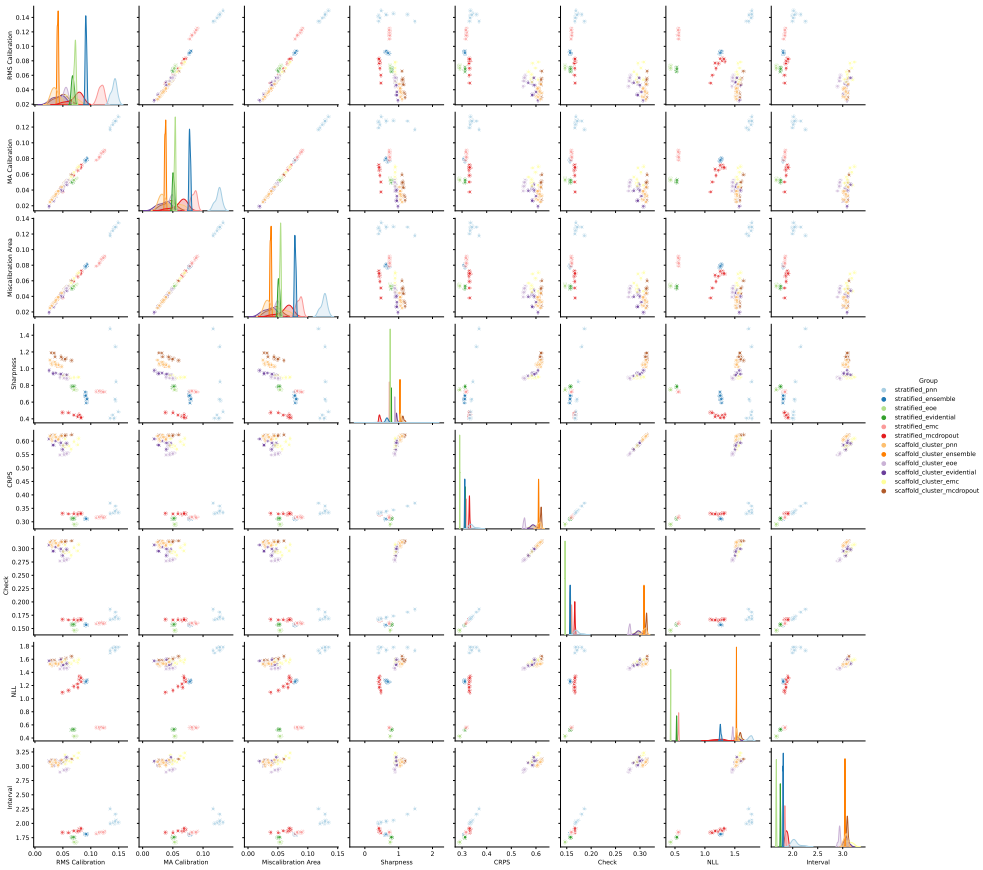
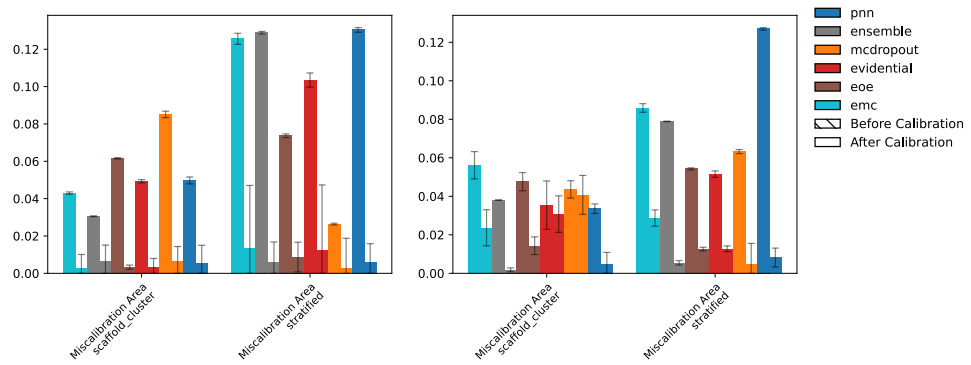


Figure S2.18: Pairplot of uncertainty metrics showing pairwise relationships for models evaluated on the K_x dataset.

S2.4.2 Comparison Before and After Isotonic Regression Recalibration

To assess the effect of post-hoc uncertainty recalibration, we applied isotonic regression on the estimated confidence intervals using the calibration (validation) set and evaluated the recalibrated outputs on the independent test set. This process adjusts the estimated cumulative distribution functions to better match the empirical calibration observed in the validation data. While we observe modest improvements in calibration-related metrics after isotonic recalibration–indicating potential value for deployment–these adjustments do not substantially alter the relative performance rankings among the evaluated models. This suggests that while post-hoc recalibration may enhance individual model reliability, it does not meaningfully change the comparative insights drawn from the main evaluation, where intrinsic uncertainty estimation quality remains the primary differentiator.



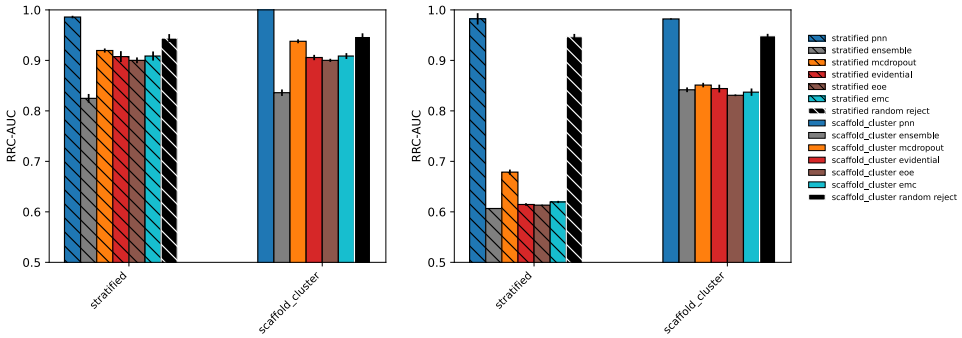
(a) xC_{50} Dataset **(b)** K_x Dataset

Figure S2.19: Comparison of Miscalibration Area for models trained on the xC_{50} and K_x datasets. Lower values indicate better calibration.

S2.4.3 RRC-AUC

To facilitate direct comparison of uncertainty-based rejection performance across models, we include bar plots of the RRC-AUC values here in the supporting figures. While the main text presents RRCs with corresponding AUC values annotated in the legends, the bar plot representation allows for a more intuitive and side-by-side comparison across datasets and splitting strategies. This visualization reinforces the relative ranking trends discussed in the main results, where EOE_{10} and Ensemble_{100} gener-

ally demonstrate superior selective estimation capabilities, particularly under high-uncertainty conditions.



(a) xC_{50} **(b)** K_x

Figure S2.20: AUC comparison for uncertainty-based rejection curves on xC_{50} and K_x datasets. Lower AUC values indicate better uncertainty-based rejection performance.

S2.5 Detailed Results Tables

Model	Activity	Aleatoric Uncertainty	Epistemic Uncertainty	Total Uncertainty
Stratified				
PNN_1	xC_{50}	.441 (.013)	N/A	.441 (.013)
$Ensemble_{100}$	xC_{50}	.444 (.021)	.010 (.005)	.454 (.026)
$MC\text{-}Dropout_{100}$	xC_{50}	.659 (.052)	.309 (.023)	.968 (.073)
$Evidential_1$	xC_{50}	.402 (.051)	2.481 (.143)	2.883 (.189)
EOE_{10}	xC_{50}	.412 (.012)	2.452 (.026)	2.864 (.036)
EMC_{10}	xC_{50}	.389 (.050)	2.382 (.138)	2.771 (.183)
PNN_1	K_x	.367 (.054)	N/A	.367 (.054)
$Ensemble_{100}$	K_x	.381 (.005)	.058 (.001)	.439 (.005)
$MC\text{-}Dropout_{100}$	K_x	.392 (.018)	.237 (.007)	.629 (.025)
$Evidential_1$	K_x	.657 (.007)	2.547 (.039)	3.204 (.043)
EOE_{10}	K_x	.608 (.002)	2.376 (.012)	2.984 (.013)
EMC_{10}	K_x	.581 (.005)	2.329 (.038)	2.910 (.041)
Scaffold-cluster				
PNN_1	xC_{50}	.755 (.021)	N/A	.755 (.021)
$Ensemble_{100}$	xC_{50}	.776 (.023)	.055 (.005)	.831 (.028)
$MC\text{-}Dropout_{100}$	xC_{50}	.705 (.016)	.234 (.017)	.939 (.027)
$Evidential_1$	xC_{50}	.992 (.014)	5.431 (.136)	6.422 (.132)
EOE_{10}	xC_{50}	1.000 (.004)	5.463 (.033)	6.463 (.036)
EMC_{10}	xC_{50}	.988 (.023)	5.221 (.102)	6.209 (.116)
PNN_1	K_x	1.005 (.020)	N/A	1.005 (.020)
$Ensemble_{100}$	K_x	.986 (.003)	.010 (.000)	.996 (.003)
$MC\text{-}Dropout_{100}$	K_x	1.048 (.030)	.147 (.010)	1.196 (.038)
$Evidential_1$	K_x	.909 (.029)	5.276 (.500)	6.185 (.519)
EOE_{10}	K_x	.840 (.008)	4.590 (.100)	5.429 (.106)
EMC_{10}	K_x	.862 (.017)	4.771 (.462)	5.633 (.474)

Table S2.2: Average Aleatoric, Epistemic, and Total Uncertainties across Models and Splits. This table reports the mean values of estimated aleatoric, epistemic, and total uncertainty for each model, evaluated on xC_{50} and K_x datasets under both stratified and scaffold-cluster splits. Evidential-based models (**$Evidential_1$**, **EOE_{10}**, and **EMC_{10}**) tend to produce substantially higher epistemic uncertainty estimates compared to Bayesian or probabilistic baselines, particularly under scaffold-cluster settings. These elevated values reflect their sensitivity to distributional shifts and capacity to express model uncertainty in regions of low training support.

Model	Activity	Performance Metrics (σ)		
		RMSE ↓	R^2 ↑	PCC ↑
Stratified				
PNN_1	xC_{50}	.732 (.004)	.632 (.004)	.796 (.003)
$Ensemble_{100}$	xC_{50}	.728 (.003)	.636 (.003)	.798 (.002)
$MC\text{-}Dropout_{100}$	xC_{50}	.763 (.008)	.600 (.009)	.776 (.006)
$Evidential_1$	xC_{50}	.682 (.010)	.681 (.009)	.826 (.005)
EOE_{10}	xC_{50}	**.646 (.002)**	**.714 (.002)**	**.845 (.001)**
EMC_{10}	xC_{50}	.693 (.007)	.670 (.007)	.819 (.004)
PNN_1	K_x	.639 (.011)	.753 (.009)	.869 (.005)
$Ensemble_{100}$	K_x	.595 (.001)	.786 (.001)	.887 (.000)
$MC\text{-}Dropout_{100}$	K_x	.641 (.004)	.752 (.003)	.868 (.002)
$Evidential_1$	K_x	.623 (.002)	.765 (.001)	.876 (.001)
EOE_{10}	K_x	**.573 (.001)**	**.801 (.001)**	**.896 (.000)**
EMC_{10}	K_x	.626 (.003)	.763 (.002)	.874 (.001)
Scaffold-cluster				
PNN_1	xC_{50}	.907 (.006)	.400 (.008)	.638 (.005)
$Ensemble_{100}$	xC_{50}	.878 (.001)	.437 (.001)	.665 (.001)
$MC\text{-}Dropout_{100}$	xC_{50}	.928 (.006)	.371 (.008)	.613 (.006)
$Evidential_1$	xC_{50}	.897 (.003)	.413 (.004)	.647 (.003)
EOE_{10}	xC_{50}	**.871 (.001)**	**.447 (.002)**	**.670 (.001)**
EMC_{10}	xC_{50}	.909 (.003)	.397 (.004)	.636 (.003)
PNN_1	K_x	1.102 (.012)	.300 (.016)	.567 (.009)
$Ensemble_{100}$	K_x	1.097 (.002)	.306 (.002)	.572 (.001)
$MC\text{-}Dropout_{100}$	K_x	1.101 (.003)	.301 (.004)	.587 (.002)
$Evidential_1$	K_x	1.061 (.017)	.351 (.021)	.611 (.014)
EOE_{10}	K_x	**.998 (.005)**	**.426 (.006)**	**.663 (.004)**
EMC_{10}	K_x	1.061 (.023)	.350 (.029)	.616 (.011)

Table S2.3: Performance Metrics (RMSE, R^2, PCC) for xC_{50} and K_x activities across Stratified and Scaffold-cluster Splits. Bold values indicate the best-performing model within each combination of activity type and data split. Values in parentheses represent the standard deviation across 10 independent runs to ensure reproducibility. Across all experimental settings, EOE_{10} consistently achieves superior performance compared to $Evidential_1$, EMC_{10}, $Ensemble_{100}$, and MC $Dropout_{100}$, demonstrating its robustness and efficiency in modeling.

Model	Activity	Uncertainty Metrics (σ)							
		Miscalib. Area ↓	RMS Calib. ↓	MA Calib. ↓	NLL ↓	CRPS ↓	Check ↓	Interval ↓	Sharpness ↓
Stratified									
PNN_1	xC_{50}	.133 (.009)	.146 (.010)	.132 (.009)	1.478 (.063)	.409 (.003)	.207 (.001)	2.264 (.031)	.465 (.012)
$Ensemble_{100}$	xC_{50}	.129 (.011)	.142 (.012)	.128 (.011)	1.441 (.065)	.406 (.000)	.205 (.000)	2.234 (.030)	.469 (.021)
$MC\text{-}Dropout_{100}$	xC_{50}	.026 (.016)	.029 (.017)	.026 (.016)	1.140 (.014)	.418 (.005)	.211 (.002)	2.111 (.015)	.669 (.052)
$Evidential_1$	xC_{50}	.103 (.035)	.118 (.037)	.102 (.035)	1.005 (.015)	.378 (.009)	.191 (.004)	2.247 (.121)	.414 (.053)
EOE_{10}	xC_{50}	.074 (.008)	.086 (.009)	.073 (.008)	.943 (.003)	.352 (.002)	.177 (.001)	2.028 (.029)	.422 (.012)
EMC_{10}	xC_{50}	.125 (.034)	.140 (.037)	.124 (.034)	1.033 (.011)	.389 (.008)	.196 (.004)	2.335 (.125)	.402 (.052)
PNN_1	K_x	.127 (.005)	.142 (.005)	.126 (.005)	1.757 (.030)	.341 (.013)	.172 (.006)	2.059 (.084)	.699 (.383)
$Ensemble_{100}$	K_x	.079 (.001)	.091 (.001)	.078 (.001)	1.261 (.010)	.311 (.001)	.157 (.000)	1.809 (.005)	.655 (.037)
$MC\text{-}Dropout_{100}$	K_x	.063 (.011)	.074 (.011)	.063 (.011)	1.230 (.083)	.330 (.001)	.167 (.001)	1.874 (.025)	.439 (.022)
$Evidential_1$	K_x	.051 (.001)	.067 (.003)	.051 (.001)	.531 (.005)	.313 (.001)	.158 (.001)	1.754 (.006)	.787 (.005)
EOE_{10}	K_x	.054 (.001)	.071 (.002)	.053 (.001)	.428 (.002)	.291 (.000)	.147 (.000)	1.673 (.005)	.749 (.002)
EMC_{10}	K_x	.086 (.004)	.119 (.005)	.085 (.004)	.560 (.005)	.317 (.002)	.160 (.001)	1.846 (.014)	.725 (.004)
Scaffold-cluster									
PNN_1	xC_{50}	.049 (.010)	.054 (.011)	.049 (.010)	1.403 (.024)	.509 (.004)	.257 (.002)	2.608 (.030)	.777 (.018)
$Ensemble_{100}$	xC_{50}	.031 (.008)	.034 (.009)	.030 (.008)	1.320 (.009)	.490 (.000)	.248 (.000)	2.474 (.009)	.795 (.023)
$MC\text{-}Dropout_{100}$	xC_{50}	.085 (.008)	.093 (.009)	.084 (.008)	1.522 (.031)	.525 (.004)	.265 (.002)	2.735 (.030)	.738 (.017)
$Evidential_1$	xC_{50}	.049 (.005)	.054 (.005)	.048 (.005)	1.360 (.004)	.502 (.002)	.253 (.001)	2.514 (.006)	.992 (.013)
EOE_{10}	xC_{50}	.062 (.001)	.067 (.001)	.061 (.001)	1.323 (.001)	.488 (.001)	.247 (.000)	2.455 (.005)	1.000 (.004)
EMC_{10}	xC_{50}	.042 (.008)	.046 (.008)	.042 (.008)	1.383 (.004)	.508 (.001)	.257 (.001)	2.545 (.007)	.989 (.022)
PNN_1	K_x	.034 (.006)	.036 (.006)	.033 (.006)	1.531 (.014)	.614 (.006)	.310 (.003)	3.060 (.034)	1.059 (.022)
$Ensemble_{100}$	K_x	.038 (.001)	.041 (.001)	.038 (.001)	1.527 (.002)	.611 (.001)	.308 (.000)	3.045 (.005)	1.038 (.004)
$MC\text{-}Dropout_{100}$	K_x	.044 (.010)	.047 (.011)	.043 (.010)	1.596 (.022)	.620 (.002)	.313 (.001)	3.092 (.017)	1.136 (.031)
$Evidential_1$	K_x	.035 (.009)	.042 (.010)	.035 (.009)	1.539 (.027)	.589 (.010)	.297 (.005)	3.072 (.041)	.939 (.018)
EOE_{10}	K_x	.047 (.004)	.053 (.005)	.047 (.004)	1.464 (.009)	.553 (.004)	.279 (.002)	2.929 (.017)	.889 (.006)
EMC_{10}	K_x	.056 (.009)	.063 (.010)	.056 (.009)	1.555 (.027)	.591 (.013)	.298 (.007)	3.133 (.065)	.904 (.011)

Table S2.4: Uncertainty Metrics for xC_{50} and K_x activities across Stratified and Scaffold-cluster Splits. The values include measures of calibration, sharpness, and proper scoring metrics. Lower values indicate better performance in calibration and uncertainty estimation.

Model	Activity	ρ_{rank} ↑	RMV vs RMSE			RRC-AUC ↓
			R^2 ↑	Slope	Intercept	
Stratified						
PNN_1	xC_{50}	.247 (.007)	.864 (.020)	1.022 (.045)	.263 (.028)	.986 (.003)
$Ensemble_{100}$	xC_{50}	.259 (.003)	.865 (.011)	1.043 (.012)	.235 (.019)	**.825 (.009)**
$MC\text{-}Dropout_{100}$	xC_{50}	.164 (.014)	.781 (.021)	.615 (.051)	.162 (.019)	.920 (.004)
$Evidential_1$	xC_{50}	.017 (.021)	.078 (.057)	.023 (.011)	.612 (.025)	.907 (.011)
EOE_{10}	xC_{50}	.021 (.010)	.077 (.048)	.027 (.008)	.563 (.023)	.900 (.006)
EMC_{10}	xC_{50}	.026 (.016)	.090 (.065)	.024 (.012)	.624 (.030)	.909 (.009)
Random Rejection	xC_{50}	-	-	-	-	.945 (.008)
PNN_1	K_x	.495 (.012)	.423 (.150)	.400 (.261)	.453 (.085)	.982 (.011)
$Ensemble_{100}$	K_x	.529 (.002)	.277 (.010)	.190 (.016)	.459 (.007)	**.607 (.001)**
$MC\text{-}Dropout_{100}$	K_x	.471 (.007)	.755 (.023)	.649 (.053)	.190 (.031)	.679 (.005)
$Evidential_1$	K_x	.590 (.002)	.903 (.011)	.107 (.001)	.229 (.005)	.615 (.003)
EOE_{10}	K_x	.588 (.002)	.865 (.005)	.098 (.001)	.232 (.003)	.613 (.001)
EMC_{10}	K_x	.572 (.003)	.853 (.009)	.104 (.002)	.272 (.004)	.620 (.002)
Random Rejection	K_x	-	-	-	-	.949 (.005)
Scaffold-cluster						
PNN_1	xC_{50}	.119 (.011)	.510 (.089)	.347 (.040)	.641 (.031)	1.008 (.003)
$Ensemble_{100}$	xC_{50}	.167 (.003)	.618 (.035)	.513 (.014)	.443 (.025)	**.836 (.006)**
$MC\text{-}Dropout_{100}$	xC_{50}	.150 (.010)	.690 (.049)	.399 (.030)	.546 (.021)	.938 (.004)
$Evidential_1$	xC_{50}	.086 (.006)	.307 (.049)	.048 (.005)	.587 (.030)	.906 (.005)
EOE_{10}	xC_{50}	.098 (.003)	.312 (.029)	.052 (.003)	.530 (.020)	.900 (.003)
EMC_{10}	xC_{50}	.075 (.004)	.309 (.054)	.045 (.005)	.628 (.034)	.909 (.006)
Random Rejection	xC_{50}	-	-	-	-	.948 (.006)
PNN_1	K_x	.237 (.006)	.860 (.013)	.541 (.019)	.540 (.019)	.982 (.001)
$Ensemble_{100}$	K_x	.240 (.001)	.860 (.004)	.563 (.003)	.518 (.001)	.842 (.005)
$MC\text{-}Dropout_{100}$	K_x	.227 (.005)	.908 (.011)	.425 (.012)	.573 (.014)	.851 (.004)
$Evidential_1$	K_x	.238 (.015)	.778 (.041)	.056 (.005)	.706 (.018)	.844 (.008)
EOE_{10}	K_x	.258 (.006)	.835 (.025)	.062 (.002)	.652 (.011)	**.831 (.002)**
EMC_{10}	K_x	.246 (.018)	.826 (.051)	.062 (.006)	.707 (.026)	.837 (.007)
Random Rejection	K_x	-	-	-	-	.950 (.003)

Table S2.5: Rank-Based Metrics for xC_{50} and K_x activities across Stratified and Scaffold-cluster Splits. Higher ρ_{rank} and R^2 values indicate better correlation and calibration. The slope should ideally be close to 1, and the intercept should be close to 0

Chapter 3

Reading Between the Assays

Assay-Aware Modeling
for Robust Bioactivity Prediction and Uncertainty Quantification

Based on the publication

PCAM: Proteochemometric Assay-aware Modeling for Robust Bioactivity Estimation and Uncertainty Quantification

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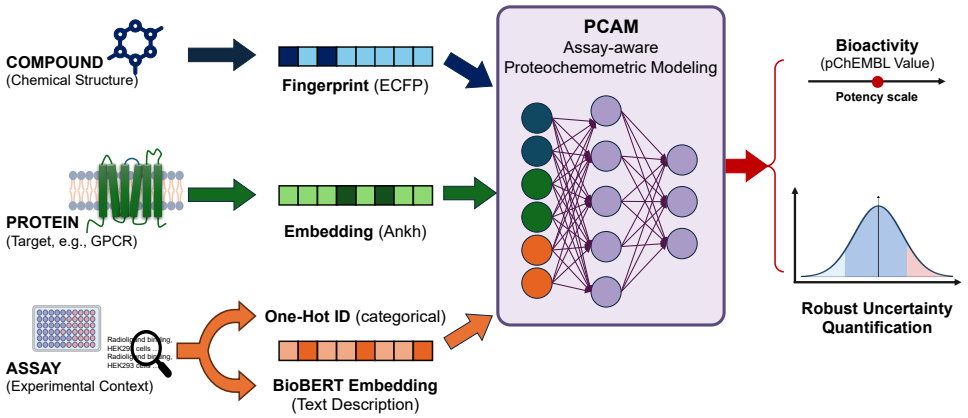


Figure 3.1: Graphical Abstract

Public bioactivity databases aggregate measurements from heterogeneous experimental assays, introducing systematic variability that can degrade model performance and distort uncertainty estimates. We present proteochemometric assay-aware modeling (PCAM), an extension of proteochemometrics that integrates assay descriptors alongside compound and protein features to capture assay context. Using a curated ChEMBL-derived benchmark spanning 167 G protein-coupled receptor targets, 77,081 compounds, and 5,161 assays, we compare models that omit assay information to models using either categorical assay identifiers or text-derived embeddings from assay descriptions. We evaluate generalization under three regimes of increasing distribution shift (random, scaffold-based, and assay-held-out splits) and assess both point accuracy and probabilistic quality. Overall difficulty is dominated by the split strategy, with assay-held-out evaluation inducing the strongest distribution shift. When assay identifiers are shared between training and evaluation, categorical assay descriptors provide the most consistent gains in accuracy and probabilistic scoring relative to models without assay context. Under assay-held-out evaluation, differences between assay representations contract, while text-derived embeddings show a consistent but typically small advantage. This regime corresponds to practical assay-transfer scenarios, such as protocol or vendor changes, new readouts, or sparse early measurements for a newly introduced assay, where leveraging assay text/metadata can help transfer signal from related assays. In addition, assay representations can reduce reliance on brittle cross-assay normalization in heterogeneous public datasets by allowing the model to learn assay-dependent shifts directly; however, we observed optimization

instability for one configuration (a probabilistic neural network using BioBERT embeddings under the random split).

3.1 Introduction

In drug discovery, modeling ligand–protein interactions is important for guiding experimental design. Traditional quantitative structure–activity relationship (QSAR) models relate chemical structure to bioactivity, but are typically confined to a single target. Proteochemometric (PCM) modeling generalizes QSAR by training on multiple proteins and ligands concurrently within a unified machine learning framework [76, 77]. This is achieved by integrating descriptors for both compounds and protein targets, enabling extrapolation to novel target-ligand pairs with support for multi-target drug design workflows [78]. PCM has been successfully applied across various protein families, including G protein-coupled receptors, kinases, and viral proteases, where performance benefits from cross-target information sharing [17, 79, 80]. These findings demonstrate PCM's suitability for large-scale bioactivity analysis.

Despite their promise, PCM models still face limitations in uncertainty awareness and generalization beyond the training domain. Standard PCM models typically yield point estimates without quantifying uncertainty, leaving users without explicit information about the reliability of each output [29, 78]. Without uncertainty quantification (UQ) and with imperfect calibration, models can appear more reliable than they are and may mislead decision-making. Furthermore, PCM models often struggle with domain generalization: performance can degrade significantly when confronted with new chemical scaffolds or protein targets not represented in the training data. In practice, such out-of-distribution (OOD) inputs lead to unreliable outputs, underscoring the insufficient robustness of current models. Although ensemble-based methods [31], Bayesian neural networks, and stochastic regularization techniques such as Monte Carlo (MC) dropout [32] have been proposed to address these issues, achieving reliable and well-calibrated uncertainty estimates remains an open challenge in proteochemometric settings [81].

In addition, the heterogeneity of experimental assays in public bioactivity datasets is often neglected. Public bioactivity databases such as ChEMBL aggregate results from different laboratories and assay protocols, which can introduce systematic shifts and inconsistencies between measurements. As a result, the same compound–target pair may display different potency values depending on the assay conditions, even when measured in the same units. Combining measurements across distinct assays without accounting for assay differences can add substantial noise: for instance, nearly half of repeated compound measurements in separate IC_{50} assays deviate by more than 0.3 log units (i.e., a factor of 2) between assays, even after careful curation [14]. The

inter-assay variability imposes an effective upper bound on achievable model accuracy (the maximum Pearson R^2 is estimated around 0.8 for large-scale K_i datasets) [82]. Ignoring assay context not only reduces performance but also compromises uncertainty estimates, as confidence may be miscalibrated across heterogeneous experimental settings.

To address these challenges, we build on recent work demonstrating that assay embeddings improved model performance for GPCRs [83]. This leads to our introduction of *proteochemometric assay-aware modeling* (PCAM) modeling combined with UQ as an extension of the PCM paradigm, using GPCRs as a validated test case. PCAM augments the traditional compound–protein feature space by incorporating descriptors of the experimental assay for each data point. By explicitly modeling assay context, the PCAM approach can capture inter-assay systematic effects, aiming to improve performance and calibration across heterogeneous bioactivity data. Two types of assay representations are explored: a dense embedding derived from textual assay descriptions using a pre-trained biomedical language model (BioBERT), and a sparse one-hot encoding of assay identifiers as a baseline for comparison [83, 84]. These representations differ fundamentally in their inductive biases: one-hot encodings emphasize assay-specific effects, whereas language-based embeddings allow for information sharing across related experimental protocols. The PCAM framework is evaluated on large-scale bioactivity regression tasks (pChEMBL value estimation) using a CAPRICHO-curated GPCR dataset containing both IC_{50} and K_i measurements [85]. We compare PCAM to PCM and evaluate both when applied to state-of-the-art UQ methods – namely, deep ensembles, and Monte Carlo dropout [31, 32] – and also assess the impact of different protein encoding strategies (contrasting a modern protein language model embedding, e.g., Ankh-Large [1], with a simple one-hot protein representation). The inclusion of assay descriptors into proteochemometric modeling constitutes the key methodological contribution of this work. In practice, semantic assay representations become relevant when assay context is new or evolving, for example, when a program introduces a new readout, changes vendor or protocol, or has sparse early measurements for a newly defined assay. In such settings, assay text and metadata can enable transfer from related assays even when the assay identifier itself has not been seen during training. A second motivation is robustness in heterogeneous public data: rather than forcing one-size-fits-all cross-assay normalization across mixed protocols, assay descriptors allow the model to learn assay-dependent shifts directly, addressing known inter-assay variability issues reported in large-scale curation studies [14]. Our results demonstrate that PCAM with one-hot encoded as-

say identifiers significantly outperforms standard PCM across performance metrics (RMSE, PCC) and uncertainty quality (CRPS) under interpolation conditions, while BioBERT embeddings enable extrapolation to novel assay types. These findings pave the way for more reliable and translatable machine learning models in drug discovery.

Beyond descriptive performance comparisons, reliable assessment of modeling advances requires careful consideration of both statistical consistency and practical relevance. In experimental settings involving repeated training runs, multiple data splits, and correlated evaluation blocks, small numerical differences may appear consistent yet offer limited practical value, while larger effects may prove unstable across repetitions. To address this, we complement aggregate performance summaries with formal statistical testing under a repeated-measures design and explicitly distinguish statistical detectability from effect magnitude and stability. This distinction is important for interpreting improvements in applied modeling settings, particularly when models are intended to inform experimental decisions rather than optimize benchmark scores [60, 81].

3.2 Results

We compared the PCM and PCAM approaches to determine whether explicit assay context improves bioactivity modeling and uncertainty quantification. We evaluated three modeling approaches: PCM (compound and protein descriptors only), PCAM-OneHot (adds one-hot encoded assay identifiers), and PCAM-BioBERT (adds BioBERT embeddings from assay descriptions). These represent a progression from ignoring assay context (PCM), to memorizing assay-specific effects (PCAM-OneHot), to capturing semantic relationships between protocols (PCAM-BioBERT). We assessed model performance under three data splitting strategies of increasing difficulty: random splits (interpolation), scaffold splits (chemical generalization), and assay splits (assay-transfer extrapolation, approximating scenarios such as protocol or vendor changes, new readouts, or limited early measurements for a newly introduced assay definition).

3.2.1 Data and Task

We identified 365 human GPCR targets for bioactivity data collection by cross-referencing UniProt [86] with ChEMBL [12] identifiers (see Section 3.5). The G-protein coupled receptor 1 family comprised the largest subset, with 275 targets, followed by 33 from

the G-protein coupled receptor 2 family, 25 from the T2R taste receptor family, 21 from the G-protein coupled receptor 3 family, and 9 from the Frizzled/Smoothened family, among others. Two non-GPCR proteins (G protein gamma and WD repeat G protein beta subunits) were excluded as G protein signaling components rather than receptors.

We assessed the quality of the curated GPCR bioactivity data by evaluating cross-assay activity concordance for compound-target activities measured in multiple assays through pairwise comparisons. Data concordance was quantified using Kendall's τ, Spearman's ρ, and R^2 as comparability metrics. This enables evaluating how different data curation steps affect measurement reliability using the Python package CAPRICHO (Figure 3.2). Each panel in Figure 3.2 isolates entries marked with a specific quality flag, revealing how that data subset deviates from perfect concordance. Panels 1 to 7 summarize dropping flags that indicate potential quality concerns, while panels 8 to 10 summarize processing flags that document transformations applied during curation. Data points with activity comments (panel 1) added by ChEMBL curators substantially reduced cross-assay comparability and were removed. Additionally, entries marked as "potential duplicate" in ChEMBL (panel 2) were removed to prevent distributional shifts on the pChEMBL value mean, used as the target variable for the modeling experiments. Following part of the max curation standards indicated by Landrum and Riniker 2024 [14], data points reported in assays with more than 100 or fewer than 20 compounds (panels 5 and 6, respectively) and activities with detected unit annotation errors (panel 7) were removed from the dataset. Calculated pChEMBL values derived from nM, µM, or mM measurements (panel 9) were also removed from the dataset, as those did not show sufficient cross-assay concordance. Finally, six assays reporting activities on GPR6 exhibited systematic unit annotation errors observed after salt-stripping, with shared compounds differing by exactly 6.0 log units between assays (Table S3.3), which were removed from the final dataset.

Although all assays contribute additional measurements, we applied lower and upper assay-size thresholds under the hypothesis that smaller assays may be performed to answer specific research questions and have, therefore, lower comparability with other assays reported in ChEMBL. Conversely, assays testing a higher number of compounds can be derived from sources other than scientific literature (e.g., large screening data deposition) and, while robust for its specific conditions, may have lower comparability with other lower-throughput assays, as observed in Figure 3.2 and indicated by Landrum and Riniker 2024 [14]. After removal of the flagged entries, a final curated all-GPCR dataset was obtained, containing 167 GPCR targets with 77,081 unique com-

3.2. Results

pounds across 5,161 distinct assays to be used for PCAM modeling, including IC_{50}, EC_{50}, AC_{50}, K_i, K_d. Pairwise cross-assay comparability of the bioactivity points showed $\tau = 0.51$, $\rho = 0.68$, and $R^2 = 0.35$. While higher correlation coefficients have been reported under previous max curation standards [14], we believe our modeling approach to distinguish between different assay types, potentially benefiting from the presence of multiple assays reporting activities on the same compound-target pair.

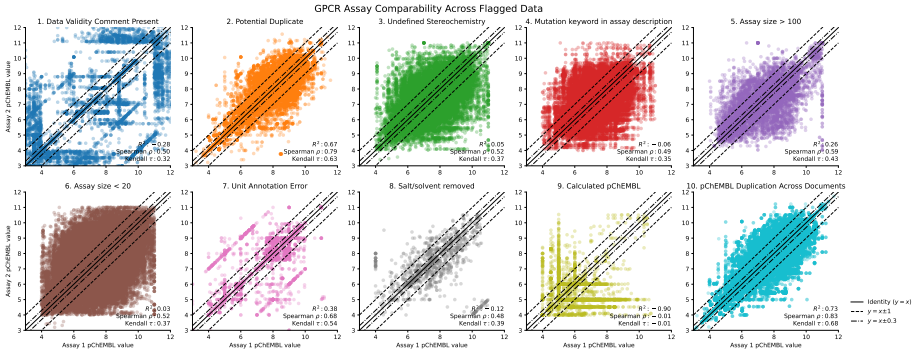


Figure 3.2: Effect of different CAPRICHO data dropping (1-7, indicating potential quality concerns) and data processing flags (8-10, indicating processing steps taken by the curation workflow) on cross-assay comparability. Data dropping steps: 1. ChEMBL entries containing a `data_validity_comment`. 2. Entries containing a `potential_duplicate` flag from ChEMBL. 3. ChEMBL entries with undefined stereocenters. 4. Entries containing a mutation keyword in the `assay_description`. 5. Activities reported in assays testing more than 100 compounds. 6. Activities reported in assays testing fewer than 20 compounds. 7. Entries with potential unit annotation error, differing in exactly three or six log units. Data processing flags: 8. Cross-assay comparability of same-compound activities identified post salt removal. 9. Cross-assay comparability of calculated pChEMBL values from activity values reported in nM, µM or mM `standard_units`. 10. Duplicated pChEMBL value across different documents based on the same-value lookups.

3.2.2 Metric Selection Through Correlation Analysis

Before comparing modeling approaches, we first identified which evaluation metrics provide non-redundant information. We examined correlations among six candidate performance metrics and eight candidate UQ metrics across all models and split strategies (Figure 3.3).

Performance metrics

For performance evaluation, we examined correlations between RMSE, coefficient of determination (R^2), mean absolute error (MAE), median absolute error (MDAE), mean absolute relative percent difference (MARPD), and Pearson Correlation Coefficient (PCC) (Figure 3.3). Traditional error metrics (RMSE, R^2, MAE, MDAE, MARPD) showed near-perfect correlation ($r \geq 0.99$), indicating redundancy of these metrics for our modeling results. In contrast, PCC exhibited moderate correlation with the other error metrics ($r \in [0.81, 0.87]$), indicating it measures a complementary aspect of performance as it represents the strength of linear association rather than absolute error. We selected RMSE and PCC as the primary performance metrics.

Uncertainty quantification (UQ) metrics

For UQ evaluation, we examined correlations among eight commonly used metrics (Figure 3.3). We observed substantial redundancy: three calibration metrics (RMS Calibration, MA Calibration, Miscalibration Area) are perfectly correlated ($r = 1.0$), as are CRPS and Check metrics, with Interval Score also showing near-perfect correlation with CRPS ($r = 0.99$). We therefore selected four non-redundant metrics capturing distinct aspects of uncertainty quality: Miscalibration Area (calibration accuracy), NLL (probabilistic estimation quality), CRPS (distribution compatibility combining sharpness and calibration), and Sharpness (output confidence). These metrics collectively capture complementary aspects of uncertainty quality.

3.2.3 Statistical Significance of Model Differences

With six non-redundant evaluation metrics established, we next assessed whether differences between modeling approaches (PCM, PCAM-OneHot, PCAM-BioBERT) are statistically significant. All comparisons were conducted under a repeated-measures design where each cross-validation fold provides matched observations for all methods, enabling paired comparisons that account for fold-specific variability.

Non-parametric statistical testing

To determine appropriate statistical tests, we examined the distribution of paired performance differences across cross-validation cycles (Supplementary Figure S3.1). Histograms and Q–Q plots revealed departures from Gaussian behavior, including skewness, heavy tails, and outliers. These patterns indicate that parametric assump-

3.2. Results

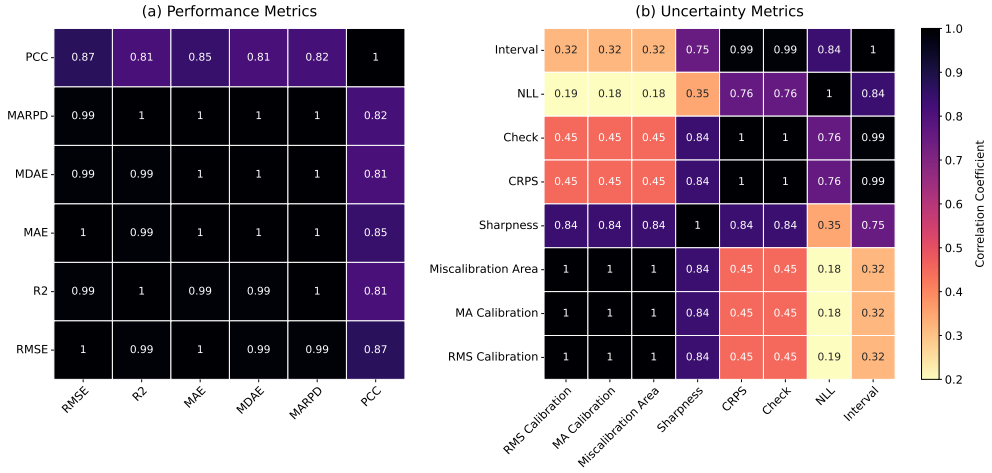


Figure 3.3: Correlation matrices for (a) performance metrics and (b) uncertainty metrics. Color scales range from light (low correlation) to dark (high correlation). Lower values indicate better performance for all selected metrics except PCC.

tions are not uniformly supported, motivating rank-based repeated-measures inference as the primary analysis for method comparison [60].

Pairwise significance patterns

To identify which modeling approaches differ significantly from each other, we performed pairwise statistical comparisons for each metric and split strategy. We present these comparisons as sign plots (Figure 3.4), where each cell represents a pairwise test between two methods. Cell shading indicates the strength of evidence against the null hypothesis of no difference: darker shading corresponds to smaller p-values (stronger evidence of difference), while "NS" indicates no significant difference.

Across all three split strategies (Figure 3.4), we observe consistent differences in pairwise separability that depend on metric type. For performance metrics (RMSE and PCC), the representation settings are typically well separated, with frequent statistically significant differences across splits. This indicates that changes in representation are associated with systematic shifts in held-out performance that persist across cross-validation cycles rather than being driven by a small subset of folds.

For uncertainty-related metrics, the strength and direction of separability depend on the specific criterion. Proper scoring rules and dispersion-oriented summaries (e.g., CRPS and Sharpness) tend to exhibit clear pairwise separation in multiple splits, while

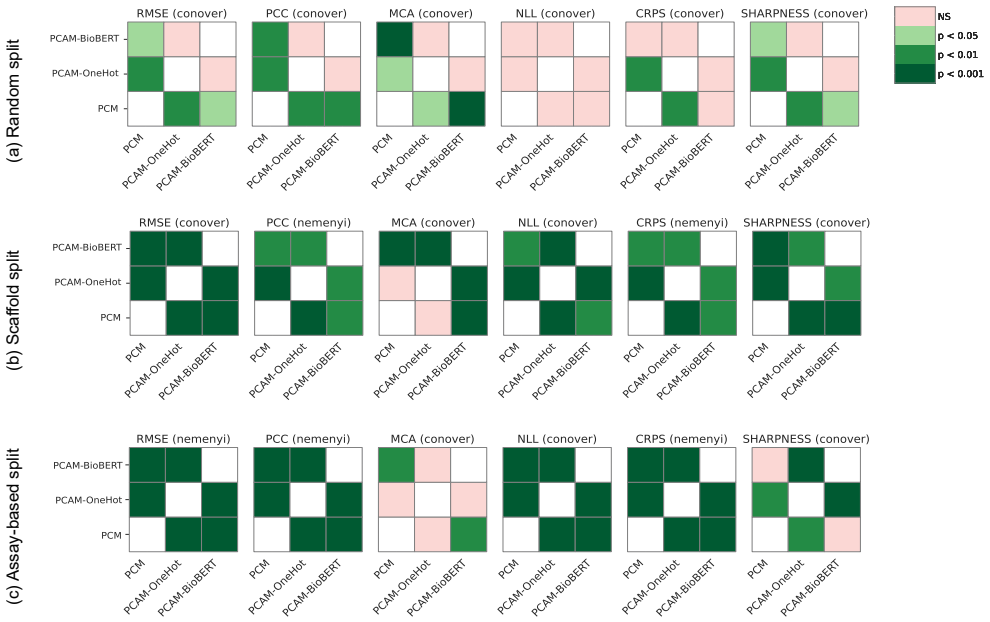


Figure 3.4: Split-wise sign plots for paired post hoc comparisons between representation settings across metrics. Darker shading indicates stronger evidence against the null (smaller p-values); "NS" denotes not significant.

calibration-sensitive criteria (e.g., Miscalibration Area and, in some settings, NLL) show more split-dependent behavior, including cases where differences are weaker or not significant for particular method pairs. This split dependence is consistent with the fact that calibration- and likelihood-based summaries can be more sensitive to the distribution shift induced by the split strategy.

The scaffold split often yields the strongest separation across methods, as reflected by the prevalence of highly significant cells across several metrics. This indicates that representation choice is particularly consequential under the more stringent chemical generalization regime, where systematic differences between methods are less likely to average out across folds.

Average-rank summaries via critical difference diagrams for all split strategies are provided in the Supplementary Information (Supplementary Figures S3.4 to S3.6).

3.2.4 Magnitude and Direction of Performance Differences

Building on the significance analysis, we characterized the direction and relative magnitude of performance differences across experimental conditions. We compared three task formulations (PCM, PCAM-OneHot, PCAM-BioBERT) across three split strategies (random, scaffold-based, assay-based) and three uncertainty quantification (UQ) methods (PNN, MC Dropout, Ensemble). Results across estimation accuracy (RMSE, PCC) and uncertainty metrics (Miscalibration Area (MCA), NLL, Sharpness, CRPS) are summarized in Figure 3.5. A complementary normalized view that highlights multi-metric trade-offs across splits and UQ methods is provided in the Supplementary Information (see Supplementary Figure S3.10).

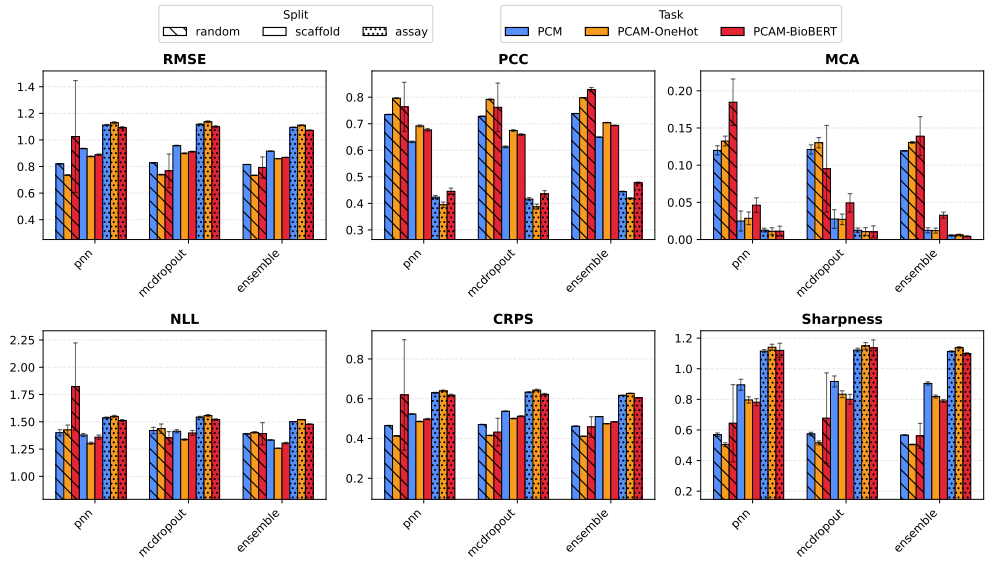


Figure 3.5: Estimation and uncertainty metrics across task variants (PCM, PCAM-OneHot, PCAM-BioBERT) and split strategies (random, scaffold, assay). Each panel shows one metric (RMSE, PCC, Miscalibration Area, NLL, Sharpness, CRPS). Error bars indicate standard deviation across five runs.

Estimation accuracy (RMSE, PCC)

Split strategy drives RMSE and PCC across all task formulations and UQ methods (Figure 3.5). Random splits yield the lowest RMSE and highest PCC, scaffold splits are intermediate, and assay splits are consistently the most challenging. This pattern matches increasing distribution shift: random splits preserve substantial overlap be-

tween training and test settings, scaffold splits enforce chemical novelty, and assay splits enforce novelty in the experimental context.

Within the random and scaffold splits, PCAM-OneHot shows the strongest accuracy overall, attaining lower RMSE and higher PCC than PCM across UQ methods. PCAM-BioBERT is more variable, and its performance depends more strongly on the UQ method. Most notably, under the random split with PNN, PCAM-BioBERT shows worse RMSE and PCC, together with higher run-to-run variability. Under the assay-based split, differences between task formulations shrink, and PCAM-BioBERT shows a small advantage over PCAM-OneHot in RMSE and PCC in most settings.

Across UQ methods, differences in RMSE and PCC are generally modest compared with the split-induced shifts. The relative ordering of task formulations is largely preserved across PNN, MC Dropout, and Ensemble, indicating that the observed assay descriptor effects are not driven by a single UQ approach.

Uncertainty quantification metrics (Miscalibration Area, NLL, Sharpness, CRPS)

Miscalibration Area is substantially lower under the assay-based split across task formulations and UQ methods (Figure 3.5), despite degraded RMSE and PCC in the same regime. This indicates that holding out assays induces less overconfident output distributions, consistent with the markedly increased Sharpness under the assay split. Under random and scaffold splits, Miscalibration Area is higher overall, and differences between task formulations are more configuration-dependent, including a clear degradation for PCAM-BioBERT with PNN under the random split.

For NLL, most conditions fall in a relatively narrow range with substantial overlap across task formulations (Figure 3.5). The clearest exception is PCAM-BioBERT with PNN under the random split, which shows higher NLL and markedly larger run-to-run variability, indicating instability for this specific combination. Outside this outlier, NLL tends to be higher under assay splits than under random or scaffold splits across task formulations and UQ methods.

Sharpness varies strongly with both split strategy and UQ method. Assay splits yield markedly higher Sharpness across all task formulations, indicating broader output distributions when assays are held out. In addition, ensemble methods consistently show higher Sharpness than PNN and MC Dropout across splits (Figure 3.5), consistent with aggregation of variability across ensemble members.

CRPS mirrors several of the accuracy trends. Under random and scaffold splits, PCAM-OneHot generally achieves the lowest CRPS across UQ methods. Under assay

splits, CRPS increases for all task formulations and the gap between them narrows; PCAM-BioBERT is marginally lower in most settings (Figure 3.5), consistent with the normalized radar profiles in the Supplementary Information (Supplementary Figure S3.10). Overall, the split strategy has the largest and most consistent effect on distributional metrics, while assay descriptor choice has its clearest impact under random and scaffold splits, and a smaller but consistent advantage for PCAM-BioBERT under assay-held-out evaluation.

3.3 Discussion

This work introduces PCAM as a framework for handling assay heterogeneity in bioactivity estimation by incorporating assay descriptors alongside compound and protein features, and by evaluating UQ behavior across increasingly challenging generalization settings. Experiments were conducted on a curated ChEMBL-derived GPCR dataset and compared three task formulations (PCM, PCAM-OneHot, PCAM-BioBERT) under three split strategies (random, scaffold-based, assay-based) and three UQ methods (PNN, MC Dropout, Ensemble). Across metrics capturing estimation accuracy (RMSE, PCC), calibration (Miscalibration Area), and distributional quality (NLL, Sharpness, CRPS), two patterns are most prominent: (i) the split strategy is the largest driver of performance differences, and (ii) the benefit of assay descriptors depends on whether evaluation occurs within previously observed assay settings or under assay-held-out conditions (Figure 3.5; Supplementary Figure S3.10).

Data curation and assay heterogeneity

To support robust model comparisons, data curation and aggregation were carried out with CAPRICHO [85], which provides a transparent workflow aligned with recently described curation standards.[14] During cross-assay comparisons on processed activity points, we identified a systematic issue affecting a small set of ChEMBL assays, consistent with annotation inconsistencies across assays (for example, differing salt forms and associated structure annotations). This observation reinforces the importance of diagnostic tooling with explicit flagging, enabling targeted review of potentially unreliable measurements rather than silent removal.

Cross-assay concordance in the final dataset remained modest, even after curation, indicating substantial residual heterogeneity in measured bioactivities. This heterogeneity both leads to assay-aware modeling and constrains the attainable accuracy

ceiling in any approach that combines measurements across experimental contexts. Prior work has reported benefits from including assay information on related target sets,[83], and our experimental design directly tested this hypothesis under controlled shifts induced by the split strategy.

Split strategy dominates estimation difficulty

Across all task formulations and UQ methods, random splits yield the lowest RMSE and highest PCC, scaffold splits are intermediate, and the assay-based split is the most challenging (Figure 3.5). This ordering is consistent with increasing distribution shift: random splits preserve substantial overlap between training and test conditions, scaffold splits enforce chemical novelty, and the assay-based split additionally enforces novelty in the experimental context. This hierarchy is observed for estimation metrics and is broadly reflected in the proper-scoring metrics; calibration behaves differently, with Miscalibration Area decreasing under the assay-based split.

When assay descriptors matter and which representation is favored

In regimes where assay identifiers are present during training (random and scaffold splits), PCAM-OneHot consistently improves RMSE and PCC relative to PCM across UQ methods (Figure 3.5). The same setting also shows lower CRPS for PCAM-OneHot, indicating improved distributional quality alongside improved pointwise agreement. In contrast, PCAM-BioBERT is more sensitive to the UQ method and exhibits a clear instability for the PNN under the random split, where RMSE, PCC, and NLL degrade, and variance across runs increases substantially (Figure 3.5). Outside this specific combination, PCAM-BioBERT is competitive but does not uniformly exceed PCAM-OneHot under random and scaffold splits.

Under assay-held-out evaluation, the gap between task formulations narrows across metrics, and PCAM-BioBERT is favored over PCAM-OneHot in RMSE, PCC, and CRPS in most settings (Figure 3.5). This pattern supports a practical distinction between two regimes. First, within an established assay panel (assay identifiers observed during training), categorical assay identifiers are sufficient and consistently beneficial. Second, when assay context changes or new assays are introduced, semantic assay representations can support assay transfer. Concretely, PCAM-BioBERT is most relevant in at least two common situations: (i) *New or changing assay contexts:* a new readout, vendor, or protocol change, or sparse early measurements for a newly de-

fined assay, where assay text/metadata can help transfer signal from related assays. (ii) *Heterogeneous public datasets:* reducing reliance on brittle cross-assay normalization by allowing the model to learn assay-dependent shifts directly from descriptors, consistent with documented inter-assay variability [14]. Supplementary radar profiles provide an aggregated normalized view of these trade-offs across metrics and splits (Supplementary Figure S3.10).

Effects on uncertainty metrics and trade-offs

Across most conditions, NLL varies within a relatively narrow range and shows substantial overlap across task formulations, limiting its ability to sharply separate approaches in this study (Figure 3.5). The main exception is the PNN with PCAM-BioBERT under the random split, which shows elevated NLL and high run-to-run variability, consistent with optimization instability for that configuration.

Sharpness increases markedly under the assay-based split across all task formulations and UQ methods, indicating broader output distributions under assay-held-out conditions (Figure 3.5). In addition, ensemble methods generally show higher Sharpness than PNN and MC Dropout across splits, consistent with the aggregation of variability across ensemble members. These observations highlight an important practical consideration: broader distributions can reflect appropriate caution under strong distribution shift, but they also interact with CRPS (uncertainty quality), which jointly reflects calibration and concentration. In this study, CRPS most clearly differentiates approaches under random and scaffold splits, where PCAM-OneHot consistently attains the lowest values (i.e., is most certain), while the assay-based split compresses differences across task formulations.

Calibration under distribution shift

Calibration analysis using the miscalibration area shows lower miscalibration under the assay-based split despite degraded RMSE and PCC (see Results). This pattern, together with the substantially higher Sharpness under the assay-based split in Figure 3.5, indicates that assay-held-out evaluation elicits broader output distributions and less overconfident behavior. In contrast, random and scaffold splits show stronger accuracy but higher miscalibration, consistent with residual noise sources that are not fully captured by the uncertainty estimates. These findings emphasize that calibration behavior depends strongly on the evaluation regime, and that conclusions drawn from random splits can overstate reliability in settings where assays differ from those seen

during training.

Limitations and outlook

The reported trends are specific to the dataset composition (ChEMBL-derived GPCR subset), the evaluated assay representations, and the three UQ methods considered. In particular, differences between assay representations contract under assay-held-out evaluation, and NLL shows substantial overlap across many conditions (Figure 3.5). Moreover, the elevated variance observed for the PNN with PCAM-BioBERT under the random split indicates that representation choice and optimization robustness should be assessed jointly rather than in isolation.

3.4 Conclusion

We introduced PCAM as an extension of proteochemometric modeling that incorporates assay descriptors alongside compound and protein features to address assay heterogeneity in public bioactivity data. Using a curated ChEMBL-derived GPCR dataset, we compared three task formulations (PCM, PCAM-OneHot, PCAM-BioBERT) across three split strategies (random, scaffold-based, and assay-based) and three UQ methods (PNN, MC Dropout, and Ensemble), evaluated with RMSE, PCC, Miscalibration Area, NLL, Sharpness, and CRPS (Figure 3.5; Supplementary Figure S3.10).

Three findings are most consistent across experimental conditions. First, the split strategy is the dominant driver of difficulty: random splits yield the lowest RMSE and highest PCC, scaffold splits are intermediate, and the assay-based split is the most challenging for all task formulations and UQ methods (Figure 3.5). Second, assay descriptor integration matters most when assay identifiers are shared between training and evaluation. Under random and scaffold splits, PCAM-OneHot achieves lower RMSE, higher PCC, and lower CRPS than PCM across UQ methods, indicating that explicit assay context can improve both estimation accuracy and distributional quality in within-assay-panel settings (Figure 3.5). Third, under assay-held-out evaluation, performance differences between task formulations contract and PCAM-BioBERT are marginally favored over PCAM-OneHot in RMSE, PCC, and CRPS in most settings (Figure 3.5), consistent with the normalized multi-metric profiles (Supplementary Figure S3.10).

Across uncertainty-related metrics, the assay-based split yields markedly higher Sharpness for all task formulations, indicating broader output distributions under

assay-held-out conditions (Figure 3.5). Calibration analysis (miscalibration area; see Results) shows lower miscalibration under the assay-based split despite worse RMSE and PCC, consistent with less overconfident behavior under a stronger distribution shift. NLL values generally occupy a narrow range with substantial overlap across task formulations; the clearest deviation is the PNN with PCAM-BioBERT under the random split, which exhibits elevated variance and worse NLL, indicating configuration-specific instability (Figure 3.5).

Taken together, these results support a practical distinction between two regimes. For applications constrained to established assay panels, one-hot assay identifiers provide the most consistent improvements relative to PCM across RMSE, PCC, and CRPS. For assay-transfer settings where assay definitions are new or evolving (for example, a protocol or vendor change, a new readout, or sparse early measurements for a newly introduced assay), text-derived assay embeddings provide a small but consistent advantage across several metrics, while requiring additional attention to optimization stability in combination with certain UQ methods.

3.5 Methods

3.5.1 Target Selection and Data Retrieval

Human GPCR targets were identified by cross-referencing UniProt (*Homo sapiens*, organism ID 9606) with ChEMBL. Entries annotated as G protein signaling components rather than receptors were excluded from the target set. The bioactivities of all identified targets were collected and curated using an early version of the ChEMBL Aggregation Package with Robust Inspection and Curation Handling Options (CAPRICHO) package [85], using ChEMBL release 35 for activity curation.

3.5.2 Quality Control and Curation

All bioactivity data were collected and curated using the ChEMBL Aggregation Package with Robust Inspection and Curation Handling Options (CAPRICHO) [85]. CAPRICHO implements a flag-based curation workflow that annotates entries with quality issues rather than silently removing them, enabling post-hoc analysis of filtering decisions. The workflow operates in two phases: (1) data retrieval, standardization, and quality flagging, and (2) aggregation with statistical summarization. Data retrieval was restricted to binding (B) and functional (F) assay types, with bioactivity types limited to potency and binding affinity measurements (IC_{50}, EC_{50}, AC_{50}, K_i, K_d).

Only bioactivity measurements with confidence scores of 7, 8, or 9 were retained (corresponding to assay results on direct protein complex subunits, homologous single protein target, or direct single protein target, respectively), and the standard relation was restricted to "=" to include only exact measurements.

CAPRICHO applies several configurable quality filters to identify potentially problematic measurements, marking them as dropping flags. Suggested dropping flags include: (1) entries containing ChEMBL curator data validity comments, (2) entries flagged as potential duplicates by ChEMBL, (3) compounds with undefined stereocenters when chirality is preserved, and other criteria introduced as part of max-curated standards [14], such as (4) assays with fewer than 20 or more than 100 tested compounds, (5) assays containing mutant-related keywords in their descriptions, (6) measurements lacking a document publication date, and (7) compound-target activity pairs where pChEMBL values differ by exactly 3.0 or 6.0 log units between assays, indicative of systematic unit conversion errors. Chemical structure processing includes: (1) SMILES standardization using the ChEMBL Structure Pipeline with chirality preservation, (2) salt and solvent removal, (3) pChEMBL value calculation for measurements reported in nM, µM, or mM units where pChEMBL was not directly provided by ChEMBL, and (4) inter-document duplication detection for identical compound-target-activity combinations reported across different publications.

Same-compound readouts were identified for aggregation based on a combination of the hashed fingerprints `RDKitFP` (`minPath=1`, `maxPath=7`) and `Morgan` (`useChirality=True`, `radius=2`), as implemented in RDKit's `rdFingerprintGenerator`. Measurements on different mutant forms of the same target were aggregated together (aggregate_mutants=True). This could reduce activity concordance in some cases where the mutations are in the binding site, but this was regarded as outside the scope of this work. Measurements from different assays were kept separate. For each aggregated group, geometric mean, geometric standard deviation, median, median absolute deviation, and number of measurements were calculated based on pChEMBL values. From those, geometric mean pChEMBL values were used as target variable for PCAM modeling.

3.5.3 Data Quality Assessment and Filtering

After data collection, we assessed data quality and cross-assay comparability to identify and remove unreliable measurements. For compounds measured across multiple assays targeting the same biological target, we evaluated concordance between each

pair of distinct assays using Kendall's τ, Spearman's ρ, and the coefficient of determination (R^2). This comparison allowed us to assess whether different data quality (e.g., unit annotation error detection) and processing flags (e.g., salt and solvent removal) were associated with systematic deviations from expected measurement concordance. Comparability was evaluated separately for the complete dataset, for concentration-response measurements (XC_{50}: IC_{50}, EC_{50}, AC_{50}), and for equilibrium binding constants (K_x: K_i, K_d) to account for potential differences in measurement precision and assay characteristics between these bioactivity type families.

This comparability analysis revealed that entries flagged with calculated pChEMBL values showed reduced correlation with direct measurements compared to entries where pChEMBL was reported in the source document, indicating that calculated values introduced additional uncertainty. Additionally, four assays (CHEMBL3887514, CHEMBL3887515, CHEMBL4421868, CHEMBL4421869) were identified as containing systematic unit annotation errors, with reported pChEMBL values for a shared set of compounds differing by exactly 6.0 log units. The structures of the compounds shared by these assays varied in terms of salt and solvent annotation, as shown in Supplementary Table S3.3. These assays were excluded from all subsequent datasets. Additional filtering criteria were applied to ensure data quality and consistency between assays. All entries with data validity comments annotated by ChEMBL were removed, as these flags indicated curator-identified concerns about measurement reliability. Entries flagged as potential duplicates were excluded to prevent redundant measurements from influencing PCAM modeling. Entries with unit annotation errors were removed. Finally, all entries with calculated rather than ChEMBL-reported pChEMBL values were excluded upon observing low cross-assay readout concordance of these values.

Upon visual inspection, compounds with molecular weight above an 850 Da threshold showed a higher incidence of peptide-like or macrocyclic structures that fall outside the typical small molecule drug space and were therefore removed from the dataset. We then introduced a target-level filtering to ensure a minimum data density for model training and evaluation. Only targets with more than 50 unique chemical connectivities (the first 14 characters of an InChI key) were retained to provide adequate chemical diversity for learning structure-activity relationships. Additionally, we required each retained target to possess at least 10 unique compounds with pChEMBL values greater than 6.5 to ensure representation of high-affinity ligands necessary for modeling potency relationships. These combined selection criteria identified 167 GPCR targets with sufficient data quality and quantity for downstream modeling applications. The

distribution of data points per target varied substantially among these 167 targets.

3.5.4 PCAM-Compatible Dataset Generation

Following quality filtering and target selection, the curated bioactivity measurements were re-aggregated, including the `assay_chembl_id` and `assay_description` fields in the aggregation identifier to ensure same-compound aggregation only for matching assay identifiers, as required for PCAM modeling. For each aggregated entry, the geometric mean of pChEMBL values was calculated as the primary aggregated activity measure, with geometric standard deviation computed to quantify the variability across individual measurements. With this approach, we generated three final datasets. The comprehensive dataset included all bioactivity types without restriction, containing measurements from 167 GPCR targets with 77,081 unique compounds across 5,161 distinct assays. The XC_{50}-specific dataset was generated by filtering to concentration-response measurements (IC_{50}, EC_{50}, and AC_{50}) prior to aggregation, yielding data for 155 targets with 44,260 unique compounds and 2,484 distinct assays. The K_x-specific dataset was generated by restricting to equilibrium binding constants (K_i and K_d) before aggregation, producing data for 103 targets with 37,071 unique compounds across 2,661 assays. All datasets maintained target-level representation with at least 51 unique compound measurements per target, ensuring sufficient data density for downstream modeling applications. The aggregated datasets preserve complete provenance information through concatenated assay identifiers and metadata fields, while the unique compound-target pairs per row facilitate direct use in machine learning workflows without additional deduplication steps.

3.5.5 Data Splitting Strategies

To evaluate model generalization under progressively stronger distribution shifts, we used three complementary data splitting strategies: random, scaffold-based, and assay-based splits. Each strategy partitions the dataset into training (70%), validation (15%), and test (15%) subsets. Across strategies, splits were constructed at the level of individual data points, but with additional grouping constraints for scaffold and assay splits as described below.

All splitting utilities implement deterministic behavior via explicit seed control across the relevant random number generators, ensuring that the same configuration yields identical partitions. In addition, the framework supports caching of precomputed splits so that repeated experiments reuse exactly the same train/validation/test

assignments.

Random split

Random splitting serves as an i.i.d.-style baseline that estimates performance under minimal distribution shift. Data points are randomly assigned to training, validation, and test sets using scikit-learn's `train_test_split` [87] to match the target proportions. Because compounds, targets, and assays can appear in more than one partition, this strategy primarily evaluates interpolation to closely related chemical and experimental settings. It also provides a reference point for comparison with prior work that reports results under random partitioning.

Scaffold split

Scaffold-based splitting evaluates performance on structurally novel compounds by ensuring that molecules sharing the same core scaffold are assigned exclusively to a single partition. Scaffolds are extracted using the Bemis–Murcko decomposition [66], which retains core ring systems and linkers while removing peripheral substituents. We compute scaffold identifiers with RDKit's `MurckoScaffoldSmiles`, excluding chirality information from the scaffold definition. Unique scaffolds are then randomly shuffled and distributed across training, validation, and test partitions according to the target proportions. This procedure ensures that the test set contains scaffolds absent from training, providing a stringent evaluation of generalization across chemical series rather than within-series interpolation.

Assay-based split

Assay-based splitting evaluates robustness to novel experimental contexts by enforcing disjoint assay identifiers across partitions. Specifically, all data points associated with a given ChEMBL assay identifier are assigned to exactly one of the training, validation, or test sets. To obtain approximately balanced set sizes while preserving assay integrity, assay identifiers are assigned using a greedy procedure: iterating over assays, each assay is placed into the split that currently has the largest deficit relative to its target proportion. This strategy is particularly relevant for assessing whether incorporating assay descriptors supports transfer across assay conditions when entire assay identifiers are held out, approximating real-world assay drift or assay introduction scenarios.

3.5.6 Model Architectures and Variants

PCM modeling extends classical QSAR by incorporating descriptors for both small molecules and protein targets within a single model framework [76]. In this work, we further generalize PCM to *proteochemometric assay-aware modeling* (PCAM), where assay descriptors are explicitly integrated as a third modality. This design enables models to leverage not only compound and protein information, but also contextual details from the experimental assay in which the bioactivity was measured.

Model Variants

By combining representation choices for compounds, proteins, and assays, we obtain three main model classes:

1. **PCM (protein–compound):** Uses Ankh-Large embeddings for proteins and ECFP fingerprints for compounds. No assay descriptor is included.

2. **PCAM-OneHot (protein–compound–assay):** Uses Ankh-Large for proteins, ECFP for compounds, and one-hot encoded categorical assay identifiers.

3. **PCAM-BioBERT (protein–compound–assay):** Uses Ankh-Large for proteins, ECFP for compounds, and BioBERT embeddings for assay descriptions.

This setup allows for systematic evaluation of (i) the impact of including assay context at all (PCM vs. PCAM), and (ii) the choice of assay representation (one-hot encoding vs. semantic embeddings). Each variant is trained as a regression model to estimate continuous bioactivity values (pChEMBL) for compound–protein–assay combinations.

Input Representations

All models share the same molecular input representation: RDKit's implementation of the Extended Connectivity Fingerprints (ECFP4) of radius 2 and length 2048 [21], a standard fingerprinting method that captures substructural features of chemical compounds. Proteins were described using Ankh-Large embeddings (1536-bits), a transformer-based language model that provides high-dimensional sequence embeddings capturing evolutionary and structural information [1].

The models differ in their treatment of assay inputs. PCM models include no assay descriptor. PCAM-OneHot uses one-hot encodings capped at a maximum length

of 2048, with the assays having the most data points being given unique identifiers. PCAM-BioBERT uses BioBERT [84] embeddings obtained as described in [83]: ChEMBL assay descriptions were encoded using BioBERT v1.2, with the resulting embeddings consisting of 768 dimensions. This comparison isolates the contribution of sophisticated text-derived assay embeddings relative to simple categorical identifiers.

Neural Network Architecture

The neural architecture follows a modular design. Each modality (compound, protein, assay) is passed through a dedicated feedforward encoder to project its representation into a shared latent space. These encoders consist of multiple fully connected layers with ReLU activations and dropout regularization. The modality-specific embeddings are then concatenated and passed into a joint regression head to produce the estimated bioactivity value $\hat{y}$:

$$\hat{y} = f_\theta\big(\phi_c(\mathbf{x}_c),\ \phi_p(\mathbf{x}_p),\ \phi_a(\mathbf{x}_a)\big) \tag{3.1}$$

where $\mathbf{x}_c$, $\mathbf{x}_p$, and $\mathbf{x}_a$ denote compound, protein, and assay features, respectively; ϕ_c, ϕ_p, ϕ_a are modality-specific encoders; and f_θ is the shared regression head parameterized by θ.

Uncertainty Quantification Methods

The core network architecture was extended with three uncertainty quantification approaches, following prior work [31, 32, 81]:

PNN: The regression head outputs two parameters (μ, σ) that define a Gaussian distribution over the target value. The network is trained by minimizing the negative log-likelihood of observed values under the estimated distribution, which encourages well-calibrated uncertainty estimates that reflect aleatoric (data) uncertainty.

MC dropout: Dropout layers remain active during inference, and multiple forward passes with different dropout masks produce a distribution of outputs. The mean serves as the point estimate while the variance across passes captures epistemic (model) uncertainty. We used 50 forward passes for uncertainty estimation.

Deep Ensembles: Multiple models with identical architecture but different random initializations are trained independently. The ensemble mean provides the point estimate, and the variance across ensemble members captures both aleatoric and epistemic uncertainty. We trained ensembles of 5 members.

For detailed formulations and comparative analysis of these methods, see Khalil et al. [81].

3.5.7 Hyperparameter Optimization

The model's hyperparameters were optimized using a 200-run Bayesian sweep. Tuned parameters included architectural choices (the depth of feature-specific input layers and post-concatenation regressor layers, dropout rate) and optimization settings (learning rate, optimizer, batch size, weight decay). The final model was selected based on the lowest validation loss, after filtering out runs exhibiting training instability, excessive parameter growth, or premature convergence, which were identified by monitoring gradient norms, parameter norms, and training times, respectively. The test set was held out throughout the optimization process and used only for final evaluation, ensuring no information leakage from test data into hyperparameter selection.

3.5.8 Statistical testing and practical relevance

The statistical evaluation protocol adopted in this work is primarily inspired by the practical significance framework proposed by Ash et al. [60], which emphasizes paired experimental designs, non-parametric inference, and rank-based comparisons to ensure robust and interpretable model assessment in applied machine learning settings. In line with these recommendations, all analyses were conducted under a repeated-measures design to explicitly account for correlations induced by evaluating multiple representation settings on matched experimental replicates.

Within each data split (`split`), a statistical block identifier (`cv_cycle`) was defined to capture matched repetitions across representation settings. In the analyses reported here, `cv_cycle` was constructed by concatenating the base model family and the random seed (i.e., `cv_cycle` = `model` + "+" + `seed`), such that each block corresponds to a shared stochastic realization across all representation settings. The within-block factor `method` denotes the representation strategy under comparison (PCM, PCAM-OneHot, PCAM-BioBERT), implemented by renaming the `task` column. This paired design ensures that differences between representation settings are evaluated relative to identical replicate identities, thereby reducing variance and preventing inflated significance due to unpaired reuse of stochastic runs.

Upstream evaluation tables occasionally contained multiple rows for a given `(cv_cycle,method,split)` combination, for example, due to repeated logging or descriptor-specific variants. To enforce a single observation per block and condition, such dupli-

cates were deterministically aggregated by their mean value for each metric prior to statistical testing. For each metric and split, analyses requiring balanced blocks were applied only after restricting the data to the subset of `cv_cycle` blocks for which all representation settings were present. The number of total blocks, retained complete blocks, and discarded blocks was recorded per metric and split as part of a quality-control procedure.

For each split and evaluation metric, global differences between representation settings were first assessed using the Friedman test, a non-parametric alternative to repeated-measures ANOVA based on within-block ranks [75, 88]. The Friedman test was chosen because distributional assumptions such as normality of residuals are frequently violated for model evaluation metrics, and because it aligns naturally with the rank-based perspective advocated by Ash et al. The resulting p-values were treated as omnibus indicators of whether systematic performance differences across representation settings were present.

When pairwise comparisons were required, post-hoc analysis was performed using the Conover–Friedman procedure with Holm correction to control the family-wise error rate across all method pairs [61, 62]. In rare cases where numerical instabilities arose (e.g., undefined values or runtime warnings), the Nemenyi–Friedman test was used as a robust fallback to obtain a well-defined pairwise comparison matrix [75, 89]. All post-hoc tests were conducted within split and metric after enforcing complete blocks.

To summarize relative performance in a manner that is both statistically grounded and practically interpretable, within-block ranks were computed for each metric and averaged across blocks. Metric-specific optimization direction was respected, with lower values indicating better performance for error and calibration metrics, and higher values indicating better performance for correlation-based metrics. These average ranks were visualized using critical-difference-style diagrams, in which methods are positioned according to their mean rank and grouped according to non-significant pairwise differences implied by the post-hoc comparison matrix at the chosen significance level.

Finally, statistical significance was interpreted jointly with practical relevance rather than in isolation. In addition to rank-based summaries, effect-oriented visualizations based on repeated-measures ANOVA error variance and Tukey-style multiplicity adjustment were used to report pairwise mean differences together with adjusted p-values and confidence intervals for the mean difference. Across all analyses, conclusions were guided by the magnitude and direction of observed differences, the exclusion of zero by confidence intervals, and the consistency of rankings across blocks, in line with recent recommendations to move beyond sole reliance on p-values when

comparing machine learning models in applied scientific contexts [60, 81].

Supplementary Information

S3.1 Supplementary Methods

This Supplementary Information provides (i) additional methodological detail to support reproducibility, and (ii) diagnostic visualizations that complement the main-text statistical comparison workflow and multi-metric interpretation.

S3.1.1 Experimental Design

This section summarizes the experimental factors varied in the study and clarifies what is held constant across comparisons.

- **Task formulations (assay descriptor integration).**
  - PCM: compound and protein descriptors only.
  - PCAM-OneHot: PCM plus categorical assay identifiers (one-hot).
  - PCAM-BioBERT: PCM plus text-derived assay embeddings.
- **Split strategies (generalization regimes).**
  - Random split (baseline, minimal distribution shift).
  - Scaffold split (chemical series held out via scaffold disjointness).
  - Assay-based split (assay identifiers held out, novel experimental contexts).
- **Uncertainty quantification methods.**
  - PNN.
  - MC dropout.
  - Ensemble.
- **Endpoints / subsets evaluated.**
  - Activity subsets and endpoint groupings used in the main study (e.g., xC50, Kx), consistent with the main-text definitions.
- **Repetitions and reproducibility.**
  - Number of independent runs (random seeds) per configuration.
  - Deterministic split generation and cached split reuse, as described in the main text.

S3.1.2 Training and Optimization

This section summarizes training choices that affect both estimation metrics and distributional metrics, with an emphasis on the probabilistic neural network (PNN) configuration and its hyperparameter optimization.

Hyperparameter optimization (PNN)

PNN hyperparameters were tuned using a Weights & Biases sweep configured for Bayesian optimization, minimizing the validation loss (`val/loss`). The sweep explored architectural depth and width for the chemical, protein, assay, and regressor branches, as well as optimization and noteworthy regularization controls (dropout, weight decay, gradient norm clipping). The search space is summarized in Table S3.1. The selected hyperparameters for each evaluated setting (endpoint and split strategy) are summarized in Table S3.2.

Table S3.1: PNN hyperparameter search space. Search ranges used in the Bayesian sweep that minimized validation loss.

Parameter	Values
Chemical MLP layers	[1024,512,256], [1024,512], [1024,256], [1024], [512], [256]
Protein MLP layers	[1024,512,256], [512,256,128], [512,128], [512,256], [512], [256]
Assay MLP layers	[1024,512,256], [512,256,128], [512,128], [512,256], [512], [256]
Regressor layers	[256,128], [128,64], [256], [128], [64]
Dropout	0.1, 0.2
Batch size	128, 256, 512
Optimizer	Adagrad, AdamW, Adam, SGD
Learning rate	1e-2, 1e-3, 5e-4
Weight decay	0, 1e-2, 1e-4
Max grad norm	`null`, 50.0, 100.0
Epoch budget	3000
Early stopping	patience = 100

Method-specific settings (sampling and aggregation) For UQ methods that require sampling or multiple members, the study used fixed inference settings recorded alongside the tuned hyperparameters (e.g., number of Monte Carlo samples and ensemble size, where applicable).

Table S3.2: Selected PNN hyperparameters for the `all_gpcr / all / all` setting. Best configurations obtained via Bayesian hyperparameter optimization are shown for each split strategy. The assay-based split corresponds to holding out `ChEMBL assay_id` groups ("`column_by_assay_chembl_id`" in configuration files).

Hyperparameter	Random	Scaffold	Assay-based
Assay descriptor	`BioBERT`	`BioBERT`	`BioBERT`
Protein descriptor	`ankh-large`	`ankh-large`	`ankh-large`
Chemical descriptor	`ecfp2048`	`ecfp2048`	`ecfp2048`
Chemical MLP layers	`[256]`	`[1024]`	`[256]`
Protein MLP layers	`[512,256]`	`[512]`	`[256]`
Assay MLP layers	`[512,128]`	`[512]`	`[256]`
Regressor layers	`[256]`	`[64]`	`[256]`
Dropout	0.2	0.2	0.1
Batch size	512	512	128
Optimizer	`adagrad`	`adagrad`	`adagrad`
Learning rate	0.0005	0.0005	0.0005
Weight decay	0.01	0.0001	0.0001
Max grad norm	50.0	50.0	50.0
Epoch budget	3000	3000	3000
Early stopping patience	100	100	100

Objective function (Gaussian negative log-likelihood)

All PNN models were trained with a Gaussian negative log-likelihood objective (`gaussnll`) with mean reduction. Concretely, targets are treated as samples from a Gaussian distribution whose mean and variance are parameterized by the network. For a target y, network mean μ, and (positive) variance σ^2, the per-sample objective (up to an additive constant) is:

$$\mathcal{L}_{\mathrm{GNLL}}(y;\mu,\sigma^2) = \frac{1}{2}\left[\log(\sigma^2) + \frac{(y-\mu)^2}{\sigma^2}\right]. \tag{3.2}$$

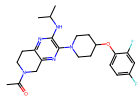
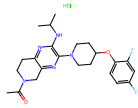
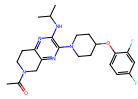
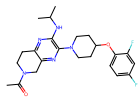
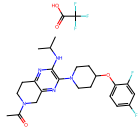
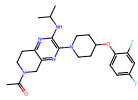
In practice, the implementation uses numerical safeguards to avoid degenerate variance values, consistent with standard Gaussian NLL formulations.

S3.2 Supplementary Tables

S3.2.1 Data curation diagnostics

The table below provides an example of systematic annotation issues detected during post-processing, illustrating the need for explicit diagnostic tooling during curation.

Table S3.3: Sample of activities with systematic unit annotation errors detected post salt removal step. Molecule and assay IDs represent the ChEMBL identifiers of the molecules and the assays for which the data were reported. Relation represents the `standard_relation` field in ChEMBL, indicating exact measurements for the reported values (=), whereas pChEMBL is the placeholder name for the $-log_{10}$-transformed activities (Molar). All activity values were reported for target GPR6 (ID: CHEMBL3714130).

Molecule ID	Structure	Relation	pChEMBL	Year	Assay ID	Target
CHEMBL3938087		=	11.02	2015	CHEMBL3887514	GPR6
CHEMBL4524556		=	5.02	2016	CHEMBL4421868	GPR6
CHEMBL3938087		=	5.02	2018	CHEMBL4419149	GPR6
CHEMBL3938087		=	10.13	2015	CHEMBL3887515	GPR6
CHEMBL4569858		=	4.13	2016	CHEMBL4421869	GPR6
CHEMBL3938087		=	4.13	2018	CHEMBL4419150	GPR6

S3.3 Supplementary Figures

S3.3.1 Normality diagnostics for paired within-fold differences

These figures assess distributional assumptions for paired within-fold differences across metrics and split strategies. They lead to the emphasis on rank-based and paired nonparametric comparisons when Gaussian assumptions are not supported.

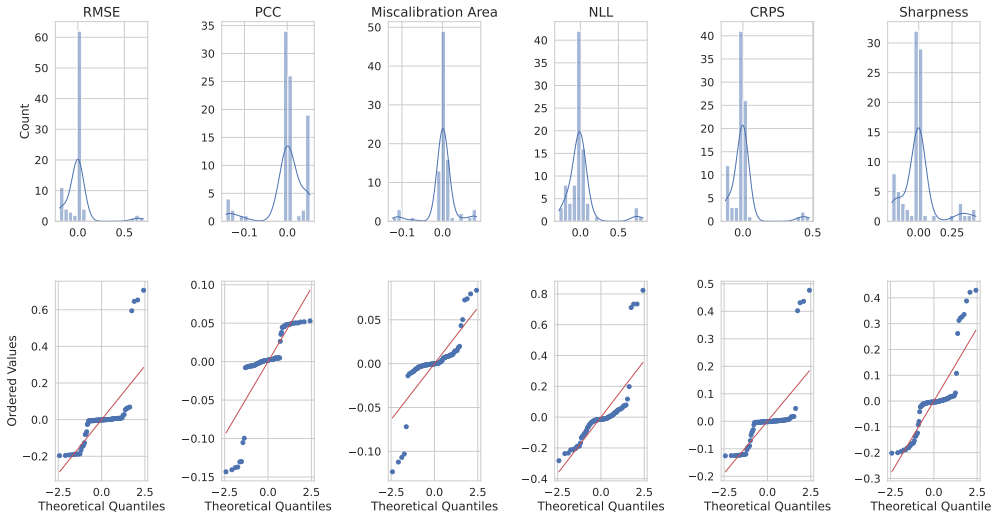


Figure S3.1: Normality diagnostics for paired, within-fold performance differences (random split). Histograms (top row) and Q–Q plots (bottom row) summarize the empirical distribution of paired performance differences across cross-validation cycles for each metric. Deviations from a bell-shaped distribution and systematic departures from the reference line in the Q–Q plots indicate that Gaussian assumptions are not uniformly supported, motivating rank-based, repeated-measures comparisons as the primary analysis [60]. Analogous diagnostics for the scaffold and assay-based splits are provided below.

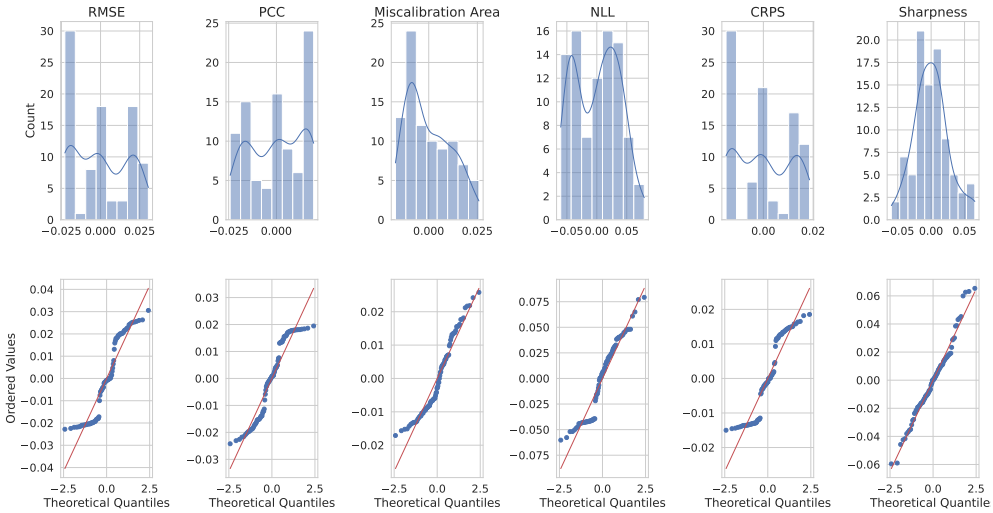


Figure S3.2: Normality diagnostics for paired, within-fold performance differences (scaffold split). Histograms (top row) and Q–Q plots (bottom row) summarize the empirical distribution of paired differences across cross-validation cycles for each metric. The scaffold split induces pronounced distributional irregularities for several metrics (e.g., skewness, heavy tails, and outliers), which helps explain why rank-based repeated-measures inference is emphasized for method comparison and why conclusions should be drawn from split-wise robustness rather than from a single parametric summary [60].

S3.3. Supplementary Figures

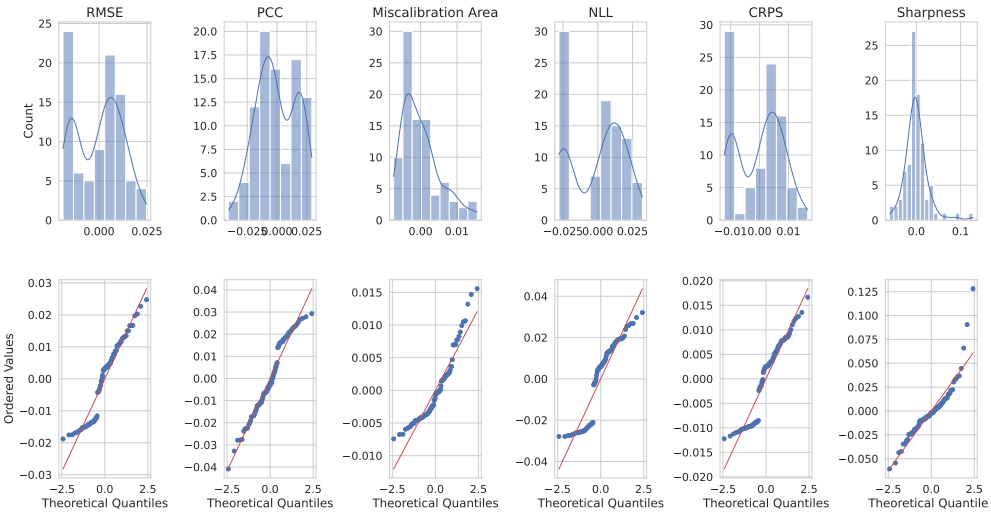


Figure S3.3: Normality diagnostics for paired, within-fold performance differences (assay-based split). Histograms (top row) and Q–Q plots (bottom row) show the distribution of paired differences across cross-validation cycles for each metric under assay-based splitting. Relative to the random split example in the main text, several metrics exhibit additional asymmetry and tail behavior, indicating that Gaussian assumptions are not uniformly supported across split strategies. This reinforces the value of the practical-significance workflow and nonparametric, repeated-measures comparisons to summarize method separability in a way that is less sensitive to distributional departures [60].

S3.3.2 Average-rank summaries via critical difference diagrams

These figures summarize method separability per metric using average ranks across cross-validation cycles, with multiple-comparison control represented by critical-difference groupings.

S3.3. Supplementary Figures

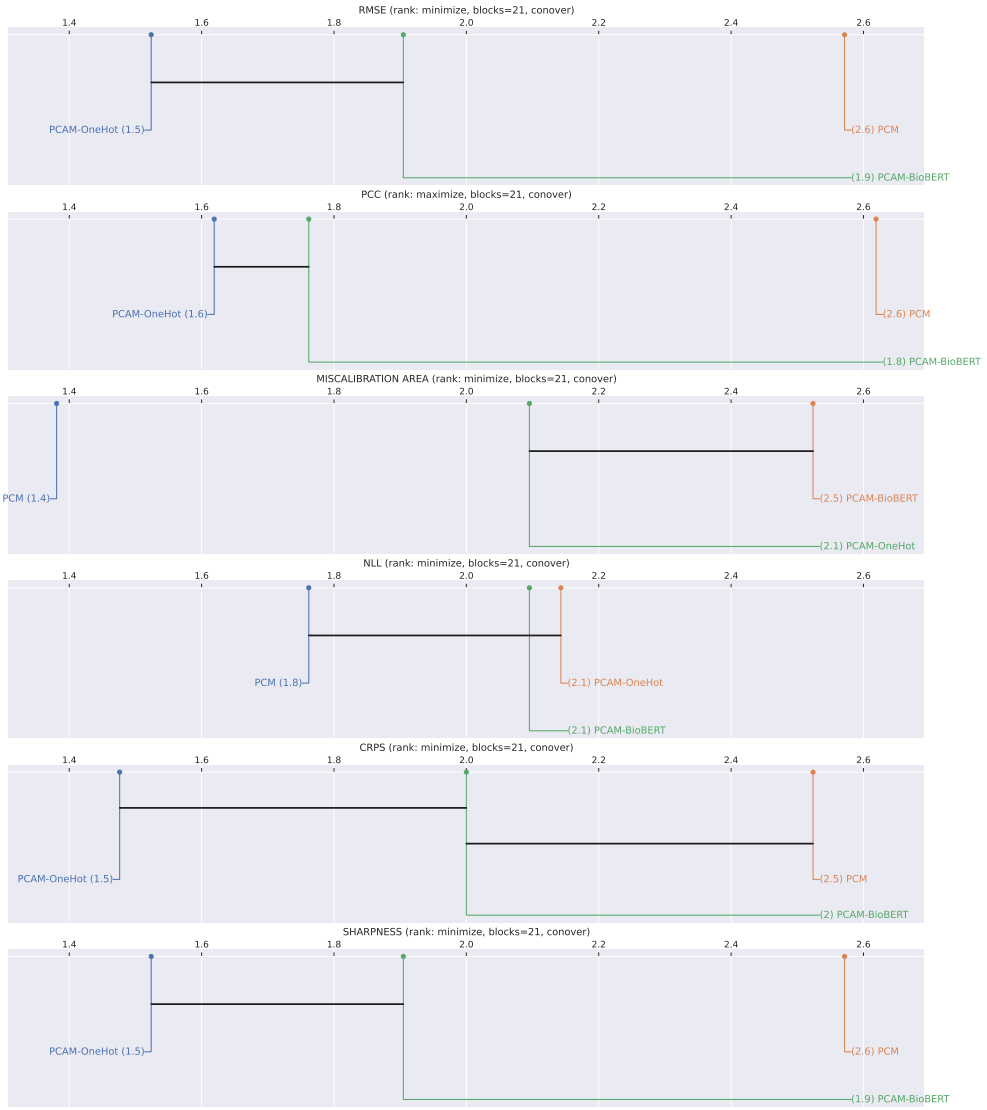


Figure S3.4: Average-rank comparison via critical difference (random split). For each metric, methods are ordered by their average rank across cross-validation cycles (lower is better for metrics minimized; higher is better for metrics maximized, as stated within each panel). Horizontal connectors denote sets of methods that are not distinguishable under the multiple-comparison control used for the critical difference analysis [60].

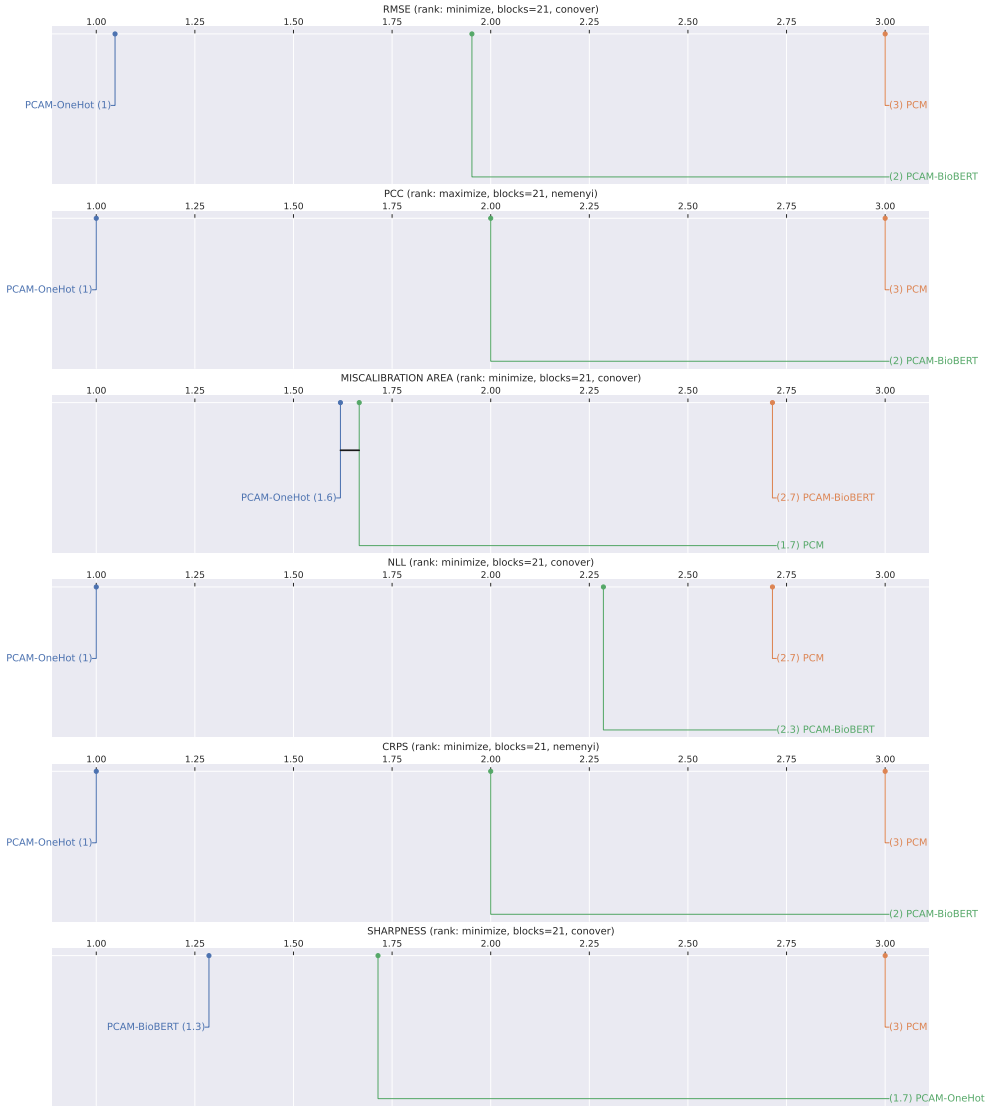


Figure S3.5: Average-rank comparison via critical difference (scaffold split). For each metric, methods are ordered by their average rank across cross-validation cycles (lower is better for metrics minimized; higher is better for metrics maximized, as stated within each panel). Horizontal connectors denote sets of methods that are not distinguishable under the multiple-comparison control used for the critical difference analysis [60]. Comparing this figure with the random split example in the main text highlights how the scaffold split can alter both the ordering and the extent of rank separability, emphasizing the importance of reporting split-specific robustness.

S3.3. Supplementary Figures

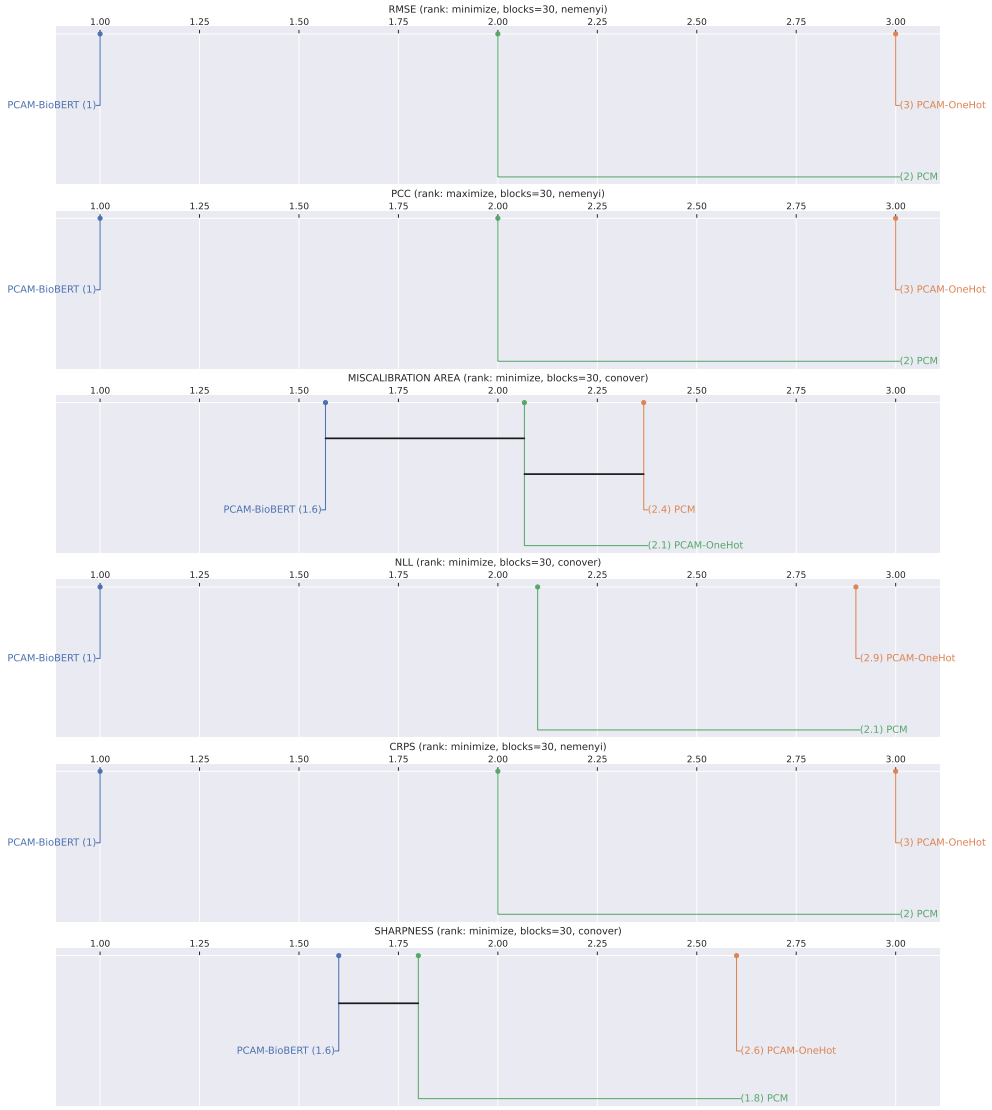


Figure S3.6: Average-rank comparison via critical difference (assay-based split). Methods are ranked per metric by average rank across cross-validation cycles, and connectors indicate groups without statistically supported separation after controlling for multiple comparisons [60]. The assay-based split provides an additional robustness check under a different form of distribution shift; in this setting, some metrics display tighter grouping (weaker rank separability), while others retain clearer separation. This split-wise view complements the main-text sign plots by summarizing which rank differences persist across folds for each metric.

S3.3.3 Mean-centered score (MCS) diagnostic plots

These plots visualize block-wise deviations from the mean for each metric and support inspection of whether observed differences are driven by a small number of blocks or reflect more consistent behavior across cycles.

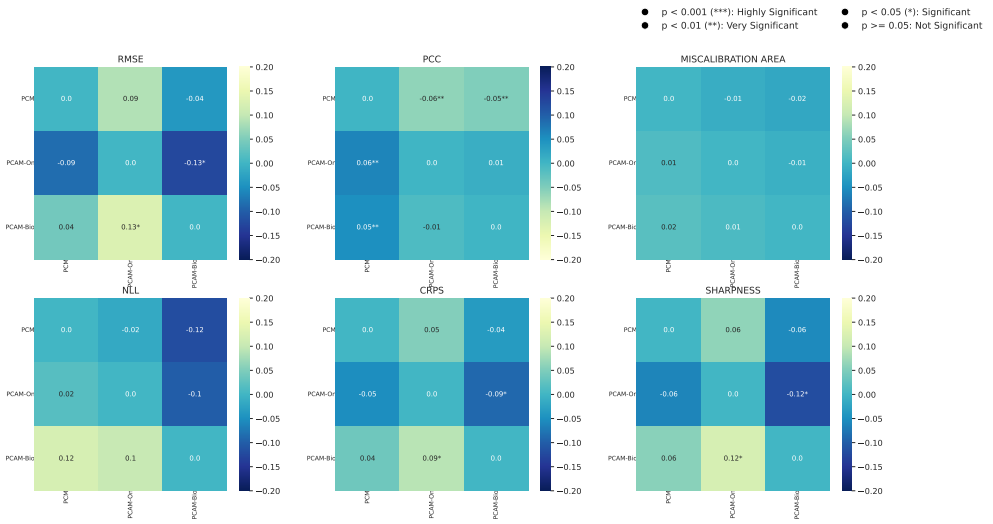


Figure S3.7: Mean-centered score (MCS) diagnostic plots for the random split. Each panel shows deviations from the block-wise mean for a given metric, allowing visual inspection of relative method behavior across cross-validation cycles. The absence of consistent extreme deviations supports the robustness of the paired comparison framework and indicates that observed differences are not driven by a small number of dominant blocks.

S3.3. Supplementary Figures

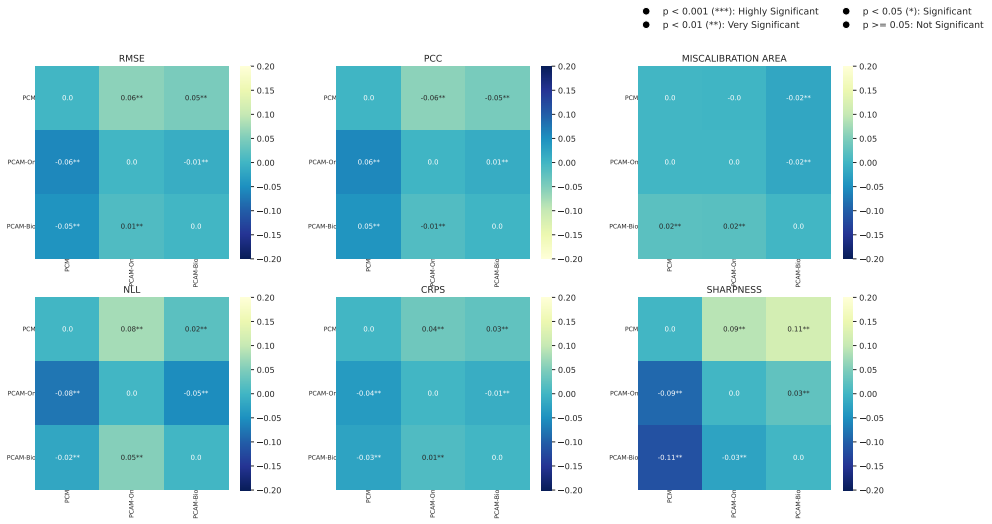


Figure S3.8: Mean-centered score (MCS) diagnostic plots for the scaffold split. Compared to the random split, increased dispersion is observed across several metrics, reflecting the higher variability induced by scaffold-based partitioning. Despite this increased spread, relative ordering between representation settings remains stable across blocks, supporting the validity of the paired non-parametric analyses.

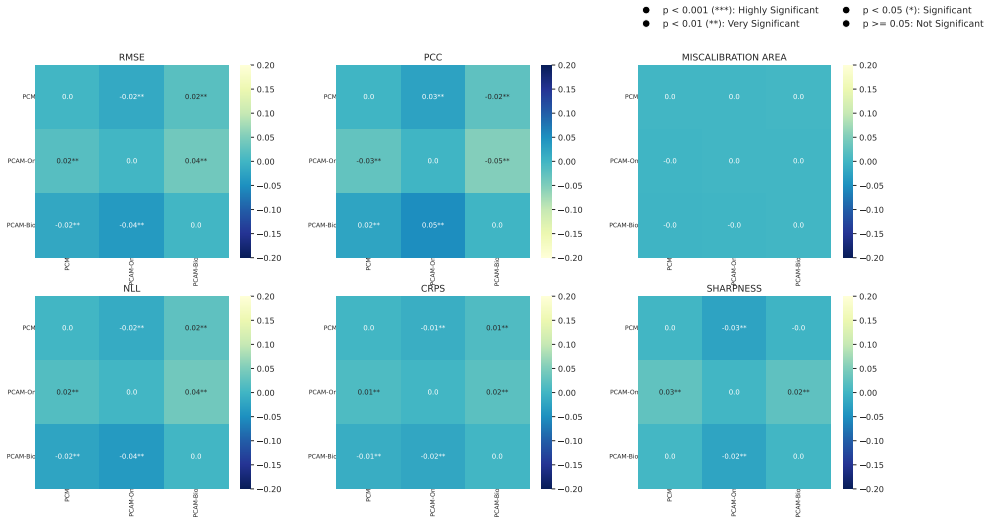


Figure S3.9: Mean-centered score (MCS) diagnostic plots for the assay-based split. The plots reveal pronounced heterogeneity across evaluation blocks, consistent with differences in assay conditions and data composition. This variability leads to the use of rank-based and sign-based analyses and highlights the limitations of relying solely on aggregate summary statistics in assay-stratified evaluations.

S3.3.4 Normalized multi-metric profiles (radar plots)

Radar plots provide a compact multi-metric summary across task variants within each UQ method and split strategy, and are intended to complement the absolute-scale summaries in Figure 3.5. For readability, radar profiles summarize RMSE, PCC, NLL, Sharpness, and CRPS; calibration (Miscalibration Area) is reported in the main multi-panel metrics figure and in the statistical comparison plots.

We report a column-normalized view (Figure S3.10) to emphasize within-split comparisons: each split column is normalized independently, so radii are comparable only within the same split. For cross-split context, we additionally report a global-normalized view (Figure S3.11) using a single normalization across all conditions; this enables cross-column comparisons but compresses within-assay differences due to the substantially different absolute metric ranges under assay splitting. For interpretability, metrics where lower values indicate better performance (RMSE, NLL, Sharpness, CRPS) are inverted so that larger radii consistently represent better performance.

S3.3. Supplementary Figures

Figure S3.10: Column-normalized radar plots summarizing the five evaluation metrics (RMSE, PCC, NLL, Sharpness, CRPS) for the three task variants (PCM, PCAM-OneHot, PCAM-BioBERT) across model architectures (rows) and split strategies (columns). Normalization is performed within each split column; absolute values are reported in Figure 3.5.

Figure S3.11: Global-normalized radar plots as in Figure S3.10. Global normalization enables cross-split comparison of overall metric levels, but can compress within-split differences when absolute metric ranges differ (notably under assay splitting). Absolute values are reported in Figure 3.5.

Chapter 4

Thinking in Fragments

Graph-Based Molecular Representations
for Cyclic Peptide Permeability Prediction

Based on the publication

GSGE: Group-SELFIES Graph Embeddings for Cyclic Peptide Membrane Permeability

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`(Under revision)`

[*]These authors contributed equally and share first authorship.

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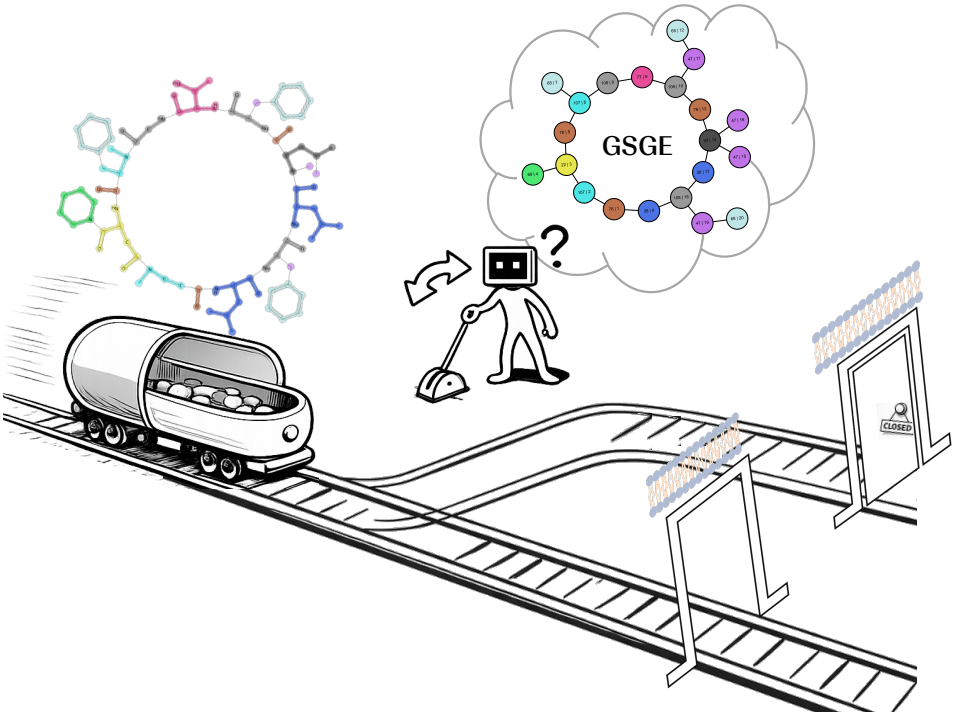

Figure 4.1: Graphical Abstract

Computational approaches for cyclic peptide membrane permeability often rely on hand-curated monomer vocabularies and proprietary molecular descriptors, limiting their portability to chemically diverse scaffolds. We introduce Group-SELFIES Graph Embeddings (GSGE), an open-source framework that derives a data-driven fragment vocabulary from any SMILES corpus and constructs fragment-level molecular graphs amenable to topology-aware modeling. On the CycPeptMPDB PAMPA benchmark (6,889 cyclic peptides), a graph convolutional network operating on GSGE graphs and fused with seven open-source 2D/3D descriptors achieves a test-set MAE of 0.329 $\log_{10}$ units ($R^2 = 0.802$), improving on the CycPeptMP baseline without proprietary features or SMILES enumeration; complementary experiments further show that graph and recurrent architectures consistently outperform Transformers for this modality and that PAMPA-trained models transfer competitively to Caco-2, MDCK, and RRCK endpoints. GSGE is released as a modular Python package alongside full

benchmarking code and data to support reproducible cyclic peptide property modeling.

4.1 Introduction

Cyclic peptides are an increasingly important modality for intracellular targets that are often difficult to modulate with small molecules or biologics [38, 90]. Their macrocyclic topology can improve proteolytic stability and residence time, and selected scaffolds achieve oral exposure [39, 40]. Effective intracellular action, however, generally requires permeation across lipid bilayers, and passive membrane diffusion remains a central bottleneck for many series [41, 42]. Experimental platforms such as PAMPA provide valuable readouts but are throughput-limited and resource-intensive. In silico approaches link structure to permeability and support chemical-space triage [43, 44, 46]. Mechanistic studies further indicate that passive diffusion is governed not only by bulk physicochemical properties but also by chameleonic conformational switching between solvent-exposed and membrane-shielded states, often associated with transient intramolecular hydrogen-bond networks [41–43].

Curation efforts have recently made this area more tractable. CycPeptMPDB aggregates 7,334 cyclic peptides from 45 publications and 2 patents, with permeability measurements spanning PAMPA, Caco-2, MDCK, and RRCK assays [44]. Because PAMPA contributes most entries, it has become the primary endpoint for comparative modeling [44, 45]. Building on this resource, Li *et al.* proposed CycPeptMP, a multi-channel pipeline/framework for PAMPA permeability modelling, which fuses three information channels—(i) atom-level attention with graph/conformer-distance modulation, (ii) whole-molecule 2D/3D features and Morgan fingerprints via a multilayer perceptron, and (iii) monomer-level descriptors via a cyclic 1D CNN—together with extensive SMILES enumeration [45], generating multiple equivalent SMILES strings for each molecule as a form of data augmentation, which can reduce a sequence model's sensitivity to arbitrary atom ordering [91]. While strong results were reported, the pipeline depends on hand-crafted monomer vocabularies and conformer-specific descriptor calculation (including proprietary software), and it incurs computational overhead from three augmentation channels—SMILES enumeration, cyclic-sequence rearrangement, and generation of 60 conformers per peptide/monomer for 3D feature calculation [45]. As chemical diversity increases to include non-canonical amino acids, N-methylation, peptidomimetic linkers, and hybrid macrocycles, such monomer vocabularies are insufficient and require ad hoc extension, which reduces portability.

A central methodological question is which inductive bias –that is, which structural assumptions built into a model's architecture (e.g., sequential ordering vs. graph connectivity)– best aligns with the structure–property relationships that govern mem-

brane translocation in macrocycles. Standard Transformers emphasize pairwise contextual relations in sequences [92]; they perform especially well in domains with strong sequence-context regularities and evolutionary constraints [26, 93, 94]. By contrast, cyclic peptides exhibit graph-like dependencies, chameleonic conformational shifts, and relatively weak linear positional regularities [41, 95]. Consequently, sequence-only architectures often require additional graph or conformer-distance information to perform well [45], leading to topology-aware encoders and representations that more directly capture local chemistry and global connectivity [95–97].

Here we advance a fragment-centric graph representation that natively encodes molecular topology while remaining flexible to non-canonical building blocks. We introduce *Group-SELFIES Graph Embeddings* (GSGE), which leverage Group-SELFIES to derive a data-driven fragment vocabulary and construct compound graphs whose nodes correspond to chemically meaningful fragments rather than individual atoms [25]. This abstraction captures local environments and higher-order connectivity without relying on fixed monomer lists that require manual extension as chemistry diversifies. On top of these graphs, we employ graph convolutional networks (GCNs) as lightweight, topology-aware encoders [98] and study a simple fusion with open-source whole-molecule 2D/3D descriptors selected from prior work [45]. In parallel, we examine an alternative to sparse one-hot fragment inputs by training a graph autoencoder over a broad fragment corpus to obtain dense, structurally informed fragment embeddings [99–101]. The benchmark comprises 24 model variants spanning (i) graph-based GSGE models with and without descriptor fusion and with alternative fragment encodings, (ii) sequence baselines (Transformer [92], LSTM [102], xLSTM [103]) across tokenisation schemes, and (iii) literature baselines [45] included to enable direct comparison under the same split protocol. We include these families to separate effects of representation (graphs vs sequences) from architectural inductive bias (attention vs recurrence).

To disentangle whether performance differences in sequence-only models arise from the sequential representation itself or from the inductive bias of attention, we benchmark recurrent architectures on SELFIES [24] and SMILES segmentations (including SMILES Pair Encoding, a substructure-aware tokenization) [104, 105]. In particular, we evaluate LSTM and the recently proposed xLSTM, which augments classical gating with exponential gating, memory mixing, and matrix memory to handle longer-range dependencies and irregular patterns more efficiently [103]. This comparison isolates how architectural bias (graph vs. sequence; attention vs. recurrence) and tokenization granularity interact for modeling membrane permeability behavior in macrocycles.

Rigorous assessment is essential beyond aggregate error metrics, we follow recent guidance for method comparison in chemical machine learning by reporting not only statistical significance but also practical relevance of observed differences across repeated fits and data splits [60]. Accordingly, our evaluation emphasizes effect sizes and interval estimates alongside standard hypothesis testing, to ensure that observed gains are meaningful for downstream triage.

In this work, we introduce GSGE, a fragment-centric graph representation for cyclic peptides, and benchmark topology-aware graph models against sequence baselines under a repeated-fit evaluation protocol. We further examine transfer across permeability endpoints and provide open-source code and data for full reproducibility.

4.2 Results

We benchmarked all models on CycPeptMPDB [44] and adopted the split protocol of Li *et al.* to enable direct comparison with CycPeptMP [45]. Unless otherwise noted, reported values summarize repeated fits on these predefined splits. We first evaluate the quality of the learned fragment representation, then compare single-modality and fused model variants, and finally assess the statistical and practical relevance of observed between-model differences.

4.2.1 Coherent Fragment Embedding Space via Graph Autoencoding

To learn dense representations of the GSGE vocabulary, we trained a graph autoencoder on a broad corpus of non-generalized molecular fragments (see Methods, Graph Autoencoder for Fragment Embeddings). At epoch 300, the model reconstructed training-set fragments with strong reconstruction metrics: atom-type macro F1 reached 0.945, edge-type macro F1 0.971, and adjacency macro F1 0.930 (Figure S4.5). On the held-out validation split, edge and adjacency reconstruction remained robust (F1 $\geq$ 0.83), whereas atom-type F1 dropped to 0.445, indicating overfitting of the atom-type reconstruction head on fragments outside the vocabulary.

Analyses of the learned embedding space indicate that it captures chemically meaningful structure. Fragments assigned to the same Maximum Common Substructure (MCS) cluster co-localize in the t-SNE projection (Figure 4.2), and corpus fragments (grey) occupy contiguous regions around the vocabulary anchors. Complementary PCA and UMAP projections (Figures S4.2 and S4.3) show similar global organization

across dimensionality-reduction methods, and representative fragments for each MCS cluster are shown in Figure S4.4.

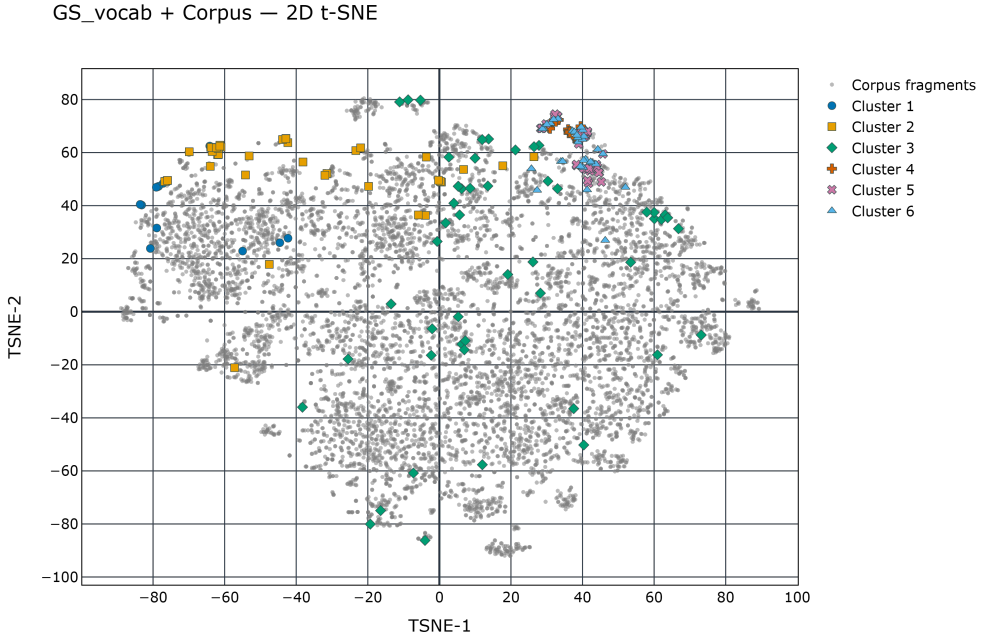

Figure 4.2: Two-dimensional t-SNE projection of graph autoencoder (GAE) fragment embeddings for the vocabulary overlaid on the GSGE training corpus. Grey points represent 8,713 corpus-only fragments embedded jointly with the vocabulary. Colored markers ($N_{\text{vocab}} = 201$ vocabulary fragments) are grouped by MCS-based hierarchical cluster membership (cophenetic coefficient = 0.95). Marker shape additionally encodes cluster identity. The GAE encoder (AttentiveFP, 300 epochs) produces 128-dimensional latent representations; t-SNE was computed with perplexity 30 and random seed 42.

Despite producing a structurally interpretable latent space, the GAE embeddings did not yield a measurable improvement in downstream estimation accuracy relative to one-hot encodings (OHE) with learned lookup tables. Both input strategies show comparable performance when paired with the same GCN and descriptor fusion architecture, indicating that the GCN message-passing layers already capture most useful fragment-level context for this dataset.

4.2.2 Test-Set Performance and Model Comparison

All models were evaluated on a fixed holdout test set of 344 cyclic peptides using the cross-validation (CV) protocol of Li *et al.* [45]. For graph-based models and non-sequence baselines trained in this work, the three CV splits were each refitted with four random seeds (12 total fits per model). In contrast, the sequence-model ablation (Transformer, LSTM, xLSTM) –i.e., systematically removing or replacing individual model components to isolate their contribution [106]– used a single seed per fold ($n_{\text{rep}} = 3$ total), matching the n_{rep} values in Table 4.1. For each graph-based model, we trained 3 CV folds $\times$ 4 random seeds = 12 independent fits. We then constructed two ensemble tiers: 4-Ens, which averages the outputs of the 4 seeds within each fold (yielding 3 fold-level ensemble outputs), and 12-Ens, which averages all 12 fits into a single output per compound. Ensembling reduces estimation variance by combining independently initialized models. Sequence-based models (Transformer, LSTM, xLSTM) were trained with a single seed per fold ($n_{\text{rep}} = 3$), so ensemble variants are not available for these architectures. Figure 4.3 ranks all 24 evaluated models by mean absolute error (MAE), including the sequence baselines, which are discussed in detail in the dedicated 'Sequence Models' subsection below. Table 4.1 reports MAE, R^2, and MSE for the principal model families.

Graph–Descriptor Fusion Models Are Among the Best Performing

Among single-model architectures, GCN+7desc (MAE = 0.358 $\pm$ 0.012, R^2 = 0.766) is numerically comparable to the CycPeptMP baseline reported by Li *et al.* (MAE = 0.355 $\pm$ 0.007, R^2 = 0.772) [45], while relying on only seven open-source descriptors, no SMILES enumeration, and no conformer-specific proprietary features. Replacing the seven descriptors with Morgan fingerprints in the single-model GCN+FPS setting (MAE = 0.423 $\pm$ 0.013, R^2 = 0.700 $\pm$ 0.016) or omitting descriptors entirely (GCN, MAE = 0.492) results in higher error, indicating that the 7-descriptor (2D/3D) descriptor channel contributes information not captured by the fragment-level GCN alone. Conversely, the descriptor-only baseline (7desc-MLP, MAE = 0.465) and fingerprint-only baseline (FPS-MLP, MAE = 0.463) both perform worse than the fused variants, indicating that the GSGE graph representation and molecular descriptors provide complementary information.

GAE-GCN+7desc performs slightly below GCN+7desc across all metrics (MAE = 0.368 $\pm$ 0.013, R^2 = 0.746). The difference is small, therefore we conclude that OHE-based lookup tables and GAE-derived embeddings carry similar information when

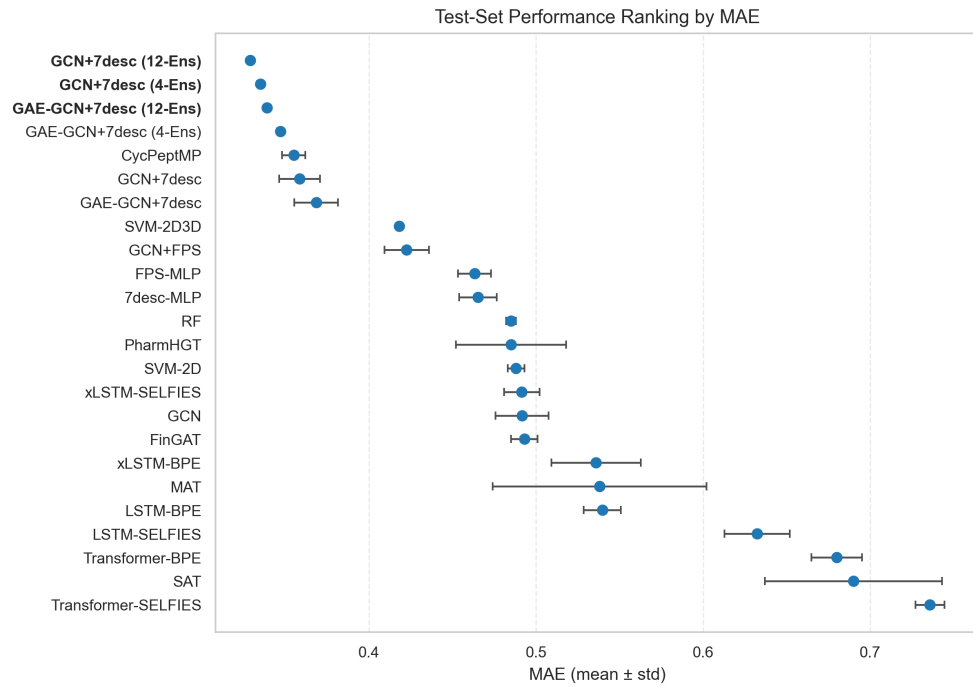

Figure 4.3: Ranked mean absolute error (MAE) for all 24 evaluated models on the held-out test set. Points represent the cross-validated mean MAE across folds and seeds; error bars indicate ±1 standard deviation where available. Models are sorted from lowest (best) to highest (worst) MAE, with the top three highlighted in bold. All outputs are in $\log_{10}(P_{\mathrm{app}} / [\mathrm{cm\,s^{-1}}])$ units, clipped to $[-8.0, -4.0]$.

trained jointly with the GCN encoder.

Ensembles Reduce Variance and Improve Accuracy

Averaging the four seed-specific fits within each CV fold (4-Ens) reduces both error and variance. GCN+7desc (4-Ens) reaches MAE = 0.335 ± 0.002 and R^2 = 0.796 ± 0.003, improving on the corresponding single-model variant and showing numerically lower MAE than CycPeptMP. Extending the ensemble to all 12 fits across folds and seeds, GCN+7desc (12-Ens) achieves the lowest overall error (MAE = 0.329, R^2 = 0.802, MSE = 0.219). The GAE counterparts follow the same trend: GAE-GCN+7desc (4-Ens) attains MAE = 0.347 and GAE-GCN+7desc (12-Ens) attains MAE = 0.339.

Figure 4.4 compares per-sample estimated versus experimental permeability for three representative GCN-family models for which per-sample outputs were retained

across all seeds and folds, enabling residual and error-dispersion diagnostics. Consistent with the consolidated metrics in Table 4.1, GCN+7desc (12-Ens) shows the tightest alignment with the identity line ($R^2 = 0.802$, MAE = 0.329), followed by GAE-GCN+7desc (12-Ens; $R^2 = 0.781$, MAE = 0.339), while the single-model GCN+FPS variant is clearly weaker ($R^2 = 0.700 \pm 0.016$, MAE = 0.423 ± 0.013). Residual analyses, absolute-error distributions, and output-dispersion plots for these three models are provided in Figures S4.9–S4.11.

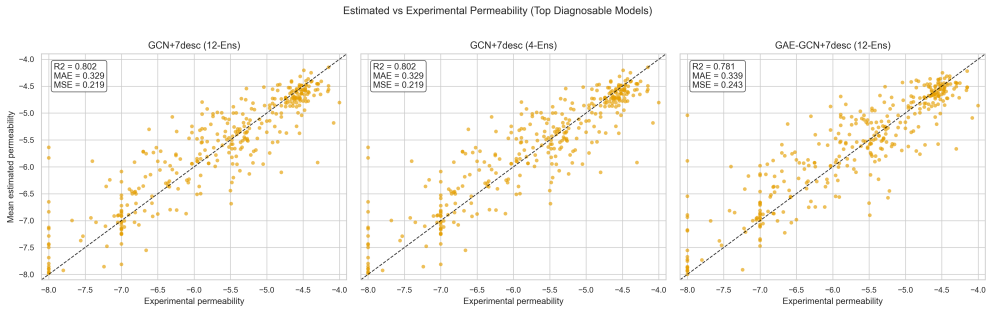

Figure 4.4: Estimated versus experimental $\log_{10}(P_{\mathrm{app}} / [\mathrm{cm\,s^{-1}}])$ for three representative GCN-family models for which per-compound outputs from every seed-fold run were available for diagnostic analysis: GCN+7desc (12-Ens), GCN+7desc (4-Ens), and GAE-GCN+7desc (12-Ens). Each point is one test compound; estimated values are the mean across all seed–fold runs. The dashed line is the identity. Inset boxes report R^2, MAE, and MSE for the plotted outputs; cross-model comparisons in the main text follow Table 4.1.

Sequence Models: Recurrent Architectures Outperform Transformers

To probe the role of architectural inductive bias, we benchmarked Transformer, LSTM, and xLSTM models on SELFIES and SMILES Byte-Pair Encoding (BPE or SPE) representations (single-seed evaluation). Among these, xLSTM-SELFIES achieved the highest R^2 (0.608 ± 0.022) and an MAE of 0.491 ± 0.011, followed by LSTM-BPE ($R^2 = 0.507$) and xLSTM-BPE ($R^2 = 0.496$). Standard transformers showed lower performance in this setup: Transformer-BPE reached $R^2 = 0.233$ and Transformer-SELFIES $R^2 = 0.103$. These results are directionally consistent with the lower performance of SAT ($R^2 = 0.207$) and the moderate recovery of Molecule Attention Transformer (MAT) ($R^2 = 0.547$), which incorporates pairwise distance modulation into its attention mechanism [45]. Given the single-seed evaluation and unequal hyperparameter-optimization budgets, this comparison is best interpreted as an architectural trend in this benchmark rather than a definitive ranking.

Table 4.1: Performance on the held-out test set (344 cyclic peptides). Models are grouped by family. All outputs are in $\log_{10}(P_{\mathrm{app}} / [\mathrm{cm\,s^{-1}}])$ units, clipped to $[-8.0, -4.0]$. Li *et al.* baselines are taken from Ref. 45.

	Model	MAE	R^2	MSE
Graph-based	GCN+7desc (12-Ens)	0.329	0.802	0.219
	GCN+7desc (4-Ens)	0.335 (±0.002)	0.796 (±0.003)	0.227 (±0.004)
	GAE-GCN+7desc (12-Ens)	0.339	0.781	0.243
	GAE-GCN+7desc (4-Ens)	0.347 (±0.002)	0.773 (±0.006)	0.252 (±0.007)
	GCN+7desc	0.358 (±0.012)	0.766 (±0.014)	0.259 (±0.015)
	GAE-GCN+7desc	0.368 (±0.013)	0.746 (±0.016)	0.282 (±0.018)
	GCN+FPS	0.423 (±0.013)	0.700 (±0.016)	0.332 (±0.018)
	GCN	0.492 (±0.016)	0.574 (±0.030)	0.472 (±0.033)
Sequence	xLSTM-SELFIES	0.491 (±0.011)	0.608 (±0.022)	0.435 (±0.025)
	LSTM-BPE	0.540 (±0.011)	0.507 (±0.028)	0.546 (±0.032)
	xLSTM-BPE	0.536 (±0.027)	0.496 (±0.045)	0.559 (±0.050)
	LSTM-SELFIES	0.632 (±0.019)	0.398 (±0.025)	0.667 (±0.027)
	Transformer-BPE	0.680 (±0.015)	0.233 (±0.035)	0.851 (±0.039)
	Transformer-SELFIES	0.736 (±0.009)	0.103 (±0.049)	0.995 (±0.054)
Baselines	CycPeptMP [45]	0.355 (±0.007)	0.772 (±0.011)	0.253 (±0.013)
	SVM-2D3D [45]	0.418 (±0.001)	0.689 (±0.002)	0.345 (±0.002)
	FPS-MLP	0.463 (±0.010)	0.653 (±0.015)	0.384 (±0.017)
	7desc-MLP	0.465 (±0.011)	0.615 (±0.020)	0.427 (±0.022)
	RF [45]	0.485 (±0.003)	0.657 (±0.003)	0.380 (±0.004)
	PharmHGT [45]	0.485 (±0.033)	0.622 (±0.029)	0.420 (±0.032)
	SVM-2D [45]	0.488 (±0.005)	0.595 (±0.012)	0.449 (±0.014)
	FinGAT [45]	0.493 (±0.008)	0.626 (±0.018)	0.415 (±0.021)
	MAT [45]	0.538 (±0.064)	0.547 (±0.091)	0.503 (±0.102)
	SAT [45]	0.690 (±0.053)	0.207 (±0.140)	0.879 (±0.155)

4.2.3 Statistical and Practical Significance

Aggregate metrics can overstate or obscure real differences between methods when evaluated on a single data split. Using the protocol described in Methods, Statistical Significance Analysis, we assessed between-model differences across MAE, MSE, Pearson r, and R^2, focusing on models with fold-level replicates. Full 16-model visualizations are provided in the Supporting Information; the main-text figures focus on a reduced set of top contenders and representative baselines for readability.

Normality diagnostics on the residualized performance values (metric value minus method mean) showed approximately symmetric, unimodal distributions for all four metrics (Figure S4.21), and Levene's test for homogeneity of variances was non-

4.2. Results

significant ($p > 0.05$), supporting the parametric assumptions. The omnibus repeated-measures ANOVA was highly significant for every metric ($p < 10^{-13}$), and all four Friedman tests likewise rejected the null hypothesis that methods share a common rank distribution ($p < 2 \times 10^{-4}$). Given only three CV folds, these omnibus tests are interpreted as directional group-level evidence and are not used as standalone proof of pairwise superiority.

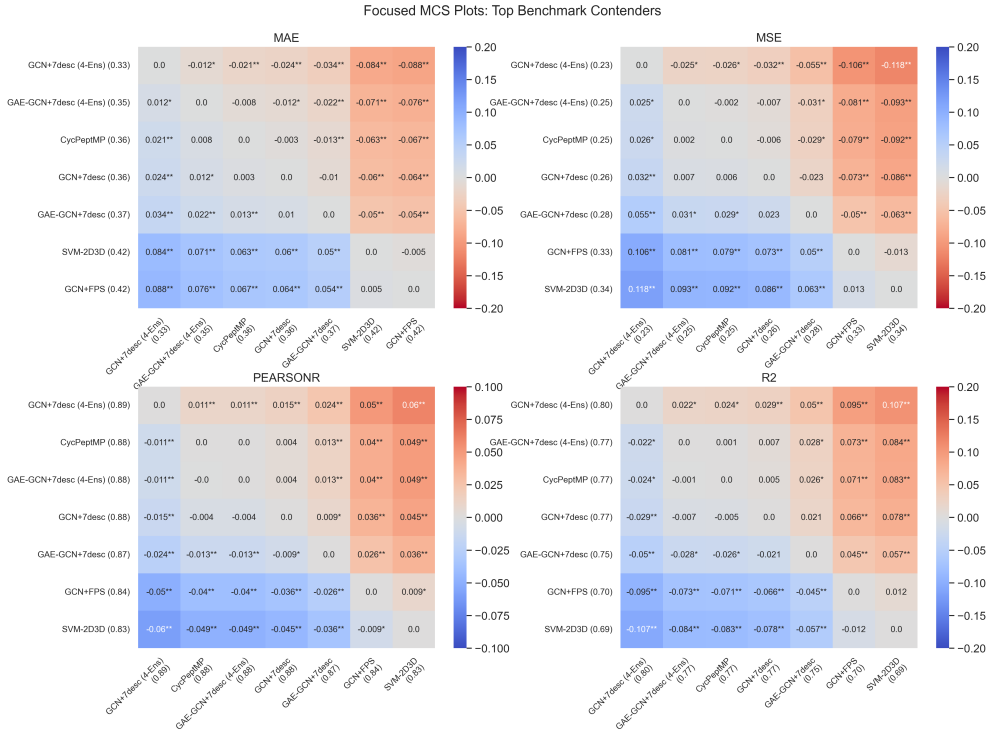

Figure 4.5: Focused Multiple Comparison of means with Significance (MCS) plots for MAE, MSE, Pearson r, and R^2 from repeated-measures ANOVA with Tukey HSD. The reduced model set improves legibility of pairwise effects among the strongest contenders. Each cell shows the mean difference (row minus column); stars denote significance ($^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$). Color intensity encodes practical significance relative to predefined thresholds (0.1 for MAE, MSE, R^2; 0.05 for Pearson r).

In the focused pairwise contrasts (Figure 4.5), GCN+7desc (4-Ens) is consistently strongest. For MAE, it outperforms CycPeptMP ($\Delta = -0.021$, 95% CI $[-0.031, -0.011]$, adjusted $p = 0.001$) and GAE-GCN+7desc (4-Ens) ($\Delta = -0.012$, 95% CI $[-0.022, -0.002]$, adjusted $p = 0.014$). CycPeptMP and GCN+7desc remain statistically non-separable on MAE and MSE in this focused view, whereas lower-ranked baselines (SVM-2D3D

and GCN+FPS) are clearly worse across metrics. Pearson r and R^2 panels follow the same ordering trend, with smaller absolute effect sizes among the top models.

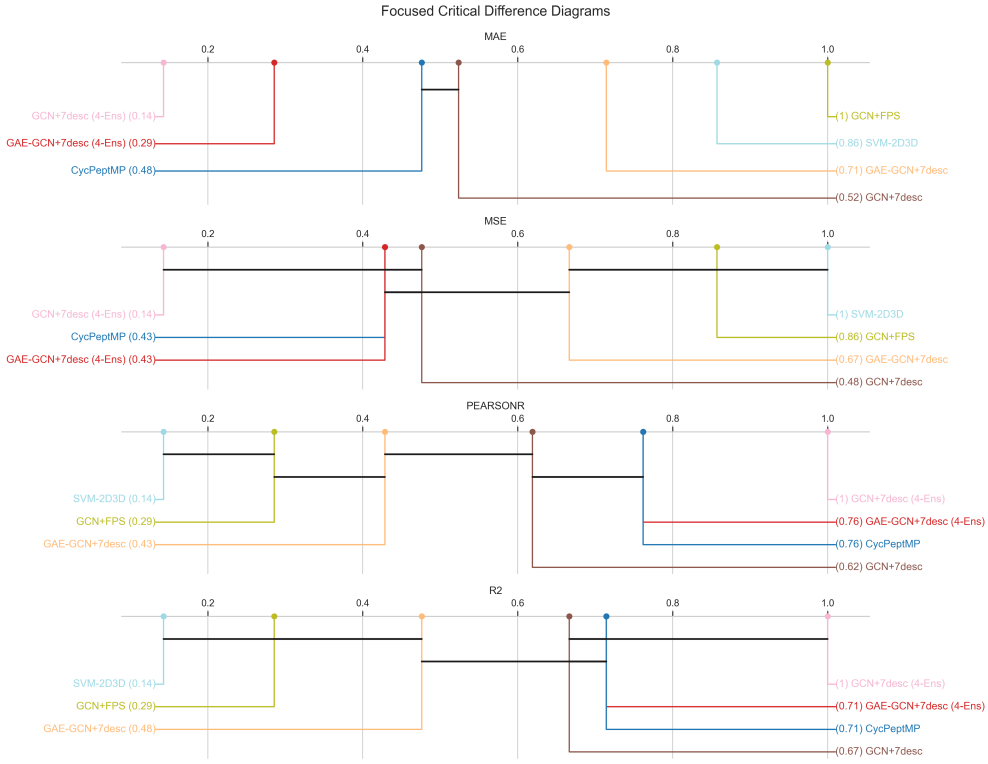

Figure 4.6: Focused critical difference (CD) diagrams from Conover–Friedman post hoc testing with Holm adjustment for MAE, MSE, Pearson r, and R^2. The reduced method set makes rank ordering and overlapping non-significant groups among top contenders easier to read. Horizontal bars connect models that are not significantly different at the 0.05 level.

Critical difference (CD) diagrams summarise rank-based post hoc comparisons; methods connected by a horizontal bar are not significantly different at the chosen α level, making it easy to see which models form statistically indistinguishable groups. The focused critical difference diagrams (Figure 4.6) provide a complementary rank-based view: GCN+7desc (4-Ens) remains at the favorable end of each metric, while CycPeptMP and GAE-GCN+7desc (4-Ens) fall into overlapping non-significant groups across several comparisons. This overlap, despite significant Tukey contrasts for selected pairs, reflects the conservative behavior of rank-based post hoc testing with only three folds. Full 16-model MCS and critical-difference diagrams are provided in Figures S4.19 and S4.20 (Supporting Information), together with nonparametric sign plots

(Figure S4.24), omnibus diagnostics (Figures S4.22 and S4.23), and a CycPeptMP-centered forest view (Figure S4.26).

An important caveat is that three CV splits provide limited statistical power. Following the guidance of Ash *et al.* [60], who recommend 25 or more replicates for robust pairwise separation, the present analysis should be read as directional evidence rather than definitive proof of equivalence within the top tier. The concordance between parametric and nonparametric pathways across all four metrics provides convergent evidence for the same broad ranking pattern.

4.2.4 Cross-Endpoint Transfer

To assess how well PAMPA-trained models generalize beyond the training endpoint, we evaluated all models on three external permeability assays: Caco-2 ($N_{\text{cmpd}} = 381$), MDCK ($N_{\text{cmpd}} = 17$), and RRCK ($N_{\text{cmpd}} = 56$), following Li *et al.* [45]. Evaluation followed the protocol described in Methods, Cross-Endpoint Transfer Evaluation.

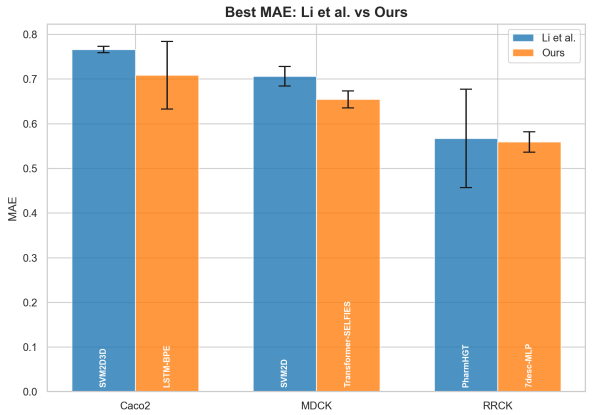

Figure 4.7: Cross-endpoint transfer MAE for the best Li *et al.* model versus the best model from this work at each endpoint. Error bars show ±1 standard deviation over the reported replicate aggregates. In this evaluation, models from this work have numerically lower mean MAE on all three endpoints, with the largest differences on Caco-2 and MDCK.

Figure 4.7 compares the best-performing Li *et al.* model and the best model from this work at each endpoint. On Caco-2, LSTM-BPE (MAE $= 0.708 \pm 0.075$) shows lower mean MAE than the best Li *et al.* baseline SVM-2D3D (MAE $= 0.766 \pm 0.007$). On MDCK, Transformer-SELFIES (MAE $= 0.654 \pm 0.019$) shows lower mean MAE than SVM-2D (MAE $= 0.706 \pm 0.022$), although this endpoint has a small sample size ($N_{\text{cmpd}} = 17$). On RRCK, 7desc-MLP (MAE $= 0.559 \pm 0.023$) is only marginally

lower than PharmHGT (MAE = 0.567 ± 0.110). Overall, these comparisons should be interpreted as directional (numerical) differences rather than formal significance claims.

Full per-model ranked MAE dot plots for each endpoint are provided in Figure S4.28, and a grouped bar chart in Figure S4.27. A complete numerical summary (MAE and Pearson r for all 24 models across the three endpoints) is given in the Supporting Information (Table S4.5).

4.3 Discussion

This work introduced GSGE, a data-driven fragment graph representation for cyclic peptides. In the benchmark setting used here, a GCN operating on GSGE compound graphs and fused with seven open-source 2D/3D molecular descriptors achieved low test-set error and a modest pairwise MAE advantage for the 4-Ens variant over CycPeptMP ($\Delta = -0.021$, 95% CI $[-0.031, -0.011]$). Corresponding Tukey-adjusted MAE/MSE contrasts were significant, while nonparametric rank tests retained overlapping top groups at three-fold resolution [45]. With three folds, these inferential results are best read as directional rather than definitive. Below we discuss the findings in relation to the reported evidence.

4.3.1 Inductive Bias: Graph versus Sequence Architectures

A consistent observation across our experiments and those of Li *et al.* [45] is that standard Transformer models showed lower performance for cyclic peptide permeability estimation ($R^2 \leq 0.23$ for our Transformer variants; $R^2 = 0.21$ for SAT). This pattern is consistent with a mismatch between the inductive bias of global pairwise attention and the structure–property relationships in macrocycles. Unlike linear protein sequences, which can exhibit evolutionarily conserved positional semantics tied to secondary and tertiary structure [26, 93], cyclic peptides often show weaker sequential constraints and substantial conformational plasticity [38, 41, 43]. Under these conditions, sequence-only attention models may capture less task-relevant structure than topology-aware encoders [95, 107].

Molecule Attention Transformer (MAT) partially recovers by injecting pairwise conformer and graph distances into its attention weights [45], which shifts the model toward topology-aware representations. In our sequence ablation, LSTM and xLSTM reached higher R^2 values than Transformers ($R^2 \geq 0.40$), with xLSTM-SELFIES

reaching $R^2 = 0.61$. The lower performance of LSTM-SELFIES relative to LSTM-BPE is consistent with longer token sequences under SELFIES. At the same time, this sequence-model comparison should be interpreted cautiously because it used single-seed evaluation and non-uniform hyperparameter-optimization budgets. Within those constraints, the results indicate a topology-favoring trend in this benchmark [95, 96, 98]. This trend is consistent with a graph-based, position-agnostic encoder being well suited to the graph-like connectivity and comparatively weak sequential regularities of cyclic peptides.

4.3.2 Descriptor Fusion and Open-Source Reproducibility

Despite retaining only 7 of the 16 descriptors used by Li *et al.* [45], the single-model GCN+7desc fusion model is numerically close to CycPeptMP (MAE 0.358 vs 0.355), and the 4-Ens variant shows lower MAE (0.335) than CycPeptMP in this benchmark. In focused Tukey comparisons, the MAE contrast versus CycPeptMP is $\Delta = -0.021$ (95% CI $[-0.031, -0.011]$, adjusted $p = 0.001$), indicating a modest but consistent advantage. The rank-based Conover–Friedman view still shows overlap among the strongest models, consistent with limited resolution at three folds. The descriptor channel is complementary to the GSGE graph channel: removing descriptors (GCN-only, MAE = 0.492) or removing the graph (7desc-MLP, MAE = 0.465) increases error relative to the fusion (GCN+7desc, MAE = 0.358). Morgan fingerprints, by contrast, contributed less when combined with the GCN.

A practical consideration for cyclic peptide modeling is the sensitivity of 3D descriptors to conformer generation. The descriptors used here are computed from a single conformer, which may not capture the conformational ensemble relevant to membrane translocation. Even so, the improved MAE of the fusion model over graph-only and descriptor-only baselines indicates that these descriptors contribute global physicochemical information not fully represented in the fragment graph alone. Conformational plasticity is not explicitly represented in the fragment graph or in the single-conformer descriptors used here.

4.3.3 Fragment Embeddings: One-Hot versus Graph Autoencoder

The GAE-based fragment embeddings yield a chemically coherent latent space in which MCS-similar fragments co-localize (Figure 4.2), yet they do not improve downstream estimation accuracy relative to OHE lookup tables. Two factors contribute: (i) the

GCN message-passing layers already contextualize fragment identities through neighborhood aggregation, and (ii) vocabulary overlap between training and test sets is high, which reduces the frequency of unseen-fragment cases.

4.3.4 Ensemble Modeling and Estimation Stability

Averaging across repeated fits (4 seeds per CV fold) consistently reduced both error and variance, with the 12-Ens variant achieving the lowest overall MAE (0.329) and R^2 (0.802) on the test set. Error-dispersion plots indicate that averaging is most beneficial in regions where individual fits show larger spread. The narrow standard deviations of the 4-Ens metrics ($\pm$0.002 for MAE) indicate stable behavior across CV folds.

4.3.5 Cross-Endpoint Transferability

Evaluating PAMPA-trained models on Caco-2, MDCK, and RRCK endpoints showed that the lowest-MAE model differs by assay: LSTM-BPE on Caco-2, Transformer-SELFIES on MDCK, and 7desc-MLP on RRCK. Endpoint-dependent ranking reflects both assay differences (cell-free PAMPA versus cell-based readouts) and dataset properties such as sample size and noise (e.g., MDCK has a small sample size ($N_{\mathrm{cmpd}} = 17$)), which can destabilise rank order.

4.3.6 Limitations

The statistical significance analysis is constrained by the use of three CV splits, which provides limited power for pairwise separation. Following the guidance of Ash *et al.* [60], our results should be interpreted as directional evidence rather than definitive proof of equivalence within the top tier. The sequence-model comparison is additionally limited by single-seed evaluation and non-uniform hyperparameter-optimization budgets. The GSGE representation currently lacks explicit encoding of bond orientation or wildcard position between fragments, which may limit its expressiveness for certain chemistries. Additionally, the GAE decoder reconstructs graphs in a single forward pass, which may constrain reconstruction quality for the most complex fragments. The four loss terms in the GAE objective were equally weighted; alternative weighting schemes were not evaluated in this work.

4.4 Conclusion

We presented GSGE, a monomer-agnostic fragment graph representation for cyclic peptides built on data-driven Group-SELFIES vocabularies. In the benchmark used here, a GCN operating on GSGE graphs and fused with seven open-source molecular descriptors achieved the lowest test-set MAE (0.329 as a 12-Ens, $R^2 = 0.802$). The corresponding single-model variant is numerically close to CycPeptMP, and focused pairwise contrasts indicate a modest MAE advantage for the 4-Ens GCN+7desc model over CycPeptMP ($\Delta = -0.021$, 95% CI $[-0.031, -0.011]$); Tukey-adjusted MAE/MSE contrasts are significant, while nonparametric rank tests still indicate overlapping top groups at the available three-fold resolution [45, 60]. Given the three-fold replicate design, these inferential results are best interpreted as directional evidence rather than definitive pairwise separation. A graph autoencoder trained on the GSGE fragment corpus yields a structurally interpretable embedding space, while one-hot fragment encodings yield comparable downstream estimation performance in this setting. Sequence-model ablations showed higher R^2 for recurrent models than for Transformers in this benchmark, but this comparison remains limited by single-seed evaluation and non-uniform hyperparameter optimization. Cross-endpoint evaluation (Caco-2, MDCK, RRCK) further showed endpoint-dependent ranking among model families, with numerically lower or similar MAE relative to the strongest Li *et al.* baselines depending on assay. Taken together, these results position GSGE as a flexible, transferable, and reproducible foundation for cyclic peptide permeability modeling.

The main contributions of this work are fivefold: (i) it introduces GSGE, a modular Group-SELFIES-derived fragment graph representation for cyclic peptides that is designed to accommodate non-canonical building blocks [25]; (ii) it shows that a GCN fused with seven open-source 2D/3D descriptors performs competitively with CycPeptMP while avoiding proprietary descriptors and SMILES enumeration [45, 98]; (iii) it establishes that graph-autoencoder fragment embeddings form a chemically coherent latent space but do not improve downstream estimation relative to one-hot lookup embeddings in this benchmark [99–101]; (iv) it provides a controlled comparison of Transformer, LSTM, and xLSTM sequence models on SELFIES and SMILES segmentations to examine architectural bias for macrocyclic membrane permeability [24, 92, 103–105]; and (v) it applies a repeated-fit evaluation protocol with statistical and practical significance analysis, complemented by cross-endpoint assessment on Caco-2, MDCK, and RRCK assays [45, 60].

4.5 Methods

4.5.1 Dataset and Evaluation Protocol

All primary benchmark experiments used the CycPeptMPDB PAMPA subset [44], comprising 6,889 cyclic peptides with experimentally measured membrane permeability. Because PAMPA is a cell-free, artificial-membrane assay, the resulting apparent permeability (P_{app}) reflects passive transcellular transport and does not capture carrier-mediated or paracellular pathways [44]. The modeling target is $\log_{10}(P_{\mathrm{app}} \,/\, [\mathrm{cm\,s^{-1}}])$, clipped to the range $[-8.0, -4.0]$ following the preprocessing of Li *et al.* [45]; all MAE, RMSE, and related error metrics reported in this work are therefore in these $\log_{10}$ units unless stated otherwise. We adopted the split protocol provided by Li *et al.* [45]: three cross-validation folds (6,201 training and 344 validation peptides per fold) plus a fixed holdout test set of 344 peptides shared across folds. For each fold, graph-based models and non-sequence baselines trained in this work were refit with four random seeds (12 total fits per model), and ensemble variants averaged outputs from either the four within-fold fits (4-Ens) or all 12 fits across folds (12-Ens). Sequence-model ablations (Transformer, LSTM, xLSTM) were instead evaluated with one seed per fold ($n_{\mathrm{rep}} = 3$ total), as reflected in Table 4.1. For contextual comparison, CycPeptMP and additional baselines from Li *et al.* were used as reported in Ref. 45. Per-compound output files required for residual and output-dispersion diagnostics were available only for models trained in this work.

4.5.2 Molecular Descriptors and Fingerprints

In addition to fragment-graph inputs, we used whole-molecule descriptors and fingerprint baselines. Li *et al.* [45] identified 16 descriptors informative for cyclic peptide permeability, nine of which require the proprietary Molecular Operating Environment (MOE) package [108]. To maintain full open-source reproducibility, we retained the seven descriptors computable with RDKit and Mordred [22, 109]: MolLogP and fr_Al_OH (RDKit 2D), VSA_EState9 (Mordred 2D), and FNSA2, FNSA4, FNSA5, RNCS (Mordred 3D), the latter four computed from a single conformer generated internally from each input SMILES using RDKit. As a separate baseline, we also evaluated Morgan fingerprints (radius 2, 2048 bits) [21, 110]. Further details on the original descriptor-selection rationale are given in Ref. 45.

4.5.3 Group-SELFIES Graph Embeddings (GSGE)

Standard molecular representations for cyclic peptides typically rely on predefined monomer vocabularies that must be manually curated for each peptide chemistry [45]. As chemical diversity increases to include non-canonical amino acids, chemical modifications, and non-peptidic macrocyclic scaffolds, such vocabularies are insufficient and require ad hoc extension, so residue assignments are not always unique or straightforward. To remove this dependency, we developed the Group-SELFIES Graph Embeddings (GSGE) framework, an open-source Python package that derives a data-driven fragment vocabulary from any SMILES corpus and uses it to construct fragment-level molecular graphs [25].

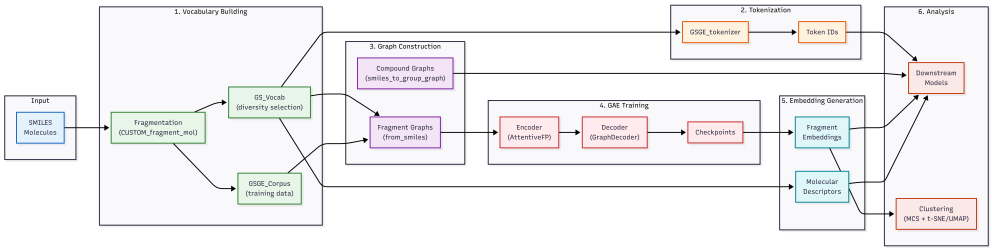

Figure 4.8: Data flow pipeline of the GSGE framework. Input SMILES are fragmented and processed through a diversity-based vocabulary selection step (GS_Vocab) and corpus construction (GSGE_Corpus). Tokenization maps fragments to integer IDs. Fragment graphs are fed into a graph autoencoder (AttentiveFP encoder + GraphDecoder) for unsupervised representation learning. The resulting fragment embeddings, together with computed molecular descriptors, serve as feature inputs for downstream models and clustering analyses.

GSGE builds on the Group-SELFIES formalism [25] and extends it in several ways (Figure 4.8). We define a fragment as a recurring molecular substructure selected in a data-driven manner from a training corpus on the basis of frequency and structural diversity, rather than as a predefined chemical group; every atom in a molecule is guaranteed to be assigned to such a fragment. Fragmentation begins with a set of user-specified SMARTS bond-cleavage rules; in this work we cleave outgoing ring bonds, amide/peptide bonds, and disulfide bonds. The resulting fragment pool is scored by frequency and structural diversity, and a greedy selection procedure (GS_Vocab) retains the most informative fragments up to a configurable vocabulary size. Because fragments are retained with their full substitution patterns rather than abstracted into Markush-style R-group templates, we refer to these as 'non-generalized fragments' throughout. Common small groups and single atoms are appended automatically so that every atom in a molecule is covered, a key extension over the original Group-

SELFIES implementation, which generally groups only a subset of the molecule [25]. To support robust fragmentation of large cyclic peptides (which exceed the typical size of small molecules), we forked the original Group-SELFIES package and resolved limitations related to serialization and a hard-coded cap on SMARTS pattern matching (see Supporting Information for details).

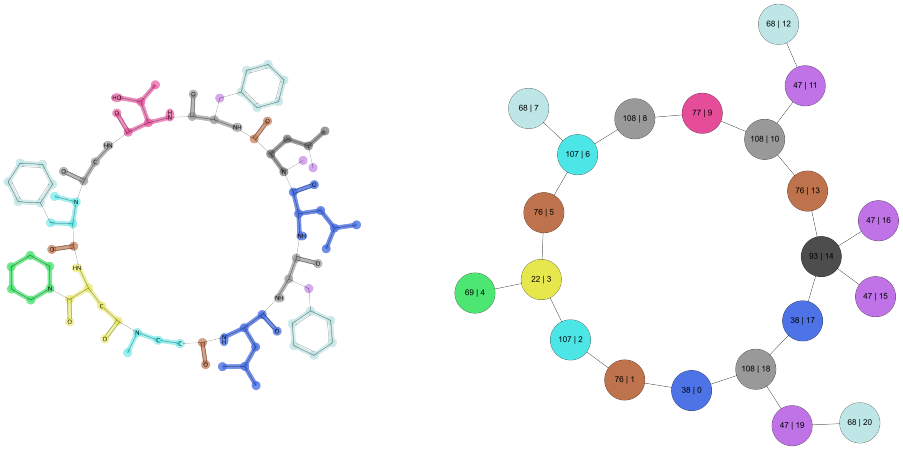

Figure 4.9: Fragmentation of a cyclic peptide into a compound graph. Left: a cyclic peptide with atoms color-coded by fragment identity. Right: the corresponding compound graph, with nodes representing fragments and edges representing inter-fragment bonds. Node labels indicate vocabulary indices. The representation captures both larger recurring substructures and residual single-atom fragments, ensuring full molecular coverage.

Given a molecule, GSGE tokenizes it by matching vocabulary fragments in order of decreasing size and constructs a compound graph whose nodes are fragments and whose edges are inter-fragment bonds (Figure 4.9). Because larger fragments are matched first, an atom that participates in a larger recurring substructure in one local context may instead fall back to a single-atom fragment elsewhere, depending on whether the surrounding structure matches an entry in the vocabulary; this is an intended consequence of greedy, size-ordered matching rather than an inconsistency in fragment definition. Each node carries either a one-hot encoding (OHE) with a learned lookup table or a dense embedding from the graph autoencoder described below. The architecture overview of the GSGE package is provided in Figure S4.1.

For this study, we assembled approximately 200,000 SMILES from PubChem, SureChEMBL, SmProt, and RandPept (from PeptideCLM) [111]. This pool spans linear and cyclic peptides with non-natural amino acids, chemical modifications, and small

molecules. From this pool, we randomly sampled 50,000 SMILES and derived a data-driven core vocabulary of 185 fragments with GSGE (maximum GS_Vocab size 200; maximum fragment size 15 heavy atoms). After appending common small groups and single-atom tokens for complete molecular coverage, the final modeling vocabulary comprised 201 fragments. The same 50,000-SMILES pool served as the source corpus for graph autoencoder training. A worked example illustrating the concrete token-level form of a Group-SELFIES string is provided in the Supporting Information (Table S4.3).

4.5.4 Graph Autoencoder for Fragment Embeddings

One-hot fragment encodings are inherently sparse and do not encode structural similarity between fragments. To obtain dense, structurally informed representations, we trained a graph autoencoder (GAE) over a broad fragment corpus, drawing on the principle that structurally related fragments should map to neighboring regions of a continuous latent space, analogous to word embeddings in natural language processing [100, 101].

The GAE consists of an AttentiveFP encoder [97] and a custom one-shot GraphDecoder, both implemented within the GSGE package [99]. The encoder processes fragment graphs with 9-dimensional atom features and 3-dimensional bond features through three graph convolutional layers (256 hidden channels each) and two attention-based message-passing steps, yielding a 128-dimensional latent vector per fragment. The decoder reconstructs the molecular graph in a single forward pass, outputting atom-type logits (22 types $\times$ up to 20 nodes), edge-type logits (5 bond types plus a "no bond" class for up to 190 edges in the upper-triangular adjacency matrix), and scalar counts for atoms and edges. Further architectural details are provided in the Supporting Information.

To construct a diverse training corpus, we applied a modified fragmentation that retains all unique core structures with their specific substituents rather than merging R-group variants. This yielded 8,713 non-generalized fragments (the "GSGE corpus"), which were split 80:20 into training and validation sets; all 201 vocabulary fragments were placed in the training partition. Training used the Adam optimizer [112] (learning rate 10^{-3}, batch size 64, 300 epochs) with the following composite reconstruction loss:

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}}^{\mathrm{atom}} + \mathcal{L}_{\mathrm{CE}}^{\mathrm{bond}} + \mathcal{L}_{\mathrm{BCE}}^{\mathrm{edge}} + \mathcal{L}_{\mathrm{MSE}}^{\mathrm{count}}, \tag{4.1}$$

where $\mathcal{L}_{\mathrm{CE}}^{\mathrm{atom}}$ and $\mathcal{L}_{\mathrm{CE}}^{\mathrm{bond}}$ are cross-entropy losses over atom-type and bond-type logits,

$\mathcal{L}_{\text{BCE}}^{\text{edge}}$ is a binary cross-entropy loss for edge existence in the upper-triangular adjacency matrix, and $\mathcal{L}_{\text{MSE}}^{\text{count}}$ is the mean squared error for atom and edge count regression. All four terms are equally weighted.

To characterize the resulting embedding space, we applied agglomerative hierarchical clustering on pairwise MCS Tanimoto similarity to the vocabulary fragments [67] and projected the joint vocabulary-plus-corpus embeddings with t-SNE, PCA, and UMAP [113, 114].

4.5.5 Model Architectures

Using the feature representations above, we benchmarked both graph-based and sequence-based estimators.

GCN–Descriptor Fusion

The primary architecture combines a graph convolutional network (GCN) operating on GSGE compound graphs with a multilayer perceptron (MLP) branch for whole-molecule descriptors or fingerprints. The GCN consists of stacked GCNConv layers with batch normalization, ReLU activations, and dropout, with optional residual connections [98]. At each layer l, node representations are updated by aggregating messages from the one-hop neighborhood:

$$\mathbf{h}_i^{(l+1)} = \sigma\Big(\sum_{j \in \mathcal{N}(i) \cup \{i\}} \frac{1}{\sqrt{\hat{d}_i \, \hat{d}_j}} \mathbf{W}^{(l)} \, \mathbf{h}_j^{(l)} \Big), \tag{4.2}$$

where $\mathcal{N}(i)$ denotes the neighbors of node i, $\hat{d}_i = 1 + |\mathcal{N}(i)|$ is the augmented degree, $\mathbf{W}^{(l)}$ is a learnable weight matrix, and σ is a nonlinear activation function [98]. Node features $\mathbf{h}_i^{(0)}$ are initialized as either OHE lookup embeddings or precomputed GAE embeddings. Mean pooling over node representations yields a fixed-size graph-level vector, which is concatenated with the MLP-encoded descriptor vector and passed through a final MLP head to produce a scalar permeability estimate. Both single-modality baselines (GCN-only, descriptor-only, fingerprint-only) and fusion variants (GCN+7desc, GCN+FPS, GAE-GCN+7desc) were evaluated. The full hyperparameter search space and selected configurations are listed in the Supporting Information.

Sequence Models

To isolate the effect of architectural inductive bias, we benchmarked three sequence model families on SELFIES token sequences [24] and SMILES Byte-Pair Encoding (BPE) – a subword tokenization algorithm that iteratively merges the most frequent character pairs into single tokens, sometimes termed SMILES Pair Encoding (SPE) [104, 105] (576-token vocabulary): a standard Transformer [92], LSTM [102], and the recently proposed xLSTM with exponential gating and matrix memory [103]. Results for the attention-augmented MAT and SAT architectures are taken from Li *et al.* [45].

4.5.6 Hyperparameter Optimization

Hyperparameter optimization (HPO) was performed with Bayesian optimization in Weights & Biases Sweeps [115, 116]. For the GCN-based models, each trial trained for 200 epochs per CV fold, and the best epoch was selected from the window [50, 200] based on validation MAE on unbounded outputs. Early stopping was deliberately avoided: the censored target distribution causes high variance in validation curves even after convergence, and premature stopping can bias performance toward the upper-bound region of the permeability range (see Figure S4.29 for representative MAE trajectories). Outputs from the selected epoch were then clipped to $[-8.0, -4.0]$ $\log_{10}(P_{\mathrm{app}} \,/\, [\mathrm{cm\,s^{-1}}])$, and the mean MAE across the three CV folds defined the HPO objective.

For the sequence model ablation, we ran 30 HPO trials per model with an epoch window of [10, 50] (Figures S4.30–S4.31). These experiments used a single seed per fold to match the evaluation protocol of Li *et al.* [45]. Due to resource constraints, HPO was not performed for the xLSTM; it was also the only model that could not be run in fully deterministic mode. Detailed search spaces and final hyperparameter values are reported in the Supporting Information.

4.5.7 Statistical Significance Analysis

Method comparisons followed the protocol recommended by Ash *et al.* [60]. Two complementary testing pathways were applied to per-fold metric values (MAE, MSE, Pearson r, R^2) across $M = 16$ models with matched fold-level replicates. Here, M denotes the number of compared models, while n_{rep} denotes replicate aggregates used for summary statistics elsewhere in this manuscript. The parametric pathway used repeated-measures ANOVA with Tukey HSD post hoc tests; the nonparametric path-

way used the Friedman test with Conover–Friedman post hoc tests and Holm p-value adjustment. Practical significance was assessed by comparing observed mean differences against predefined thresholds (0.1 for MAE, MSE, R^2; 0.05 for Pearson r). Full diagnostic plots, including normality checks and homogeneity-of-variance tests, are provided in the Supporting Information.

4.5.8 Cross-Endpoint Transfer Evaluation

To evaluate transfer beyond the PAMPA endpoint, we applied PAMPA-trained models (without endpoint-specific retraining) to the Caco-2 ($N_{\mathrm{cmpd}} = 381$), MDCK ($N_{\mathrm{cmpd}} = 17$), and RRCK ($N_{\mathrm{cmpd}} = 56$) assay subsets from CycPeptMPDB, following Li *et al.* [45]. Here, N_{cmpd} denotes the number of compounds per endpoint, and n_{rep} denotes replicate aggregates used to summarize model metrics. For each seed–fold run, outputs were clipped to $[-8.0, -4.0]$ $\log_{10}(P_{\mathrm{app}} \,/\, [\mathrm{cm\,s^{-1}}])$, after which endpoint-level MAE and Pearson r were computed. For multi-seed models, these run-level metrics were averaged across folds within each seed and then summarized across seeds (most models: $n_{\mathrm{rep}} = 4$; GCN: $n_{\mathrm{rep}} = 3$). For single-seed sequence models ($n_{\mathrm{rep}} = 1$), SD reflects fold-to-fold variation. In this cross-endpoint evaluation, 4-Ens and 12-Ens values are metric-level aggregates of constituent runs rather than output-level ensemble metrics. Cross-endpoint baseline values from Li *et al.* were used as reported.

Supplementary Information

S4.1 Mathematical Notation

To reduce symbol overload, we use the following notation conventions throughout the main text and Supporting Information.

S4.1. Mathematical Notation

Symbol	Meaning
N_{cmpd}	Number of compounds in an endpoint dataset (e.g., Caco-2, MDCK, RRCK)
N_{vocab}	Number of vocabulary fragments in embedding-space projection figures
n_{rep}	Number of replicate aggregates used for mean ± SD summaries
M	Number of compared models in a significance analysis
i	Index for an individual compound (cross-endpoint evaluation) or graph node (GCN equations)
j	Neighbor-node index used in GCN message aggregation
f	Cross-validation fold index
s	Random-seed index
l	GCN layer index
P_{app}	Apparent permeability measured by the PAMPA assay ($\text{cm}\,\text{s}^{-1}$)
$\log_{10}(P_{\text{app}} \,/\, [\text{cm}\,\text{s}^{-1}])$	Modeling target; clipped to $[-8.0, -4.0]$
$\hat{y}_{i,f,s}$	Estimated $\log_{10} P_{\text{app}}$ for compound i from fold f and seed s
$\mathbf{h}_i^{(l)}$	Node representation of node i at GCN layer l
$\mathcal{N}(i)$	One-hop neighborhood of node i in a compound graph
$\hat{d}_i$	Augmented node degree, $\hat{d}_i = 1 + \lvert\mathcal{N}(i)\rvert$
$\mathbf{W}^{(l)}$	Learnable GCN weight matrix at layer l
σ	Nonlinear activation function in the GCN update
$\mathcal{L}$	Total graph-autoencoder reconstruction loss
$\mathcal{L}_{\text{CE}}^{\text{atom}}$	Atom-type cross-entropy component of $\mathcal{L}$
$\mathcal{L}_{\text{CE}}^{\text{bond}}$	Bond-type cross-entropy component of $\mathcal{L}$
$\mathcal{L}_{\text{BCE}}^{\text{edge}}$	Edge-existence binary cross-entropy component of $\mathcal{L}$
$\mathcal{L}_{\text{MSE}}^{\text{count}}$	Atom/edge-count mean-squared-error component of $\mathcal{L}$
Δ	Pairwise mean difference reported in statistical contrasts
r	Pearson correlation coefficient
ρ	Spearman rank-correlation coefficient
p	p-value from statistical hypothesis tests
R^2	Coefficient of determination

Table S4.1: Core notation used across the main text and Supporting Information.

S4.2 Supplementary Figures

S4.2.1 GSGE Package Architecture

The GSGE framework is implemented as a modular Python package following a Facade design pattern, in which a central GSGE class delegates operations to specialized manager components. Figure S4.1 provides an overview of the package architecture. Core modules handle vocabulary construction (GS_Vocab), corpus assembly (GSGE_Corpus), and tokenization, while graph neural network modules implement the AttentiveFP encoder and GraphDecoder used for unsupervised fragment representation learning. Output modules encapsulate the resulting embeddings, compound graphs, and clustering results. This separation of concerns allows each component to be developed, tested, and extended independently.

S4.2. Supplementary Figures

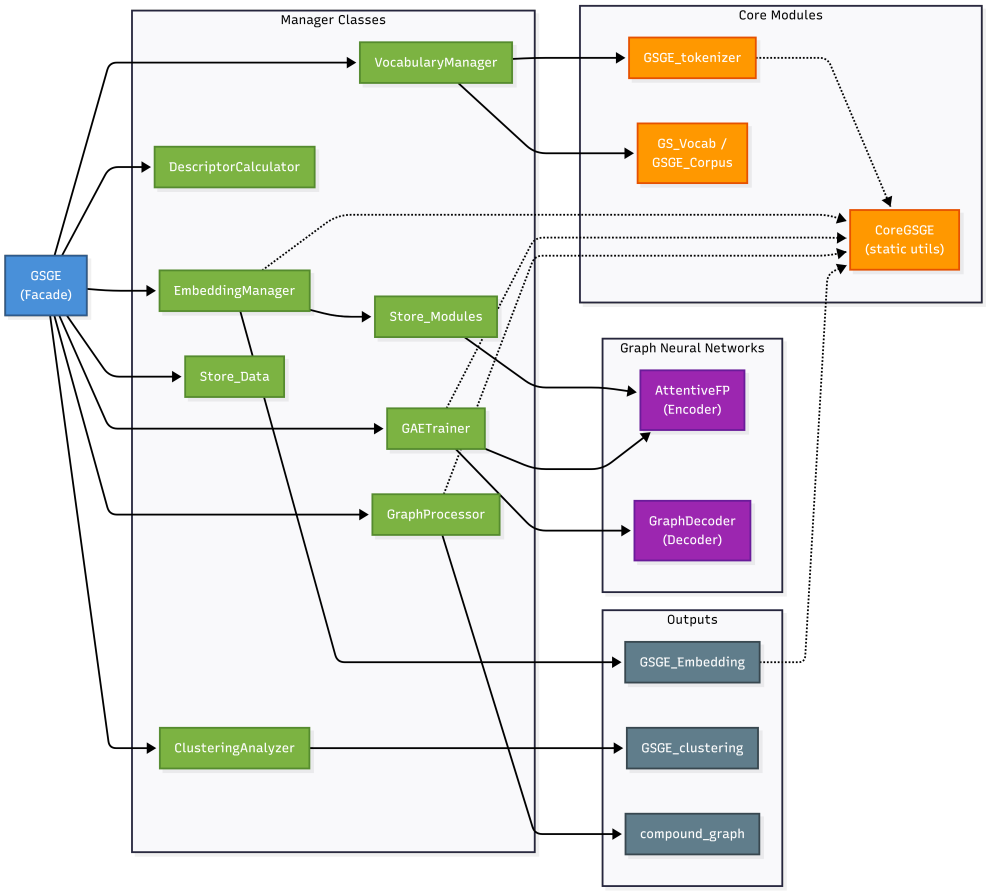

Figure S4.1: Architecture overview of the GSGE package. The central GSGE class (blue) delegates functionality to specialized manager classes (green), which orchestrate core preprocessing modules (orange) for tokenization, vocabulary construction, and shared utilities. Graph neural network components (purple) comprise an AttentiveFP encoder and GraphDecoder for learning fragment-level latent representations. Output modules (gray) encapsulate the resulting embeddings, clustering analyses, and compound graph structures. Solid arrows denote direct delegation; dashed arrows indicate shared utility dependencies.

S4.2.2 Fragment Embedding Space Visualizations

The main text presents a t-SNE projection of the GAE fragment embeddings (Figure 3 therein). Here we provide complementary PCA and UMAP projections of the same embedding space. Because PCA is a linear method, it captures the directions of maximum variance and reveals the dominant axes of chemical diversity, while UMAP preserves more of the global topology than t-SNE. Taken together, the three projections confirm that the cluster structure is robust across dimensionality-reduction methods: fragments assigned to the same MCS cluster consistently co-localize, and corpus fragments distribute continuously around the vocabulary anchors.

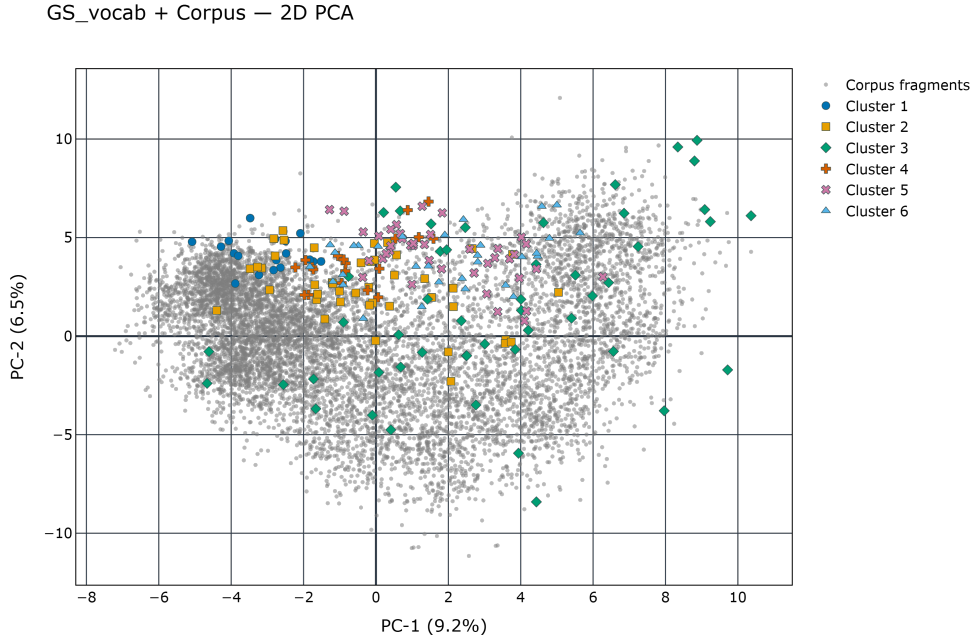

Figure S4.2: Two-dimensional PCA projection of GAE fragment embeddings for the GS_Vocab vocabulary overlaid on the GSGE training corpus. Grey points represent 8,713 corpus-only fragments projected jointly with the vocabulary onto the first two principal components. Colored markers ($N_{\text{vocab}} = 201$ vocabulary fragments) indicate cluster membership from agglomerative hierarchical clustering on MCS Tanimoto similarity (cophenetic coefficient = 0.95); marker shape additionally encodes cluster identity. As a linear method, PCA reflects the directions of maximum variance in the embedding space, providing a complementary view to the nonlinear t-SNE and UMAP projections.

S4.2. Supplementary Figures

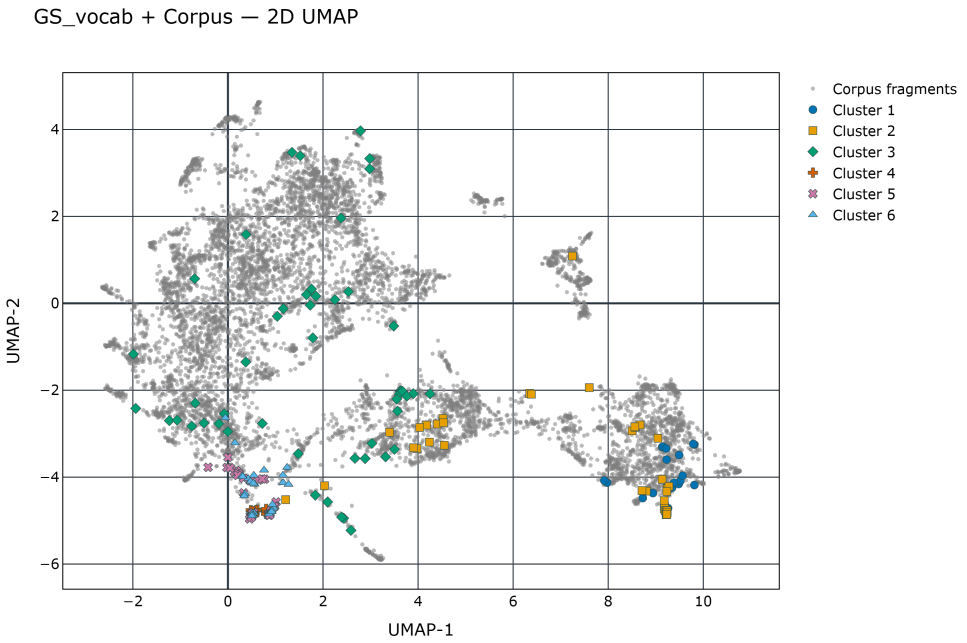

Figure S4.3: Two-dimensional UMAP projection of GAE fragment embeddings for the GS_Vocab vocabulary overlaid on the GSGE training corpus. Grey points represent 8,713 corpus-only fragments jointly embedded with the vocabulary. Colored markers ($N_{\text{vocab}} = 201$ vocabulary fragments) denote cluster membership from agglomerative hierarchical clustering on MCS Tanimoto similarity (cophenetic coefficient = 0.95); marker shape additionally encodes cluster identity. Compared to t-SNE, UMAP better preserves global topology, revealing the broader organizational structure of chemical space alongside fine-grained local clustering.

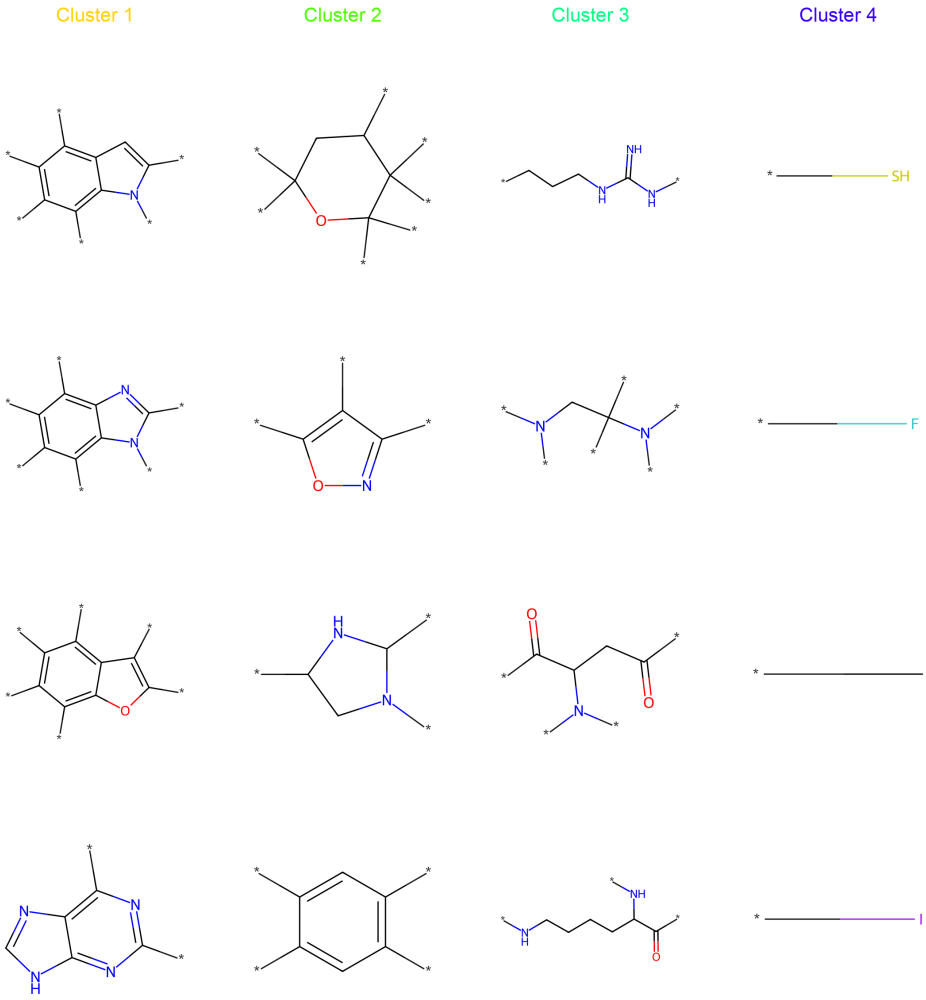

Figure S4.4: Representative molecular structures from four MCS-based hierarchical clusters, illustrating the characteristic structural motifs within each group. Fragment diversity ranges from large peptidic substructures to minimal single-atom groups, illustrating the structural spectrum captured by the GSGE vocabulary.

S4.2.3 Graph Autoencoder Training Diagnostics

The GAE was trained for 300 epochs on 8,713 non-generalized molecular fragments. Figure S4.5 tracks reconstruction quality over the course of training for both the training and validation partitions. On the training set, atom-type, edge-type, and adjacency reconstruction all converge to high macro F1 and balanced accuracy values, indicating that the encoder–decoder pipeline faithfully reconstructs the fragment graphs it has seen. On the held-out validation split, edge and adjacency metrics remain robust, while atom-type F1 drops more noticeably, reflecting the greater combinatorial diversity of unseen fragments. Figure S4.6 provides a complementary view of the best-performing downstream model (GCN+7desc 12-Ens), showing the estimated-versus-experimental scatter with per-compound uncertainty bars derived from the 12 constituent fits.

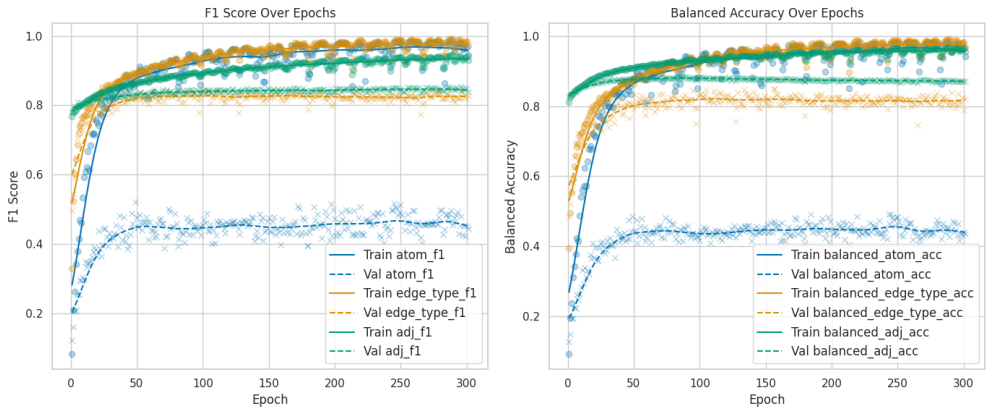

Figure S4.5: Reconstruction metrics over 300 training epochs for the molecular fragment graph autoencoder. Left: macro F1 scores for atom type (blue), edge type (orange), and adjacency (green) on training (solid) and validation (dashed) sets. Right: corresponding balanced accuracies. Training-set metrics converge rapidly and plateau above 0.93 for all three targets. Validation edge-type and adjacency F1 remain above 0.83, whereas atom-type F1 stabilizes near 0.45, consistent with the combinatorial diversity of unseen fragment structures.

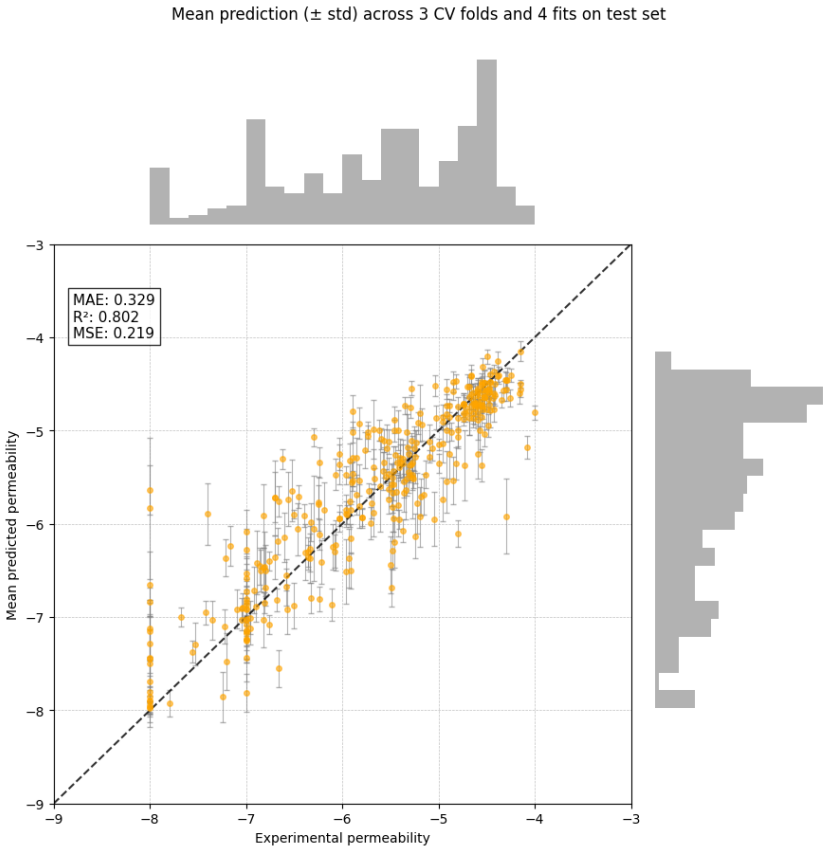

Figure S4.6: Estimated versus experimental PAMPA permeability for GCN+7desc (12-Ens), the highest-ranked model in the benchmark. Each point shows the mean estimated value ± one standard deviation across all 12 constituent fits (4 seeds × 3 CV folds). Marginal distributions of both estimated and experimental values are displayed alongside. The narrow uncertainty bars for most compounds indicate high inter-run agreement.

S4.2.4 Additional Test-Set Diagnostics

The main text reports the MAE leaderboard (Figure 4 therein) and scatter plots for the top three models (Figure 5 therein). This section provides supplementary test-set diagnostics organized into three groups: (i) alternative metric leaderboards (R^2, MSE), (ii) residual and error-distribution analyses for the top models, and (iii) output-dispersion analysis to evaluate whether run-to-run variability serves as a proxy for estimation error. Together, these figures provide a more complete picture of model behavior beyond aggregate MAE.

Alternative Metric Leaderboards

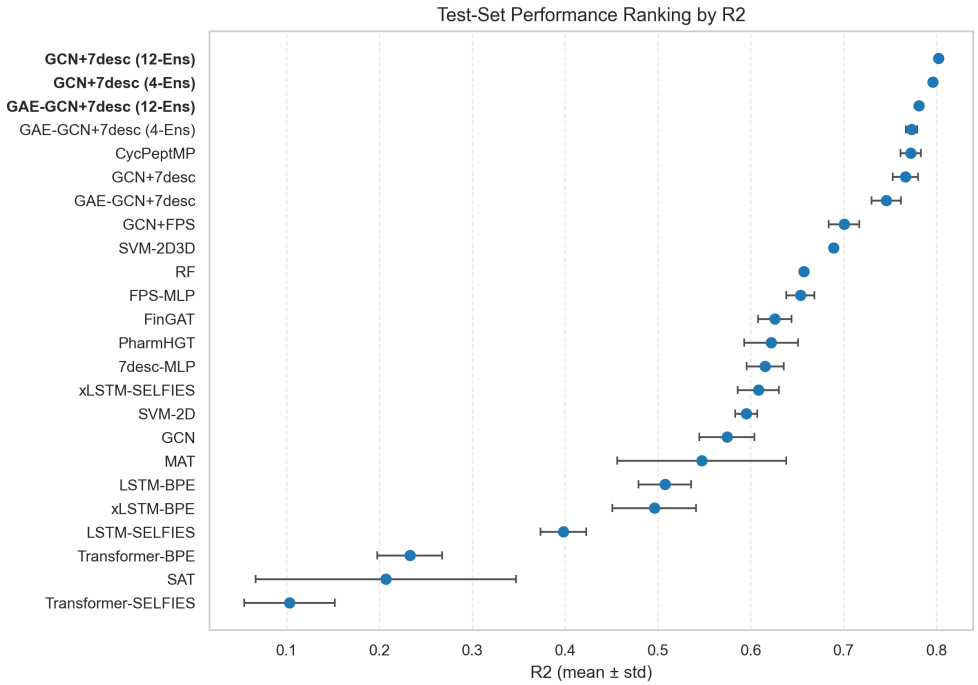

Figure S4.7: Ranked R^2 for all 24 evaluated models on the held-out test set, sorted from highest (best) to lowest. The overall ranking mirrors the MAE leaderboard presented in the main text. GCN+7desc (12-Ens) achieves $R^2 = 0.802$. Error bars indicate ± 1 standard deviation where available.

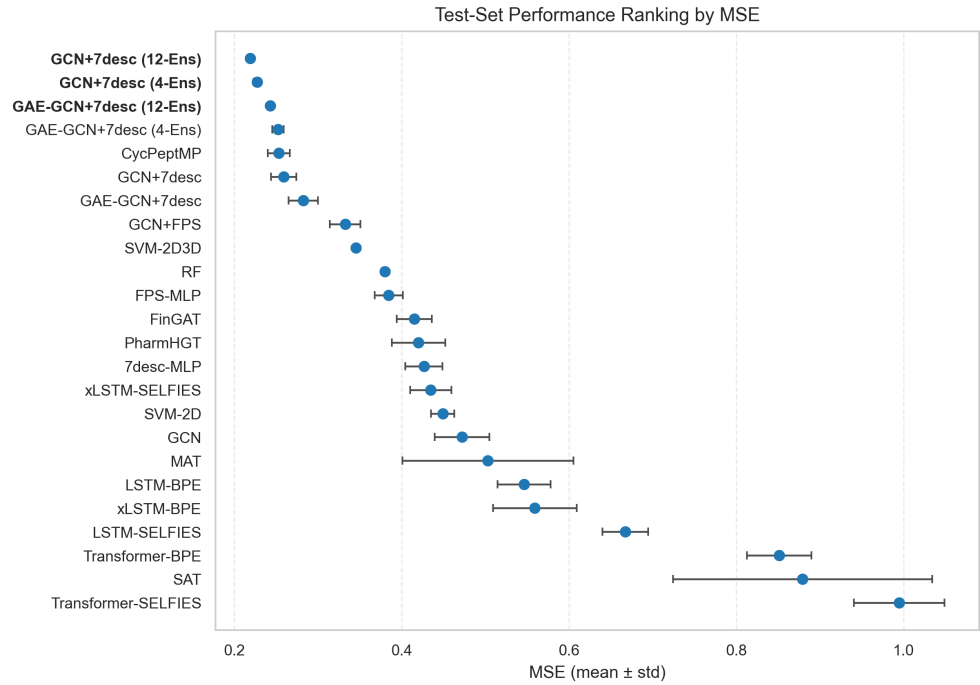

Figure S4.8: Ranked MSE for all 24 evaluated models on the held-out test set, sorted from lowest (best) to highest. The ranking is consistent with both the MAE and R^2 leaderboards. Because MSE penalizes large errors more heavily than MAE, the stable relative ordering across all three metrics indicates that no model is disproportionately affected by outliers at the aggregate level.

Residuals, Error Distributions, and Output Dispersion

The following figures examine the test-set behavior of the three highest-ranked diagnosable models (GCN+7desc, GAE-GCN+7desc, GCN+FPS) in greater detail. Residual plots reveal whether estimation bias varies systematically across the permeability range. Absolute-error distributions characterize the spread and tails of per-compound errors. Output-dispersion plots assess whether the standard deviation of a model's outputs across its 12 seed–fold runs is informative of actual estimation error, a property that would make ensemble variance a useful indicator for flagging unreliable compounds.

S4.2. Supplementary Figures

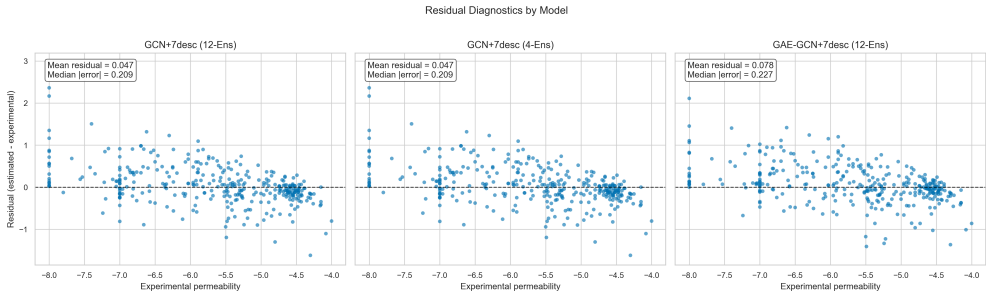

Figure S4.9: Residual plots (estimated minus experimental $\log_{10}(P_{\mathrm{app}} / [\mathrm{cm\,s^{-1}}])$) as a function of experimental $\log_{10}(P_{\mathrm{app}} / [\mathrm{cm\,s^{-1}}])$ for GCN+7desc, GAE-GCN+7desc, and GCN+FPS on the test set. The dashed horizontal line marks zero residual. Inset boxes report the mean residual (directional bias) and median absolute error. All three models show a modest positive mean residual, smallest for GCN+7desc (0.047), indicating slight upward bias in the lower-permeability region.

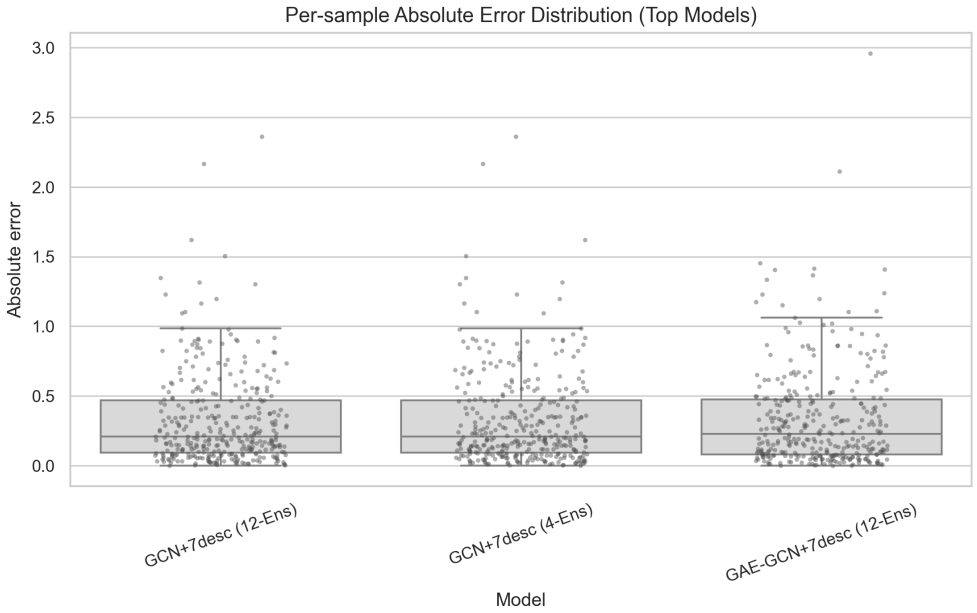

Figure S4.10: Per-sample absolute error distributions for the three top diagnosable models on the test set, displayed as boxplots overlaid with jittered individual compounds. GCN+7desc shows the most compact and lowest-error profile. The box spans the interquartile range (IQR) with the median line inside; individual data points are shown as jittered dots to reveal the full error distribution including tail behavior.

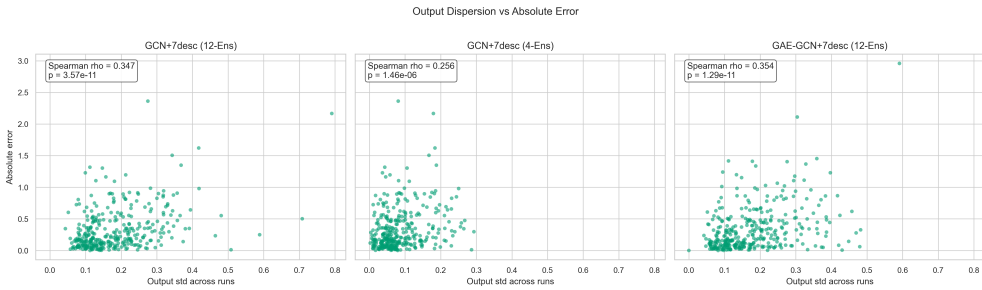

Figure S4.11: Run-to-run output dispersion (standard deviation across 12 seed–fold runs) versus per-sample absolute error for the three top diagnosable models on the test set. Spearman ρ and p-values are reported in the inset. All three models show a statistically significant positive monotonic association ($\rho \approx 0.35$, $p < 10^{-10}$), indicating that ensemble variance is informative of actual estimation error and can serve as a proxy for flagging less reliable compounds.

S4.2.5 Validation-Set Diagnostics

This section mirrors the test-set analysis but reports results on the validation partition of each CV fold. Validation-set diagnostics serve two purposes: they allow assessment of whether the model ranking is stable across data splits, and they provide insight into the degree to which hyperparameter optimization (which used validation MAE as its objective) may have biased model selection. Overall, the validation rankings are broadly consistent with the test-set rankings, supporting the stability of our model comparisons.

Validation Leaderboards

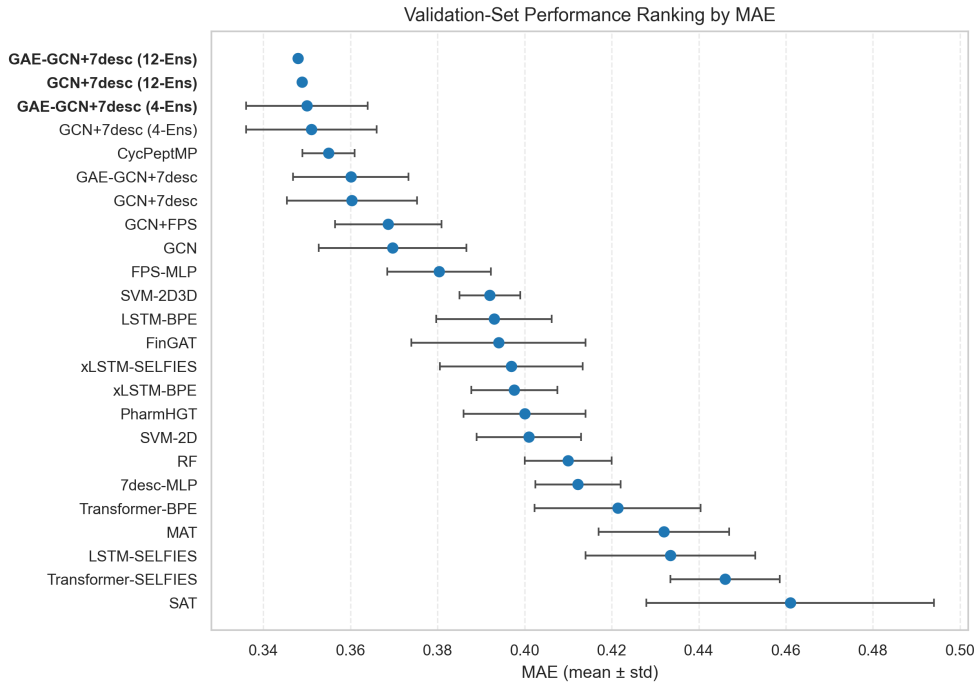

Figure S4.12: Ranked MAE for all evaluated models on the validation set (3-fold CV; graph/non-sequence models use 4 seeds per fold, sequence-model ablations use one seed per fold), sorted from lowest to highest. Error bars show ±1 standard deviation. The ranking is broadly consistent with the test-set leaderboard, supporting the stability of model comparisons across data splits. Validation MAE values are generally higher than test MAE, reflecting the role of the validation split in hyperparameter selection.

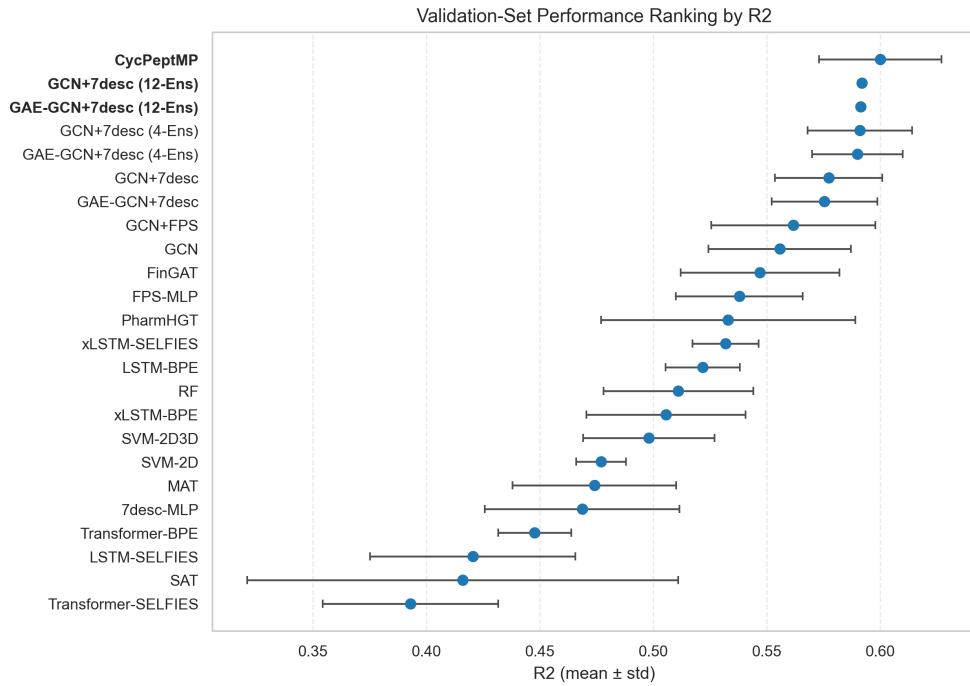

Figure S4.13: Ranked R^2 for all evaluated models on the validation set. The ranking mirrors the validation MAE leaderboard. R^2 values on the validation set are generally lower than on the test set.

S4.2. Supplementary Figures

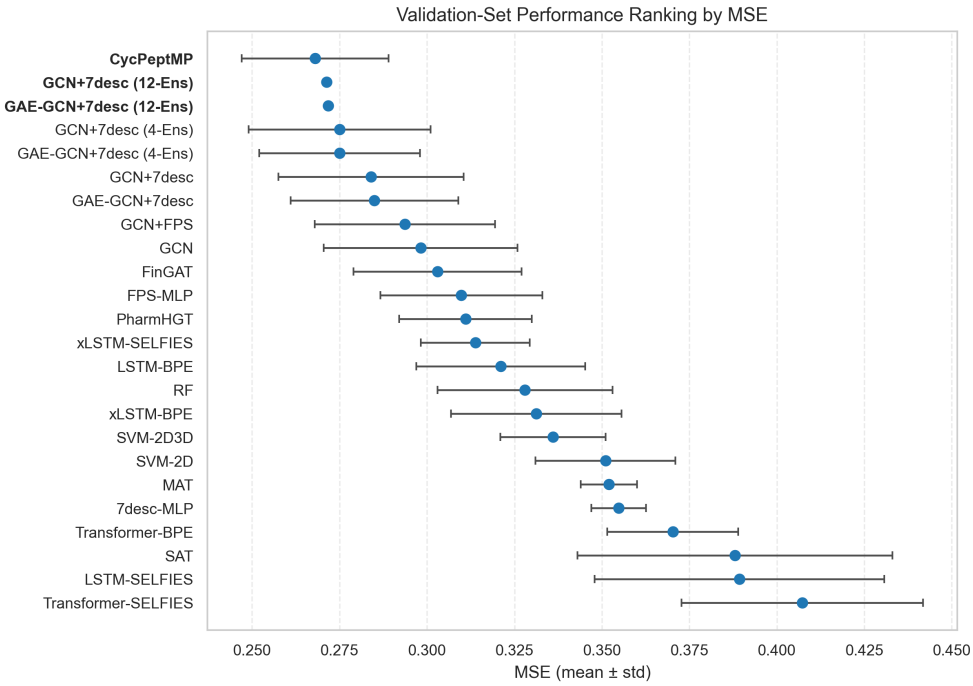

Figure S4.14: Ranked MSE for all evaluated models on the validation set, sorted from lowest to highest. The ranking is consistent with the validation MAE and R^2 leaderboards.

Validation Scatter, Residuals, Error Distributions, and Output Dispersion

S4.2. Supplementary Figures

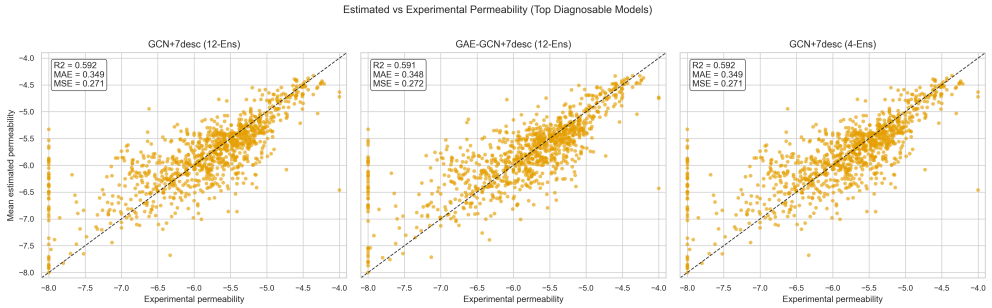

Figure S4.15: Estimated versus experimental $\log_{10}(P_{\mathrm{app}} / [\mathrm{cm\,s^{-1}}])$ for the three top diagnosable models on the validation set. Each point is one compound; estimated values are the mean across all seed–fold runs. The dashed line is the identity. Inset boxes report R^2, MAE, and MSE. Comparing this figure to the test-set scatter (main text, Figure 5) allows assessment of consistency across data splits.

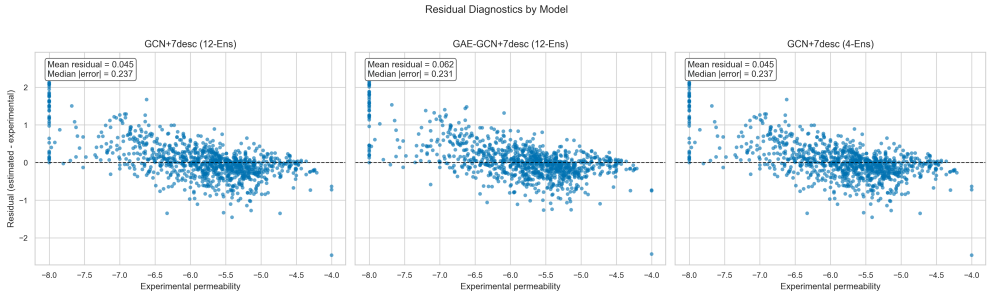

Figure S4.16: Residual plots for the three top diagnosable models on the validation set. Each point is one compound; the dashed line marks zero residual. Inset boxes report mean residual and median absolute error. The residual structure is qualitatively similar to the test set (Figure S4.9), with modest positive bias across models.

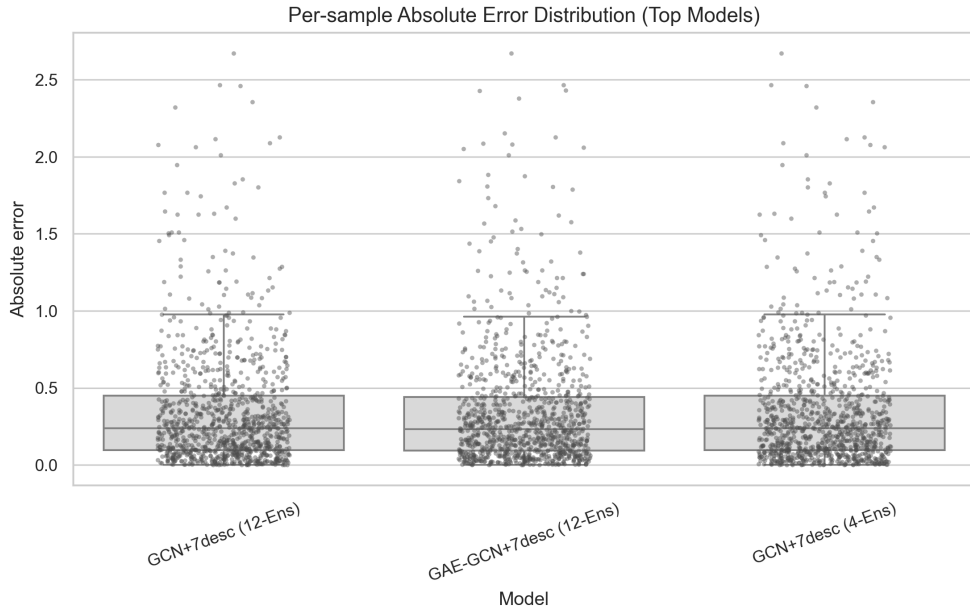

Figure S4.17: Per-sample absolute error distributions for the three top diagnosable models on the validation set, shown as overlaid box and strip plots. Error profiles are broadly similar across the three graph-based models, with only modest differences in spread and central tendency.

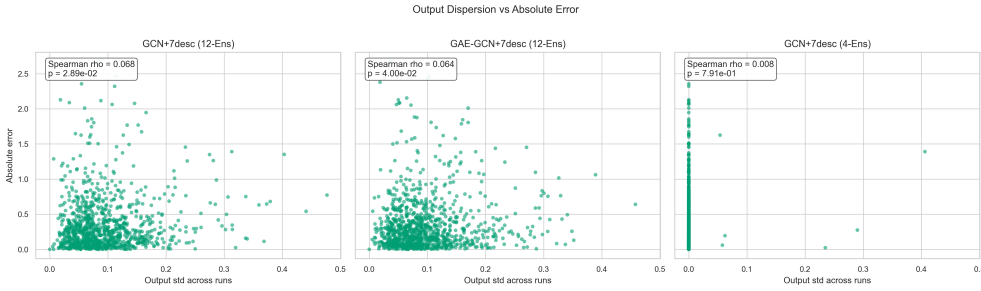

Figure S4.18: Output dispersion (standard deviation across seed–fold runs) versus per-sample absolute error on the validation set, with Spearman ρ and p-value in the inset. The correlations are weaker than on the test set (Figure S4.11), suggesting that the relationship between ensemble variance and estimation error is less pronounced for the data partition used during hyperparameter tuning.

S4.2.6 Statistical and Practical Significance: Supplementary Diagnostics

The main text presents focused MCS and critical-difference views for a readability-oriented subset of top contenders. This section provides the corresponding full 16-model visualizations together with the underlying assumption checks and additional diagnostics. The figures are organized into four groups: (i) full-scope pairwise/rank summaries, (ii) normality and omnibus diagnostics, (iii) nonparametric post hoc sign plots, and (iv) pairwise confidence-interval summaries. All analyses follow the protocol recommended by Ash *et al.*[60] and cover 16 models with independent per-fold replicates across four metrics (MAE, MSE, Pearson r, R^2).

Full-Scope Pairwise and Rank Summaries

Normality and Omnibus Diagnostics

Normality of the residualized performance values is a prerequisite for the parametric pathway (repeated-measures ANOVA with Tukey HSD). Figure S4.21 shows histograms and QQ-plots for the pooled residuals. With 48 values per metric (16 models $\times$ 3 CV splits), these checks remain coarse, but the residuals appear approximately unimodal and symmetric. Levene's test confirmed homogeneity of variances ($p > 0.05$) for all metrics. Figures S4.22 and S4.23 show the raw performance distributions with omnibus p-values from the parametric and nonparametric pathways, respectively. Both pathways reject the null hypothesis that all methods share a common performance distribution, but with only three per-method replicates these omnibus tests are interpreted as directional evidence and read alongside pairwise effect-size and confidence-interval summaries (Figures S4.25 and S4.26).

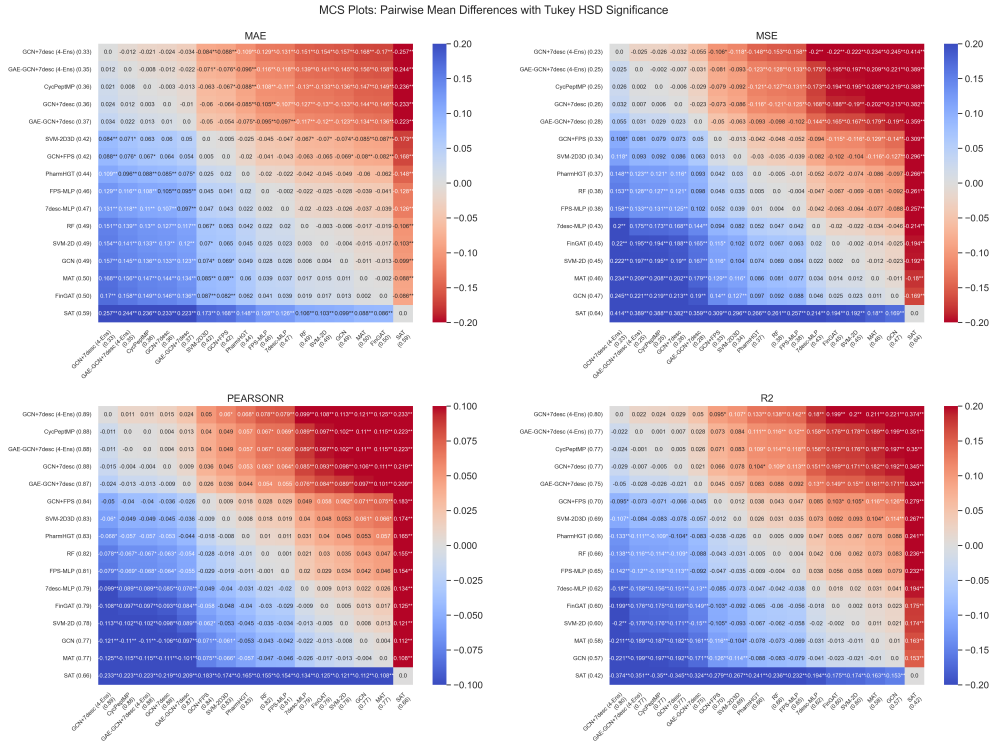

Figure S4.19: Full MCS heatmaps for all 16 models included in the repeated-measures analysis (MAE, MSE, Pearson r, R^2). Compared with the focused main-text version, this matrix preserves complete pairwise context but is substantially more crowded. Cell entries report mean differences (row minus column), stars denote adjusted significance, and color intensity encodes practical effect-size magnitude.

S4.2. Supplementary Figures

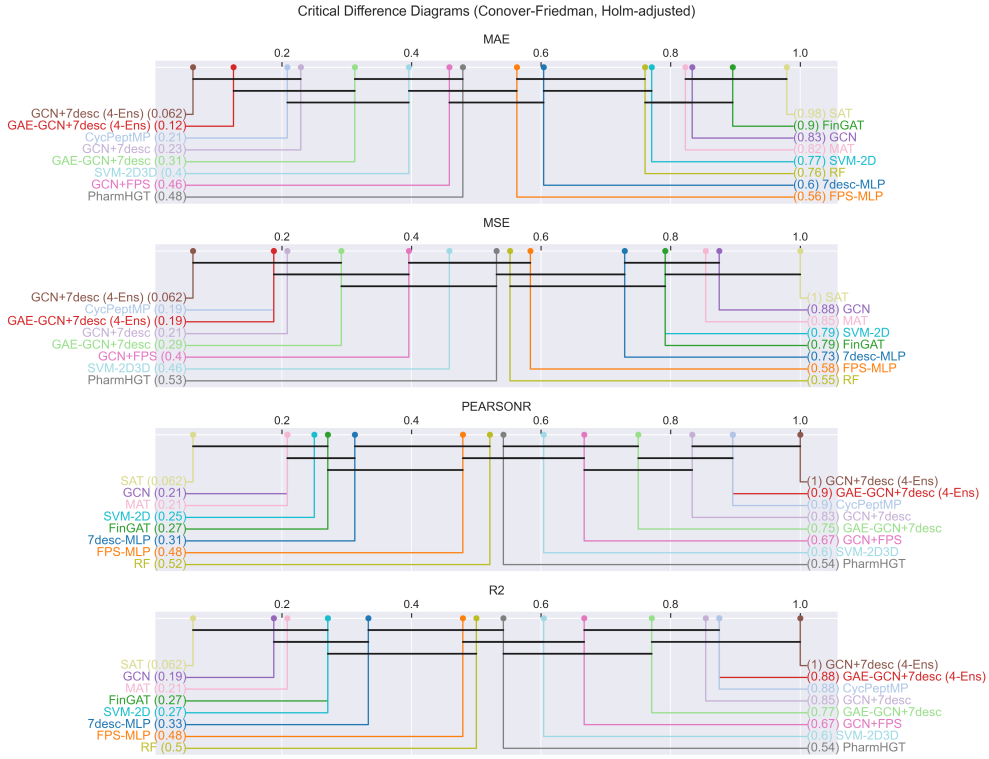

Figure S4.20: Full critical-difference diagrams for the complete 16-model comparison under Conover–Friedman post hoc testing with Holm adjustment. Relative to the focused main-text view, these diagrams show the full benchmark ordering and broader overlap structure among mid- and lower-tier methods. Horizontal bars connect models that are not significantly different at the 0.05 level.

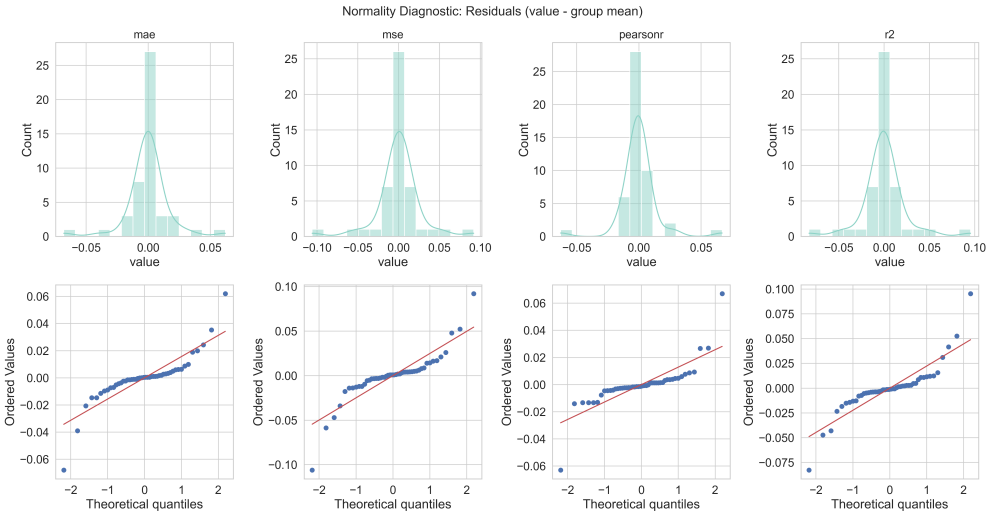

Figure S4.21: Normality diagnostics for residualized performance values (metric value minus method mean) across 16 compared methods and four metrics. Top row: histograms of pooled residuals. Bottom row: QQ-plots against the theoretical normal distribution. With 48 residual values per metric, these diagnostics are coarse but indicate approximately symmetric, unimodal distributions. Levene's test for homogeneity of variances was non-significant for all metrics ($p > 0.05$).

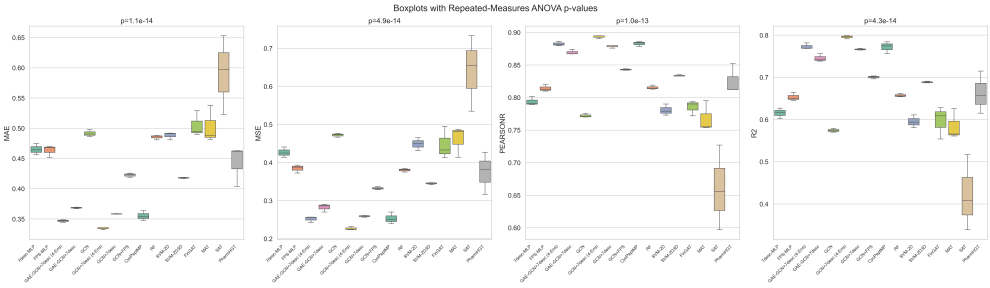

Figure S4.22: Boxplots of model performance across the three CV splits for each metric, with omnibus repeated-measures ANOVA p-values. All four omnibus tests are highly significant ($p < 10^{-13}$), indicating between-method differences at the group level. With three folds, pairwise interpretation should rely primarily on effect sizes and Tukey confidence intervals. The box spans the IQR across folds; the median line is shown inside.

S4.2. Supplementary Figures

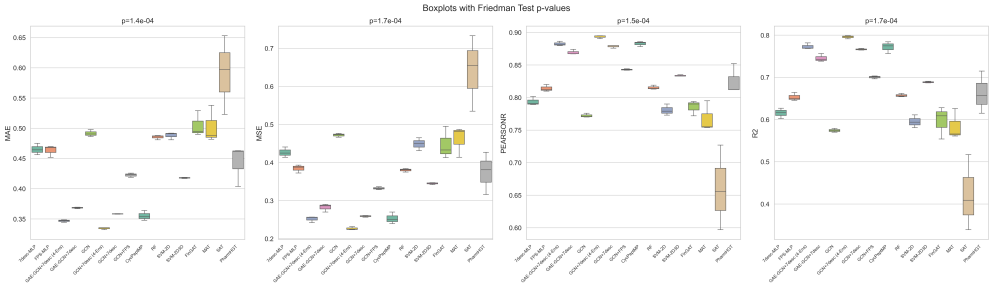

Figure S4.23: Boxplots of model performance across the three CV splits for each metric, with omnibus Friedman test p-values. All four rank-based omnibus tests remain significant ($p < 2 \times 10^{-4}$), providing nonparametric directional support for the same broad ranking pattern under weaker distributional assumptions.

S4.2.7 Nonparametric Post Hoc Sign Plots

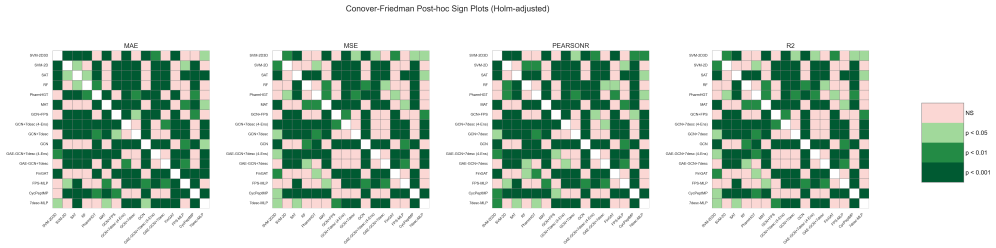

Figure S4.24: Conover–Friedman post hoc sign plots with Holm adjustment for MAE, MSE, Pearson r, and R^2. The matrices reproduce the same broad tiered structure seen in the focused and full MCS views: a cluster of non-significant pairwise relations among top-performing methods and increasing numbers of significant contrasts toward lower-ranked methods.

Pairwise Confidence-Interval Plots

The following figures display Tukey HSD confidence intervals for pairwise differences in mean performance, sorted by effect size. Intervals that cross zero indicate non-significant pairwise differences. These plots complement the MCS heatmaps by showing the actual magnitude and uncertainty of each pairwise contrast. A baseline-centered focused forest view versus CycPeptMP is additionally provided to make the top-contender comparisons easier to scan.

S4.2. Supplementary Figures

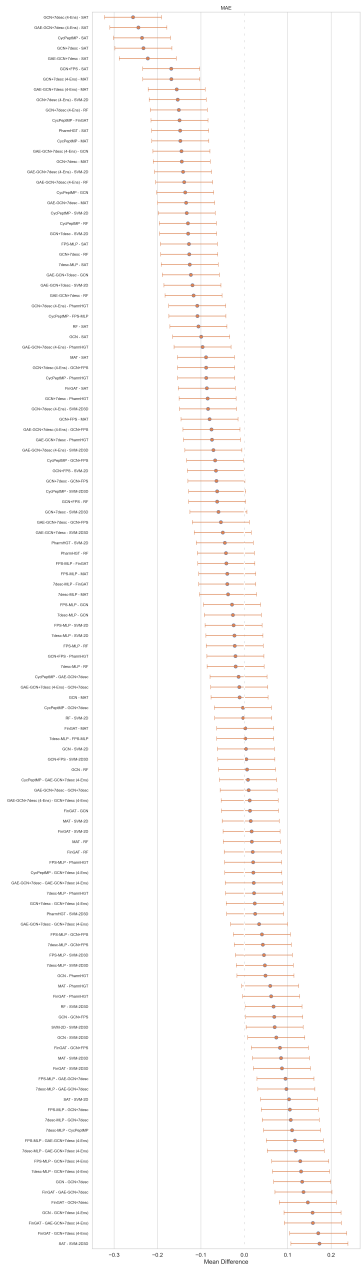

Figure S4.25: Pairwise Tukey HSD confidence intervals for differences in mean MAE, sorted by effect size. Intervals crossing zero indicate non-significant pairwise differences. The largest mean MAE advantages are observed for contrasts between top-tier models and SAT, with GCN+7desc (4-Ens) showing the largest mean MAE advantage over SAT (-0.257 $\log_{10}(P_{\mathrm{app}} / [\mathrm{cm\,s^{-1}}])$ units).

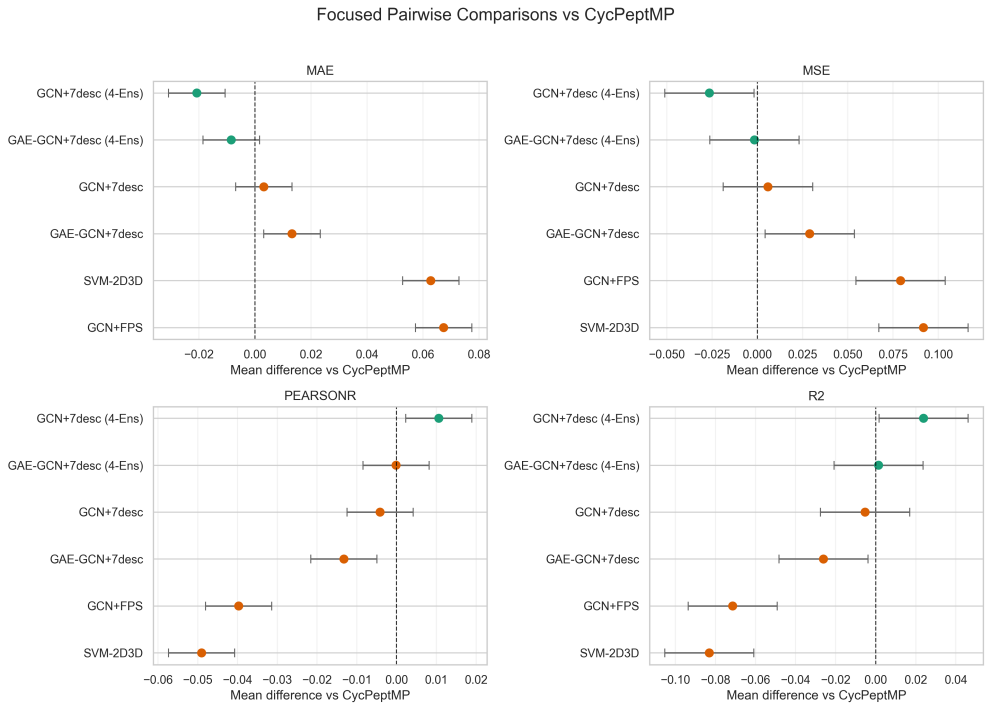

Figure S4.26: Baseline-centered forest plots comparing each focused contender directly against CycPeptMP for MAE, MSE, Pearson r, and R^2. Points show mean differences, horizontal bars show Tukey-HSD confidence intervals, and the vertical dashed line marks zero difference. This view emphasizes the direction and uncertainty of each head-to-head contrast relative to the CycPeptMP baseline.

S4.2.8 Cross-Endpoint Transfer: Additional Results

The main text reports a head-to-head comparison of the best Li *et al.* model and the best model from this work at each external endpoint (Figure 8 therein). Here we provide two complementary views of the full cross-endpoint results: a grouped bar chart that facilitates within-model comparison across endpoints (Figure S4.27), and ranked dot plots that show the complete model ordering at each endpoint (Figure S4.28). A key observation is that the identity of the best-transferring model changes by assay: sequence-based BPE models lead on Caco-2, while descriptor-based models are strongest on RRCK. This endpoint specificity underscores the value of maintaining a diverse model portfolio when the target assay is not known in advance.

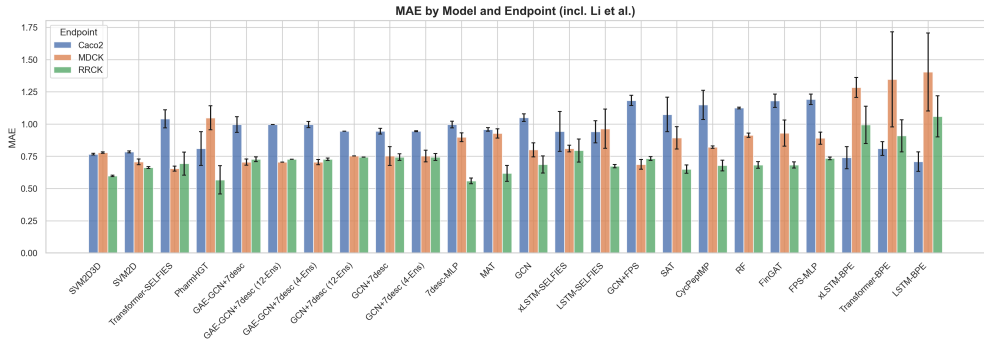

Figure S4.27: Grouped bar chart of MAE by model and endpoint for the combined Li *et al.* and in-house benchmark set. The chart facilitates direct cross-endpoint comparison within each model and shows that models with low error on one endpoint often do not retain that advantage on the others, underscoring the endpoint-specific nature of transfer performance.

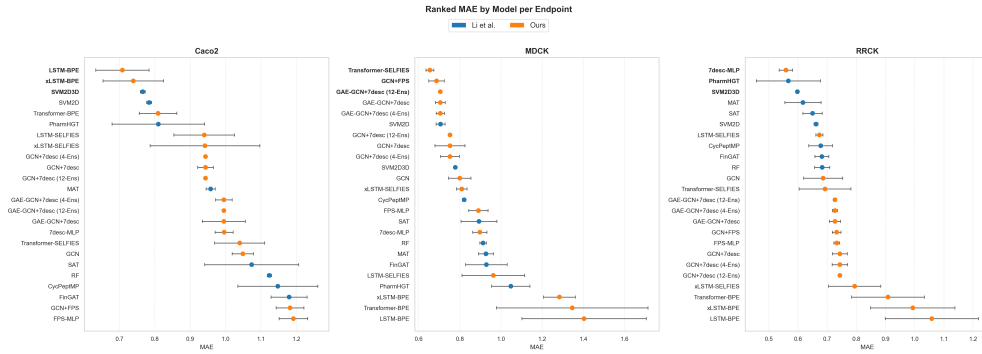

Figure S4.28: Ranked MAE dot plots for all models on Caco-2, MDCK, and RRCK. Li *et al.* models are shown in blue; our models in orange. Error bars represent ±1 standard deviation. The leading model changes by endpoint: in-house BPE sequence models lead on Caco-2, Transformer-SELFIES and graph hybrids lead on MDCK, and RRCK is topped by 7desc-MLP with several Li *et al.* baselines remaining competitive.

S4.2.9 Validation MAE Trajectories During Training

Figures S4.29–S4.31 show validation MAE as a function of training epoch for the best HPO trial of three representative model architectures: GCN+7desc (graph-based), Transformer-BPE (attention-based), and LSTM-BPE (recurrent). These trajectories motivate our choice of a fixed epoch selection window rather than conventional early stopping. The censored permeability target distribution causes persistent high-frequency fluctuations in validation MAE even after convergence, and premature stopping risks biasing performance toward the upper-bound region of the permeability range. The selection windows used were [50, 200] for graph-based models and [10, 50] for sequence models.

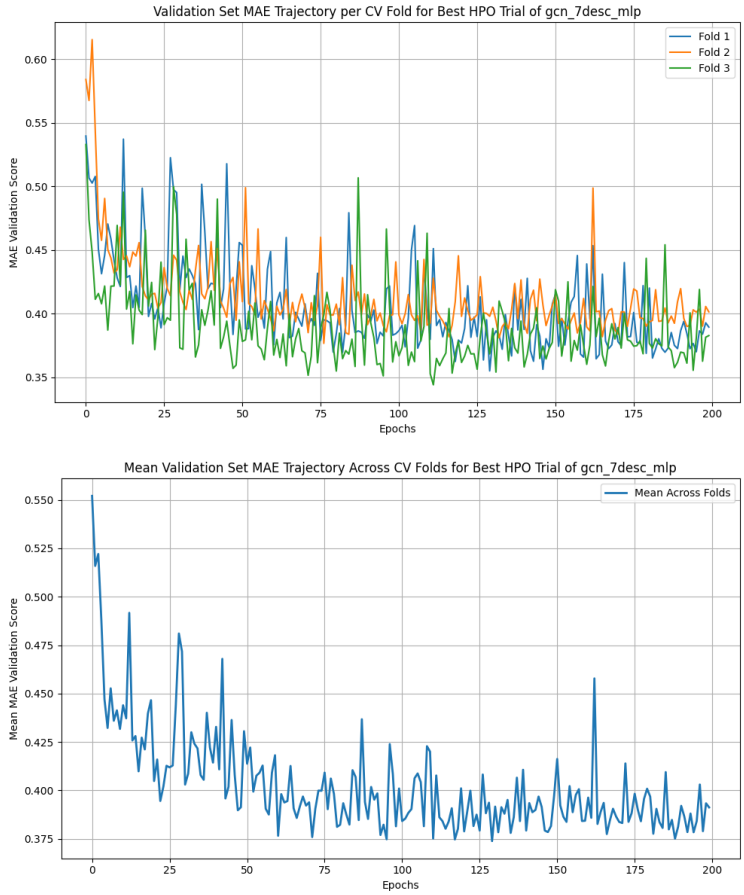

Figure S4.29: Validation MAE trajectories for the best HPO trial of GCN+7desc. Top: per-fold MAE curves across 200 training epochs, showing relatively high early variance that stabilizes after epoch 50. Persistent fluctuations after convergence are attributed to the challenging censored lower-bound region of the CycPeptMPDB PAMPA distribution. Bottom: mean MAE across folds. These results support the use of a fixed epoch selection window [50, 200] rather than early stopping.

S4.2. Supplementary Figures

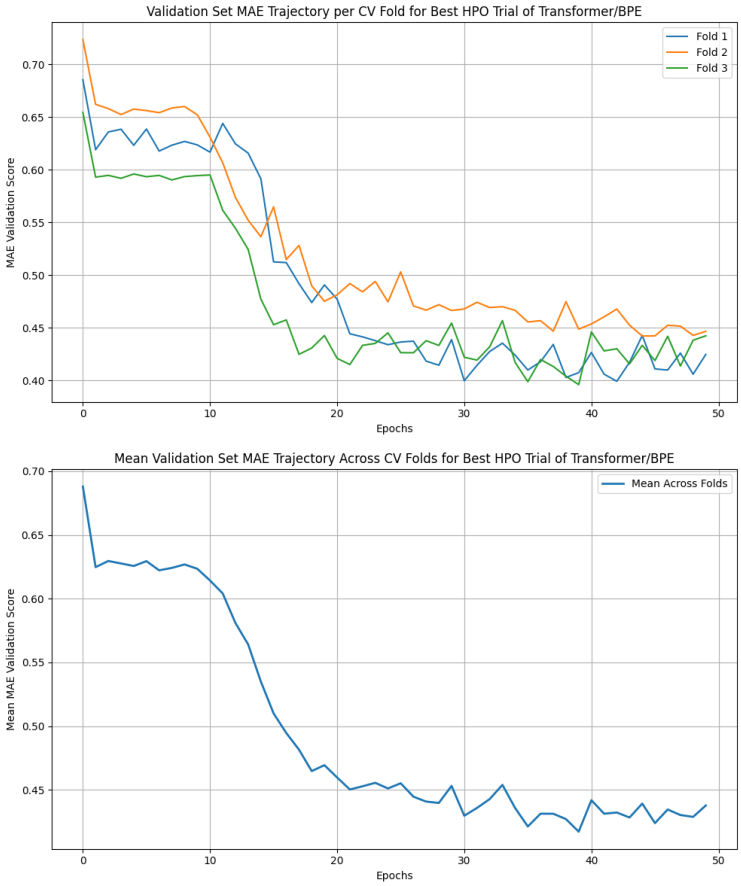

Figure S4.30: Validation MAE trajectories for the best HPO trial of Transformer-BPE. Top: per-fold MAE curves over 50 training epochs. Bottom: mean MAE across folds. Results support the use of a fixed epoch selection window [10, 50].

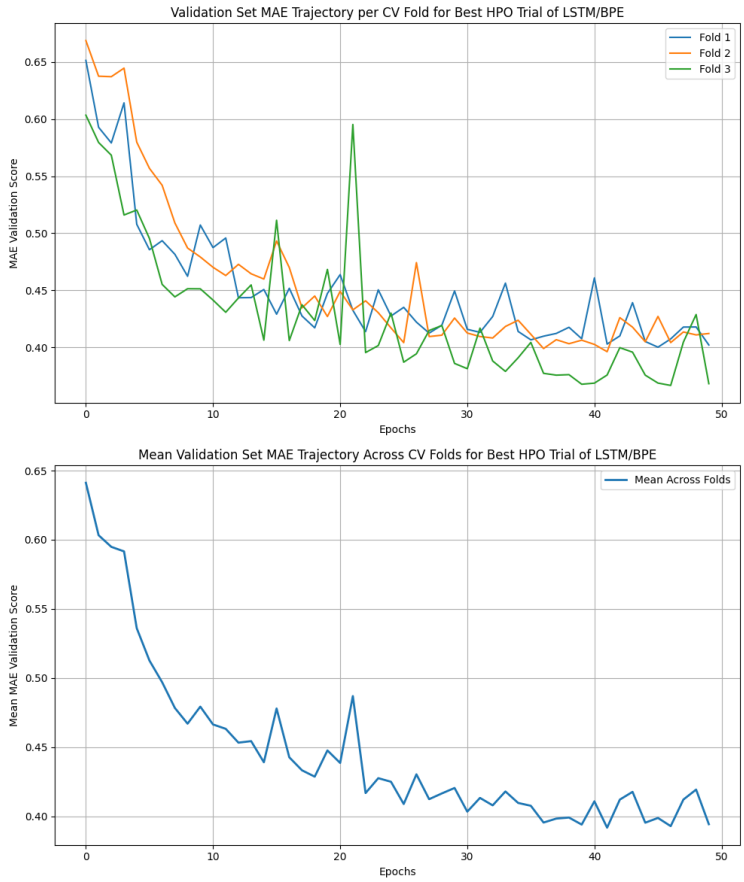

Figure S4.31: Validation MAE trajectories for the best HPO trial of LSTM-BPE. Top: per-fold MAE curves over 50 training epochs. Bottom: mean MAE across folds. Results support the use of a fixed epoch selection window [10, 50].

S4.3 Supplementary Tables

S4.3.1 Validation-Set Performance Across All Models

Table S4.2 reports validation-set performance for all 24 evaluated models, grouped by family using the same naming convention as the test table. The validation ranking is broadly consistent with the test-set ordering: graph–descriptor fusion ensembles and CycPeptMP remain top performers, and recurrent sequence models generally outperform transformer-based sequence variants. Absolute metric values differ from the held-out test set because validation data were used during hyperparameter optimization.

Table S4.2: Validation-set performance for all evaluated models, grouped by family to mirror the main-text test-set table. Values are reported as mean ± standard deviation where available; sequence-model variants use $n_{\text{rep}} = 3$ (one seed per CV fold), and 12-Ens models report a single aggregate value ($n_{\text{rep}} = 1$). Best value across all listed models for each metric is shown in bold.

	Model	n_{rep}	MAE	MSE	R^2
Graph-based	GCN+7desc (12-Ens)	1	0.349	0.271	0.592
	GCN+7desc (4-Ens)	3	0.351 (±0.015)	0.275 (±0.026)	0.591 (±0.023)
	GAE-GCN+7desc (12-Ens)	1	**0.348**	0.272	0.592
	GAE-GCN+7desc (4-Ens)	3	0.350 (±0.014)	0.275 (±0.023)	0.590 (±0.020)
	GCN+7desc	12	0.360 (±0.015)	0.284 (±0.026)	0.578 (±0.024)
	GAE-GCN+7desc	12	0.360 (±0.013)	0.285 (±0.024)	0.576 (±0.023)
	GCN+FPS	12	0.369 (±0.012)	0.294 (±0.026)	0.562 (±0.036)
	GCN	12	0.370 (±0.017)	0.298 (±0.028)	0.556 (±0.031)
Sequence	xLSTM-SELFIES	3	0.397 (±0.016)	0.314 (±0.016)	0.532 (±0.015)
	LSTM-BPE	3	0.393 (±0.013)	0.321 (±0.024)	0.522 (±0.016)
	xLSTM-BPE	3	0.398 (±0.010)	0.331 (±0.024)	0.506 (±0.035)
	LSTM-SELFIES	3	0.434 (±0.020)	0.389 (±0.041)	0.420 (±0.045)
	Transformer-BPE	3	0.421 (±0.019)	0.370 (±0.019)	0.448 (±0.016)
	Transformer-SELFIES	3	0.446 (±0.013)	0.407 (±0.035)	0.393 (±0.039)
Baselines	CycPeptMP [45]	3	0.355 (±0.006)	**0.268 (±0.021)**	**0.600 (±0.027)**
	SVM-2D3D [45]	3	0.392 (±0.007)	0.336 (±0.015)	0.498 (±0.029)
	FPS-MLP	12	0.380 (±0.012)	0.310 (±0.023)	0.538 (±0.028)
	7desc-MLP	12	0.412 (±0.010)	0.355 (±0.008)	0.469 (±0.043)
	RF [45]	3	0.410 (±0.010)	0.328 (±0.025)	0.511 (±0.033)
	PharmHGT [45]	3	0.400 (±0.014)	0.311 (±0.019)	0.533 (±0.056)
	SVM-2D [45]	3	0.401 (±0.012)	0.351 (±0.020)	0.477 (±0.011)
	FinGAT [45]	3	0.394 (±0.020)	0.303 (±0.024)	0.547 (±0.035)
	MAT [45]	3	0.432 (±0.015)	0.352 (±0.008)	0.474 (±0.036)
	SAT [45]	3	0.461 (±0.033)	0.388 (±0.045)	0.416 (±0.095)

S4.3.2 Worked Example: Group-SELFIES Encoding

To illustrate the concrete form of a Group-SELFIES string (main text, Methods), we encode one representative cyclic peptide from the training corpus using the trained 201-fragment GSGE vocabulary (Table S4.3). Grammar tokens (e.g. `Ring1`, `Branch`, `pop`) encode molecular topology; fragment tokens (`GS_frag_N`) each represent one vocabulary fragment; the single bare `C` token is an atom not captured by any fragment and is covered individually, as described in Methods. Table S4.4 decodes a subset of the fragment tokens used in this example into their corresponding substructures.

Table S4.3: Worked example of SMILES-to-Group-SELFIES encoding for one representative cyclic peptide, using the 201-fragment GSGE vocabulary described in Methods.

SMILES	CC(C)CC1NC(=O)CCN(C)C(=O)CC(C(=O)N2CCCCC2)NC(=O)C(Cc2ccccc2)N(C)C(=O)CNC(=O)C(C(C)O)NC(=O)C(Cc2ccccc2)NC(=O)C(CC(C)C)N(C)C(=O)C(CC(C)C)NC(=O)C(Cc2ccccc2)NC1=O
Group-SELFIES	[:0][GS_frag_38][C][:0][GS_frag_76][Ring1][:3][GS_frag_107][O][:3][GS_frag_22][Ring1][:0][GS_frag_69][pop][Ring1][:0][GS_frag_76][Ring1][:6][GS_frag_107][Branch][:4][GS_frag_108][Ring1][:4][GS_frag_77][Ring1][:4][GS_frag_108][Branch][:0][GS_frag_47][Ring1][:0][GS_frag_68][pop][pop][Branch][:0][GS_frag_76][Ring1][:9][GS_frag_93][Ring1][:0][GS_frag_47][pop][Ring1][:1][GS_frag_38][Ring1][:4][GS_frag_108][Branch][:0][GS_frag_47][Ring1][:0][GS_frag_68][pop][pop][Branch][Ring2][#Branch][C][pop][pop][pop][=Branch][:0][GS_frag_47][pop][pop][pop][pop][pop][pop][Ring1][:0][GS_frag_68][pop][pop][pop][pop][pop][pop]

Table S4.4: Chemical identity of selected fragment tokens appearing in the worked example of Table S4.3; * denotes an attachment point to a neighboring fragment.

Token	Substructure (SMILES) and description
`GS_frag_38`	`[1*]NC(CC(C)C)C([2*])=O` leucine-type backbone unit
`GS_frag_76`	`[1*]C([2*])=O` carbonyl unit
`GS_frag_107`	`[1*]C([2*])N([3*])C([7*])C([4*])([5*])[6*]` branched backbone unit
`GS_frag_108`	`[1*]N([2*])C([3*])([4*])C([5*])=O` backbone amide unit
`GS_frag_77`	`[1*]NC(C([5*])=O)C(O[2*])C([3*])[4*]` threonine-type backbone unit
`GS_frag_69`	`[1*]N1C([2*])([3*])C([4*])([5*])C([6*])([7*])C([8*])([9*])C1([10*])[11*]` piperidine/proline ring unit

S4.3.3 Cross-Endpoint Transfer: Full Numerical Results

Table S4.5 provides the complete cross-endpoint transfer results (MAE and Pearson r) for all evaluated models across Caco-2 ($N_{\mathrm{cmpd}} = 381$), MDCK ($N_{\mathrm{cmpd}} = 17$), and RRCK ($N_{\mathrm{cmpd}} = 56$) endpoints. Models were trained on PAMPA and evaluated without retraining; outputs are clipped to $[-8.0, -4.0]$ $\log_{10}(P_{\mathrm{app}} \,/\, [\mathrm{cm\,s^{-1}}])$. Values are reported as mean ± standard deviation across seed-level aggregates for single-model variants ($n_{\mathrm{rep}} = 4$), as mean ± standard deviation across fold-level ensemble aggregates for 4-Ens variants ($n_{\mathrm{rep}} = 3$), and as a single aggregate scalar (no SD and no ± notation) for 12-Ens variants ($n_{\mathrm{rep}} = 1$). The small sample sizes for MDCK and RRCK warrant caution when interpreting correlation values.

Table S4.5: Cross-endpoint transfer performance: Li et al. baselines + our models. Endpoint dataset size is denoted by N_{cmpd}, and replicate-aggregate count by n_{rep}. Metrics are reported as mean ± std where replicate aggregates exist; 12-Ens entries are single aggregate metrics and are therefore reported without ± values. Li et al. values are from their published benchmark; our single-model values are seed-level cross-endpoint aggregates (mean over folds within each seed, then mean ± std across seeds). Best value per endpoint and metric across all listed models is in bold.

Metric	Source	Model	Caco-2	MDCK	RRCK
MAE	Li et al.	RF	1.124 ± 0.006	0.913 ± 0.016	0.683 ± 0.026
	Li et al.	SVM-2D	0.784 ± 0.007	0.706 ± 0.022	0.662 ± 0.007
	Li et al.	SVM-2D3D	0.766 ± 0.007	0.778 ± 0.005	0.598 ± 0.005
	Li et al.	MAT	0.958 ± 0.013	0.927 ± 0.036	0.617 ± 0.062
	Li et al.	SAT	1.074 ± 0.133	0.893 ± 0.087	0.650 ± 0.033
	Li et al.	PharmHGT	0.810 ± 0.131	1.048 ± 0.093	0.567 ± 0.110
	Li et al.	FinGAT	1.180 ± 0.051	0.929 ± 0.101	0.682 ± 0.024
	Li et al.	CycPeptMP	1.148 ± 0.113	0.821 ± 0.009	0.678 ± 0.041
	Ours	Transformer-SELFIES	1.040 ± 0.071	**0.654 ± 0.019**	0.693 ± 0.089
	Ours	GAE-GCN+7desc (12-Ens)	0.995	0.705	0.727
	Ours	GAE-GCN+7desc (4-Ens)	0.995 ± 0.024	0.705 ± 0.020	0.727 ± 0.008
	Ours	GAE-GCN+7desc	0.995 ± 0.061	0.705 ± 0.024	0.727 ± 0.019
	Ours	GCN+7desc (12-Ens)	0.943	0.752	0.744
	Ours	GCN+7desc	0.943 ± 0.022	0.752 ± 0.073	0.744 ± 0.026
	Ours	GCN+7desc (4-Ens)	0.943 ± 0.004	0.752 ± 0.045	0.744 ± 0.027
	Ours	7desc-MLP	0.996 ± 0.025	0.897 ± 0.035	**0.559 ± 0.023**
	Ours	GCN	1.049 ± 0.030	0.800 ± 0.055	0.686 ± 0.067
	Ours	xLSTM-SELFIES	0.942 ± 0.155	0.810 ± 0.026	0.794 ± 0.090
	Ours	LSTM-SELFIES	0.940 ± 0.086	0.963 ± 0.152	0.673 ± 0.012
	Ours	GCN+FPS	1.183 ± 0.039	0.687 ± 0.037	0.733 ± 0.015
	Ours	FPS-MLP	1.192 ± 0.040	0.890 ± 0.047	0.733 ± 0.010
	Ours	xLSTM-BPE	0.739 ± 0.086	1.284 ± 0.077	0.994 ± 0.145
	Ours	Transformer-BPE	0.809 ± 0.053	1.347 ± 0.369	0.909 ± 0.125
	Ours	LSTM-BPE	**0.708 ± 0.075**	1.404 ± 0.302	1.059 ± 0.160
Pearson r	Li et al.	RF	0.181 ± 0.002	0.283 ± 0.021	-0.044 ± 0.045
	Li et al.	SVM-2D	0.279 ± 0.016	0.377 ± 0.072	0.245 ± 0.006
	Li et al.	SVM-2D3D	0.290 ± 0.004	0.618 ± 0.011	**0.291 ± 0.016**
	Li et al.	MAT	0.175 ± 0.043	0.415 ± 0.107	0.078 ± 0.066
	Li et al.	SAT	0.107 ± 0.023	0.388 ± 0.144	0.242 ± 0.082
	Li et al.	PharmHGT	0.195 ± 0.015	0.675 ± 0.037	0.160 ± 0.119
	Li et al.	FinGAT	0.128 ± 0.040	0.358 ± 0.080	-0.057 ± 0.010
	Li et al.	CycPeptMP	0.209 ± 0.064	0.570 ± 0.044	-0.181 ± 0.027
	Ours	Transformer-SELFIES	0.137 ± 0.047	0.324 ± 0.232	-0.009 ± 0.076
	Ours	GAE-GCN+7desc (12-Ens)	0.275	**0.732**	0.167
	Ours	GAE-GCN+7desc (4-Ens)	0.275 ± 0.016	**0.732 ± 0.026**	0.167 ± 0.015
	Ours	GAE-GCN+7desc	0.275 ± 0.017	**0.732 ± 0.017**	0.167 ± 0.020
	Ours	GCN+7desc (12-Ens)	0.211	0.722	0.129
	Ours	GCN+7desc	0.211 ± 0.018	0.722 ± 0.023	0.129 ± 0.035
	Ours	GCN+7desc (4-Ens)	0.211 ± 0.011	0.722 ± 0.036	0.129 ± 0.051
	Ours	7desc-MLP	**0.316 ± 0.014**	0.498 ± 0.056	0.203 ± 0.021
	Ours	GCN	0.273 ± 0.007	0.526 ± 0.083	0.143 ± 0.107
	Ours	xLSTM-SELFIES	0.241 ± 0.021	0.517 ± 0.060	-0.220 ± 0.124
	Ours	LSTM-SELFIES	0.279 ± 0.012	0.236 ± 0.151	-0.259 ± 0.060
	Ours	GCN+FPS	0.242 ± 0.029	0.675 ± 0.018	-0.098 ± 0.013
	Ours	FPS-MLP	0.215 ± 0.009	0.489 ± 0.062	-0.108 ± 0.025
	Ours	xLSTM-BPE	-0.042 ± 0.027	0.298 ± 0.249	-0.084 ± 0.135
	Ours	Transformer-BPE	0.012 ± 0.062	0.087 ± 0.371	-0.070 ± 0.121
	Ours	LSTM-BPE	0.073 ± 0.062	0.222 ± 0.544	-0.428 ± 0.048

S4.3.4 Hyperparameter Optimization Details

This section reports the hyperparameter search spaces and selected configurations for all models evaluated in this work. Bayesian optimization was performed using Weights & Biases [115]. For graph-based models, the general training settings are given in Table S4.6, followed by model-specific HPO tables. For sequence models, a separate set of general settings applies (Table S4.13), reflecting the shorter training budget and different evaluation protocol used for the ablation study. xLSTM hyperparameters were set manually rather than optimized due to resource constraints.

Graph-Based Model Hyperparameters

Table S4.6: General training settings shared by all graph-based models.

Setting	Value
Epochs	200
Minimal Epochs	50
Loss Function	MSELoss
Validation Score Function	MAE
Use Best Valid Score Epoch Model	True

Table S4.7: Hyperparameter search space and selected values for GCN+7desc.

Name	Search Space	Selected Value
Training		
Optimizer	[Adam, AdamW]	AdamW
Batch Size	[64, 128]	128
Learning Rate	[Min: 0.0001, Max: 0.001]	0.00062
Weight Decay	[0.00001, 0]	0
Desc MLP		
Activation	—	ReLU
Dropout	—	0
Hidden Dim	—	128
Num Layers	—	1
GNN Encoder		
Hidden Channels	[256, 512]	512
Num Layers	[2, 3, 4]	4
Dropout	[0.1, 0.2, 0.3]	0.2
Use Residual	[True, False]	True
Fusion MLP		
Num Layers	[2, 3, 4]	4
Hidden Dim	[256, 512]	256

Table S4.8: Hyperparameter search space and selected values for GAE-GCN+7desc.

Name	Search Space	Selected Value
Training		
Optimizer	[Adam, AdamW]	Adam
Batch Size	[64, 128]	64
Learning Rate	[Min: 0.0001, Max: 0.001]	0.00041
Weight Decay	[0.00001, 0]	0
Desc MLP		
Activation	—	ReLU
Dropout	—	0
Hidden Dim	—	128
Num Layers	—	1
GNN Encoder		
Hidden Channels	[256, 512]	512
Num Layers	[2, 3, 4]	4
Dropout	[0.1, 0.2, 0.3]	0.1
Use Residual	[True, False]	True
Fusion MLP		
Num Layers	[2, 3, 4]	4
Hidden Dim	[256, 512]	512

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Table S4.9: Hyperparameter search space and selected values for GCN+FPS.

Name	Search Space	Selected Value
Training		
Optimizer	[Adam, AdamW]	Adam
Batch Size	[64, 128]	128
Learning Rate	[Min: 0.0001, Max: 0.001]	0.00058
Weight Decay	[0.00001, 0]	0
Desc MLP		
Activation	—	ReLU
Dropout	[0, 0.1]	0.1
Hidden Dim	[256, 512, 1024]	256
Num Layers	[1, 2, 3]	2
GNN Encoder		
Hidden Channels	[256, 512]	256
Num Layers	[2, 3, 4]	3
Dropout	[0.1, 0.2, 0.3]	0.1
Use Residual	[True, False]	False
Fusion MLP		
Activation	—	ReLU
Num Layers	[2, 3, 4]	4
Hidden Dim	[256, 512]	512
Dropout	[0, 0.1]	0

Table S4.10: Hyperparameter search space and selected values for GCN (graph-only baseline).

Name	Search Space	Selected Value
Training		
Optimizer	[Adam, AdamW]	Adam
Batch Size	[64, 128]	128
Learning Rate	[Min: 0.0001, Max: 0.001]	0.00099
Weight Decay	[0.00001, 0]	0.00001
GNN Encoder		
Hidden Channels	[256, 512]	512
Num Layers	[2, 3, 4]	4
Dropout	[0.1, 0.2, 0.3]	0.2
Use Residual	[True, False]	True
Fusion MLP		
Activation	—	ReLU
Num Layers	[2, 3, 4]	4
Hidden Dim	[256, 512]	512
Dropout	—	0

Table S4.11: Hyperparameter search space and selected values for 7desc-MLP.

Name	Search Space	Selected Value
Training		
Optimizer	[Adam, AdamW]	AdamW
Batch Size	[64, 128]	64
Learning Rate	[Min: 0.0001, Max: 0.001]	0.00077
Weight Decay	[0.00001, 0]	0
MLP		
Num Layers	[1, 2, 3]	3
Hidden Dim	[64, 128, 256]	128
Dropout	[0, 0.1]	0
Activation	[ReLU, LeakyReLU]	ReLU

Table S4.12: Hyperparameter search space and selected values for FPS-MLP.

Name	Search Space	Selected Value
Training		
Optimizer	[Adam, AdamW]	Adam
Batch Size	[64, 128]	64
Learning Rate	[Min: 0.0001, Max: 0.001]	0.00027
Weight Decay	[0.00001, 0]	0.00001
MLP		
Num Layers	[1, 2, 3, 4]	4
Hidden Dim	[256, 512, 1024]	256
Dropout	[0, 0.1]	0.1
Activation	[ReLU, LeakyReLU]	ReLU

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Sequence Model Hyperparameters

Table S4.13: General training settings for sequence-based models.

Setting	Value
Epochs	50
Minimal Epochs	10
Loss Function	MSELoss
Validation Score Function	MAE
Use Best Valid Score Epoch Model	True

Table S4.14: Hyperparameter search space and selected values for LSTM models with BPE and SELFIES encodings.

Component	Hyperparameter	BPE	SELFIES
Training			
Optimizer	[Adam, AdamW]	Adam	AdamW
Batch Size	[64, 128]	64	64
Learning Rate	[Min: 0.0001, Max: 0.001]	0.00085	0.00036
Weight Decay	[0.00001, 0]	0.00001	0.00001
LSTM			
Num Layers	[2, 4]	2	2
Hidden Dim	[128, 256, 512]	256	256
Dropout	[0, 0.1]	0.1	0.1
MLP			
Num Layers	—	1	1
Hidden Dim	—	128	128
Dropout	[0, 0.1]	0.1	0.1
Activation	[ReLU]	ReLU	ReLU

Table S4.15: Hyperparameter search space and selected values for Transformer models with BPE and SELFIES encodings.

Component	Hyperparameter	BPE	SELFIES
Training			
Optimizer	[Adam, AdamW]	AdamW	AdamW
Batch Size	[64, 128]	128	128
Learning Rate	[Min: 0.0001, Max: 0.001]	0.00027	0.00020
Weight Decay	[0.00001, 0]	0	0.00001
Transformer			
Num Layers	[2, 4]	4	4
Num Heads	[2, 4]	4	4
Hidden Dim	[128, 256]	256	256
Dim ff ratio	[2, 4]	4	2
Dropout	[0, 0.1]	0	0.1
MLP			
Num Layers	—	1	1
Hidden Dim	—	128	128
Dropout	[0, 0.1]	0	0.1
Activation	[ReLU]	ReLU	ReLU

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Table S4.16: Hyperparameter values for xLSTM (both SMILES BPE and SELFIES). These values were set manually; HPO was not performed for xLSTM due to resource constraints.

Component	Hyperparameter	Selected Value
Training		
Optimizer	—	AdamW
Batch Size	—	64
Learning Rate	—	0.0001
Weight Decay	—	0
xLSTM		
Num Layers	—	2
Num Heads	—	8
Hidden Dim	—	512
Dropout	—	0.1
Conv1d Kernel Size	—	4
Proj Factor	—	1.3
Activation	—	gelu
Bias Init	—	powerlaw_blockdependent
SLSTM At	—	[1]
MLP		
Num Layers	—	1
Hidden Dim	—	128
Dropout	—	0.1
Activation	—	ReLU

Chapter 5

Discussion

This chapter synthesizes the findings from the three studies presented in this thesis, reflects on their shared themes and methodological considerations, and identifies open questions for future research. The first section recapitulates the main conclusions of each chapter and highlights the connections between them. The second section discusses broader implications and future perspectives organized around five thematic areas that cut across the individual contributions.

5.1 Conclusions

5.1.1 Hybrid Uncertainty Quantification for Bioactivity Modeling

Chapter 2 addressed the challenge of obtaining reliable uncertainty estimates in bioactivity prediction by proposing hybrid approaches that combine Bayesian and evidential paradigms. Deep ensembles and MC dropout, which approximate Bayesian uncertainty through multiple model evaluations, were compared with evidential deep learning, which estimates aleatoric and epistemic uncertainty in a single forward pass. Two hybrid architectures were introduced: the ensemble of evidential models (EOE), which aggregates multiple evidential networks, and the evidential MC dropout model (EMC), which combines evidential learning with stochastic dropout sampling.

The results demonstrated that EOE consistently achieved the best overall performance across the Papyrus++ dataset. On both xC_{50} and K_x bioactivity endpoints and under both stratified and scaffold cluster splitting strategies, EOE obtained the

lowest RMSE while simultaneously leading on the Continuous Ranked Probability Score (CRPS) and Interval Score. These findings held even under the most challenging scaffold cluster splits, where distributional shift between training and test data was most severe. Notably, EOE accomplished this with only 10 ensemble members, requiring substantially less computation than the 100-member deep ensembles that served as the primary Bayesian baseline. The hardware-agnostic compute analysis confirmed that EOE occupies a favorable position in the performance-efficiency trade-off, making it a practical default for uncertainty-aware bioactivity modeling.

Statistical significance testing using both parametric (repeated-measures ANOVA with Tukey HSD) and non-parametric (Friedman test with Conover-Holm post-hoc) procedures confirmed that the observed advantages of EOE were not attributable to random variation across training seeds. The complementary analysis of calibration curves revealed that evidential-based models, including EOE, tended toward slight overconfidence under scaffold splits, while MC dropout was consistently underconfident. This directional information about miscalibration is practically relevant, as the consequences of overconfidence and underconfidence differ in downstream decision-making.

A broader takeaway from this chapter is that combining paradigms with complementary strengths can yield uncertainty estimates that are both more accurate and more informative than those obtained from either paradigm alone. The evidential component provides structured decomposition of aleatoric and epistemic uncertainty in a computationally efficient manner, while the ensemble component introduces the diversity needed to capture model-level epistemic uncertainty more faithfully.

5.1.2 Assay-Aware Proteochemometric Modeling

Chapter 3 introduced proteochemometric assay-aware modeling (PCAM), an extension of the proteochemometric (PCM) framework that incorporates assay descriptors as a third input modality alongside compound and protein features. The motivation for this extension was the well-documented heterogeneity of bioactivity measurements aggregated from different experimental assays in public databases such as ChEMBL. Even after careful curation, cross-assay concordance in the GPCR dataset remained modest, reflecting the systematic variability that arises from combining measurements across laboratories, protocols, and readout technologies.

Three task formulations were compared: standard PCM (compound and protein features only), PCAM-OneHot (adding categorical assay identifiers), and PCAM-

BioBERT (adding text-derived assay embeddings from assay descriptions). These were evaluated under three splitting strategies of increasing difficulty: random, scaffold-based, and assay-held-out splits. A key finding was that the split strategy dominated the overall difficulty hierarchy across all task formulations and UQ methods. Random splits yielded the lowest RMSE and highest PCC, scaffold splits were intermediate, and assay-held-out splits were consistently the most challenging. This ordering reflects the increasing degree of distribution shift imposed by each splitting strategy.

When assay identifiers were shared between training and evaluation (random and scaffold splits), PCAM-OneHot provided the most consistent gains in both point accuracy and probabilistic scoring relative to standard PCM. Under assay-held-out evaluation, where entire assay identifiers were absent from the training data, the gap between task formulations narrowed, and PCAM-BioBERT showed a small but consistent advantage in most settings. This supports a practical distinction between two operational regimes: within an established assay panel, categorical assay identifiers are sufficient and reliably beneficial; when assay context changes or new assays are introduced, semantic assay representations can support knowledge transfer from related assays.

The study also highlighted a configuration-specific instability: the combination of probabilistic neural networks (PNN) with BioBERT embeddings under the random split exhibited elevated variance and degraded performance. This observation underscores the importance of evaluating representation choices jointly with the optimization method, rather than in isolation.

An important methodological contribution of this chapter was the systematic data curation performed with CAPRICHO, which provided a transparent workflow for quality assessment and filtering. The diagnostic analysis of cross-assay concordance identified systematic issues, including unit annotation errors affecting specific ChEMBL assays, reinforcing the value of curation tools that flag potential problems rather than silently discarding data.

5.1.3 Fragment-Level Molecular Representations for Cyclic Peptides

Chapter 4 addressed the problem of molecular representation for complex therapeutic modalities by introducing Group-SELFIES Graph Embeddings (GSGE), a novel approach that constructs molecular graphs from data-driven fragment vocabularies rather than atom-level features. The method was evaluated on the prediction of cyclic

peptide membrane permeability, a property critical for the development of peptide therapeutics that target intracellular proteins.

GSGE leverages the Group-SELFIES framework to derive a fragment vocabulary from a chemical corpus, constructing compound graphs whose nodes correspond to chemically meaningful fragments. This abstraction captures local chemical environments and higher-order connectivity patterns without relying on predefined monomer lists, making the approach applicable to peptides containing non-canonical amino acids and other chemical modifications. A graph autoencoder (GAE) was additionally explored to learn dense, structurally informed fragment embeddings, though in practice one-hot fragment encodings yielded comparable predictive performance.

The fusion model combining a graph convolutional network (GCN) operating on GSGE graphs with open-source 2D/3D molecular descriptors achieved strong benchmark performance while avoiding proprietary software dependencies and heavy data augmentation through SMILES enumeration. On the held-out PAMPA test set, the 12-member GCN+7desc ensemble achieved an MAE of 0.329 and an R^2 of 0.802, compared to CycPeptMP (MAE of 0.355, R^2 of 0.772). The corresponding single-model variant was numerically close to CycPeptMP, and focused pairwise contrasts indicated a modest MAE advantage for the 4-member ensemble over CycPeptMP ($\Delta = -0.021$, 95% CI $[-0.031, -0.011]$). Tukey-adjusted contrasts supported this difference, whereas non-parametric rank tests still indicated overlapping top groups at the available threefold resolution.

A controlled comparison of architectural inductive biases provided important insights. Standard transformer models underperformed on this task, consistent with a mismatch between attention-only sequence priors and the topology-driven structure-property relationships governing membrane permeability in cyclic peptides. Recurrent models such as LSTM and xLSTM performed better than transformers in this benchmark, but this comparison should be interpreted cautiously because the sequence-model study used single-seed evaluation and non-uniform hyperparameter-optimization budgets. Cross-endpoint transfer from PAMPA to Caco-2, MDCK, and RRCK further showed endpoint-dependent ranking among model families, with numerically lower or similar MAE relative to the strongest Li *et al.* baselines depending on assay; the very small MDCK subset ($N_{\text{cmpd}} = 17$) limits certainty for that endpoint.

5.1.4 Connecting Themes

Although the three studies address distinct methodological questions, several themes connect them. First, all three chapters emphasize the importance of evaluation rigor. Each study employs multiple splitting strategies to test generalization under progressively more challenging conditions, uses non-redundant evaluation metrics selected through correlation analysis, and applies formal statistical testing to distinguish meaningful differences from noise. This commitment to rigorous benchmarking is essential in a field where small numerical improvements can be reported without adequate statistical support [60].

Second, a recurring insight is that the data context surrounding a prediction matters as much as the model architecture. In Chapter 2, the choice of splitting strategy (stratified versus scaffold) substantially affected relative model rankings and the magnitude of uncertainty estimates. In Chapter 3, the split strategy was the single largest driver of difficulty, and the benefit of assay descriptors depended critically on whether evaluation occurred within or outside observed assay contexts. In Chapter 4, the choice of molecular representation and architectural inductive bias determined whether a model could capture the topology-driven properties of macrocyclic molecules.

Third, all three chapters demonstrate the value of combining complementary information sources. EOE combines evidential and Bayesian uncertainty paradigms. PCAM combines compound, protein, and assay descriptors. GSGE combines fragment-level graph representations with whole-molecule descriptors. In each case, the integration of complementary perspectives yielded more robust and informative models than relying on a single approach.

5.2 Future Perspectives

5.2.1 Data Quality, Assay Heterogeneity, and Standardization

The findings of this thesis underscore a fundamental tension in computational drug discovery: the desire for large training datasets must be balanced against the quality and consistency of the data they contain. Chapter 3 demonstrated that cross-assay concordance in curated GPCR bioactivity data remained modest even after extensive quality filtering, and that this residual heterogeneity imposed an effective ceiling on achievable model accuracy. The inter-assay variability documented by Landrum and Riniker, where nearly half of repeated compound measurements across separate assays deviate by more than a factor of two [14], represents a floor of irreducible noise that

affects all models trained on aggregated public data.

Several directions warrant further investigation. The CAPRICHO curation tool introduced in Chapter 3 provides a transparent, flag-based workflow that makes filtering decisions explicit and auditable. Extending such tooling to additional target families beyond GPCRs and integrating it into standard data preparation pipelines could benefit the broader cheminformatics community. More fundamentally, the field would benefit from community-wide efforts to develop standardized curation protocols and quality benchmarks, analogous to the role that the Papyrus dataset [3] has played in providing a curated resource for bioactivity modeling.

The PCAM framework itself opens the door to more sophisticated approaches to handling assay heterogeneity. Rather than treating assay information solely as an additional input feature, future work could explore hierarchical or multi-task formulations that explicitly model assay-specific biases as latent variables. Such approaches could disentangle the true compound-target interaction signal from assay-specific measurement offsets, potentially improving both prediction accuracy and transferability across experimental contexts.

5.2.2 From Uncertainty Benchmarks to Real-World Deployment

Chapter 2 established EOE as a robust and computationally practical default for uncertainty quantification in bioactivity modeling. However, benchmark performance on curated datasets is a necessary but not sufficient condition for real-world utility. Several gaps remain between the current state of UQ methods and their deployment in active drug discovery programs.

One important gap is the connection between uncertainty estimates and experimental decision-making. In practice, uncertainty is valuable only insofar as it informs actionable decisions, such as which compounds to synthesize, which assays to run, or which predictions to trust. Active learning frameworks, where model uncertainty guides the selection of the next experiments, represent a natural application of the methods developed in this thesis. The calibration analysis in Chapter 2 showed that EOE exhibited slight overconfidence under scaffold splits, which would need to be addressed for reliable active learning in chemical spaces that differ structurally from the training data.

Another open question concerns the behavior of UQ methods under more extreme distribution shifts than those tested here. The scaffold cluster splits used in Chap-

ter 2 and the assay-held-out splits in Chapter 3 introduce controlled forms of novelty, but real-world drug discovery often involves predictions for target classes, chemical series, or assay technologies that are entirely absent from historical data. Evaluating UQ methods under such settings, for instance through temporal splits that simulate prospective use, would provide a more realistic assessment of their deployment readiness.

Finally, the computational cost of uncertainty estimation, while substantially reduced by the EOE approach compared to large ensembles, remains a consideration for high-throughput virtual screening campaigns where millions of compounds must be scored. Further work on amortized or distilled uncertainty estimates, where a single model is trained to approximate the uncertainty of an ensemble, could make uncertainty-aware predictions feasible at the scale required for industrial applications.

5.2.3 Molecular Representations Beyond Small Molecules

The GSGE framework introduced in Chapter 4 represents a step toward molecular representations that can handle the structural complexity of macrocyclic and peptidic molecules. However, the space of complex therapeutic modalities extends well beyond cyclic peptides. Proteolysis-targeting chimeras (PROTACs), antibody-drug conjugates (ADCs), and other bifunctional molecules present representation challenges that current methods, including GSGE, do not yet fully address.

A central limitation of most existing molecular representations is their development and validation on small-molecule drug-like chemical space. The controlled comparison in Chapter 4 revealed that standard transformers, which have been highly successful for small molecules and proteins, underperformed dramatically on cyclic peptide property prediction. This finding suggests that architectural assumptions that work well in one molecular regime may fail in another, and that the field cannot simply scale existing approaches to new modalities without carefully considering the structural priors they encode.

The fragment-level abstraction offered by GSGE provides a flexible foundation that could be extended in several directions. The data-driven vocabulary construction can, in principle, be adapted to any chemical corpus, making it applicable to non-peptidic macrocycles, natural products, or synthetic libraries with non-standard building blocks. The graph autoencoder explored in Chapter 4 for learning dense fragment embeddings showed promise but did not yet outperform simple one-hot encodings in the downstream prediction task. In addition, the current representation

does not explicitly encode bond orientation or wildcard attachment position between fragments, and the one-shot decoder can struggle with reconstruction quality for the most complex fragments. Future work could investigate whether richer fragment-graph semantics, stronger pre-training objectives, and larger corpus diversity would improve generalization to unseen chemical space.

The broader question of how to optimally represent molecules for property prediction remains open. The success of the GCN-based fusion model in Chapter 4 suggests that combining topology-aware graph representations with curated physicochemical descriptors can capture complementary aspects of molecular structure. Exploring similar fusion strategies for other complex modalities, and understanding when and why certain representations fail, represents an important direction for future research.

5.2.4 Integration of Multi-Modal Context in Predictive Models

A theme that emerges across the chapters of this thesis is the value of incorporating contextual information beyond the primary molecular structure. Chapter 3 showed that adding assay descriptors to the compound-protein feature space could improve both prediction accuracy and uncertainty calibration. Chapter 4 demonstrated that fusing graph-level representations with whole-molecule 2D/3D descriptors yielded better performance than either modality alone. These findings point toward a broader opportunity: predictive models in drug discovery could benefit from the systematic integration of diverse information sources that capture different aspects of the prediction context.

Several additional modalities could be explored. On the molecular side, 3D conformational information, quantum-chemical descriptors, and binding pose data from docking simulations each encode structural aspects that are not fully captured by 2D graph representations or fingerprints. On the biological side, gene expression profiles, pathway annotations, and protein structural features could provide richer target context than sequence-based embeddings alone. On the experimental side, the assay description embeddings explored in Chapter 3 represent only a starting point; more detailed metadata about experimental conditions, including concentrations, incubation times, cell lines, and detection technologies, could enable finer-grained modeling of assay-specific effects.

The challenge in multi-modal integration lies not only in encoding each modality effectively but also in learning meaningful interactions between them. The modular architecture used in Chapters 3 and 4, where each modality is processed by a dedi-

cated encoder before concatenation, represents a straightforward fusion strategy. More sophisticated approaches, such as cross-modal attention mechanisms or multi-modal pre-training objectives, could capture richer interactions between modalities, though they also introduce additional complexity and require careful regularization to avoid overfitting, particularly when some modalities are sparse or noisy.

5.2.5 Statistical Evaluation Practices in Computational Drug Discovery

A methodological thread running through all chapters of this thesis is the emphasis on careful statistical evaluation of model comparisons. Each study adopted formal hypothesis testing under repeated-measures designs, selected non-redundant evaluation metrics through correlation analysis, and distinguished between statistical detectability and practical relevance. These practices are not yet standard in the computational drug discovery literature, where model comparisons are frequently based on single-seed evaluations or aggregate metrics without uncertainty quantification [60].

The experience gained across the three studies highlights several practical recommendations. First, the choice of data splitting strategy can fundamentally alter the conclusions of a benchmarking study. Chapters 2 and 3 both showed that relative model rankings were not invariant to the splitting strategy, and that the most informative comparisons arose under the most challenging splits. Reporting results only under random splits, as is still common in many studies, risks overstating model performance and underestimating the difficulty of real-world prediction tasks [79].

Second, the redundancy among commonly used evaluation metrics is underappreciated. Both Chapters 2 and 3 found high correlations among standard performance and uncertainty metrics, motivating the selection of non-redundant subsets. Reporting a large number of highly correlated metrics creates an illusion of comprehensive evaluation while providing little additional information. Selecting a small set of complementary metrics and justifying the selection through correlation analysis leads to more interpretable and actionable comparisons.

Third, the distinction between statistical significance and practical significance deserves greater attention. As emphasized in recent methodological work [60], a statistically significant difference between two models may be too small to matter in practice, while a practically meaningful difference may not reach significance in underpowered experiments. The sign plots and critical difference diagrams used in Chapter 3, together with the effect-size reporting adopted throughout this thesis, provide a template

for more nuanced model comparison that could be adopted by the broader community. Ultimately, the goal of statistical evaluation should be to support informed decisions about which methods to deploy, rather than to accumulate evidence for incremental improvements that do not translate to meaningful gains in drug discovery outcomes.

Appendix A

Data and Software Availability

This thesis was developed to promote FAIR (Findable, Accessible, Interoperable, and Reusable) data management principles and open-source software development practices. Whenever possible, data were obtained from public databases and repositories, and open-source software was used. In the same spirit, datasets and code produced in the paper chapters of this thesis are made available through public repositories.

Chapter 3

Data availability

All bioactivity data used in this chapter originate from the publicly available Papyrus dataset (version 05.6, https://doi.org/10.5281/zenodo.7821775). The pipeline automatically downloads, filters, and processes this dataset to ensure reproducibility.

Software availability

The complete software used for model training, evaluation, and uncertainty quantification is available through the GitHub repository at https://github.com/CDDLeiden/uqdd.

A.0.

Chapter 4

Data availability

All data files required to reproduce the analyses reported in this chapter are available in the public GitHub repository at `https://github.com/CDDLeiden/pcam`. The repository includes (i) the curated datasets used for training and evaluation, (ii) deterministic train/validation/test split assignments for all split strategies and random seeds, and (iii) the per-run result tables used to generate all figures and statistical analyses.

Software availability

The source code used in this chapter is publicly available in the GitHub repository `https://github.com/CDDLeiden/pcam`.

Chapter 5

Data availability

All data files required to reproduce the analyses reported in this chapter are publicly available in the GitHub repository `https://github.com/CDDLeiden/GSGE-Benchmarking`. This repository contains the datasets, curated splits, and result files used throughout the chapter.

Software availability

The software developed in this study is publicly available through two complementary repositories. The GSGE package is available at `https://github.com/CDDLeiden/GSGE`. The full benchmarking workflow used for model training, comparative evaluation, and statistical analysis is available at `https://github.com/CDDLeiden/GSGE-Benchmarking`. Together, these repositories provide the code, data artifacts, and documentation required to reproduce the analyses reported in this chapter.

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Summary

This thesis explores artificial intelligence and machine learning methodologies for drug discovery, with a focus on uncertainty quantification, bioactivity modeling, and molecular representation. The research addresses key challenges in making computational predictions more reliable, context-aware, and applicable to complex therapeutic modalities. Central to this work is the development of methods that improve the trustworthiness of machine learning models for predicting ligand-protein interactions and molecular properties. The thesis bridges advances in deep learning, cheminformatics, and statistical evaluation, aiming to strengthen the foundations on which data-driven drug discovery decisions are made.

Chapter 1 provides a general introduction to the thesis. It describes the difficulties inherent in the drug discovery process, including its high cost, long timelines, and substantial attrition rates. The chapter then introduces computer-aided drug discovery and artificial intelligence as tools for improving efficiency, followed by an overview of public bioactivity databases, quantitative structure-activity relationships (QSAR), proteochemometric modeling (PCM), molecular representations, uncertainty quantification in machine learning, and G protein-coupled receptors as an important drug target family. The chapter concludes with an outline of the remaining chapters.

Chapter 2 addresses the challenge of obtaining reliable uncertainty estimates in bioactivity prediction. Two widely used paradigms for uncertainty quantification, Bayesian methods (deep ensembles, MC dropout) and evidential deep learning, are combined into novel hybrid architectures. The ensemble of evidential models (EOE) aggregates multiple evidential networks to improve epistemic uncertainty estimation, while the evidential MC dropout model (EMC) combines evidential learning with stochastic dropout sampling. Benchmarked on the Papyrus++ dataset across two bioactivity endpoints and multiple data splitting strategies, EOE consistently achieves the best overall performance in terms of prediction accuracy, probabilistic calibration,

and uncertainty informativeness. Notably, EOE accomplishes this with only 10 ensemble members, requiring a fraction of the computational cost of 100-member deep ensembles. Statistical significance testing confirms these advantages, and a hardware-agnostic compute analysis highlights the favorable performance-efficiency trade-off. These results demonstrate that combining Bayesian and evidential principles yields more accurate and informative uncertainties for bioactivity modeling, with EOE offering a robust and computationally practical default for uncertainty-aware decision-making in drug discovery.

Chapter 3 introduces proteochemometric assay-aware modeling (PCAM), an extension of the proteochemometric framework that incorporates assay descriptors alongside compound and protein features to address the heterogeneity of bioactivity measurements in public databases. Using a curated ChEMBL-derived benchmark spanning 167 G protein-coupled receptor targets, 77,081 compounds, and 5,161 assays, three task formulations are compared: standard PCM, PCAM with categorical assay identifiers (PCAM-OneHot), and PCAM with text-derived assay embeddings (PCAM-BioBERT). Evaluation under three splitting strategies of increasing difficulty (random, scaffold-based, and assay-held-out) reveals that the split strategy dominates overall difficulty, with the assay-held-out regime inducing the strongest distribution shift. When assay identifiers are shared between training and evaluation, categorical assay descriptors provide the most consistent gains. Under assay-held-out evaluation, text-derived embeddings show a small but consistent advantage, supporting a practical distinction between within-panel and assay-transfer prediction scenarios. The study also contributes a systematic data curation workflow and highlights the importance of evaluating representation choices jointly with the optimization method.

Chapter 4 introduces Group-SELFIES Graph Embeddings (GSGE), a molecular representation framework that constructs compound graphs from data-driven fragment vocabularies. Applied to cyclic peptide membrane permeability prediction, GSGE avoids predefined monomer vocabularies and supports chemically diverse scaffolds. On the CycPeptMPDB PAMPA benchmark (6,889 cyclic peptides), a 12-member GCN+descriptor ensemble reaches an MAE of 0.329 and an R^2 of 0.802 without proprietary descriptor software or SMILES enumeration. Focused pairwise contrasts indicate a modest MAE advantage over CycPeptMP ($\Delta = -0.021$, 95% CI $[-0.031, -0.011]$), although non-parametric rank tests still show overlapping top groups with only three CV splits. Sequence-model ablations show recurrent models outperforming standard transformers in this setting, but this comparison is limited by single-seed evaluation and non-uniform hyperparameter-optimization budgets. Cross-endpoint transfer to

Caco-2, MDCK, and RRCK shows endpoint-dependent ranking with generally competitive MAE relative to the strongest published baselines.

Chapter 5 synthesizes the findings from the preceding chapters and reflects on their shared themes. The discussion highlights three connecting threads: the importance of rigorous statistical evaluation across multiple splitting strategies, the critical role of data context in shaping prediction difficulty and model performance, and the value of combining complementary information sources. Future perspectives are organized around five thematic areas: data quality and assay standardization, the transition of uncertainty quantification methods from benchmarks to real-world deployment, the extension of molecular representations to complex therapeutic modalities beyond small molecules, the integration of multi-modal contextual information in predictive models, and the adoption of sound statistical evaluation practices in computational drug discovery.

After Chapter 5, the thesis closes with multilingual summaries that recapitulate the main contributions and outline future research directions.

Together, these chapters present a cohesive exploration of how machine learning can be made more reliable, context-aware, and broadly applicable in drug discovery, from foundational improvements in uncertainty quantification and data curation to novel representations for emerging therapeutic modalities.

Samenvatting

Dit proefschrift onderzoekt methoden op het gebied van kunstmatige intelligentie en *machine learning* voor geneesmiddelenontwikkeling, met een focus op onzekerheidskwantificering, bioactiviteitsmodellering en moleculaire representatie. Het onderzoek richt zich op belangrijke uitdagingen bij het betrouwbaarder, contextbewuster en breder toepasbaar maken van computationele voorspellingen. Centraal staat de ontwikkeling van methoden die de betrouwbaarheid van *machine learning*-modellen voor het voorspellen van ligand-eiwit-interacties en moleculaire eigenschappen verbeteren. Dit proefschrift combineert vooruitgang in *deep learning*, cheminformatica en statistische evaluatie, met als doel de fundamenten te versterken waarop datagedreven beslissingen in de geneesmiddelenontwikkeling worden genomen.

Hoofdstuk 1 biedt een algemene inleiding op het proefschrift. Het beschrijft de inherente moeilijkheden van het geneesmiddelenontwikkelingsproces, waaronder de hoge kosten, lange doorlooptijden en aanzienlijke uitvalpercentages. Vervolgens worden computerondersteunde geneesmiddelenontwikkeling en kunstmatige intelligentie geïntroduceerd als hulpmiddelen om de efficiëntie te verbeteren, gevolgd door een overzicht van openbare bioactiviteitsdatabases, kwantitatieve structuur-activiteitsrelaties (QSAR), proteochemometrische modellering (PCM), moleculaire representaties, onzekerheidskwantificering in *machine learning* en G-eiwit-gekoppelde receptoren als een belangrijke geneesmiddeldoelwitfamilie. Het hoofdstuk sluit af met een overzicht van de overige hoofdstukken.

Hoofdstuk 2 behandelt de uitdaging om betrouwbare onzekerheidsschattingen te verkrijgen bij bioactiviteitsvoorspelling. Twee veelgebruikte paradigma's voor onzekerheidskwantificering, Bayesiaanse methoden (*deep ensembles*, *MC-dropout*) en evidentieel *deep learning*, worden gecombineerd tot nieuwe hybride architecturen. Het ensemble van evidentiële modellen (EOE) combineert meerdere evidentiële netwerken om de schatting van epistemische onzekerheid te verbeteren, terwijl het evidentiële *MC-*

dropout-model (EMC) evidentieel leren combineert met stochastische *dropout-sampling*. Geëvalueerd op de Papyrus++-*dataset* over twee bioactiviteitseindpunten en meerdere datasplitsingsstrategieën, behaalt EOE consistent de beste algehele prestaties op het gebied van voorspellingsnauwkeurigheid, probabilistische kalibratie en informativiteit van onzekerheidsschattingen. Opmerkelijk is dat EOE dit bereikt met slechts 10 ensemble-leden, wat slechts een fractie van de rekenkosten vergt van 100-koppige *deep ensembles*. Statistische significantietesten bevestigen deze voordelen, en een *hardware*-onafhankelijke rekenanalyse onderstreept de gunstige verhouding tussen prestatie en efficiëntie. Deze resultaten tonen aan dat het combineren van Bayesiaanse en evidentiële principes nauwkeurigere en informatievere onzekerheidsschattingen oplevert voor bioactiviteitsmodellering, waarbij EOE een robuuste en rekenkundig praktische standaardmethode biedt voor onzekerheidsbewuste besluitvorming in de geneesmiddelenontwikkeling.

Hoofdstuk 3 introduceert proteochemometrische *assay*-bewuste modellering (PCAM), een uitbreiding van het proteochemometrische raamwerk die *assay*-beschrijvingen integreert naast verbindings- en eiwitkenmerken om de heterogeniteit van bioactiviteitsmetingen in openbare databases aan te pakken. Met behulp van een gecureerde ChEMBL-afgeleide *benchmark* die 167 G-eiwit-gekoppelde receptordoelwitten, 77.081 verbindingen en 5.161 *assays* omvat, worden drie taakformuleringen vergeleken: standaard PCM, PCAM met categorische *assay*-beschrijvingen (PCAM-OneHot) en PCAM met op tekst gebaseerde *assay-embeddings* (PCAM-BioBERT). Evaluatie onder drie splitsingsstrategieën van toenemende moeilijkheid (willekeurig, op *scaffold* gebaseerd en *assay*-uitgesloten) toont aan dat de splitsingsstrategie de algehele moeilijkheidsgraad domineert, waarbij het *assay*-uitgesloten regime de sterkste distributieverschuiving veroorzaakt. Wanneer dezelfde *assays* voorkomen in training en evaluatie, leveren categorische *assay*-beschrijvingen de meest consistente verbeteringen. Bij *assay*-uitgesloten evaluatie vertonen op tekst gebaseerde *embeddings* een klein maar consistent voordeel, wat een praktisch onderscheid ondersteunt tussen voorspellingsscenario's binnen een bestaand *assaypanel* en *assay*-transferscenario's. De studie draagt ook bij aan een systematische *workflow* voor datacuratie en benadrukt het belang van het gezamenlijk evalueren van representatiekeuzes en optimalisatiemethoden.

Hoofdstuk 4 introduceert *Group-SELFIES Graph Embeddings* (GSGE), een raamwerk voor moleculaire representatie dat verbindingsgrafen construeert op basis van datagedreven fragmentvocabulaires. Toegepast op de voorspelling van membraanpermeabiliteit van cyclische peptiden vermijdt GSGE voorgedefinieerde monomeervocabulaires en ondersteunt het chemisch diverse structuren. Op de CycPeptMPDB-PAMPA-

benchmark (6.889 cyclische peptiden) behaalt een 12-ledig GCN+*descriptor*-ensemble een MAE van 0.329 en een R^2 van 0.802 zonder propriëtaire *descriptorsoftware* of *SMILES*-enumeratie. Gerichte paarsgewijze contrasten wijzen op een bescheiden MAE-voordeel ten opzichte van CycPeptMP ($\Delta = -0.021$, 95% BI $[-0.031, -0.011]$), hoewel niet-parametrische rangtoetsen bij slechts drie CV-splits nog overlappende topgroepen laten zien. Ablaties met sequentiemodellen tonen dat recurrente modellen in deze setting beter presteren dan standaard *transformers*, maar deze vergelijking blijft beperkt door evaluatie met één *seed* en niet-uniforme budgetten voor hyperparameteroptimalisatie. *Transfer* naar Caco-2, MDCK en RRCK laat endpoint-afhankelijke rangordes zien met overwegend concurrerende MAE-waarden ten opzichte van de sterkste gepubliceerde *baselines*.

Hoofdstuk 5 combineert de bevindingen uit de voorgaande hoofdstukken en reflecteert op hun gedeelde thema's. De discussie belicht drie verbindende draden: het belang van rigoureuze statistische evaluatie over meerdere splitsingsstrategieën, de cruciale rol van datacontext bij het bepalen van de voorspellingsmoeilijkheid en modelprestaties, en de waarde van het combineren van complementaire informatiebronnen. Toekomstperspectieven zijn georganiseerd rond vijf thematische gebieden: datakwaliteit en *assay*standardisatie, de overgang van onzekerheidskwantificeringsmethoden van *benchmarks* naar inzet in de praktijk, de uitbreiding van moleculaire representaties naar complexe therapeutische modaliteiten voorbij kleine moleculen, de integratie van multimodale contextuele informatie in voorspellende modellen, en de adoptie van gedegen statistische evaluatiepraktijken in computationele geneesmiddelenontwikkeling.

Na Hoofdstuk 5 wordt het proefschrift afgesloten met meertalige samenvattingen die de belangrijkste bijdragen samenvatten en toekomstige onderzoeksrichtingen schetsen.

Samen vormen deze hoofdstukken een samenhangend onderzoek naar hoe *machine learning* betrouwbaarder, contextbewuster en breder toepasbaar kan worden gemaakt in de geneesmiddelenontwikkeling, van fundamentele verbeteringen in onzekerheidskwantificering en datacuratie tot nieuwe representaties voor opkomende therapeutische modaliteiten.

ملخص

يستكشف هذا البحث منهجيات الذكاء الاصطناعي والتعلم الآلي في مجال اكتشاف الأدوية، مع التركيز على قياس عدم اليقين، ونمذجة النشاط الحيوي، والتمثيل الجزيئي. ويعالج البحث تحديات جوهرية في جعل التنبؤات الحاسوبية أكثر موثوقية ووعياً بالسياق وأوسع نطاقاً في التطبيق. ويتمحور هذا العمل حول تطوير أساليب تُعزز مصداقية نماذج التعلم الآلي في التنبؤ بتفاعلات الروابط مع البروتينات وبالخصائص الجزيئية. ويجمع البحث بين التقدم في التعلم العميق، والمعلوماتية الكيميائية، والتقييم الإحصائي، بهدف تعزيز الأسس التي تُبنى عليها القرارات المعتمدة على البيانات في اكتشاف الأدوية.

الفصل 1 يقدم مقدمة عامة للبحث. ويصف الصعوبات الكامنة في عملية اكتشاف الأدوية، بما في ذلك تكلفتها العالية وجداولها الزمنية الطويلة ومعدلات الفشل المرتفعة فيها. ثم يقدم الفصل اكتشاف الأدوية بمساعدة الحاسوب والذكاء الاصطناعي كأدوات لتحسين الكفاءة، يليه استعراض لقواعد بيانات النشاط الحيوي المتاحة، والعلاقات الكمية بين البنية والنشاط (QSAR)، والنمذجة البروتيوكيميومترية (PCM)، والتمثيلات الجزيئية، وقياس عدم اليقين في التعلم الآلي، والمستقبلات المقترنة بالبروتين جي (GPCR) كعائلة مهمة من الأهداف الدوائية. ويختتم الفصل بعرض موجز للفصول المتبقية.

الفصل 2 يعالج تحدي الحصول على تقديرات موثوقة لعدم اليقين في التنبؤ بالنشاط الحيوي. ويجمع بين نموذجين مستخدمين على نطاق واسع لقياس عدم اليقين، هما الأساليب البايزية (مجموعات التعلم العميق، وإسقاط مونت كارلو) والتعلم العميق الاستدلالي، في نماذج هجينة مبتكرة. فمجموعة النماذج الاستدلالية (EOE) تجمع عدة شبكات استدلالية لتحسين تقدير عدم اليقين المعرفي، بينما يجمع نموذج الإسقاط الاستدلالي بطريقة مونت كارلو (EMC) بين التعلم الاستدلالي وأخذ عينات الإسقاط العشوائي. وبتقييمهما على مجموعة بيانات Papyrus++ مقسمة بحسب طريقتين لقياس قيمة النشاط الحيوي واستراتيجيات متعددة لتقسيم البيانات سواء لتدريب أو لتقدير أداء النماذج، يحقق EOE باستمرار أفضل أداء عام من حيث دقة التنبؤ والمعايرة الاحتمالية ومعلوماتية تقديرات عدم اليقين. والجدير بالذكر أن EOE يحقق ذلك بعشرة أعضاء (نماذج) فقط في المجموعة، مما يتطلب جزءاً يسيراً من التكلفة الحسابية لمجموعات التعلم العميق المكونة من مائة عضو. وتؤكد اختبارات الدلالة الإحصائية هذه المزايا، ويبرز تحليل حاسوبي التوازن المواتي بين الأداء والكفاءة (سرعة التعلم الآلي) بغض النظر عن الجهاز المستخدم في عملية التدريب الآلي. وتثبت هذه النتائج أن الجمع بين المبادئ البايزية والاستدلالية ينتج تقديرات أدق وأغنى بالمعلومات لعدم اليقين في نمذجة النشاط الحيوي، مع كون EOE خياراً افتراضياً متيناً وعملياً من الناحية الحاسوبية لاتخاذ القرارات الواعية بعدم اليقين في اكتشاف الأدوية.

الفصل 3 يقدم النمذجة البروتيوكيميومترية الواعية بالفحوصات التجريبية (PCAM)، وهي امتداد للإطار البروتيوكيميومتري

ملخص

يدمج وصفاً للفحوصات التجريبية إلى جانب خصائص المركبات والبروتينات لمعالجة عدم تجانس قياسات النشاط الحيوي في قواعد البيانات المتاحة. وباستخدام معيار مرجعي منظَّم مشتق من قاعدة ChEMBL يشمل 167 هدفاً من المستقبلات المقترنة بالبروتين جي، و77,081 مركباً، و5,161 فحصاً، تمت مقارنة ثلاث صيغ للمهمة: النمذجة البروتيوكيميومترية القياسية، وPCAM مع معرفات لفحوصات تصنيفية (PCAM-OneHot)، وPCAM مع تضمينات للفحوصات مشتقة من النصوص الواصفة للتجارب (PCAM-BioBERT). ويكشف التقييم تحت ثلاث استراتيجيات لتقسيم البيانات متزايدة الصعوبة (عشوائية، وقائمة على الهياكل الأساسية للمركبات، ومع حجب الفحوصات لقياس قدرة النموذج على التعلم من تجربة و التطبيق على تجربة جديدة لم يرها من قبل) أن استراتيجية التقسيم تهيمن على الصعوبة الإجمالية، حيث يُحدث نظام حجب الفحوصات أكبر إزاحة في التوزيع أو بمعنى آخر أصعب في تخمين النتيجة بدون تعلم حقيقي للنماذج. وعند مشاركة معرفات الفحوصات بين التدريب والتقييم، توفر واصفات الفحوصات التصنيفية أكثر المكاسب اتساقاً. وتحت تقييم حجب الفحوصات، تظهر التضمينات المشتقة من النصوص ميزة صغيرة لكنها متسقة ومتكررة، مما يدعم التمييز عملياً بين حالتين: التنبؤ ضمن مجموعة من الفحوصات التي يعرفها النموذج مسبقاً، والتنبؤ بنتائج لفحوصات جديدة لم يتدرب عليها. وتسهم الدراسة أيضاً بسير عمل منهجي لتنظيم البيانات، وتبرز أهمية تقييم خيارات التمثيل بالاقتران مع طريقة التحسين.

الفصل 4 يقدم تضمينات الرسوم البيانية لمجموعات سيلفيز (GSGE)، وهو إطار للتمثيل الجزيئي يبني رسوماً بيانية للمركبات (وخاصةً الببتيدات) من مفردات أجزاء مشتقة من البيانات. وعند تطبيقه على التنبؤ بنفاذية الغشاء الخلوي للببتيدات الحلقية، يتجنب GSGE مفردات المونومرات المحددة مسبقاً ويدعم هياكل كيميائية متنوعة. وعلى معيار CycPeptMPDB PAMPA المرجعي (6,889 ببتيداً حلقياً)، يبلغ تجميعٌ مكوَّن من 12 عضواً من نوع GCN+descriptor متوسط خطأ مطلق (MAE) قدره 0.329 ومعامل تحديد R^2 قدره 0.802 دون الاعتماد على برمجيات واصفات مملوكة أو تعديد SMILES. وتشير مقارنات زوجية مركزة إلى أفضلية متواضعة في MAE مقارنةً بـCycPeptMP ($\Delta = -0.021$)، وفاصل ثقة 95% مقداره $[-0.031, -0.011]$)، وإن كانت اختبارات الرتب اللامعلمية لا تزال تُظهر تداخلاً بين النماذج الأفضل أداءً، نظراً لاقتصار التقييم على ثلاثة تقسيمات فقط للتحقق المتقاطع. وتُظهر تجارب الإزالة على النماذج التسلسلية (Sequential models) تفوق النماذج التكرارية (Recurrent models) على المحولات القياسية (Transformers) في هذا الإعداد، غير أن هذه المقارنة تظل محدودة لأنها اعتمدت على تشغيل واحد فقط لكل نموذج (بذرة عشوائية واحدة)، ولأن الموارد المخصصة لضبط المعاملات الفائقة لم تكن متساوية بين النماذج. وعند تجربة استخدام النماذج المدربة على PAMPA في تنبؤ نقاط نهاية مختلفة مثل Caco-2 وMDCK وRRCK، تُظهر النتائج ترتيباً يعتمد على نقطة النهاية مع قيم MAE تنافسية إجمالاً مقارنةً بأقوى النماذج المنشورة.

الفصل 5 يجمع النتائج من الفصول السابقة ويتأمل في موضوعاتها المشتركة. وتبرز المناقشة ثلاثة خيوط رابطة: أهمية التقييم الإحصائي الدقيق عبر استراتيجيات تقسيم متعددة، والدور الحاسم لسياق البيانات في تشكيل صعوبة التنبؤ وأداء النماذج، وقيمة الجمع بين مصادر المعلومات المتكاملة. وتُنظَّم الآفاق المستقبلية حول خمسة مجالات موضوعية: جودة البيانات وتوحيد الفحوصات، وانتقال أساليب قياس عدم اليقين من المعايير المرجعية إلى التطبيق في الواقع، وتوسيع التمثيلات الجزيئية لتشمل الأساليب العلاجية المعقدة إلى ما يتجاوز المركبات الصغيرة، ودمج المعلومات السياقية متعددة الأنماط في النماذج التنبؤية، وتبني ممارسات تقييم إحصائي سليمة في اكتشاف الأدوية الحاسوبي.

ملخص

بعد الفصل 5 يختتم البحث بملخصات متعددة اللغات تلخص المساهمات الرئيسية وتستشرف اتجاهات البحث المستقبلية.

وتقدم هذه الفصول مجتمعة استكشافاً متماسكاً لكيفية جعل التعلم الآلي أكثر موثوقية ووعياً بالسياق وقابلية للتطبيق على نطاق أوسع في اكتشاف الأدوية، بدءاً من التحسينات الأساسية في قياس عدم اليقين وتنظيم البيانات وصولاً إلى تمثيلات مبتكرة للأساليب العلاجية الناشئة.

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First and foremost, I express my sincere gratitude to my supervisors, Prof.dr. Gerard J. P. van Westen, Prof.dr. Herman van Vlijmen, and Dr. Natalia Dyubankova, for their guidance, critical feedback, and steadfast support throughout this research. Their complementary expertise—bridging academia and industry—and their intellectual rigour have profoundly shaped both this work and my development as a researcher.

Having been based primarily at Johnson & Johnson Innovative Medicine, I am deeply grateful to my colleagues and teams there for their immense support, scientific input, and day-to-day encouragement. The collaborative and inspiring environment they fostered was instrumental to the completion of this dissertation. In equal measure, I thank my colleagues at the Leiden Academic Centre for Drug Research (LACDR), and in particular the Computational Drug Discovery and Medicinal Chemistry groups, whose feedback during meetings, symposia, and informal discussions consistently strengthened this work.

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Finally, in accordance with scholarly tradition, I extend brief but heartfelt thanks to my family and friends for their unwavering encouragement throughout this academic journey. A more thorough and personal expression of my gratitude is included in the insert accompanying this dissertation.

Curriculum Vitae

Bola Khalil is a computational drug discovery scientist whose path brings together pharmaceutical sciences, medicinal chemistry, and artificial intelligence. From the start, he was drawn to the question of how molecular insight can become therapeutic action, and his career has steadily moved toward building practical computational methods that help teams make better decisions in early discovery.

He completed his B.Sc. in Pharmaceutical Sciences at Ain-Shams University in Cairo (2009–2014). In parallel, he taught pharmaceutical, physical, and organic chemistry at Ain-Shams University and Cairo University until 2018, supporting more than 200 students per cohort. Those years established a core strength that still defines his work: translating complex science into clear and useful guidance.

In 2018, he broadened his scientific training at the University of Tokyo, where he studied enzymatic catalysis in bacterial natural product biosynthesis through experimental biochemical and NMR-based work. He then moved to Vienna, where his M.Sc. in Drug Discovery and Development (University of Vienna, 2019–2021) marked his transition into cheminformatics and machine learning for toxicology, including work in the IMI2 NeuroDeRisk project.

To deepen this transition, he completed a second M.Sc. in Artificial Intelligence at Johannes Kepler University (2020–2024), focusing on probabilistic modeling, calibration, and machine learning in life sciences. This combination of domain and technical training prepared him for research at the interface of model development and uncertainty-aware decision-making.

Since November 2021, Bola has been a Marie-Curie Ph.D. researcher at the Computational Drug Discovery group at Leiden University in close collaboration with the *In silico* Discovery team at Johnson & Johnson R&D in Beerse, where he also worked as a computational chemist. His work centers on uncertainty quantification, proteochemometric bioactivity modeling, peptide discovery, and deployable computational work-

flows. In addition to the fundamental reserch done for this thesis, he has supported portfolio drug discovery projects through scaffold hopping, triaging, and generative design, including prioritization of optimized peptides for alternative lead-series exploration.

A recurring theme in his Ph.D. is collaborative impact. He led a cross-institutional benchmarking initiative for calibrated uncertainty-aware bioactivity modeling and, during a visiting period at Leiden University, developed peptide permeability models that combined transformer- and graph-based representations while supervising an affiliated master's thesis to completion.

During a research secondment at Uppsala University, he worked with biologists and bioinformaticians on polycystic kidney disease. By integrating multi-omics evidence into a biomedical knowledge graph, he helped prioritize approved compounds for repositioning and experimental follow-up. This work reflects his broader research vision: connecting data, models, and biology to accelerate compound prioritization in real discovery programs.

Alongside research, he contributes to scientific leadership and community building. He served on the J&J PhD & Postdoc Global Scholars Steering Committee, coordinated collaborations with Leiden University and Johannes Kepler University, and mentored students across projects. His contributions have been recognized with multiple Best Poster awards (EuroQSAR 2024, CMCS 2024, Broad Institute MLDD Symposium 2024), J&J Inspire Awards, and a Horizon 2020 Marie Skłodowska-Curie Doctoral Fellowship through the EU DRUGtrain consortium.

List of Publications

The thesis author's name is underlined; * indicates shared first authorship.

Publications included in this thesis

- **Combining Bayesian and Evidential Uncertainty Quantification for Improved Bioactivity Modeling**. *Bola Khalil, Kajetan Schweighofer, Natalia Dyubankova, Gerard J. P. van Westen, Herman van Vlijmen.* DOI: 10.1021/acs.jcim.5c01597.
- (Submitted) **PCAM: Proteochemometric Assay-aware Modeling for Robust Bioactivity Estimation and Uncertainty Quantification**. *Bola Khalil*, David Alencar Araripe*, Linde Schoenmaker*, Natalia Dyubankova, Herman van Vlijmen, Gerard J. P. van Westen.*
- (Submitted) **GSGE: Group-SELFIES Graph Embeddings for Cyclic Peptide Membrane Permeability**. *Bola Khalil*, Jasper Durinck*, Steven Combs, Anne Mai Wasserman, Natalia Dyubankova, Herman van Vlijmen, Gerard J. P. van Westen.*

Other Publications

- **A Patient-Derived 3D Cyst Model of Polycystic Kidney Disease That Mimics Disease Development and Responds to Repurposing Candidates**. *Alina Meyer, Bola Khalil, Margarita Iljin, Hester Bange, Leo S. Price, Natalia Dyubankova, Gerard J. P. van Westen, Herman van Vlijmen, Dorien J. M. Peters, Per Artursson.* DOI: 10.1111/cts.70214.
- **COFOLDER: Transparent Co-Folding for Ligand Screening and Design**. *Remco L. van den Broek*, Bola Khalil*, Gary Tresadern, Gerard J. P. van Westen, José Carlos Gómez-Tamayo, Willem Jespers.*

List of Publications

Glossary

active learning
: Learning strategy where a model selects informative new experiments or labels based on uncertainty or utility criteria

aleatoric uncertainty
: Irreducible uncertainty arising from inherent noise in data or measurements

assay
: An experimental protocol used to measure a biological or biochemical response

assay descriptor
: Feature representation of assay context, such as categorical identifiers or text-derived embeddings

assay heterogeneity
: Systematic variability between measurements caused by differences in assay protocols, laboratories, or readout technologies

assay-aware modeling
: Modeling approach that explicitly includes assay context so models can learn assay-dependent shifts

assay-held-out split
: Partitioning strategy where assay identifiers in evaluation are absent from training, testing assay-transfer generalization

bioactivity
: Measured interaction strength between a compound and a biological target, such as potency or affinity

chameleonic switching
: Conformation-dependent adaptation of a molecule between polar solvent-exposed and membrane-shielded states

chemical scaffold
: Core structural backbone used to group compounds by shared chemotype

cross-assay concordance
: Agreement between measurements of the same compound-target pair across different assays

GLOSSARY

cyclic peptide
: Peptide whose backbone forms a closed ring structure, often with distinct permeability and conformational properties

data curation
: Process of standardizing, filtering, and quality-controlling data before model development

distribution shift
: A mismatch between training and evaluation data distributions that can reduce model reliability

drug discovery
: The multi-stage process of identifying and optimizing compounds into potential medicines

endpoint transfer
: Evaluation setting where a model trained on one assay endpoint is tested on different endpoints without endpoint-specific retraining

epistemic uncertainty
: Uncertainty caused by limited knowledge or insufficiently representative training data

extrapolation
: Prediction beyond the observed training distribution, often with increased uncertainty

fragment vocabulary
: Data-driven set of molecular fragments used as building blocks for fragment-level molecular graph construction

Group-SELFIES
: Fragment-aware extension of SELFIES that represents molecules as chemically meaningful groups while preserving valid molecular syntax

inductive bias
: Architectural assumptions that influence what patterns a model can learn efficiently

interpolation
: Prediction in regions of feature space that are represented by training data

macrocycle
: Molecule containing a large ring system, commonly associated with complex topology and conformational behavior

membrane permeability
: Ability of a molecule to cross a biological or artificial membrane, often quantified through apparent permeability coefficients

molecular descriptor
: Numerical feature capturing structural, physicochemical, or topological properties of molecules

molecular fingerprint
: Vectorized encoding of molecular substructures used in cheminformatics modeling

molecular representation
: Encoding of molecular structure used as model input, such as fingerprints, strings, or graphs

monomer vocabulary

Predefined list of building blocks used to decompose molecules into sequence units in monomer-centric models

pChEMBL scale

Negative logarithmic activity scale used to harmonize endpoints such as IC_{50}, EC_{50}, K_i, and K_d

PhD thesis

The collection of scientific work of an individual after completing their Master's degree to apply for a doctorate

protein embedding

Numerical representation of a protein sequence learned by a language model

proteochemometric modeling

Modeling framework that jointly uses compound and protein descriptors to predict bioactivity across multiple targets

random split

Data partitioning strategy that randomly assigns samples to training and evaluation sets

real space / the space of real numbers / continuous space / $\mathbb{R}^n$

The space of real vectors of dimension n

scaffold split

Data partitioning strategy that separates compounds by scaffold to test chemical novelty generalization

topology-aware modeling

Modeling strategy that explicitly encodes graph connectivity and neighborhood structure when learning structure-property relationships

uncertainty calibration

The degree to which predicted confidence or uncertainty aligns with observed outcomes

virtual screening

Computational prioritization of compounds for experimental testing

Acronyms

ADC
Antibody-Drug Conjugate

ADMET
Absorption, Distribution, Metabolism, Excretion, and Toxicity

ANOVA
Analysis of Variance

AUC
Area Under the Curve

BioBERT
Biomedical Bidirectional Encoder Representations from Transformers

BODE
Bayesian Optimized Deep Ensembles

BOHB
Bayesian Optimization and Hyperband

BPE
Byte Pair Encoding

Caco-2
Caco-2 cell permeability assay

CADD
Computer-Aided Drug Discovery

CAPRICHO
ChEMBL Aggregation Package with Robust Inspection and Curation Handling Options

CD
Critical Difference

CI
Confidence Interval

CRPS
Continuous Ranked Probability Score

CV
Cross-Validation

ACRONYMS

DL
Deep Learning

ECFP
Extended Connectivity Fingerprints

ECFP4
Extended Connectivity Fingerprints (radius 2)

EMA
European Medicines Agency

EMC
Evidential Monte Carlo dropout

EOE
Ensemble of Evidential models

ESM
Evolutionary Scale Modeling

FDA
U.S. Food and Drug Administration

GAE
Graph Autoencoder

GCN
Graph Convolutional Network

GNN
Graph Neural Network

GPCR
G Protein-Coupled Receptor

GSGE
Group-SELFIES Graph Embeddings

HPO
Hyperparameter Optimization

HSD
Honest Significant Difference

LIACS
Leiden Institute of Advanced Computer Science

LSTM
Long Short-Term Memory

MAE
Mean Absolute Error

MARPD
Mean Absolute Relative Percent Difference

MC
Monte Carlo

MCA
Miscalibration Area

MDAE
Median Absolute Error

MDCK
Madin-Darby Canine Kidney assay

ML
Machine Learning

MLP
Multilayer Perceptron

MOE
Molecular Operating Environment

MSE
Mean Squared Error

NIG
Normal Inverse-Gamma

NLL
Negative Log-Likelihood

OOD
Out-of-Distribution

PAMPA
Parallel Artificial Membrane Permeability Assay

Papp
Apparent Permeability Coefficient

PCAM
Proteochemometric Assay-Aware Modeling

PCC
Pearson Correlation Coefficient

PCM
Proteochemometric Modeling

PNN
Probabilistic Neural Network

PROTAC
Proteolysis-Targeting Chimera

QSAR
Quantitative Structure-Activity Relationship

R&D
Research and Development

RMSE
Root Mean Squared Error

ACRONYMS

RNN
Recurrent Neural Network

RRC
RMSE-Rejection Curve

RRC-AUC
RMSE-Rejection Curve Area Under the Curve

RRCK
RRCK cell permeability assay

SELFIES
Self-Referencing Embedded Strings

SGD
Stochastic Gradient Descent

SMARTS
SMILES Arbitrary Target Specification

SMILES
Simplified Molecular Input Line Entry System

SPE
SMILES Pair Encoding

UQ
Uncertainty Quantification

Symbols

$\mathrm{CI}_{95\%}$ Two-sided 95% confidence interval

D Dataset

Δ Difference between two model performance values (for example, MAE)

e Collected list of RMSE values in RRC

e_0 Initial RMSE to normalize upon in RRC

e_i Iteration step RMSE in RRC

EC_{50} The concentration of a drug that gives half-maximal response in vitro

$\gamma, \upsilon, \alpha, \beta$ Evidential model estimated parameters

$\bar{\gamma}, \bar{\upsilon}, \bar{\alpha}, \bar{\beta}$ Means of Evidential model estimated parameters

$f(\cdot)$ function

$F(x)$ Cumulative distribution of the estimated distribution $p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{\theta})$

IC_{50} The concentration of a drug that is required for 50% inhibition in vitro

$\mathbb{1}$ Indicator function

K_d The equilibrium dissociation constant, a measure of the affinity of a ligand for a protein

K_i The equilibrium dissociation constant for an inhibitor binding to an enzyme

K_x A generalized term for equilibrium constants

λ Trade-off parameter between the uncertainty inflation and the model fit

$\mathcal{L}_i^{\mathbf{NLL}}(\theta)$ NLL loss term of the evidential regression loss

$\log_{10}\left(P_{\mathbf{app}}/[\mathbf{cm\,s}^{-1}]\right)$ Modeled permeability target used for cyclic peptide prediction

$\mathcal{L}$ Loss output

$\mathcal{L}_i(\theta)$ Total evidential regression loss

$\mathcal{L}_i^{\mathbf{R}}(\theta)$ Evidence on errors loss term of the evidential regression loss

M Number of compared models in statistical testing

$\mu(\boldsymbol{x}; \boldsymbol{\theta}_i)$ estimate for the mean per network i with parameters $\boldsymbol{\theta}_i$

$N_{\mathbf{cmpd}}$ Number of compounds evaluated for a given endpoint

$n_{\mathbf{rep}}$ Number of replicate aggregates used for summary statistics

$\boldsymbol{p_i}$ Observed probabilities for the i-th bin

$P_{\mathbf{app}}$ Apparent membrane permeability coefficient measured in permeability assays

SYMBOLS

$\hat{\boldsymbol{p}}_{\boldsymbol{i}}$ Estimated probabilities for the i-th bin

r Rejection rates

R^2 Coefficient of Determination

ρ_{rank} Spearman's rank correlation coefficient between estimated uncertainties and observed absolute errors

σ standard deviation across model replicates

$\sigma^2(\boldsymbol{x}; \boldsymbol{\theta}_i)$ estimate for the variance per network i with parameters $\boldsymbol{\theta}_i$

π half of τ

$\boldsymbol{\theta}$ Network parameters

u Uncertainty column

xC_{50} A generalized term for half-maximal concentrations, where 'x' can be substituted with 'I' for inhibition or 'E' for effect

y True (observed) values

$\bar{y}$ Mean of true (observed) values

$\hat{y}$ Estimated (mean) value

$\hat{y}_i$ Estimated (mean) value

$\bar{\hat{y}}$ Mean of estimated values

y_i True (observed) values

ABOUT THIS THESIS

AI-driven drug discovery promises to accelerate how new medicines are found, but confidence in its predictions depends on how uncertainty, experimental context, and molecular structure are represented.

This thesis connects three often separated challenges: quantifying model uncertainty, interpreting assays in context, and designing structural representations that preserve chemically meaningful information. Together, they point toward computational methods that are not only more capable, but also more reliable and interpretable.

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